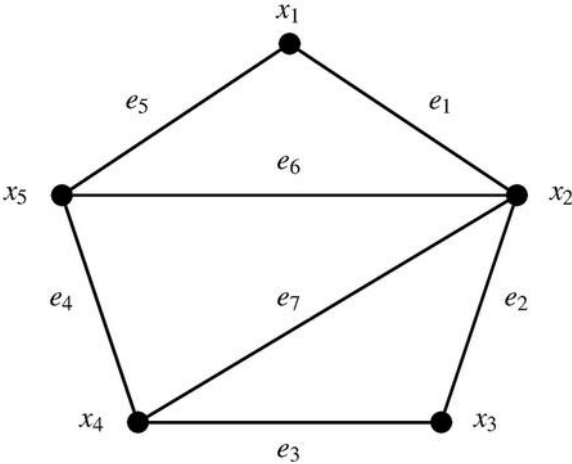


voloshin-hypergraph: formula evidence

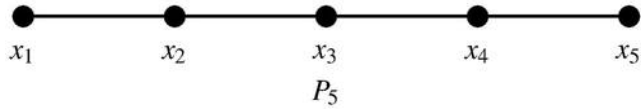
The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

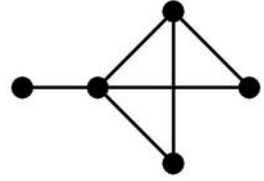
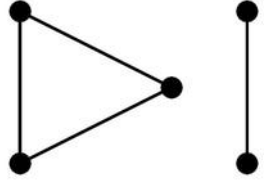
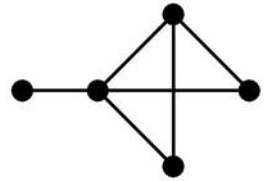
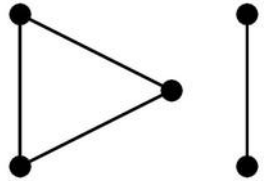
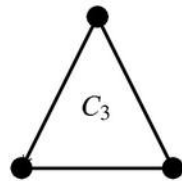
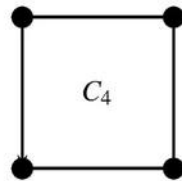
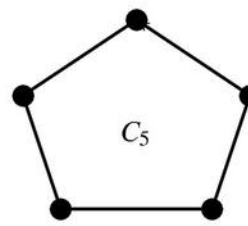
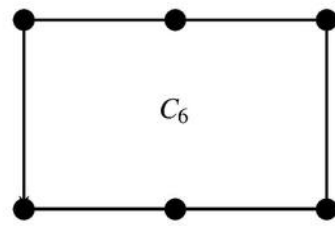
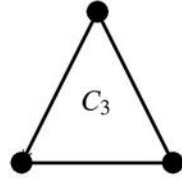
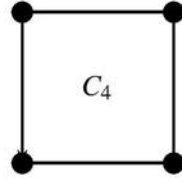
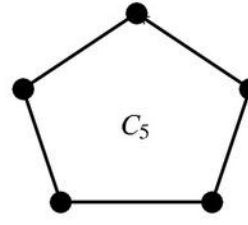
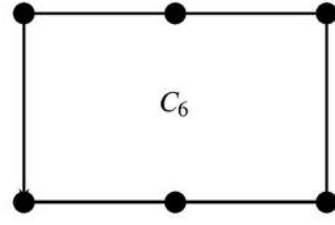
Inline formulas (first occurrence)

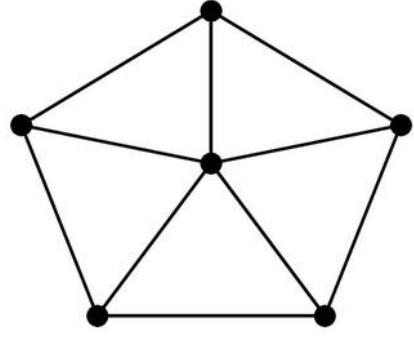
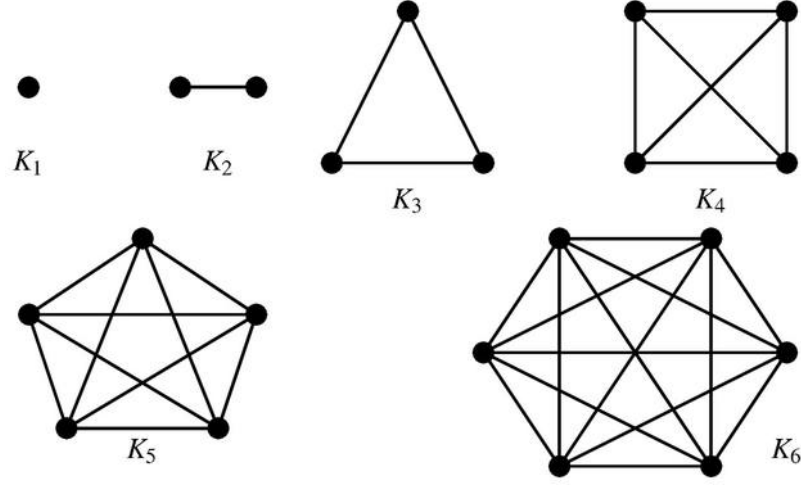
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00001	87	1.000	$K_{\{5\}}$	K_5	Theorem 4.3.1 K_5 and $K_{3,3}$ are not planar graphs.
voloshin-hypergraph_F00002	87	1.000	$K_{\{3,3\}}$	$K_{3,3}$	Theorem 4.3.1 K_5 and $K_{3,3}$ are not planar graphs.
voloshin-hypergraph_F00003	15	0.994	$\rightarrow$	$\Rightarrow$	• since the visualization is the feature of graph theory, “look $\Rightarrow$ read $\Rightarrow$ look” - rule is
voloshin-hypergraph_F00004	21	0.999	$A, B, \ldots$	$A, B, \ldots$	Throughout the book, we use the standard notation: upper case letters $A, B, \ldots$,
voloshin-hypergraph_F00005	21	1.000	X, Y, Z	X, Y, Z	X, Y, Z for sets (all sets are finite), lower case letters $a, b, \ldots, e, \ldots, x, y, z$ for the elements of
voloshin-hypergraph_F00006	21	1.000	$a, b, \ldots, e, \ldots, x, y, z$	$a, b, \ldots, e, \ldots, x, y, z$	X, Y, Z for sets (all sets are finite), lower case letters $a, b, \ldots, e, \ldots, x, y, z$ for the elements of
voloshin-hypergraph_F00007	22	—	$G=(X, E)$	$G=(X, E)$	
voloshin-hypergraph_F00008	22	1.000	A	A	to right). The number of elements in a set A is denoted by $ A $, and the empty set is denoted
voloshin-hypergraph_F00009	22	1.000	$ A $	$ A $	to right). The number of elements in a set A is denoted by $ A $, and the empty set is denoted
voloshin-hypergraph_F00010	22	0.883	$\boldsymbol{\emptyset}$	$\emptyset$	by $\emptyset$. If a set contains other sets as elements, then it is called a family ; in a family, elements
voloshin-hypergraph_F00011	22	0.967	X	X	In a graph, the set of vertices is denoted by X and is written as $X=\{x_1, x_2, \ldots, x_n\}$ where
voloshin-hypergraph_F00012	22	0.967	$X=\left\{x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}\right\}$	$X=\left\{x_1, x_2, \ldots, x_n\right\}$	In a graph, the set of vertices is denoted by X and is written as $X=\{x_1, x_2, \ldots, x_n\}$ where
voloshin-hypergraph_F00013	22	0.971	$x_{\{i\}}$	x_i	x_i is the i -th vertex and n is the number of vertices. The set of edges is denoted by E and is
voloshin-hypergraph_F00014	22	0.971	i	i	x_i is the i -th vertex and n is the number of vertices. The set of edges is denoted by E and is
voloshin-hypergraph_F00015	22	0.971	n	n	x_i is the i -th vertex and n is the number of vertices. The set of edges is denoted by E and is
voloshin-hypergraph_F00016	22	0.971	E	E	x_i is the i -th vertex and n is the number of vertices. The set of edges is denoted by E and is
voloshin-hypergraph_F00017	22	0.996	$E=\left\{e_{\{1\}}, e_{\{2\}}, \ldots, e_{\{m\}}\right\}$	$E=\left\{e_1, e_2, \ldots, e_m\right\}$	written as $E=\{e_1, e_2, \ldots, e_m\}$ where e_i is the i -th edge and m is the number of edges. Each

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00018	22	■ 0.996	$e_{\{i\}}$	e_i	written as $E = \{e_1, e_2, \dots, e_m\}$ where e_i is the i -th edge and m is the number of edges. <i>Each</i>
voloshin-hypergraph_F00019	22	■ 0.996	m	m	written as $E = \{e_1, e_2, \dots, e_m\}$ where e_i is the i -th edge and m is the number of edges. <i>Each</i>
voloshin-hypergraph_F00020	22	■ 1.000	G	G	to “invent” the last letter to denote the entire graph: G . Now we are ready to present the
voloshin-hypergraph_F00021	22	■ 1.000	$n=5$	$n = 5$	ments above. It has $n = 5$ vertices and $m = 7$ edges. We write $G = (X, E)$ where
voloshin-hypergraph_F00023	22	■ 1.000	$X=\left\{x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, x_{\{4\}}, x_{\{5\}}\right\}$	$X = \{x_1, x_2, x_3, x_4, x_5\}$	$X = \{x_1, x_2, x_3, x_4, x_5\}$ and $E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$. Since each edge is a pair of ver-
voloshin-hypergraph_F00029	22	■ 1.000	$x_{\{1\}}$	x_1	are called disjoint . For example, vertices x_1 and x_2 are adjacent, but vertices x_1 and x_3 are
voloshin-hypergraph_F00030	22	■ 1.000	$x_{\{2\}}$	x_2	are called disjoint . For example, vertices x_1 and x_2 are adjacent, but vertices x_1 and x_3 are
voloshin-hypergraph_F00031	22	■ 1.000	$x_{\{3\}}$	x_3	are called disjoint . For example, vertices x_1 and x_2 are adjacent, but vertices x_1 and x_3 are
voloshin-hypergraph_F00032	22	■ 1.000	x	x	disjoint. For a given vertex x , the number of all vertices adjacent to it is called degree of the
voloshin-hypergraph_F00033	22	■ 1.000	$d(x)$	$d(x)$	vertex x , denoted by $d(x)$. In our example, $d(x_1) = 2$, $d(x_2) = 4$, and so on. The maximum
voloshin-hypergraph_F00035	22	■ 1.000	$\Delta(G)$	$\Delta(G)$	degree over all vertices is called the maximum degree of G , denoted by $\Delta(G)$. For graph
voloshin-hypergraph_F00037	23	■ 1.000	$N(x)$	$N(x)$	noted by $N(x)$. In our graph, for example, $N(x_1) = \{x_2, x_5\}$. Evidently, the degree of a
voloshin-hypergraph_F00038	23	■ 1.000	$N\left(x_{\{1\}}\right)=\left\{x_{\{2\}}, x_{\{5\}}\right\}$	$N\left(x_1\right)=\left\{x_2, x_5\right\}$	noted by $N(x)$. In our graph, for example, $N(x_1) = \{x_2, x_5\}$. Evidently, the degree of a
voloshin-hypergraph_F00039	23	■ 1.000	$d\left(x_{\{1\}}\right)=\left N\left(x_{\{1\}}\right)\right =2$	$d\left(x_1\right)=\left N\left(x_1\right)\right =2$	vertex is the cardinality (the number of elements) of its neighborhood: $d(x_1) = N(x_1) = 2$,
voloshin-hypergraph_F00040	23	■ 1.000	$d\left(x_{\{2\}}\right)=\left N\left(x_{\{2\}}\right)\right =4$	$d\left(x_2\right)=\left N\left(x_2\right)\right =4$	<u>$d(x_2) = N(x_2) = 4$</u> , and so on.
voloshin-hypergraph_F00041	23	■ 1.000	$e_{\{1\}}$	e_1	wise. In Figure 1.2 edges e_1 and e_2 are adjacent, and edges e_1 and e_3 are disjoint. If a vertex
voloshin-hypergraph_F00042	23	■ 1.000	$e_{\{2\}}$	e_2	wise. In Figure 1.2 edges e_1 and e_2 are adjacent, and edges e_1 and e_3 are disjoint. If a vertex
voloshin-hypergraph_F00043	23	■ 1.000	$e_{\{3\}}$	e_3	wise. In Figure 1.2 edges e_1 and e_2 are adjacent, and edges e_1 and e_3 are disjoint. If a vertex
voloshin-hypergraph_F00044	23	■ 1.000	e	e	x belongs to an edge e , then we say that they are incident to each other. In the example
voloshin-hypergraph_F00045	23	■ 1.000	$x_{\{5\}}$	x_5	above, edge e_1 is incident to vertex x_1 and is not incident to vertex x_5 and so on. As one can
voloshin-hypergraph_F00046	23	■ 1.000	$V(G)$	$V(G)$	Sometimes the vertex set of a graph G is denoted by $V(G)$ and the edge set by $E(G)$.
voloshin-hypergraph_F00047	23	■ 1.000	$E(G)$	$E(G)$	Sometimes the vertex set of a graph G is denoted by $V(G)$ and the edge set by $E(G)$.
voloshin-hypergraph_F00048	23	■ 1.000	$G=(V(G), E(G))$	$G=(V(G), E(G))$	So, generally any graph <u>$G=(V(G), E(G))$</u> . The number of vertices is usually denoted
voloshin-hypergraph_F00049	23	■ 1.000	$n=n(G)$	$n=n(G)$	by $n=n(G)$, and the number of edges by $m=m(G)$. The set of edges containing a
voloshin-hypergraph_F00050	23	■ 1.000	$m=m(G)$	$m=m(G)$	by $n=n(G)$, and the number of edges by $m=m(G)$. The set of edges containing a
voloshin-hypergraph_F00051	23	■ 1.000	$E(x)$	$E(x)$	vertex x is denoted by $E(x)$. In our example, $V(G) = X = \{x_1, x_2, x_3, x_4, x_5\}$, $ X = n = 5$,
voloshin-hypergraph_F00052	23	■ 1.000	$V(G)=X=\left\{x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, x_{\{4\}}, x_{\{5\}}\right\}, X =n=5$	$V(G)=X=\left\{x_1, x_2, x_3, x_4, x_5\right\}, X =n=5$	vertex x is denoted by $E(x)$. In our example, <u>$V(G)=X=\left\{x_1, x_2, x_3, x_4, x_5\right\}, X =n=5$</u> ,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00053	23	■ 0.803	$E(G)=E=\left\{\left\{x_1, x_2\right\},\left\{x_2, x_3\right\},\left\{x_3, x_4\right\},\left\{x_4, x_5\right\},\left\{x_5, x_1\right\},\left\{x_2, x_5\right\},\left\{x_2, x_4\right\}\right\}, E =m=7$	$E(G) = E = \{\{x_1, x_2\}, \{x_2, x_3\}, \{x_3, x_4\}, \{x_4, x_5\}, \{x_5, x_1\}, \{x_2, x_5\}, \{x_2, x_4\}\}, E = m = 7$	$E(G) = E = \{\{x_1, x_2\}, \{x_2, x_3\}, \{x_3, x_4\}, \{x_4, x_5\}, \{x_5, x_1\}, \{x_2, x_5\}, \{x_2, x_4\}\}, E = m = 7$,
voloshin-hypergraph_F00054	23	■ 1.000	$E\left(x_1\right)=\left\{e_1, e_5\right\}$	$E\left(x_1\right)=\left\{e_1, e_5\right\}$	$E\left(x_1\right)=\left\{e_1, e_5\right\}$, and so on.
voloshin-hypergraph_F00055	23	—	$\square$	$\square$	—
voloshin-hypergraph_F00056	23	■ 1.000	$n-1$	$n-1$	bors is $n-1$, therefore $0 \leq d(x) \leq n-1$.
voloshin-hypergraph_F00057	23	■ 1.000	$0 \leq d(x) \leq n-1$	$0 \leq d(x) \leq n-1$	bors is $n-1$, therefore $0 \leq d(x) \leq n-1$.
voloshin-hypergraph_F00058	23	■ 0.998	$n-1.0$	$n-1.0$	of integers from 0 to $n-1$. 0 means that there is a vertex with no neighbors, and $n-1$ means
voloshin-hypergraph_F00059	23	■ 1.000	$d\left(x_1\right)=d\left(x_3\right)=2$	$d\left(x_1\right)=d\left(x_3\right)=2$	In graph G , see Figure 1.2, for example, $d\left(x_1\right)=d\left(x_3\right)=2$, and $d\left(x_4\right)=d\left(x_5\right)=3$.
voloshin-hypergraph_F00060	23	■ 1.000	$d\left(x_4\right)=d\left(x_5\right)=3$	$d\left(x_4\right)=d\left(x_5\right)=3$	In graph G , see Figure 1.2, for example, $d\left(x_1\right)=d\left(x_3\right)=2$, and $d\left(x_4\right)=d\left(x_5\right)=3$.
voloshin-hypergraph_F00061	24	■ 1.000	G_1, G_2	G_1, G_2	6. Apply degree equality (Proposition 1.1.1) to each of the graphs G_1, G_2 and G_3 .
voloshin-hypergraph_F00062	24	■ 1.000	G_3	G_3	6. Apply degree equality (Proposition 1.1.1) to each of the graphs G_1, G_2 and G_3 .
voloshin-hypergraph_F00063	24	■ 0.999	x_1, x_2, x_3, x_4	x_1, x_2, x_3, x_4	Theory. Denote chemicals A, B, C, D and E respectively by letters x_1, x_2, x_3, x_4 and x_5 and
voloshin-hypergraph_F00064	25	■ 1.000	x_4	x_4	x_2 with x_3 and x_5 , connect x_3 with x_4 and connect x_4 with x_5 . The graph obtained is shown
voloshin-hypergraph_F00065	25	■ 1.000	$\left\{x_1\right\},\left\{x_2\right\},\left\{x_3\right\}$	$\left\{x_1\right\},\left\{x_2\right\},\left\{x_3\right\}$	Looking at Figure 1.4, we can evidently partition X into five subsets: $\left\{x_1\right\},\left\{x_2\right\},\left\{x_3\right\},$
voloshin-hypergraph_F00066	25	■ 1.000	$\left\{x_4\right\},\left\{x_5\right\}$	$\left\{x_4\right\},\left\{x_5\right\}$	$\left\{x_4\right\},\left\{x_5\right\}$; this partition is feasible, i.e. good, but not optimal. We can partition X into
voloshin-hypergraph_F00067	25	■ 1.000	$\left\{x_1, x_3\right\},\left\{x_2\right\},\left\{x_4\right\},\left\{x_5\right\}$	$\left\{x_1, x_3\right\},\left\{x_2\right\},\left\{x_4\right\},\left\{x_5\right\}$	four subsets: $\left\{x_1, x_3\right\},\left\{x_2\right\},\left\{x_4\right\},\left\{x_5\right\}$, what is better but still not optimal. At last, we
voloshin-hypergraph_F00068	25	■ 1.000	$\left\{x_1, x_4\right\},\left\{x_2\right\},\left\{x_3, x_5\right\}$	$\left\{x_1, x_4\right\},\left\{x_2\right\},\left\{x_3, x_5\right\}$	can partition X into three subsets: $\left\{x_1, x_4\right\},\left\{x_2\right\},\left\{x_3, x_5\right\}$, see Figure 1.4 (the partition is
voloshin-hypergraph_F00069	25	■ 1.000	x_1, x_2	x_1, x_2	formed by vertices x_1, x_2 , and x_5 ; any partition of it into two parts (even if one part is empty)
voloshin-hypergraph_F00070	25	■ 1.000	$\left\{x_1, x_3\right\}$	$\left\{x_1, x_3\right\}$	the picture, the reader can easily find at least one another optimal solution, say $\left\{x_1, x_3\right\}$,
voloshin-hypergraph_F00071	25	■ 0.993	$\left\{x_2, x_4\right\}$	$\left\{x_2, x_4\right\}$	$\left\{x_2, x_4\right\}$, and $\left\{x_5\right\}$. How many and which optimal solutions do exist?
voloshin-hypergraph_F00072	25	■ 0.993	$\left\{x_5\right\}$	$\left\{x_5\right\}$	$\left\{x_2, x_4\right\}$, and $\left\{x_5\right\}$. How many and which optimal solutions do exist?
voloshin-hypergraph_F00073	28	■ 0.982	$x_1: x_2, x_5$	$x_1: x_2, x_5$	• For $x_1: x_2, x_5$,
voloshin-hypergraph_F00074	28	■ 0.990	$x_2: x_1, x_3, x_4, x_5$	$x_2: x_1, x_3, x_4, x_5$	• For $x_2: x_1, x_3, x_4, x_5$,
voloshin-hypergraph_F00075	28	■ 0.919	$x_3: x_2, x_4$	$x_3: x_2, x_4$	• For $x_3: x_2, x_4$,
voloshin-hypergraph_F00076	28	■ 0.999	$x_4: x_2, x_3, x_5$	$x_4: x_2, x_3, x_5$	• For $x_4: x_2, x_3, x_5$,
voloshin-hypergraph_F00077	28	■ 0.996	$x_5: x_1, x_2, x_4$	$x_5: x_1, x_2, x_4$	• For $x_5: x_1, x_2, x_4$.
voloshin-hypergraph_F00078	29	■ 1.000	$L(G)$	$L(G)$	The adjacency list, denoted by $L(G)$, is:

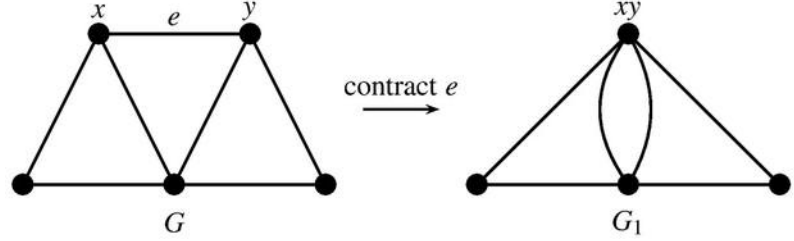
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00079	29	■ 1.000	$x_{\{j\}}$	x_j	has one row and one column for each vertex. If vertex x_i is adjacent to vertex x_j , then (i, j) -
voloshin-hypergraph_F00080	29	■ 1.000	(i, j)	(i, j)	has one row and one column for each vertex. If vertex x_i is adjacent to vertex x_j , then (i, j) -
voloshin-hypergraph_F00081	29	■ 1.000	$A(G)$	$A(G)$	For graph G , see Figure 1.5, the adjacency matrix denoted by $A(G)$ is:
voloshin-hypergraph_F00082	29	■ 1.000	$A^{\{\backslash prime\}}$	A'	2nd columns and so on), then the columns become rows, and we obtain another matrix, A'
voloshin-hypergraph_F00083	29	■ 1.000	$A=A^{\{\backslash prime\}}$	$A = A'$	the matrix. Since $A = A'$, this matrix is called symmetric . One can think about transposition
voloshin-hypergraph_F00084	30	■ 1.000	$e_{\{j\}}$	e_j	each edge of a graph. If vertex x_i is incident to edge e_j , then the (i, j) -entry in the matrix
voloshin-hypergraph_F00085	30	■ 0.789	$I(G)$	$I(G)$	is 1, otherwise it is 0. For our graph G , the incidence matrix denoted by $I(G)$ has 5 rows
voloshin-hypergraph_F00086	30	■ 0.910	I	I	tially as columns), we obtain another matrix which is different from I . We will se in Part II
voloshin-hypergraph_F00087	30	■ 1.000	$J(G)$	$J(G)$	G , the edge list, denoted by $J(G)$ is the following:
voloshin-hypergraph_F00088	31	■ 1.000	$\{a, b\}=\{b, a\}$	$\{a, b\} = \{b, a\}$	writings for the same set are equivalent: $\{a, b\} = \{b, a\}$. However, there are situations in
voloshin-hypergraph_F00089	32	■ 0.677	$\{$	$\{$	of curly braces $\{, \}$, one use parentheses $(,)$. So now $(a, b) \neq (b, a)$. In a graph, an ordered
voloshin-hypergraph_F00090	32	■ 0.677	$\}$, one use parentheses $(,)$. So now $(a, b) \neq (b, a)$	$\}, one use parentheses (,). Sonow (a, b) \neq (b, a)$	of curly braces $\{, \}$, one use parentheses $(,)$. So now $(a, b) \neq (b, a)$. In a graph, an ordered
voloshin-hypergraph_F00091	32	■ 0.984	(x, y)	(x, y)	pair of vertices is called an arc . If (x, y) is an arc, then x is called the initial vertex and
voloshin-hypergraph_F00092	32	■ 1.000	y	y	y is called the terminal vertex. A graph in which all edges are ordered pairs is called the
voloshin-hypergraph_F00093	32	■ 1.000	L	L	cency list L may contain multiple elements and empty sets, adjacency matrix A may have
voloshin-hypergraph_F00094	32	■ 1.000	J	J	matrix I may have zero rows and repeated columns, and edge list J may have repeated ele-
voloshin-hypergraph_F00095	33	■ 0.546	a, b, c, d, e, f	a, b, c, d, e, f	Notice that the columns in $I(G)$ are in the order of arcs a, b, c, d, e, f , and “ I ” in fourth
voloshin-hypergraph_F00096	33	■ 0.889	d	d	column and second row indicates that arc d is the loop at vertex 2.
voloshin-hypergraph_F00097	33	■ 0.512	$\left(x_{\{i\}}, x_{\{j\}}\right)$	(x_i, x_j)	the vertices are street crossings in the city map; crossings x_i and x_j form an arc (x_i, x_j)
voloshin-hypergraph_F00098	34	■ 1.000	$E_{\{n\}}$	E_n	treated. A graph may have no edges at all; such graphs are called empty , denoted by E_n
voloshin-hypergraph_F00099	34	■ 1.000	$E_{\{4\}}$	E_4	where n is the number of vertices. Graph E_4 is depicted in Figure 1.7. If a graph is not
voloshin-hypergraph_F00100	34	■ 0.977	$\emptyset$	$\emptyset$	empty set which is always denoted by $\emptyset$.
voloshin-hypergraph_F00101	34	—	$P_{\{5\}}$	P_5	
voloshin-hypergraph_F00102	34	■ 0.994	$x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}$	$x_1, x_2, \ldots, x_n$	$x_1, x_2, \ldots, x_n$ in such a way that there is precisely one edge connecting every two consecutive
voloshin-hypergraph_F00103	34	■ 1.000	$P_{\{n\}}$	P_n	The number of edges in a path is its length . A path on n vertices is denoted by P_n .
voloshin-hypergraph_F00104	34	■ 1.000	$m=n-1$	$m = n - 1$	Evidently, in any P_n , the number of edges $m = n - 1$. Any edge itself is a path P_2 . The
voloshin-hypergraph_F00105	34	■ 1.000	$P_{\{2\}}$	P_2	Evidently, in any P_n , the number of edges $m = n - 1$. Any edge itself is a path P_2 . The

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00106	34	■ 1.000	$x_{\{n\}}$	x_n	vertices x_1 and x_n both have degree 1 in a path P_n ; we say that the path connects vertices x_1
voloshin-hypergraph_F00107	34	■ 0.736	$\left(x_{\{1\}}, x_{\{5\}}\right)$	(x_1, x_5)	path P_5 of length 4 connects vertices x_1 and x_5 , or is (x_1, x_5) -path.
voloshin-hypergraph_F00108	34	—	$G_{\{1\}}$	G_1	 
voloshin-hypergraph_F00109	34	—	$G_{\{2\}}$	G_2	 
voloshin-hypergraph_F00110	35	■ 1.000	$G^{\{\prime\}}$	G'	same graph, say G' , and if G has k connected components, then we write: $G = kG'$.
voloshin-hypergraph_F00111	35	■ 1.000	k	k	same graph, say G' , and if G has k connected components, then we write: $G = kG'$.
voloshin-hypergraph_F00112	35	■ 1.000	$G=k\ G^{\{\prime\}}$	$G = kG'$	same graph, say G' , and if G has k connected components, then we write: $G = kG'$.
voloshin-hypergraph_F00113	35	■ 1.000	$C_{\{n\}}$	C_n	times “simple cycle”) denoted by C_n where n is the number of vertices.
voloshin-hypergraph_F00114	35	—	$C_{\{1\}}, C_{\{2\}}, C_{\{3\}}, C_{\{4\}}, C_{\{5\}}$	C_1, C_2, C_3, C_4, C_5	     
voloshin-hypergraph_F00115	35	—	$C_{\{6\}}$	C_6	     

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00116	35	■ 1.000	$C_{\{1\}}$	C_1	of the cycle, see Figure 1.12. C_1 represents an edge connecting the vertex with itself, it is
voloshin-hypergraph_F00117	35	■ 1.000	$C_{\{2\}}$	C_2	a loop. C_2 represents two parallel edges connecting the same pair of vertices. The cycle
voloshin-hypergraph_F00118	35	■ 1.000	$C_{\{3\}}$	C_3	C_3 is called triangle . In simple graphs, if there are cycles, then they must have length at
voloshin-hypergraph_F00119	35	—	$W_{\{6\}}$	W_6	
voloshin-hypergraph_F00120	35	■ 1.000	$C_{\{k\}}, k \geq 3$	$C_k, k \geq 3$	Wheels. If for any cycle $C_k, k \geq 3$, we add a new vertex and connect it to each of the
voloshin-hypergraph_F00121	36	■ 1.000	$C_{\{k\}}$	C_k	vertices of C_k , then the graph obtained is called a wheel , denoted by W_{k+1} . The wheel W_6 is
voloshin-hypergraph_F00122	36	■ 1.000	$W_{\{k+1\}}$	W_{k+1}	vertices of C_k , then the graph obtained is called a wheel , denoted by W_{k+1} . The wheel W_6 is
voloshin-hypergraph_F00123	36	■ 1.000	$K_{\{n\}}$	K_n	plete , denoted by K_n where as usually, n is the number of vertices. It is called complete
voloshin-hypergraph_F00124	36	■ 1.000	$n \geq 1$	$n \geq 1$	because we cannot add any new edge to it and obtain a simple graph. For every $n \geq 1$, the
voloshin-hypergraph_F00125	36	—	$K_{\{1\}}, K_{\{2\}}, K_{\{3\}}, K_{\{4\}}, K_{\{5\}}$	K_1, K_2, K_3, K_4, K_5	

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voloshin-hypergraph_ F00126	36	—	$K_{\{6\}}$	K_6	
voloshin-hypergraph_ F00127	36	■ 1.000	$n(n-1)=2m$	$n(n-1) = 2m$	$n(n-1) = 2m$. For example, the number of edges in K_6 is: $m = 6(6-1)/2 = 15$. If a graph
voloshin-hypergraph_ F00128	36	■ 1.000	$m=6(6-1) / 2=15$	$m = 6(6-1)/2 = 15$	$n(n-1) = 2m$. For example, the number of edges in K_6 is: $m = 6(6-1)/2 = 15$. If a graph
voloshin-hypergraph_ F00129	36	■ 1.000	$K_{\{2\}}=$	$K_2 =$	Complete graphs K_1, K_2, K_3, K_4, K_5 , and K_6 are shown in Figure 1.14. Notice that $K_2 =$
voloshin-hypergraph_ F00130	36	■ 1.000	$P_{\{2\}}, K_{\{3\}}=C_{\{3\}}$	$P_2, K_3 = C_3$	$P_2, K_3 = C_3$, and $K_4 = W_4$.
voloshin-hypergraph_ F00131	36	■ 1.000	$K_{\{4\}}=W_{\{4\}}$	$K_4 = W_4$	$P_2, K_3 = C_3$, and $K_4 = W_4$.
voloshin-hypergraph_ F00132	36	■ 1.000	$T_{\{n\}}$	T_n	vertices is denoted by T_n . In contrast to P_n, K_n, C_n and W_n there are many distinct trees on n
voloshin-hypergraph_ F00133	36	■ 1.000	$P_{\{n\}}, K_{\{n\}}, C_{\{n\}}$	P_n, K_n, C_n	vertices is denoted by T_n . In contrast to P_n, K_n, C_n and W_n there are many distinct trees on n
voloshin-hypergraph_ F00134	36	■ 1.000	$W_{\{n\}}$	W_n	vertices is denoted by T_n . In contrast to P_n, K_n, C_n and W_n there are many distinct trees on n
voloshin-hypergraph_ F00135	36	■ 1.000	$P_{\{4\}}$	P_4	just because they both are P_4 , and the 2nd is “different” from them. There are no other trees
voloshin-hypergraph_ F00136	37	■ 1.000	$X_{\{1\}}$	X_1	partitioned into two disjoint sets X_1 and X_2 , called parts, in such a way that every edge
voloshin-hypergraph_ F00137	37	■ 1.000	$X_{\{2\}}$	X_2	partitioned into two disjoint sets X_1 and X_2 , called parts, in such a way that every edge
voloshin-hypergraph_ F00138	37	■ 1.000	$G=\left(X_{\{1\}}, X_{\{2\}} ; E\right)$	$G = (X_1, X_2; E)$	write it as $G = (X_1, X_2; E)$ where X_1 is its left part and X_2 is its right part (as is the first graph
voloshin-hypergraph_ F00139	37	■ 1.000	$\left X_{\{1\}}\right =r$	$ X_1 = r$	a new edge to it and obtain another bipartite graph. If in a complete bipartite graph $ X_1 = r$
voloshin-hypergraph_ F00140	37	■ 0.987	$\left X_{\{2\}}\right =s$	$ X_2 = s$	and $ X_2 = s$, then the graph itself is denoted by $K_{r,s}$. The number of edges in $K_{r,s}$ clearly
voloshin-hypergraph_ F00141	37	■ 0.987	$K_{\{r, s\}}$	$K_{r,s}$	and $ X_2 = s$, then the graph itself is denoted by $K_{r,s}$. The number of edges in $K_{r,s}$ clearly
voloshin-hypergraph_ F00142	37	■ 1.000	$r s$	rs	equals rs .
voloshin-hypergraph_ F00143	37	■ 0.999	$K_{\{1,3\}}$	$K_{1,3}$	the second is $K_{1,3}$, and the third is C_4 . Observe that bipartition X_1, X_2 is explicitly shown for
voloshin-hypergraph_ F00144	37	■ 0.999	$C_{\{4\}}$	C_4	the second is $K_{1,3}$, and the third is C_4 . Observe that bipartition X_1, X_2 is explicitly shown for
voloshin-hypergraph_ F00145	37	■ 0.999	$X_{\{1\}}, X_{\{2\}}$	X_1, X_2	the second is $K_{1,3}$, and the third is C_4 . Observe that bipartition X_1, X_2 is explicitly shown for
voloshin-hypergraph_ F00146	37	■ 0.998	$(n-1)$	$(n-1)$	$(n-1)$ -regular graph. The 3-regular graphs are called the cubic graphs .

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voloshin-hypergraph_F00147	38	■ 0.999	$C_{\{5\}}$	C_5	obtain a graph called cube . If we do that with C_5 by drawing inner cycle differently, we
voloshin-hypergraph_F00148	39	■ 1.000	$G_{\{1\}}=\left(X_{\{1\}}, E_{\{1\}}\right)$	$G_1 = (X_1, E_1)$	graphs $G_1 = (X_1, E_1)$ and $G_2 = (X_2, E_2)$ are called isomorphic if there exists a one-to-one
voloshin-hypergraph_F00149	39	■ 1.000	$G_{\{2\}}=\left(X_{\{2\}}, E_{\{2\}}\right)$	$G_2 = (X_2, E_2)$	graphs $G_1 = (X_1, E_1)$ and $G_2 = (X_2, E_2)$ are called isomorphic if there exists a one-to-one
voloshin-hypergraph_F00150	39	■ 1.000	$G_{\{1\}} \cong G_{\{2\}}$	$G_1 \cong G_2$	G_1 is isomorphic to G_2 , and G_2 is isomorphic to G_1 , and we write $G_1 \cong G_2$.
voloshin-hypergraph_F00151	39	■ 0.999	$\left X_{\{1\}}\right =\left X_{\{2\}}\right =n$	$ X_1 = X_2 = n$	Suppose $\boxed{ X_1 = X_2 = n}$. Generally, how many one-to-one correspondences do exist?
voloshin-hypergraph_F00152	39	■ 1.000	$n-2$	$n - 2$	third vertex has $n - 2$ possibilities and so on. Total number of possibilities is $n(n - 1)(n -$
voloshin-hypergraph_F00153	39	■ 1.000	$n(n-1)(n-$	$n(n - 1)(n -$	third vertex has $n - 2$ possibilities and so on. Total number of possibilities is $n(n - 1)(n -$
voloshin-hypergraph_F00154	39	■ 0.979	$2) \cdot \cdots 3 \cdot 2 \cdot 1 = n!$	$2) \cdot \cdots 3 \cdot 2 \cdot 1 = n!$	$\boxed{2) \cdot \cdots 3 \cdot 2 \cdot 1 = n!}$. If in at least one of these $n!$ cases there is a complete “coincidence”
voloshin-hypergraph_F00155	39	■ 0.979	$n!$	$n!$	$2) \cdot \cdots 3 \cdot 2 \cdot 1 = n!$. If in at least one of these $n!$ cases there is a complete “coincidence”
voloshin-hypergraph_F00156	39	■ 0.990	$1, 2, \ldots n!$	$1, 2, \ldots n!$	phic), and $1, 2, \ldots n!$ when they are isomorphic. In Figure 1.15, for example, the first and
voloshin-hypergraph_F00157	39	■ 0.840	2^n	$2n$	is not isomorphic to P_4 . Graphs E_n and K_n have $n!$ isomorphisms, graphs C_n have $2n$ and so
voloshin-hypergraph_F00158	39	■ 1.000	$T_{\{1\}}$	T_1	in Figure 1.19. Both trees T_1 and T_2 have eight vertices, seven edges, four vertices of degree
voloshin-hypergraph_F00159	39	■ 1.000	$T_{\{2\}}$	T_2	in Figure 1.19. Both trees T_1 and T_2 have eight vertices, seven edges, four vertices of degree
voloshin-hypergraph_F00160	39	■ 1.000	$8!=40320$	$8! = 40320$	common sense tells us that there will be no match in all $8!=40320$ one-to-one correspon-
voloshin-hypergraph_F00161	40	■ 1.000	$\boldsymbol{\sigma}$	σ	by σ is the following:
voloshin-hypergraph_F00162	40	■ 1.000	a	a	are the corresponding vertices a and b in G_2 . Vertices 1 and 4 are adjacent in G_1 , so are the
voloshin-hypergraph_F00163	40	■ 1.000	b	b	are the corresponding vertices a and b in G_2 . Vertices 1 and 4 are adjacent in G_1 , so are the
voloshin-hypergraph_F00164	40	■ 0.665	$\binom{6}{2}=15$	$\binom{6}{2} = 15$	of all such comparisons is $\binom{6}{2} = 15$ which is much less than $6! = 720$, the number of all
voloshin-hypergraph_F00165	40	■ 0.665	$6!=720$	$6! = 720$	of all such comparisons is $\binom{6}{2} = 15$ which is much less than $6! = 720$, the number of all
voloshin-hypergraph_F00166	40	■ 1.000	$E_{\{n\}}, K_{\{n\}}, C_{\{n\}}, W_{\{n\}}$	E_n, K_n, C_n, W_n	graph classes that have been investigated. In the simplest classes such as E_n, K_n, C_n, W_n ,
voloshin-hypergraph_F00167	40	■ 0.987	r	r	6. For each pair of integer numbers r and s , $1 \leq r, s \leq 5$, draw a complete bipartite graph $K_{r,s}$.
voloshin-hypergraph_F00168	40	■ 0.987	$s, 1 \leq r, s \leq 5$	$s, 1 \leq r, s \leq 5$	6. For each pair of integer numbers r and $\boxed{1 < r, s < 3}$, draw a complete bipartite graph $K_{r,s}$.
voloshin-hypergraph_F00169	40	■ 1.000	$K_{\{s, r\}}$	$K_{s,r}$	7. Explain why $K_{r,s}$ and $K_{s,r}$ are isomorphic.
voloshin-hypergraph_F00170	41	■ 1.000	$L(G), A(G), I(G), J(G)$	$L(G), A(G), I(G), J(G)$	1. Given any of $\boxed{L(G), A(G), I(G), J(G)}$, recognize if G is a path.
voloshin-hypergraph_F00171	41	■ 1.000	$L\left(G_{\{1\}}\right), A\left(G_{\{1\}}\right), I\left(G_{\{1\}}\right)$	$L\left(G_1\right), A\left(G_1\right), I\left(G_1\right)$	10. Given any of $L(G_1), A(G_1), I(G_1)$, or $J(G_1)$, any of $L(G_2), A(G_2), I(G_2)$, or $J(G_2)$ and a
voloshin-hypergraph_F00172	41	■ 1.000	$J\left(G_{\{1\}}\right)$	$J\left(G_1\right)$	10. Given any of $L(G_1), A(G_1), I(G_1)$, or $J(G_1)$, any of $L(G_2), A(G_2), I(G_2)$, or $J(G_2)$ and a
voloshin-hypergraph_F00173	41	■ 1.000	$L\left(G_{\{2\}}\right), A\left(G_{\{2\}}\right), I\left(G_{\{2\}}\right)$	$L\left(G_2\right), A\left(G_2\right), I\left(G_2\right)$	10. Given any of $L(G_1), A(G_1), I(G_1)$, or $J(G_1)$, any of $L(G_2), A(G_2), I(G_2)$, or $J(G_2)$ and a
voloshin-hypergraph_F00174	41	■ 1.000	$J\left(G_{\{2\}}\right)$	$J\left(G_2\right)$	10. Given any of $L(G_1), A(G_1), I(G_1)$, or $J(G_1)$, any of $L(G_2), A(G_2), I(G_2)$, or $J(G_2)$ and a

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voloshin-hypergraph_F00175	41	■ 1.000	σ	σ	one-to-one correspondence σ between the vertices of G_1 and G_2 . Recognize if σ realizes an
voloshin-hypergraph_F00176	41	■ 1.000	$x \in X$	$x \in X$	Deletion of a vertex. Let us have a graph $G = (X, E)$ and a vertex $x \in X$. A deletion of
voloshin-hypergraph_F00177	41	■ 0.977	$X_{\{1\}} = X - \{x\}$	$X_1 = X - \{x\}$	x . Recall that $E(x)$ denotes the set of edges containing vertex x in graph G . If $X_1 = X - \{x\}$,
voloshin-hypergraph_F00178	41	■ 1.000	$E_{\{1\}} = E - E(x)$	$E_1 = E - E(x)$	and $E_1 = E - E(x)$, then deletion of x from G results in obtaining the graph $G_1 = (X_1, E_1)$,
voloshin-hypergraph_F00179	42	■ 1.000	$G_{\{1\}} = G - x$	$G_1 = G - x$	see Figure 1.22. We write this operation as $G_1 = G - x$. In G_1 , we can choose and delete
voloshin-hypergraph_F00180	42	■ 1.000	$E(x) = \{e_1, e_2, e_3\}$	$E(x) = \{e_1, e_2, e_3\}$	$E(x) = \{e_1, e_2, e_3\}$ and all these edges become loops in G_1 .
voloshin-hypergraph_F00181	42	■ 1.000	$G_{\{1\}} = G - e_{\{3\}}$	$G_1 = G - e_3$	e_3 is shown in Figure 1.24. In both cases we write: $G_1 = G - e_3$.
voloshin-hypergraph_F00182	43	■ 1.000	$e = \{x, y\} \in E$	$e = \{x, y\} \in E$	Contraction of an edge. Let us have a graph $G = (X, E)$ and an edge $e = \{x, y\} \in E$.
voloshin-hypergraph_F00183	43	■ 1.000	xy	xy	1. Identifying vertices x and y in a new vertex called xy and removing e from E ;
voloshin-hypergraph_F00184	44	■ 0.926	$N(x) \cap N(y) \neq \emptyset$	$N(x) \cap N(y) \neq \emptyset$	If G is a simple graph and $N(x) \cap N(y) \neq \emptyset$, then each vertex from $N(x) \cap N(y)$ will be
voloshin-hypergraph_F00185	44	■ 0.926	$N(x) \cap N(y)$	$N(x) \cap N(y)$	If G is a simple graph and $N(x) \cap N(y) \neq \emptyset$, then each vertex from $N(x) \cap N(y)$ will be
voloshin-hypergraph_F00186	44	■ 1.000	$e^{\prime}$	$e^{\prime}$	At last, if G is not simple and had another edge $e^{\prime}$ connecting x and y , then $e^{\prime}$ becomes
voloshin-hypergraph_F00187	44	—	$G_{\{1\}} = G \cdot e$	$G_1 = G \cdot e$	
voloshin-hypergraph_F00188	44	■ 1.000	$C_{\{k\}}, k \geq 4$	$C_k, k \geq 4$	of contractions of the edges. For example, any $C_k, k \geq 4$ is contractible to $C_3 = K_3$, any tree
voloshin-hypergraph_F00189	44	■ 1.000	$C_{\{3\}} = K_{\{3\}}$	$C_3 = K_3$	of contractions of the edges. For example, any $C_k, k \geq 4$ is contractible to $C_3 = K_3$, any tree
voloshin-hypergraph_F00190	44	■ 1.000	$K_{\{2\}}$	K_2	with at least two vertices is contractible to K_2 , any K_n is contractible to K_m if $n \geq m$ and
voloshin-hypergraph_F00191	44	■ 1.000	$K_{\{m\}}$	K_m	with at least two vertices is contractible to K_2 , any K_n is contractible to K_m if $n \geq m$ and
voloshin-hypergraph_F00192	44	■ 1.000	$n \geq m$	$n \geq m$	with at least two vertices is contractible to K_2 , any K_n is contractible to K_m if $n \geq m$ and
voloshin-hypergraph_F00193	44	■ 1.000	$K_{\{4\}}$	K_4	so on. On the other hand, no C_k can be contracted to K_4 , no tree can be contracted to K_3 .
voloshin-hypergraph_F00194	44	■ 1.000	$K_{\{3\}}$	K_3	so on. On the other hand, no C_k can be contracted to K_4 , no tree can be contracted to K_3 .
voloshin-hypergraph_F00195	44	■ 1.000	$\eta(G)$	$\eta(G)$	number of the graph G , denoted by $\eta(G)$.
voloshin-hypergraph_F00196	44	■ 1.000	$ X = n$	$ X = n$	Taking the complement. Consider a simple graph $G = (X, E)$ with $ X = n$. The com-
voloshin-hypergraph_F00197	44	■ 1.000	$\bar{G}$	$\bar{G}$	plement of G denoted by $\bar{G}$ is the graph on the same vertex set X in which two vertices are
voloshin-hypergraph_F00198	44	■ 1.000	$\bar{G} = (X, E^{\prime})$	$\bar{G} = (X, E^{\prime})$	adjacent if and only if they are disjoint in the original graph G . In other words, $\bar{G} = (X, E^{\prime})$
voloshin-hypergraph_F00199	44	■ 1.000	$E^{\prime}$	$E^{\prime}$	where $E^{\prime}$ is such an edge set that $E \cup E^{\prime}$ forms the edge set of K_n . This is why $\bar{G}$ is called the

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voloshin-hypergraph_F00200	44	■ 1.000	$E \cup E^{\prime}$	$E \cup E'$	where E' is such an edge set that $E \cup E'$ forms the edge set of K_n . This is why $\overline{G}$ is called the
voloshin-hypergraph_F00201	44	■ 1.000	$\overline{\bar{G}}=G, \bar{E}_n=K_n$	$\overline{G} = G, \bar{E}_n = K_n$	complement of G . Evidently, $\overline{G} = G, \bar{E}_n = K_n$ and $\bar{K}_n = E_n$. Notice that P_4 is isomorphic
voloshin-hypergraph_F00202	44	■ 1.000	$\bar{K}_n=E_n$	$\bar{K}_n = E_n$	complement of G . Evidently, $\overline{\bar{G}} = G, \bar{E}_n = K_n$ and $\bar{K}_n = E_n$. Notice that P_4 is isomorphic
voloshin-hypergraph_F00203	44	■ 1.000	$\overline{P_4}, C_5$	$\overline{P_4}, C_5$	to $\overline{P_4}, C_5$ is isomorphic to $\overline{C_5}$, and $\overline{K_{r,s}} = \{K_r, K_s\}$. It may happen that G is connected graph
voloshin-hypergraph_F00204	44	■ 1.000	$\overline{C_5}$	$\overline{C_5}$	to $\overline{P_4}, C_5$ is isomorphic to $\overline{C_5}$, and $\overline{K_{r,s}} = \{K_r, K_s\}$. It may happen that G is connected graph
voloshin-hypergraph_F00205	44	■ 1.000	$\overline{K_{r,s}}=\left\{K_r, K_s\right\}$	$\overline{K_{r,s}} = \{K_r, K_s\}$	to $\overline{P_4}, C_5$ is isomorphic to $\overline{C_5}$, and $\overline{K_{r,s}} = \{K_r, K_s\}$. It may happen that G is connected graph
voloshin-hypergraph_F00206	45	■ 1.000	$C_4, W_5, K_5, K_{3,3}$	$C_4, W_5, K_5, K_{3,3}$	1. Implement a weak sequential (in any order) deletion of vertices of $C_4, W_5, K_5, K_{3,3}$ and
voloshin-hypergraph_F00207	45	■ 1.000	$C_5, W_5, K_{2,3}$	$C_5, W_5, K_{2,3}$	6. Find the Hadwiger number of any tree, $C_5, W_5, K_{2,3}$.
voloshin-hypergraph_F00208	45	■ 1.000	$C_3, C_4, C_5, C_6, K_5, K_{3,5}$	$C_3, C_4, C_5, C_6, K_5, K_{3,5}$	7. Construct the complement of $C_3, C_4, C_5, C_6, K_5, K_{3,5}$, Petersen graph, cube, P_7 , any tree on 6
voloshin-hypergraph_F00209	45	■ 1.000	P_7	P_7	7. Construct the complement of $C_3, C_4, C_5, C_6, K_5, K_{3,5}$, Petersen graph, cube, P_7 , any tree on 6
voloshin-hypergraph_F00210	45	■ 1.000	$E_5, 2 C_3$	$E_5, 2C_3$	vertices, $E_5, 2C_3$.
voloshin-hypergraph_F00211	45	■ 0.999	$\bar{C}_5 \cong C_5$	$\bar{C}_5 \cong C_5$	8. Show that $\bar{C}_5 \cong C_5$.
voloshin-hypergraph_F00212	45	■ 1.000	$\bar{C}_6$	$\bar{C}_6$	9. Show that $\bar{C}_6$ is isomorphic to a prism.
voloshin-hypergraph_F00213	45	■ 1.000	$G^{\prime}=\left(X^{\prime}, E^{\prime}\right)$	$G' = (X', E')$	Subgraphs. Let us have a graph $G = (X, E)$. Any graph $G' = (X', E')$ is called a
voloshin-hypergraph_F00214	45	■ 1.000	$X^{\prime} \subseteq X$	$X' \subseteq X$	subgraph of G if and only if $X' \subseteq X$, and $E' \subseteq E$. In such case, we write $G' \subseteq G$. Since E'
voloshin-hypergraph_F00215	45	■ 1.000	$E^{\prime} \subseteq E$	$E' \subseteq E$	subgraph of G if and only if $X' \subseteq X$, and $E' \subseteq E$. In such case, we write $G' \subseteq G$. Since E'
voloshin-hypergraph_F00216	45	■ 1.000	$G^{\prime} \subseteq G$	$G' \subseteq G$	subgraph of G if and only if $X' \subseteq X$, and $E' \subseteq E$. In such case, we write $G' \subseteq G$. Since E'
voloshin-hypergraph_F00217	45	■ 1.000	$X^{\prime}$	X'	contains only the elements of E , both ends of any edge from E' must be in X' . Therefore, G'
voloshin-hypergraph_F00218	45	■ 1.000	$X-X^{\prime}$	$X - X'$	can be obtained from G by strong deletion of vertices $X - X'$ (sequentially in any order or at
voloshin-hypergraph_F00219	45	■ 1.000	$E-E^{\prime}$	$E - E'$	once) and further weak deletion of remaining edges $E - E'$ (sequentially in any order or at
voloshin-hypergraph_F00220	45	■ 1.000	$\left\{x_3, x_5\right\}$	$\{x_3, x_5\}$	of vertex x_1 and weak deletion of edge $\{x_3, x_5\}$, and G_2 is obtained by strong deletion of
voloshin-hypergraph_F00221	46	■ 1.000	$Y \subseteq X$	$Y \subseteq X$	induced by a set $Y \subseteq X$ by G_Y .
voloshin-hypergraph_F00222	46	■ 1.000	G_Y	G_Y	induced by a set $Y \subseteq X$ by G_Y .
voloshin-hypergraph_F00223	46	■ 1.000	$k \geq 3$	$k \geq 3$	every induced cycle is isomorphic to the simple cycle C_k for some $k \geq 3$. For example, the
voloshin-hypergraph_F00224	46	■ 0.996	x_2, x_3, x_4, x_5, x_2	x_2, x_3, x_4, x_5, x_2	sequence of vertices x_2, x_3, x_4, x_5, x_2 , see Figure 1.28, forms the cycle of length 4, which is
voloshin-hypergraph_F00225	46	■ 1.000	$x_1, x_2, x_3, x_4, x_5, x_1$	$x_1, x_2, x_3, x_4, x_5, x_1$	(except the first and the last) is called simple . For example, the cycle $x_1, x_2, x_3, x_4, x_5, x_1$ is
voloshin-hypergraph_F00226	46	■ 1.000	$x_1, x_2, x_5, x_3, x_4, x_5, x_1$	$x_1, x_2, x_5, x_3, x_4, x_5, x_1$	simple, but the cycle $x_1, x_2, x_5, x_3, x_4, x_5, x_1$ is not simple because the vertex x_5 is used twice.

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voloshin-hypergraph_F00227	46	■ 1.000	$K_{\{1\}}, K_{\{2\}}, K_{\{3\}}, \ldots, K_{\{n\}}$	$K_1, K_2, K_3, \dots, K_n$	of $K_1, K_2, K_3, \dots, K_n$. They are called cliques . If we have a subgraph isomorphic to K_r
voloshin-hypergraph_F00228	46	■ 1.000	$K_{\{r\}}$	K_r	of $K_1, K_2, K_3, \dots, K_n$. They are called cliques . If we have a subgraph isomorphic to K_r
voloshin-hypergraph_F00229	46	■ 1.000	$K_{\{r+1\}}$	K_{r+1}	which is not contained in a subgraph isomorphic to K_{r+1} , then we say that K_r is the maximal
voloshin-hypergraph_F00230	47	■ 0.999	$\omega(G)$	$\omega(G)$	denoted by $\omega(G)$. Simply, $\omega(G)$ is the maximum number of pairwise adjacent vertices. For
voloshin-hypergraph_F00231	47	■ 0.998	$G, 1 \leq \omega(G) \leq n$	$G, 1 \leq \omega(G) \leq n$	any graph $G, 1 \leq \omega(G) \leq n$
voloshin-hypergraph_F00232	47	■ 1.000	$x_{\{2\}}, x_{\{3\}}, x_{\{5\}}$	x_2, x_3, x_5	maximal clique since it is not maximal; it is contained in K_3 induced by vertices x_2, x_3, x_5
voloshin-hypergraph_F00233	47	■ 0.958	$\omega(G)=3$	$\omega(G) = 3$	form a maximal clique. We conclude that $\omega(G) = 3$ and $\omega(G_1) = \omega(G_2) = 2$.
voloshin-hypergraph_F00234	47	■ 0.958	$\omega(\text{left}(G_{\{1\}}\text{right})=\omega(\text{left}(G_{\{2\}}\text{right})=2$	$\omega(G_1) = \omega(G_2) = 2$	form a maximal clique. We conclude that $\omega(G) = 3$ and $\omega(G_1) = \omega(G_2) = 2$.
voloshin-hypergraph_F00235	47	■ 1.000	$E_{\{1\}}, E_{\{2\}}, \ldots, E_{\{n\}}$	$E_1, E_2, \dots, E_n$	isomorphic to some of $E_1, E_2, \dots, E_n$. In a graph G , a subset of vertices which induces
voloshin-hypergraph_F00236	47	■ 1.000	$E_{\{k\}}$	E_k	a subgraph E_k is called the stable set , or independent set . There are also maximal by
voloshin-hypergraph_F00237	47	■ 0.801	$\alpha(G)$	$\alpha(G)$	G is called the stability number , denoted by $\alpha(G)$. Simply, $\alpha(G)$ is the maximum number
voloshin-hypergraph_F00238	47	■ 1.000	$G, 1 \leq \alpha(G) \leq n$	$G, 1 \leq \alpha(G) \leq n$	of pairwise disjoint vertices. For any graph $G, 1 \leq \alpha(G) \leq n$.
voloshin-hypergraph_F00239	47	■ 1.000	$\omega(G)=\alpha(\bar{G})$	$\omega(G) = \alpha(\bar{G})$	Observe that $\omega(G) = \alpha(\bar{G})$ and $\alpha(G) = \omega(\bar{G})$ because when taking the complement
voloshin-hypergraph_F00240	47	■ 1.000	$\alpha(G)=\omega(\bar{G})$	$\alpha(G) = \omega(\bar{G})$	Observe that $\omega(G) = \alpha(\bar{G})$ and $\alpha(G) = \omega(\bar{G})$ because when taking the complement
voloshin-hypergraph_F00241	47	■ 1.000	$\alpha(G)=2$	$\alpha(G) = 2$	form another stable set. Both are maximal by inclusion, and $\alpha(G) = 2$.
voloshin-hypergraph_F00242	47	■ 1.000	$\omega(\text{left}(G_{\{1\}}\text{right})=3$	$\omega(G_1) = 3$	vertices 2, 3, and 4 form a clique, $\omega(G_1) = 3$. In G_2 , vertices a, b , and d form a maximal by
voloshin-hypergraph_F00243	47	■ 1.000	a, b	a, b	vertices 2, 3, and 4 form a clique, $\omega(G_1) = 3$. In G_2 , vertices a, b , and d form a maximal by
voloshin-hypergraph_F00244	47	■ 1.000	a, b, e	a, b, e	inclusion stable set, and vertices a, b, e , and f also form a maximal by inclusion stable set;
voloshin-hypergraph_F00245	47	■ 1.000	f	f	inclusion stable set, and vertices a, b, e , and f also form a maximal by inclusion stable set;
voloshin-hypergraph_F00246	47	■ 0.982	$\alpha(\text{left}(G_{\{2\}}\text{right})=4$	$\alpha(G_2) = 4$	the last being maximal is maximum, so $\alpha(G_2) = 4$.
voloshin-hypergraph_F00247	47	■ 1.000	$T \subseteq X$	$T \subseteq X$	Transversals (vertex covers) . In a graph $G = (X, E)$, a subset of vertices $T \subseteq X$ is
voloshin-hypergraph_F00248	47	■ 1.000	$X \setminus T$	$X \setminus T$	called a transversal (vertex cover) if its complement $X \setminus T$ is an independent set. It means
voloshin-hypergraph_F00249	47	■ 1.000	T	T	that every edge of G has at least one end in T . The minimum cardinality of a transversal
voloshin-hypergraph_F00250	47	■ 1.000	$\tau(G)$	$\tau(G)$	of a graph G is called the transversal number and denoted by $\tau(G)$. It follows from the
voloshin-hypergraph_F00251	47	■ 1.000	$G=(X, E), X =n$	$G = (X, E), X = n$	Spanning subgraphs . Let us have a graph $G=(X, E), X =n$. Any subgraph $G' \subseteq G$
voloshin-hypergraph_F00252	47	■ 0.997	$G^{\prime}=\text{left}(X, E^{\prime}\text{right})$	$G' = (X, E')$	such that $G' = (X, E')$ is called a spanning subgraph . Thus spanning subgraphs have the
voloshin-hypergraph_F00253	48	■ 0.959	$\{1, 2\}$	$\{1, 2\}$	In Figure 1.29, if we weakly delete edges $\{1, 2\}$ and $\{2, 3\}$, then we obtain a spanning
voloshin-hypergraph_F00254	48	■ 0.959	$\{2, 3\}$	$\{2, 3\}$	In Figure 1.29, if we weakly delete edges $\{1, 2\}$ and $\{2, 3\}$, then we obtain a spanning

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voloshin-hypergraph_F00255	48	0.833	$\nu(G)$	$\nu(G)$	denoted by $\nu(G)$. If a graph G has a perfect matching, then clearly, n is an even number and
voloshin-hypergraph_F00256	48	0.955	$\nu=n / 2$	$\nu = n/2$	$\nu = n/2$.
voloshin-hypergraph_F00257	48	1.000	$\{c, d\}$	$\{c, d\}$	In graph G_2 , see Figure 1.29, edge $\{c, d\}$ forms a maximal by inclusion matching.
voloshin-hypergraph_F00258	48	1.000	$\{a, c\}$	$\{a, c\}$	However, a maximum matching is formed for example by edges $\{a, c\}$ and $\{d, e\}$, and
voloshin-hypergraph_F00259	48	1.000	$\{d, e\}$	$\{d, e\}$	However, a maximum matching is formed for example by edges $\{a, c\}$ and $\{d, e\}$, and
voloshin-hypergraph_F00260	48	1.000	$\nu\left(G_2\right)=2$	$\nu\left(G_2\right)=2$	$\nu\left(G_2\right)=2$.
voloshin-hypergraph_F00261	49	1.000	$P_{\{6\}}, C_{\{3\}}, C_{\{4\}}, C_{\{5\}}, W_{\{4\}}, W_{\{5\}}, K_{\{2,3\}}$	$P_6, C_3, C_4, C_5, W_4, W_5, K_{2,3}$	2. Write down the list of all cliques of $P_6, C_3, C_4, C_5, W_4, W_5, K_{2,3}$, cube and Petersen graph.
voloshin-hypergraph_F00262	49	1.000	$C_{\{7\}}$	C_7	10. Find the longest path in C_7 , cube, W_9 , Petersen graph.
voloshin-hypergraph_F00263	49	1.000	$W_{\{9\}}$	W_9	10. Find the longest path in C_7 , cube, W_9 , Petersen graph.
voloshin-hypergraph_F00264	49	1.000	$K_{\{4,4\}}$	$K_{4,4}$	11. Find the shortest and the longest cycles in $K_{4,4}$, cube, Petersen graph.
voloshin-hypergraph_F00265	49	1.000	$K_{\{6,9\}}$	$K_{6,9}$	12. Find a spanning tree in cube, $K_{6,9}$, Petersen graph.
voloshin-hypergraph_F00266	49	0.939	$v(G)$	$v(G)$	16. Find $\nu(G)$ for every graph drawn in this section and in Figure 1.21.
voloshin-hypergraph_F00267	49	0.999	α, ω, τ	α, ω, τ	17. Find α, ω, τ , and ν of cube, Petersen graph, $K_{999}, K_{1000}, C_{20}, C_{21}, W_{99}, W_{100}, P_n, K_n, C_n, W_n$,
voloshin-hypergraph_F00268	49	0.999	ν	ν	17. Find α, ω, τ , and ν of cube, Petersen graph, $K_{999}, K_{1000}, C_{20}, C_{21}, W_{99}, W_{100}, P_n, K_n, C_n, W_n$,
voloshin-hypergraph_F00269	49	0.999	$K_{\{999\}}, K_{\{1000\}}, C_{\{20\}}, C_{\{21\}}, W_{\{99\}}, W_{\{100\}}, P_{\{n\}}, K_{\{n\}}, C_{\{n\}}, W_{\{n\}}$	$K_{999}, K_{1000}, C_{20}, C_{21}, W_{99}, W_{100}, P_n, K_n, C_n, W_n$	17. Find α, ω, τ , and ν of cube, Petersen graph, <u>$K_{999}, K_{1000}, C_{20}, C_{21}, W_{99}, W_{100}, P_n, K_n, C_n, W_n$</u>
voloshin-hypergraph_F00270	49	1.000	$n \geq 2$	$n \geq 2$	for all integers $n \geq 2$.
voloshin-hypergraph_F00271	49	1.000	$W_{\{5\}}$	W_5	18. Find all non isomorphic minors of W_5 .
voloshin-hypergraph_F00272	50	0.988	$x, y \in X$	$x, y \in X$	simple connected graph and $x, y \in X$. If vertices x and y are not adjacent, deleting the set $X -$
voloshin-hypergraph_F00273	50	0.999	$\{x, y\}$	$\{x, y\}$	$\{x, y\}$ from G leaves only vertices x and y , i.e. a disconnected graph. Any set $S \subseteq X$ of vertices
voloshin-hypergraph_F00274	50	0.999	$S \subseteq X$	$S \subseteq X$	$\{x, y\}$ from G leaves only vertices x and y , i.e. a disconnected graph. Any set $S \subseteq X$ of vertices
voloshin-hypergraph_F00275	50	1.000	S	S	set of vertices deleting of which increases the number of connected components. So, if S is
voloshin-hypergraph_F00276	50	0.448	$\kappa(G)$	$\kappa(G)$	G and denoted by $\kappa(G)$. There is only one exception here, namely graph K_n . Since it has
voloshin-hypergraph_F00277	50	0.976	$\kappa\left(K_n\right)=n-1$	$\kappa\left(K_n\right)=n-1$	no separators, it is convenient to put by definition $\kappa\left(K_n\right)=n-1$. So, for any incomplete
voloshin-hypergraph_F00278	50	0.530	$G, \kappa(G)$	$G, \kappa(G)$	connected graph G , $\kappa(G)$ is the size of the smallest separator.
voloshin-hypergraph_F00279	50	0.953	$\kappa(G) \geq k$	$\kappa(G) \geq k$	connected if connectivity $\kappa(G) \geq k$. So, all graphs are 0-connected, connected graphs are
voloshin-hypergraph_F00280	50	0.898	$\kappa(G) \geq 2$	$\kappa(G) \geq 2$	1-connected, connected graphs having $\kappa(G) \geq 2$ are 2-connected and so on. Generally, if
voloshin-hypergraph_F00281	50	1.000	$(k-1)$	$(k-1)$	a graph G is k -connected, then it is $(k-1)$ -connected, $(k-2)$ -connected, and so on. But
voloshin-hypergraph_F00282	50	1.000	$(k-2)$	$(k-2)$	a graph G is k -connected, then it is $(k-1)$ -connected, $(k-2)$ -connected, and so on. But

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voloshin-hypergraph_ F00283	50	■ 1.000	$(k+1)$	$(k+1)$	$(k+1)$ -connected. The inclusion of connectivity classes is shown in Figure 1.32.
voloshin-hypergraph_ F00284	51	■ 0.865	$\{2,3,5,6\}$	$\{2,3,5,6\}$	Figure 1.33 shows a graph G that has many separators. Set $\{2,3,5,6\}$ is a $(1,4)$ -
voloshin-hypergraph_ F00287	51	■ 0.767	$\mathrm{K}(G)=2$	$K(G) = 2$	$(1,4)$ -separators both being minimal. It is seen from the figure that for this graph $\kappa(G) = 2$.
voloshin-hypergraph_ F00288	51	■ 1.000	$X_{\{1\}} \cup S$	$X_1 \cup S$	components induced by vertex sets X_1 and X_2 . The two subgraphs induced by $X_1 \cup S$ and
voloshin-hypergraph_ F00289	51	■ 1.000	$X_{\{2\}} \cup S$	$X_2 \cup S$	$X_2 \cup S$ are called derived subgraphs of the graph G with respect to separator S . We will
voloshin-hypergraph_ F00290	51	■ 1.000	$G_{X_{\{1\}} \cup S}$	$G_{X_1 \cup S}$	denote them by $G_{X_1 \cup S}$ and $G_{X_2 \cup S}$, or simply by G_1 and G_2 respectively, see Figure 1.34.
voloshin-hypergraph_ F00291	51	■ 1.000	$G_{X_{\{2\}} \cup S}$	$G_{X_2 \cup S}$	denote them by $G_{X_1 \cup S}$ and $G_{X_2 \cup S}$, or simply by G_1 and G_2 respectively, see Figure 1.34.
voloshin-hypergraph_ F00292	51	■ 1.000	$x \in S$	$x \in S$	Proof. If some $x \in S$ has no vertex in X_1 adjacent to it, then $S - \{x\}$ is also a separator of
voloshin-hypergraph_ F00293	51	■ 1.000	$S - \{x\}$	$S - \{x\}$	Proof. If some $x \in S$ has no vertex in X_1 adjacent to it, then $S - \{x\}$ is also a separator of
voloshin-hypergraph_ F00294	52	■ 0.965	$\{3,4\}$	$\{3,4\}$	two connected components produced by this separator are induced by vertex sets $\{3,4\}$ and
voloshin-hypergraph_ F00295	52	■ 0.695	$\{1,6\}$	$\{1,6\}$	$\{1,6\}$. The two derived subgraphs produced by separator $\{2,5\}$ are induced by vertex sets
voloshin-hypergraph_ F00296	52	■ 1.000	$k \geq 2$	$k \geq 2$	deletion of S leaves any number $k \geq 2$ connected components induced by $X_1, X_2, \dots, X_k$.
voloshin-hypergraph_ F00297	52	■ 1.000	$X_{\{1\}}, X_{\{2\}}, \dots, X_{\{k\}}$	$X_1, X_2, \dots, X_k$	deletion of S leaves any number $k \geq 2$ connected components induced by $X_1, X_2, \dots, X_k$.
voloshin-hypergraph_ F00298	52	■ 1.000	$G_{\{1\}}=G_{X_{\{1\}} \cup S}, G_{\{2\}}=G_{X_{\{2\}} \cup S}, \dots$	$G_1 = G_{X_1 \cup S}, G_2 = G_{X_2 \cup S}, \dots$	The derived subgraphs then are: $G_1 = G_{X_1 \cup S}, G_2 = G_{X_2 \cup S}, \dots$, and $G_k = G_{X_k \cup S}$.
voloshin-hypergraph_ F00299	52	■ 1.000	$G_{\{k\}}=G_{X_{\{k\}} \cup S}$	$G_k = G_{X_k \cup S}$	The derived subgraphs then are: $G_1 = G_{X_1 \cup S}, G_2 = G_{X_2 \cup S}, \dots$, and $G_k = G_{X_k \cup S}$.
voloshin-hypergraph_ F00300	53	■ 0.822	x, y	x, y	and y , the minimum number of vertices in an (x,y) -separator equals the maximum number
voloshin-hypergraph_ F00301	53	■ 0.783	$\mathbf{K}(G)$	$\mathbf{K}(G)$	3. For graph G in Figure 1.36, find the connectivity $\kappa(G)$.
voloshin-hypergraph_ F00302	53	■ 1.000	$k \geq 0$	$k \geq 0$	4. For which integer $k \geq 0$, graph G in Figure 1.36 is k -connected?
voloshin-hypergraph_ F00303	53	■ 1.000	$\{1,3,4,7,8\}$	$\{1,3,4,7,8\}$	5. For separator $\{1,3,4,7,8\}$, construct the derived subgraphs.
voloshin-hypergraph_ F00304	53	■ 1.000	$\{1,8,3,4,6\}$	$\{1,8,3,4,6\}$	6. For separator $\{1,8,3,4,6\}$, construct the derived subgraphs.
voloshin-hypergraph_ F00305	53	■ 1.000	$\{5,6,7,8,2\}$	$\{5,6,7,8,2\}$	7. For separator $\{5,6,7,8,2\}$, construct the derived subgraphs
voloshin-hypergraph_ F00306	55	■ 1.000	$m(G)=n(G)-1$	$m(G) = n(G) - 1$	2. G is connected and $m(G) = n(G) - 1$;
voloshin-hypergraph_ F00307	55	■ 1.000	$1 \Rightarrow 2 \text{ . } G$	$1. \Rightarrow 2.G$	Proof. $1. \Rightarrow 2.$ G is connected by the definition of tree. Prove the equality by induction
voloshin-hypergraph_ F00308	55	■ 1.000	$n=1,2$	$n = 1, 2$	on n . For $n = 1, 2$ the statement is trivial. Assume $n(G) > 2$. Since G is a tree it has a
voloshin-hypergraph_ F00309	55	■ 1.000	$n(G)>2$	$n(G) > 2$	on n . For $n = 1, 2$ the statement is trivial. Assume $n(G) > 2$. Since G is a tree it has a
voloshin-hypergraph_ F00310	55	■ 1.000	$m\left(G_{\{1\}\backslash\mathrm{right}}\right)=n\left(G_{\{1\}\backslash\mathrm{right}}\right)-1$	$m(G_1) = n(G_1) - 1$	$m(G_1) = n(G_1) - 1$. Now return deleted vertex with an edge and reconstruct G . Evidently,
voloshin-hypergraph_ F00311	55	■ 1.000	$m(G)=m\left(G_{\{1\}\backslash\mathrm{right}}\right)+1$	$m(G) = m(G_1) + 1$	$m(G) = m(G_1) + 1$, and $n(G) = n(G_1) + 1$, and the implication follows.
voloshin-hypergraph_ F00312	55	■ 1.000	$n(G)=n\left(G_{\{1\}\backslash\mathrm{right}}\right)+1$	$n(G) = n(G_1) + 1$	$n(G) = m(G_1) + 1$, and $n(G) = n(G_1) + 1$, and the implication follows.

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voloshin-hypergraph_F00313	55	■ 1.000	$e=\{x, y\}$	$e = \{x, y\}$	Weakly delete an edge $e = \{x, y\}$ of this cycle; $G - e$ is connected since (x, y) -path remains.
voloshin-hypergraph_F00314	55	■ 1.000	$G - e$	$G - e$	Weakly delete an edge $e = \{x, y\}$ of this cycle; $G - e$ is connected since (x, y) -path remains.
voloshin-hypergraph_F00315	55	■ 0.882	$\Rightarrow 2 \cdot m = n - 1$	$\Rightarrow 2 \cdot m = n - 1$	it is connected, it is a tree, and by 1. $\Rightarrow 2 \cdot m = n - 1$, a contradiction to the fact that we
voloshin-hypergraph_F00316	55	■ 0.999	$k > 1$	$k > 1$	$k > 1$ components $G_1, G_2, \dots, G_k$, every G_i is a tree. By 1. $\Rightarrow 2 \cdot m_i = n_i - 1$ for each G_i .
voloshin-hypergraph_F00317	55	■ 0.999	$G_{\{1\}}, G_{\{2\}}, \dots, G_{\{k\}}$	$G_1, G_2, \dots, G_k$	$k > 1$ components $G_1, G_2, \dots, G_k$, every G_i is a tree. By 1. $\Rightarrow 2 \cdot m_i = n_i - 1$ for each G_i .
voloshin-hypergraph_F00318	55	■ 0.999	$G_{\{i\}}$	G_i	$k > 1$ components $G_1, G_2, \dots, G_k$, every G_i is a tree. By 1. $\Rightarrow 2 \cdot m_i = n_i - 1$ for each G_i .
voloshin-hypergraph_F00319	55	■ 0.999	$1 \cdot \Rightarrow 2 \cdot m_{\{i\}} = n_{\{i\}} - 1$	$1 \cdot \Rightarrow 2 \cdot m_i = n_i - 1$	$k > 1$ components $G_1, G_2, \dots, G_k$, every G_i is a tree. By 1. $\Rightarrow 2 \cdot m_i = n_i - 1$ for each G_i .
voloshin-hypergraph_F00320	55	■ 0.965	$m(G) = m_{\{1\}} + m_{\{2\}} + \dots + m_{\{k\}} = \left(n_{\{1\}} - 1 \right) + \left(n_{\{2\}} - 1 \right) + \dots + \left(n_{\{k\}} - 1 \right) = n - k$	$m(G) = m_1 + m_2 + \dots + m_k = (n_1 - 1) + (n_2 - 1) + \dots + (n_k - 1) = n - k$	Therefore $m(G) = m_1 + m_2 + \dots + m_k = (n_1 - 1) + (n_2 - 1) + \dots + (n_k - 1) = n - k$. So
voloshin-hypergraph_F00321	55	■ 1.000	$m = n - 1 = n - k$	$m = n - 1 = n - k$	we obtain $m = n - 1 = n - k$ which implies $k = 1$ and G is connected. It means that there
voloshin-hypergraph_F00322	55	■ 1.000	$k = 1$	$k = 1$	we obtain $m = n - 1 = n - k$ which implies $k = 1$ and G is connected. It means that there
voloshin-hypergraph_F00323	56	■ 0.996	$C_{\{l\}}$	C_l	otherwise, if we obtain two cycles C_k and C_l , the subgraph $(C_k \cup C_l) - \{x, y\}$ forms the third
voloshin-hypergraph_F00324	56	■ 0.996	$\left(C_{\{k\}} \cup C_{\{l\}} \right) - \{x, y\}$	$(C_k \cup C_l) - \{x, y\}$	otherwise, if we obtain two cycles C_k and C_l , the subgraph $(C_k \cup C_l) - \{x, y\}$ forms the third
voloshin-hypergraph_F00325	56	■ 1.000	$\Lambda(G) = m(G) -$	$\Lambda(G) = m(G) -$	Let G be a simple graph having k connected components. The number $\Lambda(G) = m(G) -$
voloshin-hypergraph_F00326	56	■ 0.516	$n(G) + k$	$n(G) + k$	$n(G) + k$ is called the cyclomatic number of G . If G is connected, then $k = 1$ and $\Lambda =$
voloshin-hypergraph_F00327	56	■ 0.516	$\wedge =$	$\wedge =$	$n(G) + k$ is called the cyclomatic number of G . If G is connected, then $k = 1$ and $\Lambda =$
voloshin-hypergraph_F00328	56	■ 1.000	$m - n + 1 = m - (n - 1)$	$m - n + 1 = m - (n - 1)$	$m - n + 1 = m - (n - 1)$. Theorem 2.1.1 in fact states that $n - 1$ is the minimum number
voloshin-hypergraph_F00329	56	■ 1.000	$m - (n - 1)$	$m - (n - 1)$	$m - (n - 1)$ shows how many extra edges graph G has. Starting from any spanning tree, one
voloshin-hypergraph_F00330	56	■ 0.743	Λ	Λ	can sequentially add remaining Λ edges to reconstruct G . Every such edge forms precisely
voloshin-hypergraph_F00331	56	■ 0.535	$\wedge(G) = 0$	$\wedge(G) = 0$	Corollary 2.1.1 $\wedge(G) = 0$ if and only if G is a forest.
voloshin-hypergraph_F00332	56	■ 0.979	$\Lambda(G) =$	$\Lambda(G) =$	edges form a spanning tree, the dashed edges show the chords. Correspondingly, $\Lambda(G) =$
voloshin-hypergraph_F00333	56	■ 1.000	$m - n + 1 = 7 - 5 + 1 = 3$	$m - n + 1 = 7 - 5 + 1 = 3$	$m - n + 1 = 7 - 5 + 1 = 3$. One can see that every chord forms exactly one elementary cycle
voloshin-hypergraph_F00334	56	■ 1.000	n^{n-2}	n^{n-2}	$n \geq 1$, equals n^{n-2} .
voloshin-hypergraph_F00335	57	■ 1.000	$n = 3$	$n = 3$	that there are total n^{n-2} trees for every n . For example, if $n = 3$, then there are three
voloshin-hypergraph_F00336	57	■ 1.000	$E_{\{n\}}, C_{\{n\}}, K_{\{n\}}, W_{\{n\}}$	E_n, C_n, K_n, W_n	1. Find the cyclomatic number of E_n, C_n, K_n, W_n , prism, cube and the Petersen graph.
voloshin-hypergraph_F00337	57	■ 1.000	$C_{\{6\}}, K_{\{4\}}, W_{\{6\}}$	C_6, K_4, W_6	3. Describe the procedure for finding all spanning trees in C_6, K_4, W_6 .
voloshin-hypergraph_F00338	57	■ 1.000	$K_{\{n\}}, W_{\{n\}}, n \geq 5$	$K_n, W_n, n \geq 5$	2. Generate all spanning trees in $K_n, W_n, n \geq 5$.
voloshin-hypergraph_F00339	57	■ 1.000	$d(x, y)$	$d(x, y)$	Let $G = (X, E)$ be a graph, $x, y \in X$. The distance from x to y denoted by $d(x, y)$ is the
voloshin-hypergraph_F00340	57	■ 0.731	$d(x, y) = \infty$	$d(x, y) = \infty$	length of the shortest (x, y) -path. If there is no such path in G , then $d(x, y) = \infty$; evidently,

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voloshin-hypergraph_F00341	57	■ 1.000	$d(x, Y) = \min_{y \in Y} d(x, y)$	$d(x, Y) = \min_{y \in Y} d(x, y)$	defined as $d(x, Y) = \min_{y \in Y} d(x, y)$. It is the shortest distance between x and any vertex of
voloshin-hypergraph_F00342	57	■ 1.000	Y	Y	—
voloshin-hypergraph_F00343	57	■ 1.000	$x, y, z \in X$	$x, y, z \in X$	For all $x, y, z \in X$, the following properties of distances hold:
voloshin-hypergraph_F00344	57	■ 1.000	$d(x, y) \geq 0$	$d(x, y) \geq 0$	1. $d(x, y) \geq 0$, and $d(x, x) = 0$;
voloshin-hypergraph_F00345	57	■ 1.000	$d(x, x) = 0$	$d(x, x) = 0$	1. $d(x, y) \geq 0$, and $d(x, x) = 0$;
voloshin-hypergraph_F00346	57	■ 1.000	$d(x, y) = d(y, x)$	$d(x, y) = d(y, x)$	2. $d(x, y) = d(y, x)$;
voloshin-hypergraph_F00347	57	■ 1.000	$d(x, y) \leq d(x, z) + d(z, y)$	$d(x, y) \leq d(x, z) + d(z, y)$	3. $d(x, y) \leq d(x, z) + d(z, y)$ (triangle inequality).
voloshin-hypergraph_F00348	57	■ 0.999	$\operatorname{diam}(G)$	$\operatorname{diam}(G)$	The diameter of G denoted by $\operatorname{diam}(G)$ is $\max_{x, y \in X} d(x, y)$; in other words it is the
voloshin-hypergraph_F00349	57	■ 0.999	$\max_{x, y \in X} d(x, y)$	$\max_{x, y \in X} d(x, y)$	The diameter of G denoted by $\operatorname{diam}(G)$ is $\max_{x, y \in X} d(x, y)$; in other words it is the
voloshin-hypergraph_F00350	57	■ 0.876	$d(x, y) = \operatorname{diam}(G)$	$d(x, y) = \operatorname{diam}(G)$	number. An (x, y) -path for which $d(x, y) = \operatorname{diam}(G)$ is called a diametral path . There may
voloshin-hypergraph_F00351	57	■ 1.000	$N_{\infty}(x)$	$N_{\infty}(x)$	be many diametral paths in a graph. Let $N_{\infty}(x)$ denote the set of farthest vertices from vertex
voloshin-hypergraph_F00352	57	■ 1.000	$y \in N_{\infty}(x)$	$y \in N_{\infty}(x)$	x . It means that if $y \in N_{\infty}(x)$ and $z \notin N_{\infty}(x)$, then $d(x, z) < d(x, y)$. The distance between
voloshin-hypergraph_F00353	57	■ 1.000	$z \notin N_{\infty}(x)$	$z \notin N_{\infty}(x)$	x . It means that if $y \in N_{\infty}(x)$ and $z \notin N_{\infty}(x)$, then $d(x, z) < d(x, y)$. The distance between
voloshin-hypergraph_F00354	57	■ 1.000	$d(x, z) < d(x, y)$	$d(x, z) < d(x, y)$	x . It means that if $y \in N_{\infty}(x)$ and $z \notin N_{\infty}(x)$, then $d(x, z) < d(x, y)$. The distance between
voloshin-hypergraph_F00355	58	■ 1.000	z	z	Proof. Let $y \in N_{\infty}(x)$. Consider (x, y) -path and a vertex z on this path adjacent to y , see
voloshin-hypergraph_F00358	58	■ 0.570	$d(x, y) = \operatorname{diam}(T)$	$d(x, y) = \operatorname{diam}(T)$	Proof. Consider x, y such that $d(x, y) = \operatorname{diam}(T)$. Evidently, $x \in N_{\infty}(y)$ and $y \in N_{\infty}(x)$,
voloshin-hypergraph_F00359	58	■ 0.570	$x \in N_{\infty}(y)$	$x \in N_{\infty}(y)$	Proof. Consider x, y such that $d(x, y) = \operatorname{diam}(T)$. Evidently, $x \in N_{\infty}(y)$ and $y \in N_{\infty}(x)$,
voloshin-hypergraph_F00360	58	■ 1.000	K_1	K_1	Theorem 2.2.2 (Jordan, 1869) <i>The center of a tree is either K_1 or K_2.</i>
voloshin-hypergraph_F00361	58	■ 0.998	$T_1, T_2, T_3, \dots$	$T_1, T_2, T_3, \dots$	obtain a sequence of trees $T_1, T_2, T_3, \dots$, the last being K_2 or K_1 what is the center.
voloshin-hypergraph_F00362	58	■ 0.941	$\operatorname{diam}(T)$	$\operatorname{diam}(T)$	prove that $\operatorname{diam}(T)$ can be found in the following two steps: 1) start at any vertex z and find
voloshin-hypergraph_F00363	58	■ 0.991	$x \in N_{\infty}(z)$	$x \in N_{\infty}(z)$	any $x \in N_{\infty}(z)$; 2) repeat the procedure for x and find any $y \in N_{\infty}(x)$: $\operatorname{diam}(T) = d(x, y)$.
voloshin-hypergraph_F00364	58	■ 0.991	$y \in N_{\infty}(x) : \operatorname{diam}(T) = d(x, y)$	$y \in N_{\infty}(x) : \operatorname{diam}(T) = d(x, y)$	any $x \in N_{\infty}(z)$; 2) repeat the procedure for x and find any $y \in N_{\infty}(x)$: $\operatorname{diam}(T) = d(x, y)$.
voloshin-hypergraph_F00365	59	■ 1.000	$\mathcal{D}$	$\mathcal{D}$	Sometimes, as in this section, we use letters $\mathcal{D}$ for an edge set and D for its elements
voloshin-hypergraph_F00366	59	■ 1.000	D	D	Sometimes, as in this section, we use letters $\mathcal{D}$ for an edge set and D for its elements
voloshin-hypergraph_F00367	59	■ 1.000	$G = (X, \mathcal{D})$	$G = (X, \mathcal{D})$	because we need letter E for another purpose. A graph $G = (X, \mathcal{D})$ is called weighted
voloshin-hypergraph_F00368	59	■ 1.000	$D \in \mathcal{D}$	$D \in \mathcal{D}$	if each edge $D \in \mathcal{D}$ is assigned a positive real number $w(D)$ called the weight of edge
voloshin-hypergraph_F00369	59	■ 1.000	$w(D)$	$w(D)$	if each edge $D \in \mathcal{D}$ is assigned a positive real number $w(D)$ called the weight of edge
voloshin-hypergraph_F00370	59	■ 1.000	$T = (X, E)$	$T = (X, E)$	capacity, resistance, probability, etc. Consider any spanning tree $T = (X, E)$ of G . The

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voloshin-hypergraph_F08371	59	■ 1.000	$w(T)$	$w(T)$	weight $w(T)$ of tree T is the sum of weights of all edges of T . Different spanning trees may
voloshin-hypergraph_F08372	60	■ 0.991	$T^* \neq T$	$T^* \neq T$	Suppose T is not an optimal spanning tree, i.e. there exist a spanning tree $T^* \neq T$ of
voloshin-hypergraph_F08373	60	■ 1.000	$w\left(T^*\right)<w(T)$	$w\left(T^*\right)<w(T)$	minimum weight such that $w(T^*) < w(T)$. Let D be the first edge in the ordering that was
voloshin-hypergraph_F08374	60	■ 1.000	T^*	T^*	chosen for T and is missing in T^* . Adding D to T^* creates a unique cycle. Since T has
voloshin-hypergraph_F08375	60	■ 1.000	$D^{\{\prime\}}$	D'	no cycles, this cycle contains an edge D' which is not in T . Construct the spanning tree
voloshin-hypergraph_F08376	60	■ 1.000	$T^*+D-D^{\{\prime\}}$	T^*+D-D'	T^*+D-D' . Now notice that $w(D) \leq w(D')$, and therefore $w(T^*+D-D') = w(T^*) +$
voloshin-hypergraph_F08377	60	■ 1.000	$w(D) \leq w\left(D^{\{\prime\}}\right)$	$w(D) \leq w\left(D'\right)$	T^*+D-D' . Now notice that $w(D) \leq w(D')$, and therefore $w(T^*+D-D') = w(T^*) +$
voloshin-hypergraph_F08378	60	■ 1.000	$w\left(T^*+D-D^{\{\prime\}}\right)=w\left(T^*\right)+$	$w\left(T^*+D-D'\right)=w\left(T^*\right)+$	T^*+D-D' . Now notice that $w(D) \leq w(D')$, and therefore $w(T^*+D-D') = w(T^*) +$
voloshin-hypergraph_F08379	60	■ 1.000	$w(D)-w\left(D^{\{\prime\}}\right) \leq w\left(T^*\right)$	$w(D)-w\left(D'\right) \leq w\left(T^*\right)$	$w(D)-w(D') \leq w(T^*)$. We obtain a new spanning tree of the minimum weight which has
voloshin-hypergraph_F08380	60	■ 1.000	$w>0$	$w>0$	4. Given weighted complete graph K_n with all weights equal $w>0$, how many minimum
voloshin-hypergraph_F08381	61	■ 0.973	B	B	Proof. $\Rightarrow$ Let G be a bipartite graph with parts A and B , and C_k be any cycle in it with a
voloshin-hypergraph_F08382	61	■ 1.000	$x \in A$	$x \in A$	vertex $x \in A$. Traverse C_k starting at x in any direction. Since G is bipartite, each time we
voloshin-hypergraph_F08383	61	■ 1.000	x, k	x, k	alternate parts A and B . Since we end at x , k is even.
voloshin-hypergraph_F08384	61	■ 0.999	$\{x\}$	$\{x\}$	any $x \in X$. Denote $\{x\}$ by N_0 , neighborhood $N(x)$ by N_1 , vertices at distance 2 from x by
voloshin-hypergraph_F08385	61	■ 0.999	$N_{\{0\}}$	N_0	any $x \in X$. Denote $\{x\}$ by N_0 , neighborhood $N(x)$ by N_1 , vertices at distance 2 from x by
voloshin-hypergraph_F08386	61	■ 0.999	$N_{\{1\}}$	N_1	any $x \in X$. Denote $\{x\}$ by N_0 , neighborhood $N(x)$ by N_1 , vertices at distance 2 from x by
voloshin-hypergraph_F08387	61	■ 0.994	$N_{\{2\}}$	N_2	N_2 , vertices at distance 3 from x by N_3 and so on, The last set in this sequence is $N_{\infty}(x)$
voloshin-hypergraph_F08388	61	■ 0.994	$N_{\{3\}}$	N_3	N_2 , vertices at distance 3 from x by N_3 and so on, The last set in this sequence is $N_{\infty}(x)$
voloshin-hypergraph_F08389	61	■ 1.000	$N_{\{k\}}$	N_k	denoted by N_k . We obtain the following partition of X :
voloshin-hypergraph_F08390	61	■ 0.979	$N_{\{i\}} \cap N_{\{j\}}=\emptyset, i \neq j$	$N_i \cap N_j = \emptyset, i \neq j$	Observe that $N_i \cap N_j = \emptyset, i \neq j$. Moreover, any edge of G connects vertices either from
voloshin-hypergraph_F08391	61	■ 1.000	$N_{\{i\}}$	N_i	the same N_i or from consecutive sets.
voloshin-hypergraph_F08392	61	■ 1.000	$X=A \cup B$	$X=A \cup B$	B all sets with odd indices, so $X=A \cup B$. It remains to prove that there are no edges of G
voloshin-hypergraph_F08393	61	■ 1.000	$e=\{y, z\}$	$e=\{y, z\}$	inside any of sets N_i . By contrary, assume there is an edge $e=\{y, z\}$, such that $y, z \in N_i$ for
voloshin-hypergraph_F08394	61	■ 1.000	$y, z \in N_{\{i\}}$	$y, z \in N_i$	inside any of sets N_i . By contrary, assume there is an edge $e=\{y, z\}$, such that $y, z \in N_i$ for
voloshin-hypergraph_F08395	61	■ 0.926	$i>0$	$i>0$	some $i>0$, see Figure 2.5. Consider the shortest (x,y) -path and (x,z) -path. They may have
voloshin-hypergraph_F08396	61	■ 0.926	(x, z)	(x, z)	some $i>0$, see Figure 2.5. Consider the shortest (x,y) -path and (x,z) -path. They may have
voloshin-hypergraph_F08397	61	■ 1.000	$x^{\{\prime\}}$	x'	path. Let x' be the last common vertex counting from x . Then (x',y) -path and (x',z) -path
voloshin-hypergraph_F08398	61	■ 1.000	$\left(x^{\{\prime\}}, y\right)$	(x', y)	path. Let x' be the last common vertex counting from x . Then (x',y) -path and (x',z) -path

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voloshin-hypergraph_F00399	61	■ 1.000	$\left(x^{\{\prime\}}, z\right)$	(x', z)	path. Let x' be the last common vertex counting from x . Then (x', y) -path and (x', z) -path
voloshin-hypergraph_F00400	61	■ 1.000	l	l	have the same length, say l , and no common vertices other than x' . Therefore, these paths
voloshin-hypergraph_F00401	61	■ 1.000	C_{2l+1}	C_{2l+1}	and edge e form a cycle C_{2l+1} , a contradiction.
voloshin-hypergraph_F00402	61	■ 0.993	$i=0$	$i = 0$	1. Start at arbitrary unmarked vertex and mark it with 0; set $i = 0$;
voloshin-hypergraph_F00403	61	■ 0.992	$i+1$	$i + 1$	2. Mark by $i + 1$ all vertices which are not marked and adjacent to vertices marked i ;
voloshin-hypergraph_F00404	62	■ 1.000	$i:=i+1$	$i := i + 1$	3. If there are unmarked vertices adjacent to marked vertices, set $i:=i+1$ and go to
voloshin-hypergraph_F00405	62	■ 1.000	$N(S)$	$N(S)$	neighborhood $N(S)$ of a subset $S \subseteq X$ is the union of all neighborhoods of the vertices
voloshin-hypergraph_F00406	62	■ 1.000	$G=(X, Y ; E)$	$G = (X, Y; E)$	If $G = (X, Y; E)$ is a bipartite graph, then the neighborhood of any subset of X is in
voloshin-hypergraph_F00407	62	■ 1.000	$N_{\{G\}}(S)$	$N_G(S)$	Y , and the neighborhood of any subset of Y is in X . We will use the notation $N_G(S)$ to
voloshin-hypergraph_F00408	63	■ 1.000	$ S $	$ S $	has at least $ S $ neighbors in Y , i.e., $ N_G(S) \geq S $.
voloshin-hypergraph_F00409	63	■ 1.000	$\left N_{\{G\}}(S)\right \geq S $	$ N_G(S) \geq S $	has at least $ S $ neighbors in Y , i.e., $ N_G(S) \geq S $.
voloshin-hypergraph_F00410	63	■ 1.000	$ X $	$ X $	induction on $ X $. If $ X = 1$, then by the definition of bipartite graph, the single vertex x of
voloshin-hypergraph_F00411	63	■ 1.000	$ X =1$	$ X = 1$	induction on $ X $. If $ X =1$, then by the definition of bipartite graph, the single vertex x of
voloshin-hypergraph_F00412	63	■ 1.000	$ X >1$	$ X > 1$	Let now $ X > 1$ and the theorem be true for all bipartite graphs with the first part on
voloshin-hypergraph_F00413	63	■ 1.000	$< X $	$< X $	$< X $ vertices. There are two cases to consider.
voloshin-hypergraph_F00414	63	■ 1.000	$S \subset X, S \neq X$	$S \subset X, S \neq X$	Case 1: for any subset $S \subset X, S \neq X$,
voloshin-hypergraph_F00415	63	■ 1.000	$G^{\{\prime\}}=\left(X^{\{\prime\}}, Y^{\{\prime\}} ; E^{\{\prime\}}\right)$	$G' = (X', Y'; E')$	the bipartite graph obtained by $G' = (X', Y'; E')$. Evidently, $ X' = X - 1 < X $. Since
voloshin-hypergraph_F00416	63	■ 1.000	$\left X^{\{\prime\}}\right = X -1< X $	$ X' = X - 1 < X $	the bipartite graph obtained by $G' = (X', Y'; E')$. Evidently, $ X' = X - 1 < X $. Since
voloshin-hypergraph_F00417	63	■ 1.000	$S^{\{\prime\}} \subseteq X^{\{\prime\}}$	$S' \subseteq X'$	$S' \subseteq X'$, the following holds:
voloshin-hypergraph_F00418	63	■ 1.000	$S_{\{0\}} \subset X, S_{\{0\}} \neq X$	$S_0 \subset X, S_0 \neq X$	Case 2: there exists a subset $S_0 \subset X, S_0 \neq X$, such that
voloshin-hypergraph_F00419	63	■ 1.000	$G_{\{A\}}$	G_A	and consider two induced bipartite subgraphs, G_A and G_B .
voloshin-hypergraph_F00420	63	■ 1.000	$G_{\{B\}}$	G_B	and consider two induced bipartite subgraphs, G_A and G_B .
voloshin-hypergraph_F00421	63	■ 1.000	$S \subseteqq S_{\{0\}}$	$S \subseteq S_0$	In G_A , for any subset $S \subseteq S_0$, we have that $N_{G_A}(S) = N_G(S)$. Therefore, $ S \leq N_{G_A}(S) $.
voloshin-hypergraph_F00422	63	■ 1.000	$N_{\{G_{\{A\}}\}}(S)=N_{\{G\}}(S)$	$N_{G_A}(S) = N_G(S)$	In G_A , for any subset $S \subseteq S_0$, we have that $N_{G_A}(S) = N_G(S)$. Therefore, $ S \leq N_{G_A}(S) $.
voloshin-hypergraph_F00423	63	■ 1.000	$ S \leq\left N_{\{G_{\{A\}}\}}(S)\right $	$ S \leq N_{G_A}(S) $	In G_A , for any subset $S \subseteq S_0$, we have that $N_{G_A}(S) = N_G(S)$. Therefore, $ S \leq N_{G_A}(S) $.
voloshin-hypergraph_F00424	63	■ 1.000	$S_{\{0\}}$	S_0	By the induction hypothesis, G_A has a matching that covers S_0 .
voloshin-hypergraph_F00425	63	■ 1.000	$S \subseteqq X-S_{\{0\}}$	$S \subseteq X - S_0$	In G_B , for any subset $S \subseteq X - S_0$, we have that

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voloshin-hypergraph_F00426	64	■ 1.000	$\left S_{\{0\}}\right =\left N_{\{G\}}\left(S_{\{0\}}\right)\right $	$ S_0 = N_G(S_0) $	Since $ S_0 = N_G(S_0) $ by (2.2), we obtain that $ S \leq N_{G_B}(S) $. Hence by the induction hy-
voloshin-hypergraph_F00427	64	■ 1.000	$ S \leq \left N_{\{G_B\}}(S)\right $	$ S \leq N_{G_B}(S) $	Since $ S_0 = N_G(S_0) $ by (2.2), we obtain that $ S \leq N_{G_B}(S) $. Hence by the induction hy-
voloshin-hypergraph_F00428	64	■ 1.000	$X-S_{\{0\}}$	$X - S_0$	pothesis, graph G_B has a matching that covers $X - S_0$.
voloshin-hypergraph_F00429	64	■ 0.746	$k \geq 1$	$k \geq 1$	Proof. Let $G = (X, Y; E)$ be a k -regular ($k \geq 1$) bipartite graph. Counting the degrees of
voloshin-hypergraph_F00430	64	■ 1.000	$k X =k Y $	$k X = k Y $	vertices in X and in Y leads to the equality $k X = k Y $ what implies $ X = Y $. It means
voloshin-hypergraph_F00431	64	■ 1.000	$ X = Y $	$ X = Y $	vertices in X and in Y leads to the equality $k X = k Y $ what implies $ X = Y $. It means
voloshin-hypergraph_F00432	64	■ 1.000	$i=k S $	$i = k S $	Let $S \subseteq X$. Since G is k -regular, the number i of edges from S to $N_G(S)$ is $i = k S $. Since
voloshin-hypergraph_F00433	64	■ 1.000	$k, i \leq k\left N_{\{G\}}(S)\right $	$k, i \leq k N_G(S) $	each vertex from $N_G(S)$ has degree k , $i \leq k N_G(S) $; therefore, for any $S \subseteq X$, $ N_G(S) \geq S $.
voloshin-hypergraph_F00434	64	■ 1.000	$S \subseteq X, \left N_{\{G\}}(S)\right \geq S $	$S \subseteq X, N_G(S) \geq S $	each vertex from $N_G(S)$ has degree k , $i \leq k N_G(S) $; therefore, for any $\boxed{S \subseteq X, N_G(S) \geq S }$.
voloshin-hypergraph_F00435	64	■ 0.987	$G, \tau(G) \geq v(G)$	$G, \tau(G) \geq v(G)$	any graph $\boxed{G, \tau(G) \geq v(G)}$.
voloshin-hypergraph_F00436	65	■ 1.000	$T \subseteq(X \cup Y)$	$T \subseteq (X \cup Y)$	Proof. Let $G = (X, Y; E)$ be a bipartite graph and $T \subseteq (X \cup Y)$ be a minimum transversal
voloshin-hypergraph_F00437	65	■ 1.000	$\tau(G)= T $	$\tau(G) = T $	of G . We will construct a matching of size $\tau(G) = T $ what proves the theorem.
voloshin-hypergraph_F00438	65	■ 0.999	$X \cap T=A$	$X \cap T = A$	Let $X \cap T = A$ and $Y \cap T = B$, and $G_1 = G_{A \cup(Y \setminus B)}$ and $G_2 = G_{B \cup(X \setminus A)}$, see Figure 2.7.
voloshin-hypergraph_F00439	65	■ 0.999	$Y \cap T=B$	$Y \cap T = B$	Let $X \cap T = A$ and $Y \cap T = B$, and $G_1 = G_{A \cup(Y \setminus B)}$ and $G_2 = G_{B \cup(X \setminus A)}$, see Figure 2.7.
voloshin-hypergraph_F00440	65	■ 0.999	$G_{\{1\}}=G_{\{A \cup(Y \backslash B)\}}$	$G_1 = G_{A \cup(Y \setminus B)}$	Let $X \cap T = A$ and $Y \cap T = B$, and $G_1 = G_{A \cup(Y \setminus B)}$ and $G_2 = G_{B \cup(X \setminus A)}$, see Figure 2.7.
voloshin-hypergraph_F00441	65	■ 0.999	$G_{\{2\}}=G_{\{B \cup(X \backslash A)\}}$	$G_2 = G_{B \cup(X \setminus A)}$	Let $X \cap T = A$ and $Y \cap T = B$, and $G_1 = G_{A \cup(Y \setminus B)}$ and $G_2 = G_{B \cup(X \setminus A)}$, see Figure 2.7.
voloshin-hypergraph_F00442	65	■ 1.000	$A \cup B$	$A \cup B$	Since $A \cup B$ is a transversal, there are no edges between $Y \setminus B$ and $X \setminus A$. For each $S \subseteq A$,
voloshin-hypergraph_F00443	65	■ 1.000	$Y \backslash B$	$Y \setminus B$	Since $A \cup B$ is a transversal, there are no edges between $Y \setminus B$ and $X \setminus A$. For each $S \subseteq A$,
voloshin-hypergraph_F00444	65	■ 1.000	$X \backslash A$	$X \setminus A$	Since $A \cup B$ is a transversal, there are no edges between $Y \setminus B$ and $X \setminus A$. For each $S \subseteq A$,
voloshin-hypergraph_F00445	65	■ 1.000	$S \subseteq A$	$S \subseteq A$	Since $A \cup B$ is a transversal, there are no edges between $Y \setminus B$ and $X \setminus A$. For each $S \subseteq A$,
voloshin-hypergraph_F00446	65	■ 1.000	$N_{\{G_{\{1\}}\}}(S)$	$N_{G_1}(S)$	consider $N_{G_1}(S)$. If $ N_{G_1}(S) < S $, then, because $N_{G_1}(S)$ “touches” all edges incident to S
voloshin-hypergraph_F00447	65	■ 1.000	$\left N_{\{G_{\{1\}}\}}(S)\right < S $	$ N_{G_1}(S) < S $	consider $N_{G_1}(S)$. If $ N_{G_1}(S) < S $, then, because $N_{G_1}(S)$ “touches” all edges incident to S
voloshin-hypergraph_F00448	65	■ 1.000	$\left N_{\{G_{\{1\}}\}}(S)\right \geq S $	$ N_{G_1}(S) \geq S $	G than T . Since T is a minimum transversal, this is impossible, and therefore, $\boxed{ N_{G_1}(S) \geq S }$
voloshin-hypergraph_F00449	65	■ 0.887	$ A + B = T =\tau(G)$	$ A + B = T = \tau(G)$	graph G that has $\boxed{ A + B = T = \tau(G)}$ edges. Hence $\tau(G) = v(G)$.
voloshin-hypergraph_F00450	65	■ 0.887	$\tau(G)=v(G)$	$\tau(G) = v(G)$	graph G that has $ A + B = T = \tau(G)$ edges. Hence $\tau(G) = v(G)$.
voloshin-hypergraph_F00451	65	■ 1.000	$m, n \geq 1$	$m, n \geq 1$	2. For integers $m, n \geq 1$, formulate the conditions when $K_{m, n}$ has a perfect matching.
voloshin-hypergraph_F00452	65	■ 1.000	$K_{\{m, n\}}$	$K_{m, n}$	2. For integers $m, n \geq 1$, formulate the conditions when $K_{m, n}$ has a perfect matching.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00453	65	■ 0.999	$\tau\left(K_{\{m, n\}}\right)$	$\tau\left(K_{m,n}\right)$	4. For integers $m, n \geq 1$, find $\tau(K_{m,n})$ and $v(K_{m,n})$.
voloshin-hypergraph_F00454	65	■ 0.999	$v\left(K_{\{m, n\}}\right)$	$v\left(K_{m,n}\right)$	4. For integers $m, n \geq 1$, find $\tau(K_{m,n})$ and $v(K_{m,n})$.
voloshin-hypergraph_F00455	65	■ 0.637	τ	τ	6. For graph in Figure 2.8, find the values of τ and v and the respective transversal and matching.
voloshin-hypergraph_F00456	67	■ 1.000	≥ 4	≥ 4	diagonal of the cycle. Graph G is called chordal if <i>every</i> cycle of length ≥ 4 has a chord.
voloshin-hypergraph_F00457	67	■ 1.000	$C_{\{1\}}, C_{\{2\}}$	C_1, C_2	of them is induced. Notice that cycles C_1, C_2 , and C_3 do not have non-consecutive vertices
voloshin-hypergraph_F00458	68	■ 1.000	$G_{\{1\}}=G_{\{X_{\{1\}} \cup S\}$	$G_1 = G_{X_1 \cup S}$	two derived subgraphs $G_1 = G_{X_1 \cup S}$ and $G_2 = G_{X_2 \cup S}$, see Figure 3.2.
voloshin-hypergraph_F00459	68	■ 1.000	$G_{\{2\}}=G_{\{X_{\{2\}} \cup S\}$	$G_2 = G_{X_2 \cup S}$	two derived subgraphs $G_1 = G_{X_1 \cup S}$ and $G_2 = G_{X_2 \cup S}$, see Figure 3.2.
voloshin-hypergraph_F00460	68	■ 1.000	$G_{\{S\}}$	G_S	We need to prove that G_S is a clique. Consider any pair of vertices $x, y \in S$. Since S is
voloshin-hypergraph_F00461	68	■ 1.000	$x, y \in S$	$x, y \in S$	We need to prove that G_S is a clique. Consider any pair of vertices $x, y \in S$. Since S is
voloshin-hypergraph_F00462	69	■ 0.997	$y \in C_{\{k\}}$	$y.C_k$	two nonadjacent vertices of C_k , say, x and y . C_k is formed by two different (x, y) -paths.
voloshin-hypergraph_F00463	69	■ 1.000	$k \geq 4$	$k \geq 4$	Since $k \geq 4$, each such path contains at least one internal vertex. Since C_k is induced, inter-
voloshin-hypergraph_F00464	69	■ 1.000	$ X =$	$ X =$	Proof. Let $G = (X, E)$ be a connected chordal graph which is not a clique; hence $ X =$
voloshin-hypergraph_F00465	69	■ 1.000	$X - \{x, y\}$	$X - \{x, y\}$	adjacent. Hence vertex set $X - \{x, y\}$ is a separator. Choose any minimal separator from
voloshin-hypergraph_F00466	69	■ 1.000	$G_{\{X_{\{1\}}\}}$	G_{X_1}	Suppose deleting of S from G leaves two connected components, G_{X_1} and G_{X_2} ; let
voloshin-hypergraph_F00467	69	■ 1.000	$G_{\{X_{\{2\}}\}}$	G_{X_2}	Suppose deleting of S from G leaves two connected components, G_{X_1} and G_{X_2} ; let
voloshin-hypergraph_F00468	69	■ 1.000	$< n$	$< n$	It is connected because G_{X_1} is connected and G_S is a clique. It has $< n$ vertices because
voloshin-hypergraph_F00469	69	■ 0.549	$X_{\{2\}} \neq \emptyset$	$X_2 \neq \emptyset$	$X_2 \neq \emptyset$. By the induction hypothesis, G_1 contains at least two simplicial vertices which are
voloshin-hypergraph_F00470	69	■ 1.000	$K_{\{n\}}, n \geq 2$	$K_n, n \geq 2$	$K_n, n \geq 2$, where all vertices are simplicial, with the exception that all they are pairwise
voloshin-hypergraph_F00471	70	■ 1.000	$G_{\{1\}}=(X, E)$	$G_1 = (X, E)$	Let $G_1 = (X, E)$ be a chordal graph; it contains a simplicial vertex, denote it by x_1 .
voloshin-hypergraph_F00472	70	■ 0.875	$G_{\{n+1\}}=\emptyset$	$G_{n+1} = \emptyset$	step in this procedure will occur when we delete the last vertex x_n and arrive to $G_{n+1} = \emptyset$.
voloshin-hypergraph_F00473	70	■ 0.999	$\boldsymbol{\sigma}=\left(x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}\right)$	$\boldsymbol{\sigma} = \left(x_1, x_2, \ldots, x_n\right)$	We obtain the ordering of vertices $x_1, x_2, \ldots, x_n$. Denote it by σ , so $\sigma = (x_1, x_2, \ldots, x_n)$. The
voloshin-hypergraph_F00474	70	■ 1.000	$x_{\{i\}}, x_{\{i+1\}}, \ldots, x_{\{n\}}$	$x_i, x_{i+1}, \ldots, x_n$	$x_i, x_{i+1}, \ldots, x_n$ and $G_{i+1} = G_i - x_i, i = 1, 2, \ldots, n$.
voloshin-hypergraph_F00475	70	■ 1.000	$G_{\{i+1\}}=G_{\{i\}}-x_{\{i\}}, i=1,2, \ldots, n$	$G_{i+1} = G_i - x_i, i = 1, 2, \ldots, n$	$x_i, x_{i+1}, \ldots, x_n$ and $G_{i+1} = G_i - x_i, i = 1, 2, \ldots, n$.
voloshin-hypergraph_F00478	70	■ 0.846	$\boldsymbol{\sigma}=(1,2,3,4,5,6,7,8)$	$\boldsymbol{\sigma} = (1, 2, 3, 4, 5, 6, 7, 8)$	cial vertices in order 1, 2, 3, ..., 8, so $\boldsymbol{\sigma} = (1, 2, 3, 4, 5, 6, 7, 8)$. Observe that there are only
voloshin-hypergraph_F00479	70	■ 0.907	$\binom{8}{4}$	$\binom{8}{4}$	graph, then we have to check $\binom{8}{4}$ subsets of size 4 for the case if any induces C_4 . Then
voloshin-hypergraph_F00480	70	■ 0.943	$\binom{8}{5}$	$\binom{8}{5}$	repeat the procedure for C_5 , it would give us $\binom{8}{5}$ subsets. Then we would have to consider
voloshin-hypergraph_F00481	70	■ 0.668	$\binom{8}{6}$	$\binom{8}{6}$	$\binom{8}{6}$ subsets for C_6 , $\binom{8}{7}$ subsets for C_8 and $\binom{8}{8}$ subsets for C_8 . Visualization of G could save

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voloshin-hypergraph_F00482	70	0.668	$C_{\{6\}, \binom{8}{7}}$	$C_6, \binom{8}{7}$	$\binom{8}{6}$ subsets for C_6 , $\binom{8}{7}$ subsets for C_8 and $\binom{8}{8}$ subsets for C_8 . Visualization of G could save
voloshin-hypergraph_F00483	70	0.668	$C_{\{8\}}$	C_8	$\binom{8}{6}$ subsets for C_6 , $\binom{8}{7}$ subsets for C_8 and $\binom{8}{8}$ subsets for C_8 . Visualization of G could save
voloshin-hypergraph_F00484	70	0.668	$\binom{8}{8}$	$\binom{8}{8}$	$\binom{8}{6}$ subsets for C_6 , $\binom{8}{7}$ subsets for C_8 and $\binom{8}{8}$ subsets for C_8 . Visualization of G could save
voloshin-hypergraph_F00485	71	1.000	$\sigma^{\prime}=(2,3,8,7,6,5,4,1)$	$\sigma' = (2, 3, 8, 7, 6, 5, 4, 1)$	ordering $\sigma' = (2, 3, 8, 7, 6, 5, 4, 1)$.
voloshin-hypergraph_F00486	71	1.000	$S^{\prime}$	S'	Proof. Since x is simplicial, every maximum independent set S' contains precisely one
voloshin-hypergraph_F00487	71	1.000	$\{x\} \cup N(x)$	$\{x\} \cup N(x)$	vertex from set $\{x\} \cup N(x)$ because otherwise the set $S' \cup \{x\}$ would be independent and
voloshin-hypergraph_F00488	71	1.000	$S^{\prime} \cup \{x\}$	$S' \cup \{x\}$	vertex from set $\{x\} \cup N(x)$ because otherwise the set $S' \cup \{x\}$ would be independent and
voloshin-hypergraph_F00489	71	1.000	$y \in N(x)$	$y \in N(x)$	have greater cardinality. If there is a vertex $y \in N(x)$ such that $y \in S'$, then we replace it by
voloshin-hypergraph_F00490	71	1.000	$y \in S^{\prime}$	$y \in S'$	have greater cardinality. If there is a vertex $y \in N(x)$ such that $y \in S'$, then we replace it by
voloshin-hypergraph_F00491	71	1.000	$S=S^{\prime} \backslash \{y\} \cup \{x\}$	$S = S' \setminus \{y\} \cup \{x\}$	x and put $S = S' \setminus \{y\} \cup \{x\}$.
voloshin-hypergraph_F00492	71	0.995	$ S =\alpha(G)$	$ S = \alpha(G)$	<i>OUTPUT: A maximum independent set $S \subseteq X$ with $S = \alpha(G)$</i>
voloshin-hypergraph_F00493	71	0.872	$S=\emptyset$	$S = \emptyset$	<i>I. Set $S = \emptyset$.</i>
voloshin-hypergraph_F00494	72	0.813	$\{1,4,5\}$	$\{1, 4, 5\}$	1 is included in S , and set $\{1,4,5\}$ is strongly deleted; on the second step, vertex 2 is
voloshin-hypergraph_F00495	72	1.000	$\{2,3,8\}$	$\{2, 3, 8\}$	included in S , and set $\{2,3,8\}$ is strongly deleted; at last, on the third step, vertex 6 is
voloshin-hypergraph_F00496	72	0.968	$S=\{1,2,6\}$	$S = \{1, 2, 6\}$	conclude that one of the maximum stable sets of graph G is $S = \{1, 2, 6\}$, and $\alpha(G) = 3$.
voloshin-hypergraph_F00497	72	0.968	$\alpha(G)=3$	$\alpha(G) = 3$	conclude that one of the maximum stable sets of graph G is $S = \{1, 2, 6\}$, and $\alpha(G) = 3$.
voloshin-hypergraph_F00498	72	1.000	$\{3,4,5,7,8\}$	$\{3, 4, 5, 7, 8\}$	Respectively, vertices $\{3,4,5,7,8\}$ form a minimum transversal and $\tau(G) = 5$. Depending
voloshin-hypergraph_F00499	72	1.000	$\tau(G)=5$	$\tau(G) = 5$	Respectively, vertices $\{3,4,5,7,8\}$ form a minimum transversal and $\tau(G) = 5$. Depending
voloshin-hypergraph_F00500	72	0.966	$\{1,4,5\}, \{2,3,8\}$	$\{1, 4, 5\}, \{2, 3, 8\}$	imum number of cliques that cover X , namely: the cliques induced by $\{1,4,5\}$, $\{2,3,8\}$
voloshin-hypergraph_F00501	72	0.999	$\theta(G)$	$\theta(G)$	In general, the clique cover number $\theta(G)$ of a graph G is the minimum number of
voloshin-hypergraph_F00502	72	0.986	$\theta(G)=3$	$\theta(G) = 3$	in graph theory. So, for the graph in Figure 3.3, $\theta(G) = 3$.
voloshin-hypergraph_F00503	73	1.000	$M(G)$	$M(G)$	As it follows from the definition, to find $M(G)$ one need to consider all induced sub-
voloshin-hypergraph_F00504	73	1.000	t	t	value t over all these minimums coincides with $M(G)$.
voloshin-hypergraph_F00505	73	1.000	$M(G)>t$	$M(G) > t$	just one subgraph from the sequence of graphs. On the other hand, if $M(G) > t$, then
voloshin-hypergraph_F00507	73	1.000	$G, M(G) \geq \omega(G)-1$	$G, M(G) \geq \omega(G) - 1$	Proposition 3.3.1 For any graph G , $M(G) \geq \omega(G) - 1$.
voloshin-hypergraph_F00508	73	0.999	$\omega(G)-1$	$\omega(G) - 1$	$\omega(G) - 1$. Since $M(G)$ is the maximum of minimum degrees over all induced subgraphs,
voloshin-hypergraph_F00509	74	0.969	$\boldsymbol{\omega}(G)-1$	$\omega(G) - 1$	As we see, the value $\omega(G) - 1$ in fact is the lower bound for $M(G)$ for all graphs. The

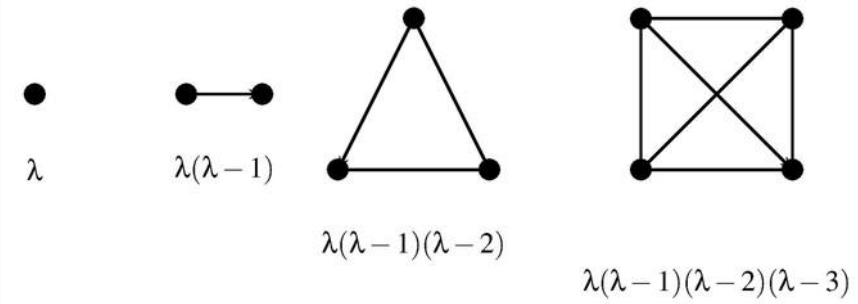
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voloshin-hypergraph_F00510	74	■ 1.000	$M(\text{G}^{\prime})=\omega(\text{G}^{\prime})-1$	$M(G') = \omega(G') - 1$	1) $M(G') = \omega(G') - 1$ for each induced subgraph $G' \subseteq G$;
voloshin-hypergraph_F00511	74	■ 1.000	$M(\text{C}_k)=2=\omega(\text{C}_k)$	$M(C_k) = 2 = \omega(C_k)$	is not chordal. Then G contains an induced cycle C_k of length ≥ 4 . But $M(C_k) = 2 = \omega(C_k)$,
voloshin-hypergraph_F00512	74	■ 1.000	$M(\text{G}) \geq \omega(\text{G})-1$	$M(G) \geq \omega(G) - 1$	chordal, without loss of generality, prove the equality for G . Note that $M(G) > \omega(G) - 1$.
voloshin-hypergraph_F00513	74	■ 1.000	$\omega(\text{G})=t+1$	$\omega(G) = t + 1$	vertex in such decomposition be t . Then the size of maximum clique in G is $\omega(G) = t + 1$.
voloshin-hypergraph_F00514	74	■ 1.000	$M(\text{G}) \leq t=\omega(\text{G})-1$	$M(G) \leq t = \omega(G) - 1$	we have $M(G) \leq t = \omega(G) - 1$. Hence $M(G) = \omega(G) - 1$.
voloshin-hypergraph_F00515	74	■ 1.000	$M(\text{G})=\omega(\text{G})-1$	$M(G) = \omega(G) - 1$	we have $M(G) \leq t = \omega(G) - 1$. Hence $M(G) = \omega(G) - 1$.
voloshin-hypergraph_F00516	74	■ 1.000	$K_{n, n}$	$K_{n,n}$	example, for complete bipartite graph $K_{n,n}$ with $n \geq 2$, we obtain $M(K_{n,n}) = n, \omega(K_{n,n}) = 2$,
voloshin-hypergraph_F00517	74	■ 1.000	$M(\text{K}_{n, n})=n, \omega(\text{K}_{n, n})=2$	$M(K_{n,n}) = n, \omega(K_{n,n}) = 2$	example, for complete bipartite graph $K_{n,n}$ with $n \geq 2$, we obtain $M(K_{n,n}) = n, \omega(K_{n,n}) = 2$,
voloshin-hypergraph_F00518	74	■ 1.000	$M(\text{G}^{\prime})>\omega(\text{G}^{\prime})-1$	$M(G') > \omega(G') - 1$	graph, find a minimal induced subgraph G' such that $M(G') > \omega(G') - 1$.
voloshin-hypergraph_F00519	74	■ 1.000	$M(\text{C}_n), M(\text{K}_n), M(\text{W}_n)$	$M(C_n), M(K_n), M(W_n)$	3. Determine $M(C_n), M(K_n), M(W_n)$, and the value of Szekeres-Wilf number of the cube, prism
voloshin-hypergraph_F00520	74	■ 1.000	$G, M(\text{G})=k$	$G, M(G) = k$	5. Prove that for any k -regular graph $G, M(G) = k$.
voloshin-hypergraph_F00521	75	■ 1.000	$G=$	$G =$	Proof. Without loss of generality, consider a connected nontrivial chordal graph $G =$
voloshin-hypergraph_F00522	75	■ 0.998	(X, E)	(X, E)	(X, E) . Prove the theorem by induction on n . The statement is evident for $n = 2$. Assume it
voloshin-hypergraph_F00523	75	■ 0.998	$n=2$	$n = 2$	(X, E) . Prove the theorem by induction on n . The statement is evident for $n = 2$. Assume it
voloshin-hypergraph_F00524	75	■ 0.999	$N_k=N_{\infty}(x)$	$N_k = N_{\infty}(x)$	where $N_k = N_{\infty}(x)$, see Figure 3.7.
voloshin-hypergraph_F00525	76	■ 1.000	$N_{\infty}(x)=X-\{x\}$	$N_{\infty}(x) = X - \{x\}$	If $k = 1$, then $N_{\infty}(x) = X - \{x\}$. Chordal graph $G - x$ has a simplicial vertex. Any
voloshin-hypergraph_F00526	76	■ 1.000	$G-x$	$G - x$	If $k = 1$, then $N_{\infty}(x) = X - \{x\}$. Chordal graph $G - x$ has a simplicial vertex. Any
voloshin-hypergraph_F00527	76	■ 1.000	N_{k-1}	N_{k-1}	Let now $k \geq 2$. The set N_{k-1} evidently is a separator in G . It contains a minimal sepa-
voloshin-hypergraph_F00528	76	■ 1.000	G_{N_k}	G_{N_k}	rator, and subgraph G_{N_k} has at least one connected component. Without loss of generality,
voloshin-hypergraph_F00529	76	■ 1.000	$G_{N_{k-1}}$	$G_{N_{k-1}}$	By Theorem 3.2, $G_{N_{k-1}}$ is a clique. If $G_{N_{k-1} \cup N_k}$ is a clique, then every vertex from N_k
voloshin-hypergraph_F00530	76	■ 1.000	$G_{N_{k-1}} \cup N_k$	$G_{N_{k-1} \cup N_k}$	By Theorem 3.2, $G_{N_{k-1}}$ is a clique. If $G_{N_{k-1} \cup N_k}$ is a clique, then every vertex from N_k
voloshin-hypergraph_F00531	76	■ 1.000	$y \in N_k$	$y \in N_k$	that every vertex $y \in N_k$ which is simplicial in $G_{N_{k-1} \cup N_k}$, is simplicial in G .
voloshin-hypergraph_F00532	76	■ 0.976	$y^{\prime} \in N_{\infty}(x)$	$y' \in N_{\infty}(x)$	such that $d(x, y) = \text{diam}(G)$, see Figure 3.8. By Theorem 3.4.1, there is a vertex $y' \in N_{\infty}(x)$
voloshin-hypergraph_F00533	76	■ 0.986	$d(x, y^{\prime})=d(x, y)=\operatorname{diam}(G)$	$d(x, y') = d(x, y) = \text{diam}(G)$	which is simplicial in G . Obviously, $d(x, y') = d(x, y) = \text{diam}(G)$. By the same reason, for
voloshin-hypergraph_F00534	76	■ 0.934	$y^{\prime}$	y'	vertex y' , there is a simplicial vertex $x' \in N_{\infty}(y')$ such that $d(y', x') = d(y', x) = \text{diam}(G)$.
voloshin-hypergraph_F00535	76	■ 0.934	$x^{\prime} \in N_{\infty}(y^{\prime})$	$x' \in N_{\infty}(y')$	vertex y' , there is a simplicial vertex $x' \in N_{\infty}(y')$ such that $d(y', x') = d(y', x) = \text{diam}(G)$.
voloshin-hypergraph_F00536	76	■ 0.934	$d(y^{\prime}, x^{\prime})=d(y^{\prime}, x)=\operatorname{diam}(G)$	$d(y', x') = d(y', x) = \text{diam}(G)$	vertex y' , there is a simplicial vertex $x' \in N_{\infty}(y')$ such that $d(y', x') = d(y', x) = \text{diam}(G)$.

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voloshin-hypergraph_F08537	76	■ 0.997	$\operatorname{diam}(G)=4$	$\operatorname{diam}(G) = 4$	$\operatorname{diam}(G) = 4$. Let us start at vertex 2. So, $N_0 = \{2\}$. Then $N_1(2) = \{1, 12, 10, 3\}$. Next we
voloshin-hypergraph_F08538	76	■ 0.997	$N_{\{0\}}=\{2\}$	$N_0 = \{2\}$	$\operatorname{diam}(G) = 4$. Let us start at vertex 2. So, $N_0 = \{2\}$. Then $N_1(2) = \{1, 12, 10, 3\}$. Next we
voloshin-hypergraph_F08539	76	■ 0.997	$N_{\{1\}}(2)=\{1, 12, 10, 3\}$	$N_1(2) = \{1, 12, 10, 3\}$	$\operatorname{diam}(G) = 4$. Let us start at vertex 2. So, $N_0 = \{2\}$. Then $N_1(2) = \{1, 12, 10, 3\}$. Next we
voloshin-hypergraph_F08540	76	■ 1.000	$N_{\{2\}}(2)=\{11, 4, 8, 9\}$	$N_2(2) = \{11, 4, 8, 9\}$	find $N_2(2) = \{11, 4, 8, 9\}$. At last, $N_3(2) = N_\infty(2) = \{5, 6, 7\}$. Observe that vertices 5 and 7
voloshin-hypergraph_F08541	76	■ 1.000	$N_{\{3\}}(2)=N_{\{\infty\}}(2)=\{5, 6, 7\}$	$N_3(2) = N_\infty(2) = \{5, 6, 7\}$	find $N_2(2) = \{11, 4, 8, 9\}$. At last, $N_3(2) = N_\infty(2) = \{5, 6, 7\}$. Observe that vertices 5 and 7
voloshin-hypergraph_F08542	77	■ 0.997	$N_{\{0\}}(5)=\{5\}$, $N_{\{1\}}(5)=\{4, 6\}$	$N_0(5) = \{5\}, N_1(5) = \{4, 6\}$	We now run the same procedure starting at, say, 5. $N_0(5) = \{5\}, N_1(5) = \{4, 6\}$.
voloshin-hypergraph_F08543	77	■ 1.000	$N_{\{2\}}(5)=\{7, 8, 9, 3\}$, $N_{\{3\}}(5)=\{2, 10\}$, $N_{\{4\}}(5)=N_{\{\infty\}}(5)=\{1, 12, 11\}$	$N_2(5) = \{7, 8, 9, 3\}, N_3(5) = \{2, 10\}, N_4(5) = N_\infty(5) = \{1, 12, 11\}$	$N_2(5) = \{7, 8, 9, 3\}, N_3(5) = \{2, 10\}, N_4(5) = N_\infty(5) = \{1, 12, 11\}$ We find two simplicial
voloshin-hypergraph_F08544	77	■ 0.993	$d(11, 5)=\operatorname{diam}(G)=4$	$d(11, 5) = \operatorname{diam}(G) = 4$	vertices, 11 and 5 such that $d(11, 5) = \operatorname{diam}(G) = 4$. There are two diametral (11,5)-paths:
voloshin-hypergraph_F08545	78	■ 1.000	$\overline{C_l}, l \geq 4$	$\overline{C_l}, l \geq 4$	induced $\overline{C_l}, l \geq 4$. Latticed graphs are invariant with respect to taking the complement in the
voloshin-hypergraph_F08546	78	■ 1.000	$C_k, k \geq 6$	$C_k, k \geq 6$	as of graphs with all vertices being “strongly” cyclic. All cycles $C_k, k \geq 6$ are latticed: each
voloshin-hypergraph_F08547	78	■ 1.000	$\overline{C_4}$	$\overline{C_4}$	vertex belongs to C_k and $\overline{C_4}$, see Figure 3.11.
voloshin-hypergraph_F08548	79	■ 0.971	$G - S$	$G - S$	induce a component of $G - S$ that does not contain x , see Figure 3.12. Choose any two
voloshin-hypergraph_F08549	79	■ 1.000	$y, z \in S$	$y, z \in S$	vertices $y, z \in S$. Since S is a minimal separator, each of y and z has a neighbor in Y . Since
voloshin-hypergraph_F08550	80	■ 1.000	$n(G)$	$n(G)$	We prove (iii) $\Rightarrow$ (i) by induction on the number of vertices $n(G)$. Let $G = (X, E)$ be a
voloshin-hypergraph_F08551	80	■ 1.000	$C_l, l \geq 4$	$C_l, l \geq 4$	any two of them as induced subgraphs, say, $C_k, k \geq 4$ and $C_l, l \geq 4$. Considered together they
voloshin-hypergraph_F08552	80	■ 0.989	$X^{\{\prime\}} \neq \emptyset$	$X' \neq \emptyset$	weakly cyclic vertex. Denote the set of all weakly cyclic vertices by X' . Hence, $X' \neq \emptyset$.
voloshin-hypergraph_F08553	80	■ 1.000	$G^{\{\prime\}}=G-X^{\{\prime\}}$	$G' = G - X'$	Let $G' = G - X'$. Subgraph G' has at least one vertex, because otherwise $X' = X$, all
voloshin-hypergraph_F08554	80	■ 1.000	$X^{\{\prime\}}=X$	$X' = X$	Let $G' = G - X'$. Subgraph G' has at least one vertex, because otherwise $X' = X$, all
voloshin-hypergraph_F08555	80	■ 1.000	$n\left(G^{\{\prime\}}\right)<n(G)$	$n(G') < n(G)$	Since $X' \neq \emptyset$ we conclude that $n(G') < n(G)$. Evidently, G' does not contain latticed
voloshin-hypergraph_F08556	80	■ 1.000	$C_k, k \geq 4, y$	$C_k, k \geq 4, y$	Since every vertex of G' lies in some $C_k, k \geq 4$, y is co-simplicial. We will prove that
voloshin-hypergraph_F08557	80	■ 1.000	$G=G_{\{1\}} \cup G_{\{2\}}, G_{\{1\}} \cap G_{\{2\}}=S$	$G = G_1 \cup G_2, G_1 \cap G_2 = S$	Let G_1, G_2 be induced subgraphs of G such that $G = G_1 \cup G_2, G_1 \cap G_2 = S$, and there is
voloshin-hypergraph_F08558	80	■ 1.000	$G_{\{1\}}-S$	$G_1 - S$	no edge between $G_1 - S$ and $G_2 - S$. If at least one of G_1 or G_2 is chordal, then there is a
voloshin-hypergraph_F08559	80	■ 1.000	$G_{\{2\}}-S$	$G_2 - S$	no edge between $G_1 - S$ and $G_2 - S$. If at least one of G_1 or G_2 is chordal, then there is a
voloshin-hypergraph_F08560	80	■ 1.000	$e_{\{1\}}, e_{\{2\}}$	e_1, e_2	of the second cycle that lies completely in $G_2 - S$. All endpoints of e_1, e_2 are in G' . Since
voloshin-hypergraph_F08561	81	■ 1.000	$S \subseteq N(x)$	$S \subseteq N(x)$	$S \subseteq N(x)$, and we obtain that y is adjacent to vertex x . Since x was chosen arbitrary from
voloshin-hypergraph_F08562	81	■ 1.000	$X^{\{\prime\}}, y$	X', y	X' , y is adjacent to all vertices from X' and therefore is co-simplicial for the entire graph G .
voloshin-hypergraph_F08563	82	■ 1.000	$C_{\{n\}}, K_{\{n\}}, W_{\{n\}}, n \geq 3$	$C_n, K_n, W_n, n \geq 3$	5. Which of the graphs $C_n, K_n, W_n, n \geq 3$, cube, prism and Petersen graph is quasi-triangulated
voloshin-hypergraph_F08564	83	■ 1.000	$f(G)$	$f(G)$	G , we will denote the number of faces by $f(G)$ or simply by f .

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voloshin-hypergraph_F00565	83	■ 1.000	$f_{\{1\}}$	f_1	first drawing, face f_1 is bounded by the cycle 1-2-4-1, while in the second by the cycle 1-2-
voloshin-hypergraph_F00566	83	■ 1.000	$f_{\{4\}}$	f_4	of K_4 is denoted by f_4 .
voloshin-hypergraph_F00567	85	■ 1.000	$K_{\{2,3\}}$	$K_{2,3}$	6. Find at least two different plane embeddings of the prism, cube, $K_{2,3}$ and W_n ($n \geq 4$).
voloshin-hypergraph_F00568	85	■ 1.000	$W_{\{n\}}(n \geq 4)$	$W_n(n \geq 4)$	6. Find at least two different plane embeddings of the prism, cube, $K_{2,3}$ and W_n ($n \geq 4$).
voloshin-hypergraph_F00569	85	■ 1.000	$m(T)=n-1, f(T)=1$	$m(T) = n - 1, f(T) = 1$	Proof. Let T be a spanning tree of G . Evidently, $m(T) = n - 1, f(T) = 1$, so $n(T) -$
voloshin-hypergraph_F00570	85	■ 1.000	$n(T) -$	$n(T) -$	Proof. Let T be a spanning tree of G . Evidently, $m(T) = n - 1, f(T) = 1$, so $n(T) -$
voloshin-hypergraph_F00571	85	■ 1.000	$m(T)+f(T)=n-(n-1)+1=2$	$m(T) + f(T) = n - (n - 1) + 1 = 2$	$m(T) + f(T) = n - (n - 1) + 1 = 2$, and the formula holds for T .
voloshin-hypergraph_F00572	85	■ 1.000	$m-n+2$	$m - n + 2$	<i>number of faces equal to $m - n + 2$.</i>
voloshin-hypergraph_F00573	85	■ 1.000	$m(G)$	$m(G)$	Proof. Indeed, since $n(G)$ and $m(G)$ are constant, Euler's formula implies that $f(G) =$
voloshin-hypergraph_F00574	85	■ 1.000	$f(G)=$	$f(G) =$	Proof. Indeed, since $n(G)$ and $m(G)$ are constant, Euler's formula implies that $f(G) =$
voloshin-hypergraph_F00575	85	■ 1.000	$3 f$	$3f$	can obtain is $3f$. In fact, each edge is counted twice, so we obtain $2m$. Therefore, $3f \leq 2m$.
voloshin-hypergraph_F00576	85	■ 1.000	$2 m$	$2m$	can obtain is $3f$. In fact, each edge is counted twice, so we obtain $2m$. Therefore, $3f \leq 2m$.
voloshin-hypergraph_F00577	85	■ 1.000	$3 f \leq 2 m$	$3f \leq 2m$	can obtain is $3f$. In fact, each edge is counted twice, so we obtain $2m$. Therefore, $3f \leq 2m$.
voloshin-hypergraph_F00578	85	■ 1.000	$f=2-n+m$	$f = 2 - n + m$	From Euler's formula, $f = 2 - n + m$. Hence, $3(2 - n + m) \leq 2m$ what gives $m \leq 3n - 6$.
voloshin-hypergraph_F00579	85	■ 1.000	$3(2-n+m) \leq 2 m$	$3(2 - n + m) \leq 2m$	From Euler's formula, $f = 2 - n + m$. Hence, $3(2 - n + m) \leq 2m$ what gives $m \leq 3n - 6$.
voloshin-hypergraph_F00580	85	■ 1.000	$m \leq 3 n - 6$	$m \leq 3n - 6$	From Euler's formula, $f = 2 - n + m$. Hence, $3(2 - n + m) \leq 2m$ what gives $m \leq 3n - 6$.
voloshin-hypergraph_F00581	86	■ 0.615	$6 n \leq 2 m$	$6n \leq 2m$	each vertex has degree at least 6. Counting edges around each vertex results in $6n \leq 2m$,
voloshin-hypergraph_F00582	86	■ 1.000	$3 n \leq m$	$3n \leq m$	or, equivalently, $3n \leq m$. Combining this inequality with Corollary 4.2.2 we obtain the
voloshin-hypergraph_F00583	86	■ 1.000	$3 n \leq m \leq 3 n - 6$	$3n \leq m \leq 3n - 6$	following: $3n \leq m \leq 3n - 6$, which leads to contradiction $0 \leq -6$.
voloshin-hypergraph_F00584	86	■ 1.000	$0 \leq -6$	$0 \leq -6$	following: $3n \leq m \leq 3n - 6$, which leads to contradiction $0 \leq -6$.
voloshin-hypergraph_F00585	86	■ 0.997	$n-m+$	$n - m +$	Corollary 4.2.4 <i>If a convex polyhedron has n vertices, m edges and f faces, then $n - m +$</i>
voloshin-hypergraph_F00586	86	■ 1.000	$f=2$	$f = 2$	$f = 2$.
voloshin-hypergraph_F00587	87	■ 1.000	$n=8, m=12, f=6$	$n = 8, m = 12, f = 6$	For example, consider the cube: $n=8, m=12, f=6$ and evidently, the formula holds.
voloshin-hypergraph_F00588	87	■ 1.000	$K_{\{n\}}, W_{\{n\}}, K_{\{m, n\}}(m \geq 1, n \geq 3)$	$K_n, W_n, K_{m,n}(m \geq 1, n \geq 3)$	4. Check the inequality of Corollary 4.2.2 for prism, cube, $K_n, W_n, K_{m,n}(m \geq 1, n \geq 3)$, and the
voloshin-hypergraph_F00589	89	■ 1.000	$X=\{1,2,3,4,5,6\}$	$X = \{1, 2, 3, 4, 5, 6\}$	Now suppose $K_{3,3}$ is planar, and vertex set $X = \{1, 2, 3, 4, 5, 6\}$. Since $K_{3,3}$ is bipartite,
voloshin-hypergraph_F00590	90	■ 1.000	$k, l \geq 3$	$k, l \geq 3$	For example, any two cycles C_k and C_l , with $k, l \geq 3$ are homeomorphic because they both
voloshin-hypergraph_F00591	90	■ 0.995	$G_{\{i\}}^{\prime}=G_{\{i\}} \cup \{x, y\}$	$G'_i = G_i \cup \{x, y\}$	derived subgraphs are G_1 and G_2 . Let $G'_i = G_i \cup \{x, y\}$. Using step 1, one prove then
voloshin-hypergraph_F00592	90	■ 1.000	$G_{\{1\}}^{\prime}, G_{\{2\}}^{\prime}$	G'_1, G'_2	that if G is not planar, then at least one of G'_1, G'_2 is not planar.

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voloshin-hypergraph_F00593	90	■ 1.000	≥ 5	≥ 5	4. At this point one use the lemma stating that any 3-connected graph on ≥ 5 vertices
voloshin-hypergraph_F00594	91	■ 1.000	$G - V(G')$	$G - V(G')$	2) a connected component of $G - V(G')$ together with edges connecting it to G' (vertices
voloshin-hypergraph_F00595	91	■ 1.000	$F(A)$	$F(A)$	of G with respect to G' . For each fragment A , determine a set of faces $F(A)$ that contain all
voloshin-hypergraph_F00596	91	■ 1.000	$F(A) = \emptyset$	$F(A) = \emptyset$	vertices of attachment. If $F(A) = \emptyset$ for some A , then G is not planar. If $ F(A) = 1$ for some
voloshin-hypergraph_F00597	91	■ 1.000	$ F(A) = 1$	$ F(A) = 1$	vertices of attachment. If $F(A) = \emptyset$ for some A , then G is not planar. If $ F(A) = 1$ for some
voloshin-hypergraph_F00598	91	■ 1.000	$ F(A) > 1$	$ F(A) > 1$	A , then select A for the next step. If $ F(A) > 1$ for all A , then select any fragment A .
voloshin-hypergraph_F00599	91	■ 1.000	$G' \neq G$	$G' \neq G$	some subgraph $G' \neq G$.
voloshin-hypergraph_F00600	91	■ 1.000	$m \geq 1$	$m \geq 1$	1. For which values of $m \geq 1$ and $n \geq 3$, graphs $K_n, K_{m,n}, W_n$ are planar?
voloshin-hypergraph_F00601	91	■ 1.000	$n \geq 3$	$n \geq 3$	1. For which values of $m \geq 1$ and $n \geq 3$, graphs $K_n, K_{m,n}, W_n$ are planar?
voloshin-hypergraph_F00602	91	■ 1.000	$K_n, K_{m,n}, W_n$	$K_n, K_{m,n}, W_n$	1. For which values of $m \geq 1$ and $n \geq 3$, graphs $K_n, K_{m,n}, W_n$ are planar?
voloshin-hypergraph_F00603	92	■ 1.000	$n \geq 4$	$n \geq 4$	4. Given a graph G and $n \geq 4$, determine if G is homeomorphic to C_n .
voloshin-hypergraph_F00604	92	■ 1.000	$m = 3n - 6$	$m = 3n - 6$	Theorem 4.5.1 For any plane triangulation, $m = 3n - 6$.
voloshin-hypergraph_F00605	92	■ 1.000	$3f = 2m$	$3f = 2m$	Proof. Indeed, if we count the edges around each face, we obtain the equality $3f = 2m$.
voloshin-hypergraph_F00606	93	■ 1.000	$m = 3n - 6 = 15 - 6 = 9$	$m = 3n - 6 = 15 - 6 = 9$	$m = 3n - 6 = 15 - 6 = 9$ edges. Thus any plane graph on five vertices is a subgraph of
voloshin-hypergraph_F00607	93	■ 1.000	G^*	G^*	G^* and called the dual of G . The rules are the following:
voloshin-hypergraph_F00608	93	■ 1.000	f_i	f_i	2. For each edge of G separating faces f_i and f_j construct an edge of G^* connecting
voloshin-hypergraph_F00609	93	■ 1.000	f_j	f_j	2. For each edge of G separating faces f_i and f_j construct an edge of G^* connecting
voloshin-hypergraph_F00610	93	■ 1.000	$n(G^*) = f, m(G^*) = m$	$n(G^*) = f, m(G^*) = m$	It is evident that $n(G^*) = f, m(G^*) = m$ and $f(G^*) = n$. The last follows from the
voloshin-hypergraph_F00611	93	■ 1.000	$f(G^*) = n$	$f(G^*) = n$	It is evident that $n(G^*) = f, m(G^*) = m$ and $f(G^*) = n$. The last follows from the
voloshin-hypergraph_F00612	93	■ 0.992	$(G^*)^*$	$(G^*)^*$	Theorem 4.5.2 If G is a plane connected graph, then $(G^*)^*$ is isomorphic to G .
voloshin-hypergraph_F00613	94	■ 1.000	$W_n, n \geq 4$	$W_n, n \geq 4$	5. Construct the dual to $W_n, n \geq 4$.
voloshin-hypergraph_F00614	96	■ 0.989	$1, 2, \dots, \lambda$	$1, 2, \dots, \lambda$	coloring the vertices of a graph, we just label them by numbers $1, 2, \dots, \lambda$ where λ is the
voloshin-hypergraph_F00615	96	■ 0.989	λ	λ	coloring the vertices of a graph, we just label them by numbers $1, 2, \dots, \lambda$ where λ is the
voloshin-hypergraph_F00616	96	■ 1.000	$\{1, 2, \dots, \lambda\}$	$\{1, 2, \dots, \lambda\}$	Let $G = (X, E)$ be a simple graph and $\{1, 2, \dots, \lambda\}$ be the set of available colors . Any
voloshin-hypergraph_F00617	96	■ 1.000	$P(G, \lambda)$	$P(G, \lambda)$	colors is denoted by $P(G, \lambda)$. We will see that $P(G, \lambda)$ is a polynomial in λ and therefore
voloshin-hypergraph_F00618	96	■ 1.000	$\chi(G)$	$\chi(G)$	$\chi(G)$. Evidently, the maximum number of colors that can be used in a proper λ -coloring is
voloshin-hypergraph_F00619	96	■ 1.000	$\min\{n(G), \lambda\}$	$\min\{n(G), \lambda\}$	$\min\{n(G), \lambda\}$.
voloshin-hypergraph_F00620	96	■ 1.000	$P(G, \lambda) = 0$	$P(G, \lambda) = 0$	Since there are no colorings with less than $\chi(G)$ colors, $P(G, \lambda) = 0$ for all integer

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voloshin-hypergraph_F00621	96	■ 1.000	$1 \leq \lambda \leq \chi(G) - 1$	$1 \leq \lambda \leq \chi(G) - 1$	values of λ such that $\boxed{1 \leq \lambda \leq \chi(G) - 1}$. Since any coloring with $\chi(G)$ colors is at the same
voloshin-hypergraph_F00622	96	■ 1.000	$\lambda \geq \chi(G)$	$\lambda \geq \chi(G)$	time a proper λ -coloring for any $\lambda \geq \chi(G)$, we conclude that $P(G, \lambda) \geq 1$ for any integer
voloshin-hypergraph_F00623	96	■ 1.000	$P(G, \lambda) \geq 1$	$P(G, \lambda) \geq 1$	time a proper λ -coloring for any $\lambda \geq \chi(G)$, we conclude that $P(G, \lambda) \geq 1$ for any integer
voloshin-hypergraph_F00624	96	■ 1.000	$X = \{x, y, z\}$	$X = \{x, y, z\}$	Consider an example. Let $G = (X, E)$ where $X = \{x, y, z\}$, and $E = \{\{x, y\},$
voloshin-hypergraph_F00625	96	■ 1.000	$E = \{\{x, y\}$	$E = \{\{x, y\}$	Consider an example. Let $G = (X, E)$ where $X = \{x, y, z\}$, and $E = \{\{x, y\},$
voloshin-hypergraph_F00626	96	■ 0.997	$\{y, z\}\}$	$\{y, z\}\}$	$\{y, z\}\}$, see Figure 5.1. There is no coloring with one color, so $\chi(G) > 1$. On the other
voloshin-hypergraph_F00627	96	■ 0.997	$\chi(G) > 1$	$\chi(G) > 1$	$\{y, z\}\}$, see Figure 5.1. There is no coloring with one color, so $\chi(G) > 1$. On the other
voloshin-hypergraph_F00628	96	■ 0.998	$\chi(G) = 2$	$\chi(G) = 2$	hand, it is easy to see a coloring with two colors, say 1-2-1, so $\chi(G) = 2$. Suppose the set
voloshin-hypergraph_F00629	96	■ 0.998	$\{1, 2, 3\}$	$\{1, 2, 3\}$	of available colors is $\{1, 2, 3\}$, consequently, $\lambda = 3$. Having totally three colors and keep-
voloshin-hypergraph_F00630	96	■ 0.998	$\lambda = 3$	$\lambda = 3$	of available colors is $\{1, 2, 3\}$, consequently, $\lambda = 3$. Having totally three colors and keep-
voloshin-hypergraph_F00631	96	■ 1.000	$i = 2$	$i = 2$	used colors: $i = 2$ and $i = 3$. If we use only 2 colors, then we have to chose them from
voloshin-hypergraph_F00632	96	■ 1.000	$i = 3$	$i = 3$	used colors: $i = 2$ and $i = 3$. If we use only 2 colors, then we have to chose them from
voloshin-hypergraph_F00633	96	■ 0.523	$\binom{\lambda}{i} = \binom{3}{2} = 3$	$\binom{\lambda}{i} = \binom{3}{2} = 3$	of choices equals $\binom{\lambda}{i} = \binom{3}{2} = 3$. For the first choice, the colorings are 1-2-1 and 2-1-2, see
voloshin-hypergraph_F00636	97	■ 1.000	$r_i(G)$	$r_i(G)$	the following inequality: $1 \leq i \leq n(G)$. Let $r_i(G)$ be the number of feasible partitions of
voloshin-hypergraph_F00637	98	■ 0.927	$X: X_1 = \{x, z\}$	$X: X_1 = \{x, z\}$	there is only one feasible partition of X : $X_1 = \{x, z\}$ (first cell) and $X_2 = \{y\}$ (second cell).
voloshin-hypergraph_F00638	98	■ 0.927	$X_2 = \{y\}$	$X_2 = \{y\}$	there is only one feasible partition of X : $X_1 = \{x, z\}$ (first cell) and $X_2 = \{y\}$ (second cell).
voloshin-hypergraph_F00639	98	■ 1.000	$r_2(G) = 1$	$r_2(G) = 1$	Therefore, $\boxed{r_2(G) = 1}$.
voloshin-hypergraph_F00640	98	■ 1.000	$X_1 = \{x\}, X_2 = \{y\}$	$X_1 = \{x\}, X_2 = \{y\}$	partition with cells $\boxed{X_1 = \{x\}, X_2 = \{y\}}$, and $X_3 = \{z\}$. Hence $r_3(G) = 1$. Since evidently,
voloshin-hypergraph_F00641	98	■ 1.000	$X_3 = \{z\}$	$X_3 = \{z\}$	partition with cells $X_1 = \{x\}, X_2 = \{y\}$, and $X_3 = \{z\}$. Hence $r_3(G) = 1$. Since evidently,
voloshin-hypergraph_F00642	98	■ 1.000	$r_3(G) = 1$	$r_3(G) = 1$	partition with cells $X_1 = \{x\}, X_2 = \{y\}$, and $X_3 = \{z\}$. Hence $r_3(G) = 1$. Since evidently,
voloshin-hypergraph_F00643	98	■ 1.000	$r_1(G) = 0$	$r_1(G) = 0$	$r_1(G) = 0$, we obtain that the chromatic spectrum of G is:
voloshin-hypergraph_F00644	98	■ 1.000	$R(G)$	$R(G)$	The chromatic spectrum $R(G)$ is called continuous (gap-free) if it does not contain
voloshin-hypergraph_F00645	98	■ 0.850	$R(G) = \left(0, 0, \dots, 0, r_\chi, r_{\chi+1}, \dots, r_n\right)$	$R(G) = (0, 0, \dots, 0, r_\chi, r_{\chi+1}, \dots, r_n)$	Proof. As we noticed, $\boxed{R(G) = (0, 0, \dots, 0, r_\chi, r_{\chi+1}, \dots, r_n)}$. By definition, $r_\chi > 0$. Consider
voloshin-hypergraph_F00646	98	■ 0.850	$r_\chi > 0$	$r_\chi > 0$	Proof. As we noticed, $R(G) = (0, 0, \dots, 0, r_\chi, r_{\chi+1}, \dots, r_n)$. By definition, $r_\chi > 0$. Consider
voloshin-hypergraph_F00647	98	■ 0.476	χ	χ	any strict coloring of G with χ colors and choose any color class which has > 1 vertices.
voloshin-hypergraph_F00648	98	■ 0.987	$\chi + 1$	$\chi + 1$	Split it into two non-empty subsets; we obtain another feasible partition using $\chi + 1$ cells
voloshin-hypergraph_F00649	98	■ 0.360	$r_{\chi+1} > 0$	$r_{\chi+1} > 0$	what implies $r_{\chi+1} > 0$. Repeating this procedure further for any other cell having > 1 ver-

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voloshin-hypergraph_F00650	98	■ 1.000	$r_{\chi+2}>0, r_{\chi+3}>0, \ldots$	$r_{\chi+2} > 0, r_{\chi+3} > 0, \dots$	tices, conclude that $r_{\gamma+2} > 0, r_{\gamma+3} > 0, \dots$. We cannot continue the splitting when all cells
voloshin-hypergraph_F00651	98	■ 1.000	$r_n=1>0$	$r_n = 1 > 0$	have precisely one vertex each, and that corresponds to $r_n = 1 > 0$.
voloshin-hypergraph_F00652	98	■ 1.000	$G, \chi(G) \leq 2$	$G, \chi(G) \leq 2$	Theorem 5.2.2 For a graph G , $\chi(G) \leq 2$ if and only if it has no odd cycles.
voloshin-hypergraph_F00653	98	■ 1.000	$\chi(G) \leq \Delta(G)$	$\chi(G) \leq \Delta(G)$	odd cycle, then $\chi(G) \leq \Delta(G)$.
voloshin-hypergraph_F00654	98	■ 1.000	$E_{\{n\}}, K_{\{n\}}, K_{\{m, n\}}$	$E_n, K_n, K_{m,n}$	1. Find the chromatic number of $E_n, K_n, K_{m,n}$, tree $T_n, C_{2n}, C_{2n+1}, W_n$, prism, cube and Petersen
voloshin-hypergraph_F00655	98	■ 1.000	$T_{\{n\}}, C_{\{2 n\}}, C_{\{2 n+1\}}, W_{\{n\}}$	$T_n, C_{2n}, C_{2n+1}, W_n$	1. Find the chromatic number of $E_n, K_n, K_{m,n}$, tree $T_n, C_{2n}, C_{2n+1}, W_n$, prism, cube and Petersen
voloshin-hypergraph_F00656	98	■ 1.000	$P\left(G^{\{\prime\}}, 3\right)$	$P(G', 3)$	2. In graph G , see Figure 5.1, weakly delete an edge and find all feasible partitions, $P(G', 3)$ and
voloshin-hypergraph_F00657	98	■ 1.000	$R\left(G^{\{\prime\}}\right)$	$R(G')$	$R(G')$.
voloshin-hypergraph_F00658	99	■ 1.000	$i=6$	$i = 6$	5. Beginning with the feasible partition into $i = 6$ cells for the graph in Figure 5.2, construct a
voloshin-hypergraph_F00659	99	■ 1.000	$i+1, i+2, \ldots, n=10$	$i + 1, i + 2, \dots, n = 10$	sequence of feasible partitions into $i+1, i+2, \dots, n=10$ cells.
voloshin-hypergraph_F00660	99	■ 0.904	$\chi, \chi+1, \chi+2, \ldots, n=10$	$\chi, \chi + 1, \chi + 2, \dots, n = 10$	partitions into $\chi, \chi+1, \chi+2, \dots, n=10$ cells.
voloshin-hypergraph_F00661	99	■ 1.000	$\lambda \geq 1$	$\lambda \geq 1$	Let us have $\lambda \geq 1$ colors and consider graph K_n . Evidently, if $\lambda < n$, then there are no proper
voloshin-hypergraph_F00662	99	■ 1.000	$\lambda < n$	$\lambda < n$	Let us have $\lambda \geq 1$ colors and consider graph K_n . Evidently, if $\lambda < n$, then there are no proper
voloshin-hypergraph_F00663	99	■ 1.000	$\lambda \geq n$	$\lambda \geq n$	$\lambda \geq n$. Any assignment of n different colors to the vertices of K_n results in a proper coloring,
voloshin-hypergraph_F00664	99	■ 1.000	$\chi\left(K_{\{n\}}\right)=n$	$\chi(K_n) = n$	so $\chi(K_n) = n$. What about $P(K_n, \lambda)$?
voloshin-hypergraph_F00665	99	■ 1.000	$P\left(K_{\{n\}}, \lambda\right)$	$P(K_n, \lambda)$	so $\chi(K_n) = n$. What about $P(K_n, \lambda)$?
voloshin-hypergraph_F00666	99	■ 0.647	$\lambda-1$	$\lambda - 1$	the second vertex, we have $(\lambda - 1)$ possibilities. For the third vertex there are $(\lambda - 2)$
voloshin-hypergraph_F00667	99	■ 0.647	$\lambda-2$	$\lambda - 2$	the second vertex, we have $(\lambda - 1)$ possibilities. For the third vertex there are $(\lambda - 2)$
voloshin-hypergraph_F00668	100	—	$K_{\{1\}}, K_{\{2\}}, K_{\{3\}}, K_{\{4\}}$	K_1, K_2, K_3, K_4	
voloshin-hypergraph_F00670	100	■ 1.000	$\lambda^{\{n\}}$	$\lambda^{(n)}$	The product $\lambda(\lambda - 1)(\lambda - 2) \cdots (\lambda - n + 1)$ is denoted by $\lambda^{(n)}$ and sometimes called the
voloshin-hypergraph_F00673	100	■ 1.000	$G \cdot e$	$G \cdot e$	$G \cdot e$ denotes the graph obtained by contraction of an edge e , see Section 1.6., Figure 1.25).
voloshin-hypergraph_F00677	101	■ 1.000	$s \geq 1$	$s \geq 1$	for some integer $s \geq 1$.
voloshin-hypergraph_F00678	101	■ 1.000	$P\left(K_{\{i_j\}}, \lambda\right)$	$P(K_{i_j}, \lambda)$	Since $P(K_{i_j}, \lambda)$ is a polynomial in λ , and the sum of polynomials is a polynomial, we
voloshin-hypergraph_F00679	101	■ 1.000	$K_{\{i_1\}}, K_{\{i_2\}}, \ldots, K_{\{i_s\}}$	$K_{i_1}, K_{i_2}, \dots, K_{i_s}$	graphs $K_{i_1}, K_{i_2}, \dots, K_{i_s}$ only one is isomorphic to K_n ; namely that which is obtained by

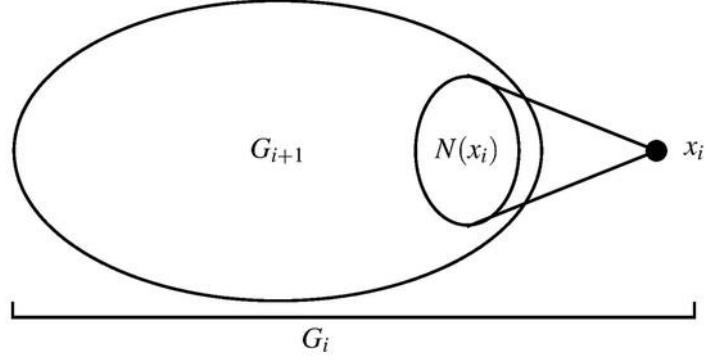
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voloshin-hypergraph_F00680	101	■ 1.000	$K_{\{i\}}$	K_i	K_i have composite names. Only the names of vertices in the unique graph K_n are exactly
voloshin-hypergraph_F00681	101	■ 0.996	$x \ y \ z$	xyz	some K_i is xyz , for example, then there is a strict i -coloring of G where vertices x, y , and z
voloshin-hypergraph_F00682	102	■ 0.998	$r_{\{1\}}=r_{\{2\}}=\cdots=r_{\{\chi-1\}}=0$	$r_1 = r_2 = \cdots = r_{\chi-1} = 0$	Since $r_1 = r_2 = \cdots = r_{\chi-1} = 0$ and $P(K_i, \lambda) = \lambda^{(i)}$, we arrive at last to the final funda-
voloshin-hypergraph_F00683	102	■ 0.998	$P(\text{left}(K_{\{i\}}, \lambda) = \lambda^{(i)}$	$P(K_i, \lambda) = \lambda^{(i)}$	Since $r_1 = r_2 = \cdots = r_{\chi-1} = 0$ and $P(K_i, \lambda) = \lambda^{(i)}$, we arrive at last to the final funda-
voloshin-hypergraph_F00684	102	■ 0.998	$\text{left}(\lambda^{(i)})$	$\left(\lambda^{(i)}\right)$	obtained from this partition by permutation of the colors $(\lambda^{(i)})$, then do that for all feasible
voloshin-hypergraph_F00685	102	■ 0.875	$\text{left}(r_{\{i\}}(G) \lambda^{(i)})$	$\left(r_i(G) \lambda^{(i)}\right)$	partitions $(r_i(G) \lambda^{(i)})$, and at last count that for all i (obtain $\sum_{i=\chi}^n r_i(G) \lambda^{(i)}$). Connection-
voloshin-hypergraph_F00686	102	■ 0.875	$\sum_{i=\mathrm{x}}^n r_{\{i\}}(G) \lambda^{(i)}$	$\sum_{i=x}^n r_i(G) \lambda^{(i)}$	partitions $(r_i(G) \lambda^{(i)})$, and at last count that for all i (obtain $\sum_{i=\chi}^n r_i(G) \lambda^{(i)}$). Connection-
voloshin-hypergraph_F00687	102	■ 1.000	$P(G, 3)=3(3-1)^2=12$	$P(G, 3) = 3(3 - 1)^2 = 12$	$P(G, 3) = 3(3 - 1)^2 = 12$. Moreover, since $P(G, \lambda) = P(K_3, \lambda) + P(K_2, \lambda) = 0 \cdot P(K_1, \lambda) +$
voloshin-hypergraph_F00688	102	■ 1.000	$P(G, \lambda)=P(\text{left}(K_{\{3\}}, \lambda)+P(\text{left}(K_{\{2\}}, \lambda)+P(\text{left}(K_{\{1\}}, \lambda)+$	$P(G, \lambda) = P(K_3, \lambda) + P(K_2, \lambda) = 0 \cdot P(K_1, \lambda) +$	$P(G, 3) = 3(3 - 1)^2 = 12$. Moreover, since $P(G, \lambda) = P(K_3, \lambda) + P(K_2, \lambda) = 0 \cdot P(K_1, \lambda) +$
voloshin-hypergraph_F00689	102	■ 1.000	$1 \cdot P(\text{left}(K_{\{2\}}, \lambda)+1 \cdot P(\text{left}(K_{\{3\}}, \lambda)+$	$1 \cdot P(K_2, \lambda) + 1 \cdot P(K_3, \lambda)$	$1 \cdot P(K_2, \lambda) + 1 \cdot P(K_3, \lambda)$, we conclude that the chromatic spectrum $R(G) = (0, 1, 1)$. The
voloshin-hypergraph_F00690	102	■ 1.000	$R(G)=(0, 1, 1)$	$R(G) = (0, 1, 1)$	$1 \cdot P(K_2, \lambda) + 1 \cdot P(K_3, \lambda)$, we conclude that the chromatic spectrum $R(G) = (0, 1, 1)$. The
voloshin-hypergraph_F00691	102	■ 1.000	$G_{\{3\}}, G_{\{4\}}, G_{\{5\}}$	G_3, G_4, G_5	on the zero level, graphs G_1 and G_2 on the first level, graphs G_3, G_4, G_5 , and G_6 on the
voloshin-hypergraph_F00692	102	■ 1.000	$G_{\{6\}}$	G_6	on the zero level, graphs G_1 and G_2 on the first level, graphs G_3, G_4, G_5 , and G_6 on the
voloshin-hypergraph_F00693	103	■ 1.000	$K_{\{i_{\{j\}}\}}$	K_{i_j}	second level and so on down. Not all of the graphs K_{i_j} are on the last level. Some of them
voloshin-hypergraph_F00694	103	■ 1.000	$P(\text{left}(C_{\{4\}}, 10)=10 \cdot 9 \cdot \text{left}(10^2-3 \cdot 10+3 \cdot 10+3)=6570$	$P(C_4, 10) = 10 \cdot 9 \cdot (10^2 - 3 \cdot 10 + 3) = 6570$	$P(C_4, 10) = 10 \cdot 9 \cdot (10^2 - 3 \cdot 10 + 3) = 6570$.
voloshin-hypergraph_F00695	104	■ 1.000	$E_{\{4\}}, E_{\{5\}}, C_{\{5\}}, C_{\{6\}}, W_{\{4\}}, W_{\{5\}}, P_{\{4\}}, P_{\{5\}}, P_{\{6\}}$	$E_4, E_5, C_5, C_6, W_4, W_5, P_4, P_5, P_6$	graphs: $E_4, E_5, C_5, C_6, W_4, W_5, P_4, P_5, P_6$ and P_n for every $n \geq 6$.
voloshin-hypergraph_F00696	104	■ 1.000	$n \geq 6$	$n \geq 6$	graphs: $E_4, E_5, C_5, C_6, W_4, W_5, P_4, P_5, P_6$ and P_n for every $n \geq 6$.
voloshin-hypergraph_F00697	104	■ 1.000	$P_{\{4\}}, P_{\{5\}}, P_{\{6\}}, C_{\{5\}}$	P_4, P_5, P_6, C_5	2. Apply the connection-contraction algorithm to find all feasible partitions for P_4, P_5, P_6, C_5
voloshin-hypergraph_F00698	105	■ 1.000	$P(\text{left}(G_{\{1\}}, \lambda)$	$P(G_1, \lambda)$	$P(G_1, \lambda)$ proper colorings, G_2 has $P(G_2, \lambda)$ proper colorings, and so on, the total number of
voloshin-hypergraph_F00699	105	■ 1.000	$P(\text{left}(G_{\{2\}}, \lambda)$	$P(G_2, \lambda)$	$P(G_1, \lambda)$ proper colorings, G_2 has $P(G_2, \lambda)$ proper colorings, and so on, the total number of
voloshin-hypergraph_F00700	105	■ 1.000	$E_{\{n\}}=n K_{\{1\}}$	$E_n = nK_1$	The equality (5.5) can be immediately applied to graph E_n : since $E_n = nK_1$, and
voloshin-hypergraph_F00701	105	■ 1.000	$P(\text{left}(K_{\{1\}}, \lambda)=\lambda$	$P(K_1, \lambda) = \lambda$	$P(K_1, \lambda) = \lambda$, we obtain
voloshin-hypergraph_F00702	105	■ 1.000	$S(n, i)=r_{\{i\}}(\text{left}(E_{\{n\}})$	$S(n, i) = r_i(E_n)$	where $S(n, i) = r_i(E_n)$ are some numbers; in other words, $S(n, i)$ equals the number of
voloshin-hypergraph_F00703	105	■ 1.000	$S(n, i)$	$S(n, i)$	where $S(n, i) = r_i(E_n)$ are some numbers; in other words, $S(n, i)$ equals the number of
voloshin-hypergraph_F00704	106	■ 1.000	$R(\text{left}(E_{\{1\}})=(1), R(\text{left}(E_{\{2\}})=(1, 1), R(\text{left}(E_{\{3\}})=(1, 3, 1)$	$R(E_1) = (1), R(E_2) = (1, 1), R(E_3) = (1, 3, 1)$	One can check, for example, that $R(E_1) = (1), R(E_2) = (1, 1), R(E_3) = (1, 3, 1)$,
voloshin-hypergraph_F00705	106	■ 1.000	$R(\text{left}(E_{\{4\}})=(1, 7, 6, 1)$	$R(E_4) = (1, 7, 6, 1)$	$R(E_4) = (1, 7, 6, 1)$ and so on.
voloshin-hypergraph_F00706	106	■ 0.890	$\lambda^{(4)}=-6 \lambda+11 \lambda^2-$	$\lambda^{(4)} = -6\lambda + 11\lambda^2 -$	which are called the Stirling numbers of the first kind . For example, $\lambda^{(4)} = -6\lambda + 11\lambda^2 -$

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voloshin-hypergraph_F00707	106	■ 1.000	$6 \lambda^3 + \lambda^4$	$6\lambda^3 + \lambda^4$	$6\lambda^3 + \lambda^4$, and therefore $s(4, 1) = -6$, $s(4, 2) = 11$, $s(4, 3) = -6$, and $s(4, 4) = 1$.
voloshin-hypergraph_F00708	106	■ 1.000	$s(4, 1)=-6, \ s(4, 2)=11, \ s(4, 3)=-6$	$s(4, 1) = -6, s(4, 2) = 11, s(4, 3) = -6$	$6\lambda^3 + \lambda^4$, and therefore $s(4, 1) = -6, s(4, 2) = 11, s(4, 3) = -6$, and $s(4, 4) = 1$.
voloshin-hypergraph_F00709	106	■ 1.000	$s(4, 4)=1$	$s(4, 4) = 1$	$6\lambda^3 + \lambda^4$, and therefore $s(4, 1) = -6$, $s(4, 2) = 11$, $s(4, 3) = -6$, and $s(4, 4) = 1$.
voloshin-hypergraph_F00710	106	■ 1.000	λ^n	λ^n	Generally, Stirling numbers serve as the coefficients to express λ^n (or, equivalently, E_n)
voloshin-hypergraph_F00711	106	■ 1.000	$\lambda^{(i)}$	$\lambda^{(i)}$	in terms of $\lambda^{(i)}$ (or, equivalently, K_i) and $\lambda^{(n)}$ in terms of λ^i , $i = 1, 2, \dots, n$. The latter in a
voloshin-hypergraph_F00712	106	■ 1.000	$\lambda^{i}, \ i=1, 2, \ \ldots, \ n$	$\lambda^i, i = 1, 2, \dots, n$	in terms of $\lambda^{(i)}$ (or, equivalently, K_i) and $\lambda^{(n)}$ in terms of $\lambda^i, i = 1, 2, \dots, n$. The latter in a
voloshin-hypergraph_F00713	106	■ 0.736	$1, -m(G), \ \ldots$	$1, -m(G), \dots$	<i>integer coefficients alternating in sign and beginning with $1, -m(G), \dots$.</i>
voloshin-hypergraph_F00714	106	■ 0.998	$m=0$	$m = 0$	Proof. We prove by induction on the number of edges $m(G)$. The theorem holds for $m = 0$
voloshin-hypergraph_F00715	106	■ 1.000	$G=E_n$	$G = E_n$	because in this case $G = E_n$ and $P(G, \lambda) = \lambda^n$. Assume now that the theorem is true for all
voloshin-hypergraph_F00716	106	■ 1.000	$P(G, \lambda)=\lambda^n$	$P(G, \lambda) = \lambda^n$	because in this case $G = E_n$ and $P(G, \lambda) = \lambda^n$. Assume now that the theorem is true for all
voloshin-hypergraph_F00717	106	■ 1.000	$m>1$	$m > 1$	graphs with m edges and let G be an n -vertex graph with $m > 1$ edges. Consider arbitrary
voloshin-hypergraph_F00718	106	■ 1.000	$\{a_i\}$	$\{a_i\}$	$n - 1$ vertices. By the induction hypothesis, there exist nonnegative integer numbers $\{a_i\}$
voloshin-hypergraph_F00719	106	■ 1.000	$\{b_i\}$	$\{b_i\}$	and $\{b_i\}$ such that
voloshin-hypergraph_F00720	107	■ 1.000	$\lambda^{(k)}$	$\lambda^{(k)}$	<i>of G can be partitioned into $\lambda^{(k)}$ classes having $P(G, \lambda)/\lambda^{(k)}$ colorings each.</i>
voloshin-hypergraph_F00721	107	■ 1.000	$P(G, \lambda) / \lambda^{(k)}$	$P(G, \lambda)/\lambda^{(k)}$	<i>of G can be partitioned into $\lambda^{(k)}$ classes having $P(G, \lambda)/\lambda^{(k)}$ colorings each.</i>
voloshin-hypergraph_F00722	107	■ 1.000	$N_1=N_2$	$N_1 = N_2$	coloring of S . Denote the number of such colorings by N_2 . We claim that $N_1 = N_2$. Indeed,
voloshin-hypergraph_F00723	107	■ 1.000	j	j	first set (and vice versa) by a permutation of colors. Permutation of colors i and j means
voloshin-hypergraph_F00724	108	■ 1.000	$t=P(\left(G_S, \lambda\right)=P(\left(K_k, \lambda\right)=\lambda^{(k)}$	$t = P(G_S, \lambda) = P(K_k, \lambda) = \lambda^{(k)}$	Let $t = P(G_S, \lambda) = P(K_k, \lambda) = \lambda^{(k)}$. Repeating the reasoning above for each of t color-
voloshin-hypergraph_F00725	108	■ 1.000	$N_1=N_2=N_3=\cdots=N_t$	$N_1 = N_2 = N_3 = \dots = N_t$	ings of S , we arrive to the conclusion that $N_1 = N_2 = N_3 = \dots = N_t$. Hence the set of all
voloshin-hypergraph_F00726	108	■ 1.000	$k-1$	$k - 1$	induced by S has two different colorings, one with k colors and another with $k - 1$ colors.
voloshin-hypergraph_F00727	108	■ 1.000	$P(\left(G_{X_1 \cup S}, \lambda\right)$	$P(G_{X_1 \cup S}, \lambda)$	Proof. Since G_S is a clique of size k in $G_{X_1 \cup S}$, by Lemma 5.4.1 the set of all $P(G_{X_1 \cup S}, \lambda)$
voloshin-hypergraph_F00728	108	■ 1.000	$P(\left(G_S, \lambda\right)=\lambda^{(k)}$	$P(G_S, \lambda) = \lambda^{(k)}$	colorings can be partitioned into $P(G_S, \lambda) = \lambda^{(k)}$ equal classes. Each class contains
voloshin-hypergraph_F00729	108	■ 1.000	$P(\left(G_{X_1 \cup S}, \lambda\right) / \lambda^{(k)}$	$P(G_{X_1 \cup S}, \lambda)/\lambda^{(k)}$	$P(G_{X_1 \cup S}, \lambda)/\lambda^{(k)}$ colorings. Similarly, the $P(G_{X_2 \cup S}, \lambda)$ colorings of $G_{X_2 \cup S}$ can be partitioned
voloshin-hypergraph_F00730	108	■ 1.000	$P(\left(G_{X_2 \cup S}, \lambda\right)$	$P(G_{X_2 \cup S}, \lambda)$	$P(G_{X_1 \cup S}, \lambda)/\lambda^{(k)}$ colorings. Similarly, the $P(G_{X_2 \cup S}, \lambda)$ colorings of $G_{X_2 \cup S}$ can be partitioned
voloshin-hypergraph_F00731	108	■ 1.000	$P(\left(G_{X_2 \cup S}, \lambda\right) / \lambda^{(k)}$	$P(G_{X_2 \cup S}, \lambda)/\lambda^{(k)}$	into $\lambda^{(k)}$ equal classes, and each such class contains exactly $P(G_{X_2 \cup S}, \lambda)/\lambda^{(k)}$ colorings.
voloshin-hypergraph_F00732	109	■ 0.947	$G_1, \ G_2, \ \ldots \ G_l$	$G_1, G_2, \dots G_l$	<i>clique of size k. Suppose $G_1, G_2, \dots, G_l$ are the derived subgraphs with respect to the sepa-</i>
voloshin-hypergraph_F00733	109	■ 0.996	$G_i, \ i=1, 2, \ \ldots, \ l$	$G_i, i = 1, 2, \dots, l$	$G_i, i = 1, 2, \dots, l$. The colorings of graphs all combine in every class of colorings generated

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voloshin-hypergraph_ F00734	109	■ 0.733	$P(\emptyset, \lambda)=1$	$P(\emptyset, \lambda) = 1$	having a separator which is an empty set, and define $P(\emptyset, \lambda) = 1$.
voloshin-hypergraph_ F00735	109	■ 1.000	$G_{\{1\}}=C_{\{4\}}, G_{\{2\}}=K_{\{3\}}$	$G_1 = C_4, G_2 = K_3$	An example to the theorem above is shown in Figure 5.11. Since $G_1 = C_4, G_2 = K_3$ and
voloshin-hypergraph_ F00736	109	■ 1.000	$ S =2$	$ S = 2$	$ S = 2$, we use the formula (5.4) for $P(C_4, \lambda)$ and $P(K_3, \lambda) = \lambda(\lambda - 1)(\lambda - 2)$ to compute
voloshin-hypergraph_ F00737	109	■ 1.000	$P(\text{left}(C_{\{4\}}, \lambda\text{right}))$	$P(C_4, \lambda)$	$ S = 2$, we use the formula (5.4) for $P(C_4, \lambda)$ and $P(K_3, \lambda) = \lambda(\lambda - 1)(\lambda - 2)$ to compute
voloshin-hypergraph_ F00738	109	■ 1.000	$P(\text{left}(K_{\{3\}}, \lambda\text{right}))=\lambda(\lambda-1)(\lambda-2)$	$P(K_3, \lambda) = \lambda(\lambda - 1)(\lambda - 2)$	$ S = 2$, we use the formula (5.4) for $P(C_4, \lambda)$ and $P(K_3, \lambda) = \lambda(\lambda - 1)(\lambda - 2)$ to compute
voloshin-hypergraph_ F00739	109	■ 0.986	$X_{\{2\}}=\{x\}$	$X_2 = \{x\}$	separator, i.e. a separator induced by a clique on k vertices. In this case $X_2 = \{x\}$, see
voloshin-hypergraph_ F00742	110	■ 1.000	$G=K_{\{k+1\}}, G-x=K_{\{k\}}$	$G = K_{k+1}, G - x = K_k$	If G is a complete graph, then $G = K_{k+1}, G - x = K_k$ and the formula follows directly.
voloshin-hypergraph_ F00743	110	■ 1.000	$G_{\{1\}}=G-x=C_{\{4\}}$	$G_1 = G - x = C_4$	and $G_1 = G - x = C_4$, compute the chromatic polynomial of G directly:
voloshin-hypergraph_ F00744	110	■ 0.999	$(i-k) r_{\{i\}}(G-x)$	$(i - k)r_i(G - x)$	colors in the neighborhood of x are different, we have $(i - k)r_i(G - x)$ such possibilities. If
voloshin-hypergraph_ F00745	110	■ 1.000	$r_{\{i-1\}}(G-x)$	$r_{i-1}(G - x)$	vertex x is colored with the color not used in $G - x$, then there are $r_{i-1}(G - x)$ possibilities.
voloshin-hypergraph_ F00746	110	■ 1.000	$E_{\{n\}}: r_{\{i\}}(\text{left}(E_{\{n\}}\text{right}))=S(n, i)$	$E_n : r_i(E_n) = S(n, i)$	Proof. Apply the theorem above to graph E_n : $r_i(E_n) = S(n, i)$ and each vertex may be
voloshin-hypergraph_ F00747	110	■ 1.000	$r_{\{1\}}(\text{left}(E_{\{1\}}\text{right}))=S(1,1)=1$	$r_1(E_1) = S(1, 1) = 1$	Since trivially $r_1(E_1) = S(1, 1) = 1$, the first five rows and columns of Stirling numbers
voloshin-hypergraph_ F00748	110	■ 1.000	$P_{\{3\}}, P_{\{4\}}, P_{\{5\}}, P_{\{n\}}$	P_3, P_4, P_5, P_n	P_3, P_4, P_5, P_n for $n \geq 6$, for C_4, C_5, C_6, C_7 , and for W_4, W_5, W_6 .
voloshin-hypergraph_ F00749	110	■ 1.000	$C_{\{4\}}, C_{\{5\}}, C_{\{6\}}, C_{\{7\}}$	C_4, C_5, C_6, C_7	P_3, P_4, P_5, P_n for $n \geq 6$, for C_4, C_5, C_6, C_7 , and for W_4, W_5, W_6 .
voloshin-hypergraph_ F00750	110	■ 1.000	$W_{\{4\}}, W_{\{5\}}, W_{\{6\}}$	W_4, W_5, W_6	P_3, P_4, P_5, P_n for $n \geq 6$, for C_4, C_5, C_6, C_7 , and for W_4, W_5, W_6 .
voloshin-hypergraph_ F00751	111	■ 1.000	$E_{\{6\}}, E_{\{7\}}$	E_6, E_7	2. Compute the Stirling numbers of the second kind corresponding to E_6, E_7 , and E_8 , and of the
voloshin-hypergraph_ F00752	111	■ 1.000	$E_{\{8\}}$	E_8	2. Compute the Stirling numbers of the second kind corresponding to E_6, E_7 , and E_8 , and of the
voloshin-hypergraph_ F00753	111	■ 1.000	$K_{\{3\}}, K_{\{4\}}, K_{\{5\}}$	K_3, K_4, K_5	first kind corresponding to K_3, K_4, K_5 , and K_6 .
voloshin-hypergraph_ F00754	111	■ 1.000	$i \leq n$	$i \leq n$	1. For any positive integers n and $i \leq n$, compute $S(n, i)$.
voloshin-hypergraph_ F00755	111	■ 0.994	$G_{\{1\}}=(X, E), X =n$	$G_1 = (X, E), X = n$	Let $G_1 = (X, E), X = n$, be a chordal graph, and $\sigma = (x_1, x_2, \dots, x_n)$ be a simplicial
voloshin-hypergraph_ F00756	111	■ 0.999	$G_{\{2\}}=G_{\{1\}}-x_{\{1\}}, G_{\{3\}}=G_{\{2\}}-x_{\{2\}}, \dots, G_{\{n\}}=G_{\{n-1\}}-x_{\{n-1\}}=\text{left}\{x_{\{n\}}\text{right}\}$	$G_2 = G_1 - x_1, G_3 = G_2 - x_2, \dots, G_n = G_{n-1} - x_{n-1} = \{x_n\}$	elimination ordering. Let $G_2 = G_1 - x_1, G_3 = G_2 - x_2, \dots, G_n = G_{n-1} - x_{n-1} = \{x_n\}$ such
voloshin-hypergraph_ F00757	111	■ 0.858	$G_{\{n+1\}}=G_{\{n\}}-x_{\{n\}}=\emptyset$	$G_{n+1} = G_n - x_n = \emptyset$	that $G_{n+1} = G_n - x_n = \emptyset$. It means that vertex x_i is a simplicial vertex of degree, say, k_i , in
voloshin-hypergraph_ F00758	111	■ 0.858	$k_{\{i\}}$	k_i	that $G_{n+1} = G_n - x_n = \emptyset$. It means that vertex x_i is a simplicial vertex of degree, say, k_i , in
voloshin-hypergraph_ F00759	111	■ 1.000	$G_{\{i\}}, i=1, \dots, n$	$G_i, i = 1, \dots, n$	the graph $G_i, i = 1, \dots, n$. Apply sequentially formula (5.10) to each of the graphs in this
voloshin-hypergraph_ F00760	111	■ 1.000	$G_{\{n\}}$	G_n	respective simplicial vertices. Since G_n consists only of one vertex x_n , conclude that $k_n = 0$
voloshin-hypergraph_ F00761	111	■ 1.000	$k_{\{n\}}=0$	$k_n = 0$	respective simplicial vertices. Since G_n consists only of one vertex x_n , conclude that $k_n = 0$
voloshin-hypergraph_ F00762	111	■ 1.000	$\lambda=0$	$\lambda = 0$	and we fix the first root $\lambda = 0$. If G_1 is disconnected, then in each component there will
voloshin-hypergraph_ F00763	112	■ 1.000	$\omega(\text{left}(G_{\{1\}}\text{right}))$	$\omega(G_1)$	By the definition of clique number, G_1 contains a complete subgraph on $\omega(G_1)$ vertices,

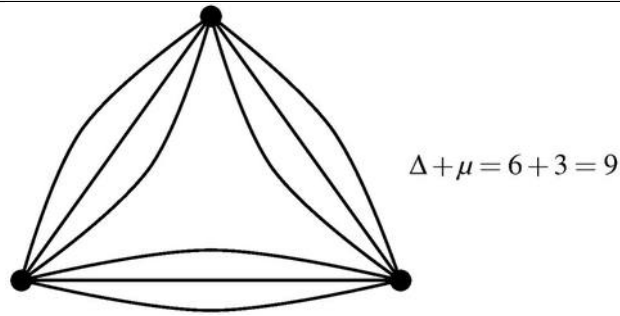
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00764	112	■ 1.000	$\omega\left(G_{\{1\}}\right)-1$	$\omega\left(G_1\right)-1$	so maximum among all k_i 's is $\omega(G_1) - 1$. We now fix the second root of the chromatic
voloshin-hypergraph_F00765	112	■ 1.000	$\lambda=\omega\left(G_{\{1\}}\right)-1$	$\lambda=\omega\left(G_1\right)-1$	polynomial, namely, $\lambda=\omega(G_1) - 1$. Recall that $\omega(G_1) \leq \chi(G_1)$ what implies that there is
voloshin-hypergraph_F00766	112	■ 1.000	$\omega\left(G_{\{1\}}\right) \leq \chi\left(G_{\{1\}}\right)$	$\omega\left(G_1\right) \leq \chi\left(G_1\right)$	polynomial, namely, $\lambda=\omega(G_1) - 1$. Recall that $\omega(G_1) \leq \chi(G_1)$ what implies that there is
voloshin-hypergraph_F00767	112	■ 0.999	ω	ω	no coloring using less than ω colors. Consequently, every integer on the closed interval
voloshin-hypergraph_F00768	112	■ 0.987	$[0, \omega-1]$	$[0, \omega-1]$	$[0, \omega-1]$ is a root of $P(G_1, \lambda)$. Therefore, all roots k_i are the integer numbers from the
voloshin-hypergraph_F00769	112	■ 1.000	$\chi\left(G_{\{1\}}\right)=\omega\left(G_{\{1\}}\right)$	$\chi\left(G_1\right)=\omega\left(G_1\right)$	Finally, let us show that $\chi(G_1) = \omega(G_1)$ by coloring the vertices in the order inverse
voloshin-hypergraph_F00770	112	■ 0.866	$x_{\{n\}}, x_{\{n-1\}}, x_{\{n-2\}}, \ldots, x_{\{2\}}, x_{\{1\}}$	$x_n, x_{n-1}, x_{n-2}, \ldots, x_2, x_1$	to σ , i.e. $x_n, x_{n-1}, x_{n-2}, \ldots, x_2, x_1$. In general case this procedure is called online coloring
voloshin-hypergraph_F00771	112	■ 0.993	$x_{\{n-1\}}$	x_{n-1}	To do that, color vertex x_n with color 1. Then, if x_{n-1} is not possible to color with 1 (i.e.,
voloshin-hypergraph_F00772	112	■ 0.699	$x_{\{n-2\}}$	x_{n-2}	x_n and x_{n-1} are adjacent), color it with color 2. For x_{n-2} try to use color 1, if not possible,
voloshin-hypergraph_F00773	112	■ 0.982	$\omega-1$	$\omega-1$	among k_i 's is equal to $\omega-1$, we eventually obtain a coloring of G_1 with ω colors. Hence,
voloshin-hypergraph_F00774	112	■ 1.000	$\omega(G)=\chi(G)$	$\omega(G)=\chi(G)$	the conclusion is that $\omega(G)=\chi(G)$. All these facts can be summarized in the next theorem.
voloshin-hypergraph_F00775	112	■ 0.995	$s_{\{i\}} \geq 1(i=0,1, \ldots, \chi-1)$	$s_i \geq 1(i=0,1, \ldots, \chi-1)$	where $s_i \geq 1(i=0,1, \ldots, \chi-1)$ is the number of simplicial vertices of degree i in the sim-
voloshin-hypergraph_F00776	112	■ 0.531	$2,2,1,0$	$2,2,1,0$	$2,2,1,0$. These are the roots of the chromatic polynomial, they all are from the interval
voloshin-hypergraph_F00777	113	■ 1.000	$\omega\left(G_{\{1\}}\right)=\chi\left(G_{\{1\}}\right)=4$	$\omega\left(G_1\right)=\chi\left(G_1\right)=4$	Vertices 3, 6, 7 and 8 form the unique maximum clique, so $\omega(G_1)=\chi(G_1)=4$. Online
voloshin-hypergraph_F00778	113	■ 1.000	$k_{\{1\}}=k_{\{2\}}=\cdots=k_{\{n-1\}}=1, k_{\{n\}}=0$	$k_1=k_2=\cdots=k_{n-1}=1, k_n=0$	i.e. $k_1=k_2=\cdots=k_{n-1}=1, k_n=0$.
voloshin-hypergraph_F00779	113	■ 1.000	$C_{\{n\}}, n \geq 1$	$C_n, n \geq 1$	Lemma 5.5.1 For every cycle $C_n, n \geq 1$
voloshin-hypergraph_F00780	113	■ 1.000	$n=1,2,3$	$n=1,2,3$	Proof. Induction on the number of vertices n . For $n=1,2,3$ verify directly (though C_1 and
voloshin-hypergraph_F00781	114	■ 0.987	$\chi^{\prime}\left(G^{\prime}\right)=\chi\left(G^{\prime}\right)$	$\chi^{\prime}=\chi\left(G^{\prime}\right)$	where $\chi^{\prime}=\chi(G')$, and $s'_i \geq 1$ ($i=0,1, \ldots, \chi'-1$) is the number of simplicial vertices of
voloshin-hypergraph_F00782	114	■ 0.987	$s_{\{i\}}^{\prime} \geq 1\left(i=0,1, \ldots, \chi^{\prime}-1\right)$	$s'_i \geq 1$ ($i=0,1, \ldots, \chi'-1$)	where $\chi^{\prime}=\chi(G')$, and $s'_i \geq 1$ ($i=0,1, \ldots, \chi'-1$) is the number of simplicial vertices of
voloshin-hypergraph_F00783	114	■ 1.000	$\chi\left(C_{\{k\}}\right)=2$	$\chi\left(C_k\right)=2$	It is easy to see that $\chi(C_k) = 2$ if k is even and $\chi(C_k) = 3$ if k is odd. Hence all the roots
voloshin-hypergraph_F00784	114	■ 1.000	$\chi\left(C_{\{k\}}\right)=3$	$\chi\left(C_k\right)=3$	It is easy to see that $\chi(C_k) = 2$ if k is even and $\chi(C_k) = 3$ if k is odd. Hence all the roots
voloshin-hypergraph_F00785	114	■ 1.000	$P\left(C_{\{k\}}, \lambda\right)$	$P\left(C_k, \lambda\right)$	of $P(C_k, \lambda)$ should be from the set $\{0,1,2\}$. But by Lemma 5.5.1 $P(C_k, \lambda) = (\lambda-1)^k +$
voloshin-hypergraph_F00786	114	■ 1.000	$\{0,1,2\}$	$\{0,1,2\}$	of $P(C_k, \lambda)$ should be from the set $\{0,1,2\}$. But by Lemma 5.5.1 $P(C_k, \lambda) = (\lambda-1)^k +$
voloshin-hypergraph_F00787	114	■ 1.000	$P\left(C_{\{k\}}, \lambda\right)=(\lambda-1)^k+$	$P\left(C_k, \lambda\right)=(\lambda-1)^k+$	of $P(C_k, \lambda)$ should be from the set $\{0,1,2\}$. But by Lemma 5.5.1 $P(C_k, \lambda) = (\lambda-1)^k +$
voloshin-hypergraph_F00788	114	■ 1.000	$(-1)^k(\lambda-1)$	$(-1)^k(\lambda-1)$	$(-1)^k(\lambda-1)$. One can prove that $P(C_k, \lambda)$ being a polynomial of degree $k \geq 4$ has at least
voloshin-hypergraph_F00789	114	■ 0.997	$A \subset X$	$A \subset X$	Let $G=(X, E)$ be a graph, $A \subset X$ be an arbitrary separator, $G_1^*=(X_1, E_1)$, $G_2^*=$
voloshin-hypergraph_F00790	114	■ 0.997	$G_{\{1\}}^*=\left(X_{\{1\}}, E_{\{1\}}\right), G_{\{2\}}^*=\left(X_{\{2\}}, E_{\{2\}}\right)$	$G_1^*=\left(X_1, E_1\right), G_2^*=\left(X_2, E_2\right)$	Let $G=(X, E)$ be a graph, $A \subset X$ be an arbitrary separator, $G_1^*=(X_1, E_1)$, $G_2^*=$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F00818	115	■ 1.000	$G_{i}^{\prime}$	G'_i	where the graphs G'_i are obtained from G_i by deletion of vertex x_0 , $i = 0, 1, \dots, k$.
voloshin-hypergraph_F00819	115	■ 1.000	$x_{\{0\}}, i=0,1, \ldots, k$	$x_0, i = 0, 1, \dots, k$	where the graphs G'_i are obtained from G_i by deletion of vertex $x_0, i = 0, 1, \dots, k$.
voloshin-hypergraph_F00820	115	■ 1.000	$i=0,2, \ldots, k$	$i = 0, 2, \dots, k$	for $i = 0, 2, \dots, k$.
voloshin-hypergraph_F00821	116	■ 1.000	$G_{\{1\}}, \ldots, G_{\{k\}}$	$G_1, \dots, G_k$	chromatic polynomial of a disconnected graph G with connected components $G_1, \dots, G_k$.
voloshin-hypergraph_F00822	116	■ 1.000	$A=\emptyset$	$A = \emptyset$	(case $A = \emptyset$).
voloshin-hypergraph_F00823	116	■ 0.999	$x \in X^{\prime}$	$x \in X'$	connected induced subgraph $G' = (X', E')$ for any vertex $x \in X'$ the following equality
voloshin-hypergraph_F00824	116	■ 0.982	$X=\{x\} \cup N(x)$	$X = \{x\} \cup N(x)$	Case 1. $X = \{x\} \cup N(x)$. Since in any coloring of G with λ colors vertex x requires a
voloshin-hypergraph_F00825	116	■ 1.000	$X \neq \{x\} \cup N(x)$	$X \neq \{x\} \cup N(x)$	Case 2. $X \neq \{x\} \cup N(x)$. Since G is connected, the subgraph $G_{N(x)}$ is a separator. Apply
voloshin-hypergraph_F00826	116	■ 1.000	$G_{N(x)}$	$G_{N(x)}$	Case 2. $X \neq \{x\} \cup N(x)$. Since G is connected, the subgraph $G_{N(x)}$ is a separator. Apply
voloshin-hypergraph_F00827	117	■ 1.000	C	C	denoted by just C . Observe that $C - x$ is a tree on $k - 1$ vertices, and $C_{N(x)} = E_2$. We see
voloshin-hypergraph_F00828	117	■ 1.000	$C - x$	$C - x$	denoted by just C . Observe that $C - x$ is a tree on $k - 1$ vertices, and $C_{N(x)} = E_2$. We see
voloshin-hypergraph_F00829	117	■ 1.000	$C_{N(x)}=E_2$	$C_{N(x)} = E_2$	denoted by just C . Observe that $C - x$ is a tree on $k - 1$ vertices, and $C_{N(x)} = E_2$. We see
voloshin-hypergraph_F00830	117	■ 0.996	$P(C-x, \lambda)=\lambda(\lambda-1)^{k-2}, W\left(C_{N(x)}, \lambda\right)=\lambda^{-1}(\lambda-1)^2$	$P(C - x, \lambda) = \lambda(\lambda - 1)^{k-2}, W(C_{N(x)}, \lambda) = \lambda^{-1}(\lambda - 1)^2$	that $P(C - x, \lambda) = \lambda(\lambda - 1)^{k-2}, W(C_{N(x)}, \lambda) = \lambda^{-1}(\lambda - 1)^2$, and by the formula
voloshin-hypergraph_F00831	117	■ 1.000	$N(x)=K_k$	$N(x) = K_k$	Proof. Indeed, $N(x) = K_k$, so
voloshin-hypergraph_F00832	117	■ 1.000	$W\left(G_{N(4)}, \lambda\right)$	$W(G_{N(4)}, \lambda)$	the function $W(G_{N(4)}, \lambda)$. Since $N(4) = \{1, 5, 6\}$, the induced subgraph is a tree on three
voloshin-hypergraph_F00833	117	■ 1.000	$N(4)=\{1,5,6\}$	$N(4) = \{1, 5, 6\}$	the function $W(G_{N(4)}, \lambda)$. Since $N(4) = \{1, 5, 6\}$, the induced subgraph is a tree on three
voloshin-hypergraph_F00834	118	■ 1.000	$\{5,6,7,8\}$	$\{5, 6, 7, 8\}$	1. For graph G in Figure 5.16, compute the chromatic polynomial using separator $\{5, 6, 7, 8\}$.
voloshin-hypergraph_F00835	118	■ 0.975	$G, \chi(G) \leq M(G)+1$	$G, \chi(G) \leq M(G) + 1$	Theorem 5.6.1 For any graph $G, \chi(G) \leq M(G) + 1$.
voloshin-hypergraph_F00836	118	■ 1.000	$G_{\{1\}}=G, G_{\{2\}}, G_{\{3\}}, \ldots, G_{\{n\}}$	$G_1 = G, G_2, G_3, \dots, G_n$	We obtain a sequence of graphs $G_1 = G, G_2, G_3, \dots, G_n$ and corresponding se-
voloshin-hypergraph_F00837	118	■ 1.000	$x_{\{n\}}, x_{\{n-1\}}, \ldots, x_{\{2\}}, x_{\{1\}}$	$x_n, x_{n-1}, \dots, x_2, x_1$	structing it in ordering $x_n, x_{n-1}, \dots, x_2, x_1$. That means we color the vertices in or-
voloshin-hypergraph_F00838	118	■ 0.988	$(M(G)+1)$	$(M(G) + 1)$	wise, we just assign to x_i the first missing color). We then assign $(M(G) + 1)th$
voloshin-hypergraph_F00839	118	■ 1.000	$M(G)+1$	$M(G) + 1$	$M(G) + 1$.

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voloshin-hypergraph_ F08840	119	—	G_{i+1}	G_{i+1}	
voloshin-hypergraph_ F08841	119	■ 0.999	$\chi(G) \leq 6$	$\chi(G) \leq 6$	Corollary 5.6.1 If G is a planar graph, then $\chi(G) \leq 6$.
voloshin-hypergraph_ F08842	119	■ 1.000	$M(G) \leq 5$	$M(G) \leq 5$	minimum degrees, the degrees do not exceed 5. Hence $M(G) \leq 5$. Applying Theorem 5.6.1
voloshin-hypergraph_ F08843	119	■ 1.000	$\chi(G) \leq 5$	$\chi(G) \leq 5$	then $\chi(G) \leq 5$.
voloshin-hypergraph_ F08844	119	■ 1.000	≤ 5	≤ 5	ing ≤ 5 vertices are 5-colorable (color all vertices differently).
voloshin-hypergraph_ F08845	119	■ 0.998	$d(x) \leq 5$	$d(x) \leq 5$	$d(x) \leq 5$. By the induction hypothesis, $\chi(G-x) \leq 5$. Consider a proper 5-coloring of $G-x$.
voloshin-hypergraph_ F08846	119	■ 0.998	$\chi(G-x) \leq 5$	$\chi(G-x) \leq 5$	$d(x) \leq 5$. By the induction hypothesis, $\chi(G-x) \leq 5$. Consider a proper 5-coloring of $G-x$.
voloshin-hypergraph_ F08847	119	■ 0.693	≤ 4	≤ 4	Case 1: the number of colors used in $N(x)$ is ≤ 4 . We assign the 5th color to x and
voloshin-hypergraph_ F08848	119	■ 0.988	$d(x)=5$	$d(x) = 5$	Case 2: the number of colors used in $N(x)$ is 5. It means that $d(x) = 5$. The main idea
voloshin-hypergraph_ F08849	119	■ 1.000	$G_i=G$	$G_i = G$	The situation is equivalent to that shown in Figure 5.17 for $G_i = G$, but now it can
voloshin-hypergraph_ F08850	119	■ 0.999	$N(x)=$	$N(x) =$	be depicted more specifically in Figure 5.18. Without loss of generality, denote $N(x) =$
voloshin-hypergraph_ F08851	119	■ 0.812	(a, c)	(a, c)	Subcase 2.1: in the 5-coloring of $G-x$, there exists an (a, c) -path P with vertices
voloshin-hypergraph_ F08852	119	■ 0.812	P	P	Subcase 2.1: in the 5-coloring of $G-x$, there exists an (a, c) -path P with vertices
voloshin-hypergraph_ F08853	119	■ 1.000	$a x$	ax	Path P along with edges ax and xc form a cycle C in the plane with all vertices colored 1
voloshin-hypergraph_ F08854	119	■ 1.000	$x c$	xc	Path P along with edges ax and xc form a cycle C in the plane with all vertices colored 1
voloshin-hypergraph_ F08855	120	■ 1.000	G_{13}	G_{13}	alternating colors 1 and 3. Let G_{13} be a subgraph of $G-x$ induced by the vertices colored
voloshin-hypergraph_ F08856	120	■ 0.725	c	c	with colors 1 and 3. G_{13} is a disconnected graph and vertices a and c are in different
voloshin-hypergraph_ F08857	120	■ 0.994	$\chi(G) \leq 4$	$\chi(G) \leq 4$	graph, then $\chi(G) \leq 4$.
voloshin-hypergraph_ F08858	120	■ 1.000	$\chi(G-x) \leq 4$	$\chi(G-x) \leq 4$	esis, $\chi(G-x) \leq 4$. Consider a proper 4-coloring of $G-x$. By this, all the vertices of $N(x)$
voloshin-hypergraph_ F08859	121	■ 0.912	≤ 3	≤ 3	Case 1: the number of colors used in $N(x)$ is ≤ 3 . We assign the 4th color to x and
voloshin-hypergraph_ F08860	121	■ 0.997	$d(x)=4$	$d(x) = 4$	Case 2: the number of colors used in $N(x)$ is 4. It means that $d(x) = 4$ or $d(x) = 5$.
voloshin-hypergraph_ F08861	121	■ 0.999	$G^{\prime}=G-e$	$G' = G - e$	Subcase 2.1: $d(x) = 4$. Apply the reasoning from Theorem 5.6.2 for $\overline{G'} = \overline{G-d}$ (in
voloshin-hypergraph_ F08862	121	■ 0.973	$N(x)=\{a, b, c, d, e\}$	$N(x) = \{a, b, c, d, e\}$	the same color. Denote $\overline{N(x)} = \{a, b, c, d, e\}$ and respective colors by 1, 2, 3, and 4. Since

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voloshin-hypergraph_F08891	126	■ 0.999	$\chi(G)=3, \omega(G)=3$	$\chi(G) = 3, \omega(G) = 3$	For example, a graph G in Figure 5.24 has $\chi(G) = 3$, $\omega(G) = 3$, and $\alpha(G) = 2$ and
voloshin-hypergraph_F08892	126	■ 1.000	$\theta(G)=2$	$\theta(G) = 2$	$\theta(G) = 2$. A proper 3-coloring is shown by numbers, and clique covering is shown by
voloshin-hypergraph_F08893	126	■ 1.000	$\chi(G)=2, \omega(G)=2, \alpha(G)=3$	$\chi(G) = 2, \omega(G) = 2, \alpha(G) = 3$	dotted ellipses. Respectively, for its complement $\overline{G}$, we have $\chi(G)=2, \omega(G)=2, \alpha(G)=3$
voloshin-hypergraph_F08894	126	■ 1.000	$C_5: \chi(C_5)=3, \omega(C_5)=2, \alpha(C_5)=2$	$C_5: \chi(C_5) = 3, \omega(C_5) = 2, \alpha(C_5) = 2$	perfect graphs. An example of an imperfect graph is C_5 : $\chi(C_5) = 3, \omega(C_5) = 2, \alpha(C_5) = 2$
voloshin-hypergraph_F08895	126	■ 0.972	$\theta(C_5)=3$	$\theta(C_5) = 3$	and $\theta(C_5) = 3$ (recall that C_5 is isomorphic to $\overline{C_5}$). In these examples graphs and their
voloshin-hypergraph_F08896	126	■ 1.000	$\theta(G') \neq \alpha(G')$	$\theta(G') \neq \alpha(G')$	$\theta(G') \neq \alpha(G')$ for at least one induced subgraph? In other words, may we have a perfect
voloshin-hypergraph_F08897	127	■ 1.000	$n=3,4,5$	$n = 3, 4, 5$	for $n=3,4,5$. Let it be true for all quasi-triangulated graphs on $< n$ vertices and $n(G) = n$.
voloshin-hypergraph_F08898	127	■ 1.000	$\chi(G-x)=$	$\chi(G-x) =$	Case 1: x is a simplicial vertex in G . Consider a proper coloring of $G-x$ by $\chi(G-x) =$
voloshin-hypergraph_F08899	127	■ 1.000	$\omega(G-x)$	$\omega(G-x)$	$\omega(G-x)$ colors. If $\omega(G) = \omega(G-x)$, then $ N(x) \leq \omega(G-x) - 1$ and we can use a color
voloshin-hypergraph_F09900	127	■ 1.000	$\omega(G)=\omega(G-x)$	$\omega(G) = \omega(G-x)$	$\omega(G-x)$ colors. If $\omega(G) = \omega(G-x)$, then $ N(x) \leq \omega(G-x) - 1$ and we can use a color
voloshin-hypergraph_F09901	127	■ 1.000	$ N(x) \leq \omega(G-x) - 1$	$ N(x) \leq \omega(G-x) - 1$	$\omega(G-x)$ colors. If $\omega(G) = \omega(G-x)$, then $ N(x) \leq \omega(G-x) - 1$ and we can use a color
voloshin-hypergraph_F09902	127	■ 0.953	$\omega(G)=\omega(G-x)+1$	$\omega(G) = \omega(G-x) + 1$	missing in $N(x)$ to color vertex x . If $\omega(G) = \omega(G-x) + 1$, then $ N(x) = \omega(G-x)$ and we
voloshin-hypergraph_F09903	127	■ 0.953	$ N(x) =\omega(G-x)$	$ N(x) = \omega(G-x)$	missing in $N(x)$ to color vertex x . If $\omega(G) = \omega(G-x) + 1$, then $ N(x) = \omega(G-x)$ and we
voloshin-hypergraph_F09904	127	■ 0.902	$\overline{G-x}$	$\overline{G-x}$	$< n$ vertices, and by the induction hypothesis is perfect. By Theorem 5.7.1, graph $\overline{G-x}$ is
voloshin-hypergraph_F09905	127	■ 0.993	C_{2k+1}	C_{2k+1}	Thomas, 2003 A graph G is perfect if and only if it contains no induced C_{2k+1} and $\overline{C_{2k+1}}$,
voloshin-hypergraph_F09906	127	■ 0.993	$\bar{C}_{2k+1}$	$\bar{C}_{2k+1}$	Thomas, 2003 A graph G is perfect if and only if it contains no induced C_{2k+1} and $\overline{C_{2k+1}}$,
voloshin-hypergraph_F09907	128	■ 0.999	P_5, C_4, C_6, K_4	P_5, C_4, C_6, K_4	3. Apply one step of Mycielski's construction starting with P_5, C_4, C_6, K_4 , prism.
voloshin-hypergraph_F09908	128	■ 0.999	$\theta(C_7) > \alpha(C_7)$	$\theta(C_7) > \alpha(C_7)$	4. Prove that $\theta(C_7) > \alpha(C_7)$.
voloshin-hypergraph_F09909	128	■ 0.829	$\Theta(G)$	$\Theta(G)$	3. Given a quasi-triangulated graph G , find the clique cover number $\theta(G)$ and the respective
voloshin-hypergraph_F09910	128	■ 1.000	$\lambda \geq 0$	$\lambda \geq 0$	loops, and $\lambda \geq 0$ be the number of available colors.
voloshin-hypergraph_F09911	128	■ 1.000	$ E $	$ E $	Since the maximum number of colors that can be used equals $ E $, we are interested in
voloshin-hypergraph_F09912	128	■ 1.000	$\chi'(G)$	$\chi'(G)$	G and denoted by $\chi'(G)$.
voloshin-hypergraph_F09913	128	■ 1.000	$G' =$	$G' =$	vertices representing the edges of G . For graph $G = (X, E)$, construct a simple graph $G' =$
voloshin-hypergraph_F09914	128	■ 1.000	(X', E')	(X', E')	(X', E') such that $X' = E$ and E' is formed in the following way: two vertices in G' are
voloshin-hypergraph_F09915	128	■ 1.000	$X' = E$	$X' = E$	(X', E') such that $X' = E$ and E' is formed in the following way: two vertices in G' are
voloshin-hypergraph_F09916	129	■ 1.000	$G' = L(G)$	$G' = L(G)$	vertex 4-coloring of $G' = L(G)$ is shown in Figure 5.25. One can see that every proper
voloshin-hypergraph_F09917	129	■ 1.000	$P(L(G), \lambda) = \lambda(\lambda-1)(\lambda-2)(\lambda-3)$	$P(L(G), \lambda) = \lambda(\lambda-1)(\lambda-2)^2(\lambda-3)$	matic polynomial $P(L(G), \lambda) = \lambda(\lambda-1)(\lambda-2)^2(\lambda-3)$ as it follows from the respective

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voloshin-hypergraph_F00918	129	■ 1.000	$\{b, c, d, e\}$	$\{b, c, d, e\}$	but not vice versa: clique $\boxed{b, c, d, e}$ in $L(G)$ does not have a corresponding vertex in G . At
voloshin-hypergraph_F00919	129	■ 1.000	$\chi^{\{\prime\}}(G)=4=\chi(L(G))$	$\chi'(G) = 4 = \chi(L(G))$	last, observe that $\boxed{\chi'(G)=4=\chi(L(G))}$, and moreover, the equality $\chi'(G) = \chi(L(G))$ holds
voloshin-hypergraph_F00920	129	■ 1.000	$\chi^{\{\prime\}}(G)=\chi(L(G))$	$\chi'(G) = \chi(L(G))$	last, observe that $\chi'(G) = 4 = \chi(L(G))$, and moreover, the equality $\chi'(G) = \chi(L(G))$ holds
voloshin-hypergraph_F00921	129	■ 1.000	$\chi^{\{\prime\}}(G) \geq \Delta(G)$	$\chi'(G) \geq \Delta(G)$	As usually, let $\Delta(G)$ denote the maximum vertex degree of G . Evidently, $\chi'(G) \geq \Delta(G)$.
voloshin-hypergraph_F00922	129	■ 1.000	Δ	Δ	Maximum degree Δ plays a similar role for the chromatic index χ' as the clique number ω
voloshin-hypergraph_F00923	129	■ 1.000	$\chi^{\{\prime\}}$	χ'	Maximum degree Δ plays a similar role for the chromatic index χ' as the clique number ω
voloshin-hypergraph_F00924	129	■ 0.996	$\boldsymbol{\chi}$	χ	plays for the chromatic number χ in vertex coloring where $\chi \geq \omega$. But how far may be the
voloshin-hypergraph_F00925	129	■ 0.996	$\boldsymbol{\chi} \geq \boldsymbol{\omega}$	$\chi \geq \omega$	plays for the chromatic number χ in vertex coloring where $\chi \geq \omega$. But how far may be the
voloshin-hypergraph_F00926	129	■ 1.000	C_{2k}	C_{2k}	Proof. Simple examples (like C_{2k} and C_{2k+1}) show that there are graphs with both values
voloshin-hypergraph_F00927	129	■ 0.990	$\Delta+1$	$\Delta + 1$	of χ' . We will show that any graph can be edge colored with $\Delta + 1$ colors.
voloshin-hypergraph_F00928	129	■ 1.000	$y_{\{0\}}$	y_0	Suppose an edge connecting vertices x and y_0 and denoted by just xy_0 , is uncolored. We will
voloshin-hypergraph_F00929	129	■ 1.000	$x y_{\{0\}}$	xy_0	Suppose an edge connecting vertices x and y_0 and denoted by just xy_0 , is uncolored. We will
voloshin-hypergraph_F00930	130	■ 1.000	$c_{\{0\}}$	c_0	c_0 be missing at x and color c_1 be missing at y_0 . If there is an edge xy_k colored c_1 with both
voloshin-hypergraph_F00931	130	■ 1.000	$c_{\{1\}}$	c_1	c_0 be missing at x and color c_1 be missing at y_0 . If there is an edge xy_k colored c_1 with both
voloshin-hypergraph_F00932	130	■ 1.000	$x y_{\{k\}}$	xy_k	c_0 be missing at x and color c_1 be missing at y_0 . If there is an edge xy_k colored c_1 with both
voloshin-hypergraph_F00933	130	■ 1.000	$c_{\{i\}}$	c_i	ends missing a color c_i , then re-color xy_k with c_i and color xy_0 with c_1 .
voloshin-hypergraph_F00934	130	■ 1.000	$x y_{\{1\}}$	xy_1	The bad case: there is no edge xy_k colored c_1 with both ends missing a color. Edge xy_1
voloshin-hypergraph_F00935	130	■ 1.000	$c_{\{2\}}$	c_2	is colored c_1 ; a color c_2 is missing at y_1 . Edge xy_2 is colored c_2 , a color c_3 is missing at
voloshin-hypergraph_F00936	130	■ 1.000	$y_{\{1\}}$	y_1	is colored c_1 ; a color c_2 is missing at y_1 . Edge xy_2 is colored c_2 , a color c_3 is missing at
voloshin-hypergraph_F00937	130	■ 1.000	$x y_{\{2\}}$	xy_2	is colored c_1 ; a color c_2 is missing at y_1 . Edge xy_2 is colored c_2 , a color c_3 is missing at
voloshin-hypergraph_F00938	130	■ 1.000	$c_{\{3\}}$	c_3	is colored c_1 ; a color c_2 is missing at y_1 . Edge xy_2 is colored c_2 , a color c_3 is missing at
voloshin-hypergraph_F00939	130	■ 1.000	$y_{\{2\}}$	y_2	y_2 . Edge xy_3 is colored c_3 , a color c_4 is missing at y_3 , and so on, see Figure 5.26 (missing
voloshin-hypergraph_F00940	130	■ 1.000	$x y_{\{3\}}$	xy_3	y_2 . Edge xy_3 is colored c_3 , a color c_4 is missing at y_3 , and so on, see Figure 5.26 (missing
voloshin-hypergraph_F00941	130	■ 1.000	$c_{\{4\}}$	c_4	y_2 . Edge xy_3 is colored c_3 , a color c_4 is missing at y_3 , and so on, see Figure 5.26 (missing
voloshin-hypergraph_F00942	130	■ 1.000	$y_{\{3\}}$	y_3	y_2 . Edge xy_3 is colored c_3 , a color c_4 is missing at y_3 , and so on, see Figure 5.26 (missing
voloshin-hypergraph_F00943	130	■ 0.999	$\bar{c}_{\{i\}}$	$\bar{c}_i$	colors are denoted by $\bar{c}_i$). Suppose this sequence continues until a color c_{l+1} is missing at
voloshin-hypergraph_F00944	130	■ 0.999	$c_{\{l+1\}}$	c_{l+1}	colors are denoted by $\bar{c}_i$). Suppose this sequence continues until a color c_{l+1} is missing at
voloshin-hypergraph_F00945	130	■ 1.000	$y_{\{l\}}$	y_l	both vertex x and y_l . Then we re-color each edge xy_i from c_i to c_{i+1} , $i = 1, 2, \dots, l$. Call this

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voloshin-hypergraph_F00946	130	■ 1.000	$x y_{\{i\}}$	xy_i	both vertex x and y_l . Then we re-color each edge xy_i from c_i to c_{i+1} , $i = 1, 2, \dots, l$. Call this
voloshin-hypergraph_F00947	130	■ 1.000	$c_{\{i+1\}}, i=1,2, \ldots, l$	$c_{i+1}, i = 1, 2, \dots, l$	both vertex x and y_l . Then we re-color each edge xy_i from c_i to $c_{i+1}, i = 1, 2, \dots, l$. Call this
voloshin-hypergraph_F00948	130	■ 1.000	$c_{\{1\}}, c_{\{2\}}, c_{\{3\}}, \ldots$	$c_1, c_2, c_3, \dots$	of an edge. Then in the sequence of colors $c_1, c_2, c_3, \dots$, sooner or later, some color c_{l+1}
voloshin-hypergraph_F00949	130	■ 1.000	$c_{\{l+1\}}=c_{\{k\}}$	$c_{l+1} = c_k$	repeats (i.e., we have a loop) because the number of colors is finite. Let $c_{l+1} = c_k$ be the
voloshin-hypergraph_F00950	130	■ 1.000	$1 \leq k \leq l$	$1 \leq k \leq l$	first such coincidence of the colors, $1 \leq k \leq l$. Observe that color c_0 is present at all colored
voloshin-hypergraph_F00951	131	■ 1.000	$c_{\{k\}}$	c_k	two alternating colors c_0 and c_k and beginning at vertex y_l . Since we color the edges, each
voloshin-hypergraph_F00952	131	■ 1.000	$y_{\{k\}}$	y_k	Subcase 1: P reaches vertex y_k and therefore ends at x . Switch colors c_0 and c_k along
voloshin-hypergraph_F00953	131	■ 1.000	$y_{\{k-1\}}$	y_{k-1}	Subcase 2: P reaches vertex y_{k-1} and therefore ends at y_{k-1} . Again, switch colors c_0
voloshin-hypergraph_F00954	131	■ 1.000	$x y_{\{k-1\}}$	xy_{k-1}	obtain edge xy_{k-1} with color c_0 missing at both ends. Perform downshifting from y_{k-1} and
voloshin-hypergraph_F00955	131	■ 1.000	x, P	x, P	Subcase 3: P never reaches y_k or y_{k-1} . Since c_k is present at x , P never reaches vertex x .
voloshin-hypergraph_F00956	131	■ 1.000	$\chi^{\{\prime\}}(G)=$	$\chi'(G) =$	The theorem above partitions all simple graphs into two classes: Class 1 if $\chi'(G) =$
voloshin-hypergraph_F00957	131	■ 0.998	$\chi^{\{\prime\}}(G)=\Delta(G)+1$	$\chi'(G) = \Delta(G) + 1$	$\Delta(G)$ and Class 2 if $\chi'(G) = \Delta(G) + 1$. Determining whether a simple graph is Class 1 or
voloshin-hypergraph_F00958	131	■ 1.000	$\mu(G)$	$\mu(G)$	by $\mu(G)$, is the maximum number of parallel edges connecting a pair of vertices. Theorem
voloshin-hypergraph_F00959	131	■ 1.000	$\Delta(G) \leq \chi^{\{\prime\}}(G) \leq \Delta(G)+\mu(G)$	$\Delta(G) \leq \chi'(G) \leq \Delta(G) + \mu(G)$	$\Delta(G) \leq \chi'(G) \leq \Delta(G) + \mu(G)$.
voloshin-hypergraph_F00960	131	—	$\Delta+\mu$	$\Delta + \mu$	
voloshin-hypergraph_F00961	131	■ 1.000	$\chi^{\{\prime\}}(G)=\Delta(G)+\mu(G)$	$\chi'(G) = \Delta(G) + \mu(G)$	The “worst” example when the upper bound is achieved (i.e. $\chi'(G) = \Delta(G) + \mu(G)$).
voloshin-hypergraph_F00962	131	■ 1.000	$\mu=1$	$\mu = 1$	generalization of Theorem 5.8.1 because for simple graphs $\mu = 1$.
voloshin-hypergraph_F00963	131	■ 1.000	$\chi^{\{\prime\}}(G)=\Delta(G)$	$\chi'(G) = \Delta(G)$	Theorem 5.8.3 (König, 1916) <i>If G is a bipartite graph, then $\chi'(G) = \Delta(G)$.</i>
voloshin-hypergraph_F00964	131	■ 1.000	$K_{\{n\}}, C_{\{n\}}, W_{\{n\}}, K_{\{m, n\}}$	$K_n, C_n, W_n, K_{m,n}$	1. Find the chromatic index and an optimal edge coloring of: $K_n, C_n, W_n, K_{m,n}$, where $m, n \geq 3$.
voloshin-hypergraph_F00965	131	■ 1.000	$m, n \geq 3$	$m, n \geq 3$	1. Find the chromatic index and an optimal edge coloring of: $K_n, C_n, W_n, K_{m,n}$, where $m, n \geq 3$.
voloshin-hypergraph_F00966	132	■ 1.000	$\chi^{\{\prime\}}=4$	$\chi' = 4$	3. Prove that for the Petersen graph $\chi' = 4$.
voloshin-hypergraph_F00967	132	■ 0.988	$k \leq \lambda$	$k \leq \lambda$	agree that if k colors are really used in a proper edge coloring, $k \leq \lambda$, then the coloring is
voloshin-hypergraph_F00968	132	■ 0.881	$\boldsymbol{k}$	$\mathbf{k}$	called strict edge k-coloring , or just k -coloring.
voloshin-hypergraph_F00969	132	■ 1.000	$\bar{\chi}^{\{\prime\}}(G)$	$\bar{\chi}'(G)$	of multigraph G is called the upper chromatic index and denoted by $\bar{\chi}'(G)$.

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voloshin-hypergraph_ F08970	132	■ 0.988	$\bar{\chi}^{\prime}(G)=3$	$\bar{\chi}^{\prime}(G) = 3$	show that it uses the maximum number of colors, i.e., $\bar{\chi}^{\prime}(G) = 3$.
voloshin-hypergraph_ F08971	132	■ 0.990	$\bar{\chi}^{\prime}(G)=1$	$\bar{\chi}^{\prime}(G) = 1$	Theorem 5.9.1 For a connected multigraph G , the upper chromatic index $\bar{\chi}^{\prime}(G) = 1$ if and
voloshin-hypergraph_ F08972	132	■ 1.000	$\Delta(G) \leq 2$	$\Delta(G) \leq 2$	only if the maximum degree $\Delta(G) \leq 2$.
voloshin-hypergraph_ F08973	132	■ 0.876	$\Delta(G) \geq 3$	$\Delta(G) \geq 3$	Proof. $\Rightarrow$ Let $\bar{\chi}^{\prime}(G) = 1$. For a contradiction, suppose $\Delta(G) \geq 3$. If G contains a cycle,
voloshin-hypergraph_ F08974	133	■ 0.999	$\bar{\chi}^{\prime}(G) \geq 2$	$\bar{\chi}^{\prime}(G) \geq 2$	For a connected multigraph G , theorem 5.9.1 immediately implies that $\bar{\chi}^{\prime}(G) \geq 2$ if and
voloshin-hypergraph_ F08975	133	■ 0.996	$\{C_1, C_2, \dots, C_k\}$	$\{C_1, C_2, \dots, C_k\}$	$\{C_1, C_2, \dots, C_k\}$ where each C_i is a set of edges colored with color i . We denote such parti-
voloshin-hypergraph_ F08976	133	■ 0.996	C_i	C_i	$\{C_1, C_2, \dots, C_k\}$ where each C_i is a set of edges colored with color i . We denote such parti-
voloshin-hypergraph_ F08977	133	■ 1.000	$G_i = (X_i, C_i)$	$G_i = (X_i, C_i)$	Let $G_i = (X_i, C_i)$ be the subgraph of G with a set of vertices X_i determined by endpoints
voloshin-hypergraph_ F08978	133	■ 1.000	X_i	X_i	Let $G_i = (X_i, C_i)$ be the subgraph of G with a set of vertices X_i determined by endpoints
voloshin-hypergraph_ F08979	133	■ 0.839	$(\bar{\chi}^{\prime}(G)+1)$	$(\bar{\chi}^{\prime}(G) + 1)$	Therefore any proper coloring of G_i with at least two colors leads to a proper $\bar{\chi}^{\prime}(G) + 1$ -
voloshin-hypergraph_ F08980	133	■ 0.996	$1 \leq i \leq$	$1 \leq i \leq$	Corollary 5.9.1 Let f be a coloring of G using $\bar{\chi}^{\prime}(G)$ colors. Then each G_i , where $1 \leq i \leq$
voloshin-hypergraph_ F08981	133	■ 0.939	G_j	G_j	2) if x is satisfied by two cycles, say G_i and G_j , then x is the only common vertex for G_i
voloshin-hypergraph_ F08982	134	■ 1.000	G_k	G_k	by more than one G_k , then all G_i s which satisfy x , are all cycles with one vertex in common.
voloshin-hypergraph_ F08983	134	■ 1.000	G_i s	G_i s	by more than one G_k , then all G_i s which satisfy x , are all cycles with one vertex in common.
voloshin-hypergraph_ F08984	134	■ 0.998	$c^{\prime}$	$c^{\prime}$	Clearly, the number $c^{\prime}$ of subgraphs in A satisfies $c^{\prime} \leq c$ where c is the maximum number
voloshin-hypergraph_ F08985	134	■ 0.998	$c^{\prime} \leq c$	$c^{\prime} \leq c$	Clearly, the number $c^{\prime}$ of subgraphs in A satisfies $c^{\prime} \leq c$ where c is the maximum number
voloshin-hypergraph_ F08986	134	■ 1.000	$ X =n, E =m$	$ X = n, E = m$	multigraph with $ X = n, E = m$, the number of pendant vertices p , and the maximum
voloshin-hypergraph_ F08987	135	■ 1.000	$\bar{\chi}^{\prime}(G) \leq c+m-n+p$	$\bar{\chi}^{\prime}(G) \leq c + m - n + p$	Proof. We prove first that $\bar{\chi}^{\prime}(G) \leq c + m - n + p$. By Corollary 5.9.1, if a coloring f uses
voloshin-hypergraph_ F08988	135	■ 0.999	$G_i, 1 \leq i \leq \bar{\chi}^{\prime}(G)$	$G_i, 1 \leq i \leq \bar{\chi}^{\prime}(G)$	$\bar{\chi}^{\prime}(G)$ colors, then subgraphs $G_i, 1 \leq i \leq \bar{\chi}^{\prime}(G)$, are either cycles or paths, while Corollary
voloshin-hypergraph_ F08989	135	■ 1.000	$v^{\prime}$	$v^{\prime}$	Let $c^{\prime}$ be the number of subgraphs, and $v^{\prime}$ be the number of satisfied vertices in class
voloshin-hypergraph_ F08990	135	■ 1.000	G_l	G_l	disjoint. Let x be a satisfied vertex in G . If x is satisfied by some G_l from class A, then it is
voloshin-hypergraph_ F08991	135	■ 0.419	G_k s	G_k s	possible to look at all the other G_k s which satisfy x , like paths with endpoints x .
voloshin-hypergraph_ F08992	135	■ 0.915	$n-v^{\prime}-p$	$n - v^{\prime} - p$	number of vertices that are satisfied by subgraphs from B equals $n - v^{\prime} - p$. They are the
voloshin-hypergraph_ F08993	135	■ 0.973	$m^{\prime}$	$m^{\prime}$	Let $m^{\prime}$ be the number of edges in classes A and B combined. Then we obtain:
voloshin-hypergraph_ F08994	135	■ 1.000	$m-m^{\prime}$	$m - m^{\prime}$	Since class C contains $m - m^{\prime}$ edges, the number of colors in f is:
voloshin-hypergraph_ F08995	135	■ 0.995	$\bar{\chi}^{\prime}(G) \geq c+m-n+p$	$\bar{\chi}^{\prime}(G) \geq c + m - n + p$	Next we prove that $\bar{\chi}^{\prime}(G) \geq c + m - n + p$. Let us choose c vertex disjoint cycles
voloshin-hypergraph_ F08996	135	■ 1.000	$C_1, \dots, C_c$	$C_1, \dots, C_c$	$C_1, \dots, C_c$ and denote the subgraph with vertex set X and edges of these cycles by A_0 . Color

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voloshin-hypergraph_F00997	135	■ 1.000	$A_{\{0\}}$	A_0	$C_1, \dots, C_c$ and denote the subgraph with vertex set X and edges of these cycles by A_0 . Color
voloshin-hypergraph_F00998	135	■ 0.999	$1, \ldots, c$	$1, \dots, c$	the edges of the cycles properly with the colors $1, \dots, c$. Since the number of edges $m(A_0) =$
voloshin-hypergraph_F00999	135	■ 0.999	$m\left(A_{\{0\}}\right)=$	$m\left(A_0\right)=$	the edges of the cycles properly with the colors $1, \dots, c$. Since the number of edges $m(A_0) =$
voloshin-hypergraph_F01000	135	■ 1.000	$n\left(A_{\{0\}}\right)$	$n\left(A_0\right)$	$n(A_0)$, and the number of pendant vertices $p(A_0) = 0$, we obtain a proper coloring of A_0 us-
voloshin-hypergraph_F01001	135	■ 1.000	$p\left(A_{\{0\}}\right)=0$	$p\left(A_0\right)=0$	$n(A_0)$, and the number of pendant vertices $p(A_0) = 0$, we obtain a proper coloring of A_0 us-
voloshin-hypergraph_F01002	135	■ 1.000	$c+m\left(A_{\{0\}}\right)-n\left(A_{\{0\}}\right)+p\left(A_{\{0\}}\right)$	$c+m\left(A_0\right)-n\left(A_0\right)+p\left(A_0\right)$	ing $c+m(A_0)-n(A_0)+p(A_0)$ colors. Therefore, $\bar{\chi}'(A_0) \geq c+m(A_0)-n(A_0)+p(A_0)$, and
voloshin-hypergraph_F01003	135	■ 1.000	$\bar{\chi}'\left(A_{\{0\}}\right) \geq c+m\left(A_{\{0\}}\right)-n\left(A_{\{0\}}\right)+p\left(A_{\{0\}}\right)$	$\bar{\chi}'\left(A_0\right) \geq c+m\left(A_0\right)-n\left(A_0\right)+p\left(A_0\right)$	ing $c+m(A_0)-n(A_0)+p(A_0)$ colors. Therefore, $\bar{\chi}'(A_0) \geq c+m(A_0)-n(A_0)+p(A_0)$, and
voloshin-hypergraph_F01004	135	■ 1.000	$x=y$	$x=y$	either a cycle or a path (if they form a cycle, then $x = y$).
voloshin-hypergraph_F01005	135	■ 1.000	$A_{\{1\}}$	A_1	Add these new vertices and edges to A_0 and denote the obtained subgraph by A_1 . Color
voloshin-hypergraph_F01006	136	■ 1.000	$A_{\{2\}}, A_{\{3\}}, \ldots$	$A_2, A_3, \dots$	graphs $A_2, A_3, \dots$ until all the edges of G are colored and all non-pendant vertices satisfied.
voloshin-hypergraph_F01007	136	■ 1.000	$c=1, m=5, n=4$	$c=1, m=5, n=4$	For our example, see Figure 5.28, it is easy to see that $c=1, m=5, n=4$ and $p=1$,
voloshin-hypergraph_F01008	136	■ 1.000	$p=1$	$p=1$	For our example, see Figure 5.28, it is easy to see that $c=1, m=5, n=4$ and $p=1$,
voloshin-hypergraph_F01009	136	■ 1.000	$c=0$	$c=0$	Proof. Indeed, for any tree $c=0$ and $m=n-1$.
voloshin-hypergraph_F01010	136	■ 1.000	$\bar{\chi}'(G)=c+m-n+p$	$\bar{\chi}'(G)=c+m-n+p$	$\bar{\chi}'(G)=c+m-n+p$ and respective edge coloring.
voloshin-hypergraph_F01011	137	■ 1.000	$C_{\{n\}}, W_{\{n\}}, K_{\{n\}}, K_{\{m, n\}}$	$C_n, W_n, K_n, K_{m, n}$	5. Find the upper chromatic index and a maximal edge coloring of: $C_n, W_n, K_n, K_{m, n}$, where
voloshin-hypergraph_F01012	137	■ 1.000	$m, n \geq 2$	$m, n \geq 2$	$m, n \geq 2$.
voloshin-hypergraph_F01013	137	■ 1.000	$K_{\{n\}}, n \geq 5$	$K_n, n \geq 5$	4. Generate a maximal edge coloring of complete graph $K_n, n \geq 5$.
voloshin-hypergraph_F01014	139	■ 1.000	$x_{\{0\}}, e_{\{1\}}, x_{\{1\}}, e_{\{2\}}, \ldots, e_{\{k\}}, x_{\{k\}}$	$x_0, e_1, x_1, e_2, \dots, e_k, x_k$	at a vertex and has the following form: $x_0, e_1, x_1, e_2, \dots, e_k, x_k$. There are no restrictions on
voloshin-hypergraph_F01015	139	■ 0.999	$x_{\{k\}}$	x_k	beginning at x_0 and ending at x_k has the length k ; it is a (x_0, x_k) -walk. A walk with $x_0 = x_k$
voloshin-hypergraph_F01016	139	■ 0.999	$\left(x_{\{0\}}, x_{\{k\}}\right)$	$\left(x_0, x_k\right)$	beginning at x_0 and ending at x_k has the length k ; it is a (x_0, x_k) -walk. A walk with $x_0 = x_k$
voloshin-hypergraph_F01017	139	■ 0.999	$x_{\{0\}}=x_{\{k\}}$	$x_0=x_k$	beginning at x_0 and ending at x_k has the length k ; it is a (x_0, x_k) -walk. A walk with $x_0 = x_k$
voloshin-hypergraph_F01019	140	■ 0.562	$\sum_{i=1}^n d\left(x_{\{i\}}\right)=2 m$	$\sum_{i=1}^n d\left(x_i\right)=2 m$	2. The degree equality $\sum_{i=1}^n d(x_i) = 2m$ (See Proposition 1.1.1) implies that in any graph G , the
voloshin-hypergraph_F01020	140	■ 1.000	$k / 2$	$k / 2$	of trails that partition the edges is $k / 2$.
voloshin-hypergraph_F01021	141	■ 1.000	$ X =n \geq 3$	$ X =n \geq 3$	Theorem 6.2.1 (Ore, 1960) <i>If in a simple graph $G=(X, E)$, with $X =n \geq 3$, for every</i>
voloshin-hypergraph_F01022	141	■ 1.000	$d\left(x_{\{1\}}\right)+d\left(x_{\{n\}}\right) \geq n$	$d\left(x_1\right)+d\left(x_n\right) \geq n$	$d(x_1)+d(x_n) \geq n$. On the other hand, since x_1 and x_n are disjoint, $d(x_1) \leq n-2$ and $d(x_n) \leq$
voloshin-hypergraph_F01023	141	■ 1.000	$d\left(x_{\{1\}}\right) \leq n-2$	$d\left(x_1\right) \leq n-2$	$d(x_1)+d(x_n) \geq n$. On the other hand, since x_1 and x_n are disjoint, $d(x_1) \leq n-2$ and $d(x_n) \leq$
voloshin-hypergraph_F01024	141	■ 1.000	$d\left(x_{\{n\}}\right) \leq$	$d\left(x_n\right) \leq$	$d(x_1)+d(x_n) \geq n$. On the other hand, since x_1 and x_n are disjoint, $d(x_1) \leq n-2$ and $d(x_n) \leq$
voloshin-hypergraph_F01025	141	■ 1.000	$n-3$	$n-3$	$n-2$. Consider all $n-3$ internal edges of path P . If for each of them the left end is not

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voloshin-hypergraph_ F01026	142	■ 0.984	$\left(x_{\{i\}}, x_{\{i+1\}}\right)$	(x_i, x_{i+1})	of P there exists one, say, (x_i, x_{i+1}) with x_i adjacent to x_n and x_{i+1} adjacent to x_1 .
voloshin-hypergraph_ F01027	142	■ 0.984	$x_{\{i+1\}}$	x_{i+1}	of P there exists one, say, (x_i, x_{i+1}) with x_i adjacent to x_n and x_{i+1} adjacent to x_1 .
voloshin-hypergraph_ F01028	142	■ 1.000	$n / 2$	$n/2$	<i>of each vertex is at least $n/2$, then G is Hamiltonian.</i>
voloshin-hypergraph_ F01029	142	■ 1.000	$d(x)+d(y) \geq n$	$d(x) + d(y) \geq n$	Proof. The statement follows directly from the theorem above since $d(x) + d(y) \geq n$ holds
voloshin-hypergraph_ F01030	142	■ 1.000	$\overline{C_{\{7\}}}$	$\overline{C_7}$	5. Which of the graphs $\overline{C_7}$ and $\overline{C_8}$ are Hamiltonian?
voloshin-hypergraph_ F01031	142	■ 1.000	$\overline{C_{\{8\}}}$	$\overline{C_8}$	5. Which of the graphs $\overline{C_7}$ and $\overline{C_8}$ are Hamiltonian?
voloshin-hypergraph_ F01032	143	■ 1.000	$N=(X, A)$	$N = (X, A)$	is called a digraph. A digraph $N = (X, A)$ is called a network , if X is a set of vertices
voloshin-hypergraph_ F01033	143	■ 1.000	$a \in A$	$a \in A$	(sometimes called nodes), A is a set of arcs, and to each arc $a \in A$ a non-negative real
voloshin-hypergraph_ F01034	143	■ 1.000	$c(a)$	$c(a)$	number $c(a)$ is assigned which is called the capacity of arc a . For any vertex $y \in X$, any
voloshin-hypergraph_ F01035	143	■ 1.000	$y \in X$	$y \in X$	number $c(a)$ is assigned which is called the capacity of arc a . For any vertex $y \in X$, any
voloshin-hypergraph_ F01036	143	■ 0.666	(y, z)	(y, z)	arc of type (x, y) is called incoming , and every arc of type (y, z) is called outcoming . In N ,
voloshin-hypergraph_ F01037	143	■ 0.666	N	N	arc of type (x, y) is called incoming , and every arc of type (y, z) is called outcoming . In N ,
voloshin-hypergraph_ F01038	143	■ 1.000	$u \in X$	$u \in X$	there are two special vertices: $u \in X$, which is not incident to any incoming arc and is called
voloshin-hypergraph_ F01039	143	■ 1.000	$v \in X$	$v \in X$	the source , and $v \in X$, which is not incident to any outcoming arc and is called the sink .
voloshin-hypergraph_ F01040	143	■ 0.958	F	F	A network flow F is an assignment to <i>each</i> arc $a \in A$ a non-negative real number $f(a)$
voloshin-hypergraph_ F01041	143	■ 0.958	$f(a)$	$f(a)$	A network flow F is an assignment to <i>each</i> arc $a \in A$ a non-negative real number $f(a)$
voloshin-hypergraph_ F01042	143	■ 0.991	$f(a) \leq c(a)$	$f(a) \leq c(a)$	1. $f(a) \leq c(a)$;
voloshin-hypergraph_ F01043	143	■ 0.999	u	u	2. for any vertex $x \in X$, except u and v , the following flow conservation law holds: <i>the</i>
voloshin-hypergraph_ F01044	143	■ 0.999	v	v	2. for any vertex $x \in X$, except u and v , the following flow conservation law holds: <i>the</i>
voloshin-hypergraph_ F01045	143	■ 1.000	$f(a)=c(a)$	$f(a) = c(a)$	An arc $a \in A$ for which $f(a) = c(a)$ is called saturated ; if $f(a) < c(a)$, then arc a is
voloshin-hypergraph_ F01046	143	■ 1.000	$f(a)<c(a)$	$f(a) < c(a)$	An arc $a \in A$ for which $f(a) = c(a)$ is called saturated ; if $f(a) < c(a)$, then arc a is
voloshin-hypergraph_ F01047	143	■ 1.000	(u, x)	(u, x)	(u, x) ; the flow conservation law implies that it equals the sum of all flows in arcs of type
voloshin-hypergraph_ F01048	143	■ 1.000	(x, v)	(x, v)	(x, v) .
voloshin-hypergraph_ F01049	143	■ 0.798	$u=x_{\{0\}} \rightarrow x_{\{1\}} \rightarrow x_{\{2\}} \rightarrow \cdots \rightarrow x_{\{k\}}$	$u = x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \cdots \rightarrow x_k$	$u = x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \cdots \rightarrow x_k$ such that either (x_i, x_{i+1}) is an unsaturated arc, or (x_{i+1}, x_i) is
voloshin-hypergraph_ F01050	143	■ 0.798	$\left(x_{\{i+1\}}, x_{\{i\}}\right)$	(x_{i+1}, x_i)	$u = x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \cdots \rightarrow x_k$ such that either (x_i, x_{i+1}) is an unsaturated arc, or (x_{i+1}, x_i) is
voloshin-hypergraph_ F01051	143	■ 0.982	$i=0,1, \ldots, k-1$	$i = 0, 1, \dots, k - 1$	an arc with a non-zero flow, $i = 0, 1, \dots, k - 1$. Denote the set of all such vertices in all such
voloshin-hypergraph_ F01052	143	■ 1.000	$u \in Y$	$u \in Y$	at least one such sequence and $u \in Y$. Let $Z = X \setminus Y$. We claim that the sink $v \in Z$.

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voloshin-hypergraph_ F01053	143	■ 1.000	$Z=X \backslash Y$	$Z = X \setminus Y$	at least one such sequence and $u \in Y$. Let $Z = X \setminus Y$. We claim that the sink $v \in Z$.
voloshin-hypergraph_ F01054	143	■ 1.000	$v \in Z$	$v \in Z$	at least one such sequence and $u \in Y$. Let $Z = X \setminus Y$. We claim that the sink $v \in Z$.
voloshin-hypergraph_ F01055	143	■ 1.000	$v \in Y$	$v \in Y$	By contradiction, assume $v \in Y$. It means that there exists a sequence $u = x_0 \rightarrow x_1 \rightarrow$
voloshin-hypergraph_ F01056	143	■ 1.000	$u=x_{\{0\}} \rightarrow x_{\{1\}} \rightarrow$	$u = x_0 \rightarrow x_1 \rightarrow$	By contradiction, assume $v \in Y$. It means that there exists a sequence $u = x_0 \rightarrow x_1 \rightarrow$
voloshin-hypergraph_ F01057	143	■ 0.975	$x_{\{2\}} \rightarrow \cdots \rightarrow x_{\{k\}}=v$	$x_2 \rightarrow \cdots \rightarrow x_k = v$	 with the property above. Choose a number $\delta > 0$ which does not exceed
voloshin-hypergraph_ F01058	143	■ 0.975	$\delta>0$	$\delta > 0$	$x_2 \rightarrow \cdots \rightarrow x_k = v$ with the property above. Choose a number $\delta > 0$ which does not exceed
voloshin-hypergraph_ F01059	144	■ 0.777	$x_{\{i\}}, x_{\{i+1\}}$	x_i, x_{i+1}	the value needed to saturate any unsaturated arc of type (x_i, x_{i+1}) and does not exceed the
voloshin-hypergraph_ F01060	144	■ 0.997	δ	δ	and decrease it in all arcs of the second type by the same value δ . It is clear that the flow
voloshin-hypergraph_ F01061	144	■ 1.000	Z	Z	Now, let S be the set of all arcs with initial vertex in Y and terminal vertex in Z . Evi-
voloshin-hypergraph_ F01062	144	■ 0.980	$(0,3)$	$(0,3)$	The pair of numbers (0,3) attached to arc (u, x_1) in the initial flow F_0 means that the
voloshin-hypergraph_ F01063	144	■ 0.980	$\left(u, x_{\{1\}}\right)$	(u, x_1)	The pair of numbers (0,3) attached to arc (u, x_1) in the initial flow F_0 means that the
voloshin-hypergraph_ F01064	144	■ 0.980	$F_{\{0\}}$	F_0	The pair of numbers (0,3) attached to arc (u, x_1) in the initial flow F_0 means that the
voloshin-hypergraph_ F01065	144	■ 0.922	$\left(x_{\{3\}}, x_{\{1\}}\right)$	(x_3, x_1)	and assigning flow 0 in all remaining arcs. It immediately saturates the arc (x_3, x_1) as the
voloshin-hypergraph_ F01066	144	■ 1.000	$(1,1)$	$(1,1)$	pair (1,1) shows.
voloshin-hypergraph_ F01067	144	■ 0.815	$F_{\{1\}}$	F_1	Flow F_1 is obtained from the flow F_0 by adding one unit in arc (u, x_1) , subtracting it in
voloshin-hypergraph_ F01068	144	■ 0.815	$u, x_{\{1\}}$	u, x_1	Flow F_1 is obtained from the flow F_0 by adding one unit in arc (u, x_1) , subtracting it in
voloshin-hypergraph_ F01069	144	■ 0.971	$\left(x_{\{1\}}, x_{\{3\}}\right)$	(x_1, x_3)	backward arc (x_1, x_3) , and adding it in arc the (x_3, v) .
voloshin-hypergraph_ F01070	144	■ 0.971	$\left(x_{\{3\}}, v\right)$	(x_3, v)	backward arc (x_1, x_3) , and adding it in arc the (x_3, v) .
voloshin-hypergraph_ F01071	144	■ 1.000	$F_{\{2\}}$	F_2	Flow F_2 is obtained from the flow F_1 by adding two units along the path
voloshin-hypergraph_ F01072	144	■ 1.000	$F_{\{3\}}$	F_3	Flow F_3 (see the next figure) is obtained from the flow F_2 by adding one unit along the
voloshin-hypergraph_ F01073	144	■ 1.000	$F_{\{4\}}$	F_4	At last maximum flow F_4 is obtained from the flow F_3 by adding one unit along the path
voloshin-hypergraph_ F01074	144	■ 0.522	$\left(u, x_{\{1\}}\right), \left(x_{\{3\}}, x_{\{1\}}\right)$	$(u, x_1), (x_3, x_1)$	We conclude that the flow value equals 6, it is maximum because the arcs $(u, x_1), (x_3, x_1)$
voloshin-hypergraph_ F01075	144	■ 0.546	$x_{\{4\}}, x_{\{2\}}$	x_4, x_2	1. In the example in Figure 6.3, add arc (x_4, x_2) with capacity 2 and find the maximum
voloshin-hypergraph_ F01076	146	■ 1.000	$W_{\{7\}}$	W_7	5. Given a cube, prism, wheel W_7 , assign direction to every edge (making them arcs) in
voloshin-hypergraph_ F01077	151	■ 0.967	$\mathcal{D}=\left\{D_{\{1\}}, D_{\{2\}}, \ldots, D_{\{m\}}\right\}$	$\mathcal{D} = \{D_1, D_2, \dots, D_m\}$	graph concepts ([6]). Let $X = \{x_1, x_2, \dots, x_n\}$ be a finite set, and let $\mathcal{D} = \{D_1, D_2, \dots, D_m\}$
voloshin-hypergraph_ F01078	151	■ 1.000	$\mathcal{H}=(X, \mathcal{D})$	$\mathcal{H} = (X, \mathcal{D})$	be a family of subsets of X . The pair $\mathcal{H} = (X, \mathcal{D})$ is called a hypergraph with vertex set
voloshin-hypergraph_ F01079	151	■ 1.000	$V(\mathcal{H})$	$V(\mathcal{H})$	X also denoted by $V(\mathcal{H})$, and with edge set $\mathcal{D}$ also denoted by $\mathcal{D}(\mathcal{H})$. Sometimes, the

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voloshin-hypergraph_F01080	151	■ 1.000	$\mathcal{D}(\mathcal{H})$	$\mathcal{D}(\mathcal{H})$	X also denoted by $V(\mathcal{H})$, and with edge set $\mathcal{D}$ also denoted by $\mathcal{D}(\mathcal{H})$. Sometimes, the
voloshin-hypergraph_F01081	152	■ 1.000	$n(\mathcal{H})$	$n(\mathcal{H})$	$ X = n$ is called the order of the hypergraph, written also as n , or $n(\mathcal{H})$. The elements
voloshin-hypergraph_F01082	152	■ 0.999	$D_{\{1\}}, D_{\{2\}}, \ldots, D_{\{m\}}$	$D_1, D_2, \dots, D_m$	$x_1, x_2, \dots, x_n$ are called the vertices and the sets $D_1, D_2, \dots, D_m$ are called the edges (hy-
voloshin-hypergraph_F01083	152	■ 1.000	$m(\mathcal{H})$	$m(\mathcal{H})$	peredges) . The number of edges is usually denoted by m or $m(\mathcal{H})$. Sometimes we will
voloshin-hypergraph_F01084	152	■ 1.000	$x_{\{i\}} \in X$	$x_i \in X$	if their intersection is not empty. If a vertex $x_i \in X$ belongs to an edge $D_j \in \mathcal{D}$, then we say
voloshin-hypergraph_F01085	152	■ 1.000	$D_{\{j\}} \in \mathcal{D}$	$D_j \in \mathcal{D}$	if their intersection is not empty. If a vertex $x_i \in X$ belongs to an edge $D_j \in \mathcal{D}$, then we say
voloshin-hypergraph_F01086	152	■ 0.996	$\mathcal{D}(x), x \in X$	$\mathcal{D}(x), x \in X$	$\mathcal{D}(x), x \in X$, will denote all the edges containing the vertex x . The number $ \mathcal{D}(x) $ is
voloshin-hypergraph_F01087	152	■ 0.996	$ \mathcal{D}(x) $	$ \mathcal{D}(x) $	$\mathcal{D}(x), x \in X$, will denote all the edges containing the vertex x . The number $ \mathcal{D}(x) $ is
voloshin-hypergraph_F01088	152	■ 1.000	$ \mathcal{D}_{\{i\}} $	$ D_i $	called the degree of the vertex x , the number $ D_i $ is called the degree (size, cardinality)
voloshin-hypergraph_F01089	152	■ 1.000	$D_{\{i\}}$	D_i	of the edge D_i . The maximum degree of the hypergraph $\mathcal{H}$ is denoted by
voloshin-hypergraph_F01090	152	■ 1.000	$\mathcal{H}$	$\mathcal{H}$	of the edge D_i . The maximum degree of the hypergraph $\mathcal{H}$ is denoted by
voloshin-hypergraph_F01091	152	■ 1.000	$r \geq 0$	$r \geq 0$	hypergraph in which all edges have the same degree $r \geq 0$ is called r-uniform . The rank
voloshin-hypergraph_F01092	152	■ 0.998	$ \mathcal{D}_{\{i\}} = 2$	$ D_i = 2$	A simple hypergraph $\mathcal{H}$ with $ D_i = 2$ for each $D_i \in \mathcal{D}$ is thus a <i>simple graph</i> , maybe
voloshin-hypergraph_F01093	152	■ 0.998	$D_{\{i\}} \in \mathcal{D}$	$D_i \in \mathcal{D}$	A simple hypergraph $\mathcal{H}$ with $ D_i = 2$ for each $D_i \in \mathcal{D}$ is thus a <i>simple graph</i> , maybe
voloshin-hypergraph_F01094	152	■ 0.998	$\mathcal{H}_1$	$\mathcal{H}_1$	Two simple hypergraphs $\mathcal{H}_1$ and $\mathcal{H}_2$ are called isomorphic if there exists a one-to-one
voloshin-hypergraph_F01095	152	■ 0.998	$\mathcal{H}_2$	$\mathcal{H}_2$	Two simple hypergraphs $\mathcal{H}_1$ and $\mathcal{H}_2$ are called isomorphic if there exists a one-to-one
voloshin-hypergraph_F01096	155	■ 1.000	$I(\mathcal{H})$	$I(\mathcal{H})$	The incidence matrix of a hypergraph $\mathcal{H} = (X, \mathcal{D})$ is a matrix $I(\mathcal{H})$ with n rows that
voloshin-hypergraph_F01097	156	■ 1.000	$\mathcal{D} = \{D_{\{1\}}, D_{\{2\}}, \ldots\}$	$\mathcal{D} = \{D_1, D_2, \dots$	Let $\mathcal{H} = (X, \mathcal{D})$ be a hypergraph with $X = \{x_1, x_2, \dots, x_n\}$ and $\mathcal{D} = \{D_1, D_2, \dots,$
voloshin-hypergraph_F01098	156	■ 1.000	$\mathcal{D}_m\}$	$D_m\}$	$D_m\}$. The dual of the hypergraph $\mathcal{H}$ is a hypergraph $\mathcal{H}^* = (Y, \mathcal{Z})$ whose vertex set is
voloshin-hypergraph_F01099	156	■ 1.000	$\mathcal{H}^* = (Y, \mathcal{Z})$	$\mathcal{H}^* = (Y, \mathcal{Z})$	$D_m\}$. The dual of the hypergraph $\mathcal{H}$ is a hypergraph $\mathcal{H}^* = (Y, \mathcal{Z})$ whose vertex set is
voloshin-hypergraph_F01100	156	■ 1.000	$Y = \{d_{\{1\}}, d_{\{2\}}, \ldots, d_{\{m\}}\}$	$Y = \{d_1, d_2, \dots, d_m\}$	$Y = \{d_1, d_2, \dots, d_m\}$, and the edge set is defined as follows:
voloshin-hypergraph_F01101	156	■ 1.000	$I(\mathcal{H}^*)$	$I(\mathcal{H}^*)$	that the incidence matrix $I(\mathcal{H}^*)$ is the transpose I^* of the incidence matrix $I(\mathcal{H})$. So,
voloshin-hypergraph_F01102	156	■ 1.000	I^*	I^*	that the incidence matrix $I(\mathcal{H}^*)$ is the transpose I^* of the incidence matrix $I(\mathcal{H})$. So,
voloshin-hypergraph_F01103	156	■ 1.000	$\mathcal{H}^*$	$\mathcal{H}^*$	the alternative definition of the dual is the following: a hypergraph $\mathcal{H}^*$ is the dual of the
voloshin-hypergraph_F01104	156	■ 1.000	$I(\mathcal{H}^*) = I^*(\mathcal{H})$	$I(\mathcal{H}^*) = I^*(\mathcal{H})$	hypergraph $\mathcal{H}$ if $I(\mathcal{H}^*) = I^*(\mathcal{H})$. Evidently, if we transpose a matrix twice, we receive the
voloshin-hypergraph_F01105	156	■ 1.000	$(\mathcal{H}^*)^* = \mathcal{H}$	$(\mathcal{H}^*)^* = \mathcal{H}$	same matrix, therefore $(\mathcal{H}^*)^* = \mathcal{H}$. Following the agreement above this holds even in the
voloshin-hypergraph_F01106	156	■ 1.000	$\Delta(\mathcal{H}) = r(\mathcal{H}^*)$	$\Delta(\mathcal{H}) = r(\mathcal{H}^*)$	$\Delta(\mathcal{H}) = r(\mathcal{H}^*)$ and the dual of a k -regular hypergraph is k -uniform. The definition above

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voloshin-hypergraph_F01107	156	0.869	$I(\mathcal{H}) \Rightarrow$	$I(\mathcal{H}) \Rightarrow$	$\Rightarrow$ construct incidence matrix $I(\mathcal{H}) \Rightarrow$ transpose it to obtain $I^*(\mathcal{H}) \Rightarrow$ draw dual $\mathcal{H}^*$.
voloshin-hypergraph_F01108	156	0.869	$I^*(\mathcal{H}) \Rightarrow$	$I^*(\mathcal{H}) \Rightarrow$	$\Rightarrow$ construct incidence matrix $I(\mathcal{H}) \Rightarrow$ transpose it to obtain $I^*(\mathcal{H}) \Rightarrow$ draw dual $\mathcal{H}^*$.
voloshin-hypergraph_F01109	156	1.000	$B(\mathcal{H}) = (X, \mathcal{D}; E)$	$B(\mathcal{H}) = (X, \mathcal{D}; E)$	bipartite graph $B(\mathcal{H}) = (X, \mathcal{D}; E)$ with the vertex set $X \cup \mathcal{D}$, where X is the left part, $\mathcal{D}$ is
voloshin-hypergraph_F01110	156	1.000	$X \cup \mathcal{D}$	$X \cup \mathcal{D}$	bipartite graph $B(\mathcal{H}) = (X, \mathcal{D}; E)$ with the vertex set $X \cup \mathcal{D}$, where X is the left part, $\mathcal{D}$ is
voloshin-hypergraph_F01111	156	1.000	$B(\mathcal{H})$	$B(\mathcal{H})$	the right part, and E is the edge set; vertex $x \in X$ is adjacent to vertex $D \in \mathcal{D}$ in $B(\mathcal{H})$ if
voloshin-hypergraph_F01112	156	1.000	$B(\mathcal{H}^*)$	$B(\mathcal{H}^*)$	It is seen that $B(\mathcal{H}^*)$ is obtained from $B(\mathcal{H})$ simply by interchanging the right part and
voloshin-hypergraph_F01113	156	1.000	D_1	D_1	Notice that edge D_1 with $ D_1 = 1$ (singleton) is drawn as the circle containing only vertex
voloshin-hypergraph_F01114	156	1.000	$ D_1 = 1$	$ D_1 = 1$	Notice that edge D_1 with $ D_1 = 1$ (singleton) is drawn as the circle containing only vertex
voloshin-hypergraph_F01115	156	1.000	D_2	D_2	5 are incident to the same edges, edge D_2 is included in edge D_3 ; in $I(\mathcal{H})$, column D_1
voloshin-hypergraph_F01116	156	1.000	D_3	D_3	5 are incident to the same edges, edge D_2 is included in edge D_3 ; in $I(\mathcal{H})$, column D_1
voloshin-hypergraph_F01117	156	0.817	d_1	d_1	contains one 1, row 6 is a zero row, rows 3 and 5 are identical; in $\mathcal{H}^*$, vertex d_1 is of degree
voloshin-hypergraph_F01118	157	—	$B(\mathcal{H})([6])$	$B(\mathcal{H})([6])$	
voloshin-hypergraph_F01119	158	1.000	$B(\mathcal{H}) =$	$B(\mathcal{H}) =$	Proof. Consider the bipartite representation of $\mathcal{H}$, i.e. the bipartite graph $B(\mathcal{H}) =$

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voloshin-hypergraph_F01120	158	■ 1.000	$(X, \mathcal{D})$	$(X, \mathcal{D}; E)$	$(X, \mathcal{D}; E)$. If we sum the degrees of vertices in the first part, we obtain the left side of
voloshin-hypergraph_F01121	159	■ 1.000	$n \times n$	$n \times n$	Adjacency matrix. An $n \times n$ matrix $A = (a_{ij})$ is an adjacency matrix of a hypergraph
voloshin-hypergraph_F01122	159	■ 1.000	$A = (a_{ij})$	$A = (a_{ij})$	Adjacency matrix. An $n \times n$ matrix $A = (a_{ij})$ is an adjacency matrix of a hypergraph
voloshin-hypergraph_F01123	159	■ 1.000	$A(\mathcal{H})$	$A(\mathcal{H})$	$\mathcal{H}$, denoted by $A(\mathcal{H})$, if
voloshin-hypergraph_F01124	159	■ 1.000	$(0, 1)$	$(0, 1)$	4. Write down an arbitrary $(0,1)$ -matrix and draw the respective hypergraph and its dual.
voloshin-hypergraph_F01125	159	■ 1.000	$B(B(\mathcal{H}))$	$B(B(\mathcal{H}))$	8. Construct bipartite representations $B(B(\mathcal{H}))$ (Figure 7.3), and $B(G)$ (Figure 7.4).
voloshin-hypergraph_F01126	159	■ 1.000	$B(G)$	$B(G)$	8. Construct bipartite representations $B(B(\mathcal{H}))$ (Figure 7.3), and $B(G)$ (Figure 7.4).
voloshin-hypergraph_F01127	159	■ 1.000	E_n, K_n, P_n, C_n	E_n, K_n, P_n, C_n	9. Construct the dual of: E_n, K_n, P_n, C_n , and W_n , $n = 3, 4, 5, 6, 7$.
voloshin-hypergraph_F01128	159	■ 1.000	$W_n, n = 3, 4, 5, 6, 7$	$W_n, n = 3, 4, 5, 6, 7$	9. Construct the dual of: E_n, K_n, P_n, C_n , and W_n , $n = 3, 4, 5, 6, 7$.
voloshin-hypergraph_F01129	160	■ 1.000	$0 \leq r \leq n$	$0 \leq r \leq n$	Complete hypergraphs. For $0 \leq r \leq n$, we define the complete r-uniform hypergraph to
voloshin-hypergraph_F01130	160	■ 1.000	$K_n^r = (X, \mathcal{D})$	$K_n^r = (X, \mathcal{D})$	be the simple hypergraph $K_n^r = (X, \mathcal{D})$ such that $ X = n$ and $\mathcal{D}(K_n^r)$ coincides with all the
voloshin-hypergraph_F01131	160	■ 1.000	$\mathcal{D}(K_n^r)$	$\mathcal{D}(K_n^r)$	be the simple hypergraph $K_n^r = (X, \mathcal{D})$ such that $ X = n$ and $\mathcal{D}(K_n^r)$ coincides with all the
voloshin-hypergraph_F01132	160	■ 0.999	K_n^2	K_n^2	r -subsets of X . Thus a complete graph on n vertices is a complete 2-uniform hypergraph K_n^2
voloshin-hypergraph_F01133	160	■ 1.000	K_n^r	K_n^r	are denoted by K_n^r , i.e. $K_n^r = (X, K_n^r)$, where $ X = n$.
voloshin-hypergraph_F01134	160	■ 1.000	$K_n^r = (X, K_n^r)$	$K_n^r = (X, K_n^r)$	are denoted by K_n^r , i.e. $K_n^r = (X, K_n^r)$, where $ X = n$.
voloshin-hypergraph_F01135	160	■ 1.000	K_4^0	K_4^0	All five complete hypergraphs on four vertices are shown in Figure 7.5. For K_4^0 , we
voloshin-hypergraph_F01136	160	■ 1.000	$1+4+6+4+1=16=2^4=2^n$	$1 + 4 + 6 + 4 + 1 = 16 = 2^4 = 2^n$	edges is $1+4+6+4+1=16=2^4=2^n$ what is the number of all subsets on the set of
voloshin-hypergraph_F01137	160	■ 1.000	$n \geq 0$	$n \geq 0$	four vertices. This holds for any $n \geq 0$ and all r such that $0 \leq r \leq n$. It is important to notice
voloshin-hypergraph_F01138	160	■ 1.000	K_4^3	K_4^3	For example, in K_4^3 we have
voloshin-hypergraph_F01139	160	■ 1.000	K_4^2	K_4^2	Among all complete hypergraphs on four vertices, only K_4^2 is a simple graph. At last,
voloshin-hypergraph_F01140	160	■ 1.000	$\binom{n-1}{r-1}$	$\binom{n-1}{r-1}$	observe that all complete hypergraphs are r -uniform and $\binom{n-1}{r-1}$ -regular hypergraphs.
voloshin-hypergraph_F01141	160	■ 1.000	$x_0, x_1, x_2, \dots, x_{t-1}$	$x_0, x_1, x_2, \dots, x_{t-1}$	of distinct vertices $x_0, x_1, x_2, \dots, x_{t-1}$ and distinct edges $D_0, D_1, D_2, \dots, D_{t-1}$ satisfying
voloshin-hypergraph_F01142	160	■ 1.000	$D_0, D_1, D_2, \dots, D_{t-1}$	$D_0, D_1, D_2, \dots, D_{t-1}$	of distinct vertices $x_0, x_1, x_2, \dots, x_{t-1}$ and distinct edges $D_0, D_1, D_2, \dots, D_{t-1}$ satisfying
voloshin-hypergraph_F01143	160	■ 1.000	$x_i, x_{i+1} \in D_i, i = 0, 1, \dots, t-1$	$x_i, x_{i+1} \in D_i, i = 0, 1, \dots, t-1$	$x_i, x_{i+1} \in D_i, i = 0, 1, \dots, t-1$, is called a path connecting the vertices x_0 and x_t , or, equiv-
voloshin-hypergraph_F01144	160	■ 1.000	x_t	x_t	$x_i, x_{i+1} \in D_i, i = 0, 1, \dots, t-1$, is called a path connecting the vertices x_0 and x_t , or, equiv-
voloshin-hypergraph_F01145	160	■ 1.000	(x_0, x_t)	(x_0, x_t)	alently, (x_0, x_t) - path ; it is called a cycle if $x_t = x_0$. The value of t is called the length of the
voloshin-hypergraph_F01146	160	■ 1.000	$x_t = x_0$	$x_t = x_0$	alently, (x_0, x_t) - path ; it is called a cycle if $x_t = x_0$. The value of t is called the length of the

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voloshin-hypergraph_F01147	160	■ 0.831	$\mu_1=1 \ D_{\{2\}} \ 2 \ D_{\{4\}} \ 3 \ D_{\{5\}} \ 5$	$\mu_1 = 1D_22D_43D_55$	For example, in Figure 7.3, the sequence $\overline{u_1 = 1D_52D_43D_55}$ is a (1,5)-path of length 3,
voloshin-hypergraph_F01148	160	■ 1.000	$\mu_2=1 \ D_{\{2\}} \ 2 \ D_{\{4\}} \ 3 \ D_{\{5\}} \ 4 \ D_{\{3\}} \ 1$	$\mu_2 = 1D_22D_43D_54D_31$	and the sequence $\overline{u_2 = 1D_52D_43D_54D_31}$ is a cycle of length 4.
voloshin-hypergraph_F01149	160	■ 1.000	$\mathcal{H}, \ \mathcal{H}^{\ast}$	$\mathcal{H}, \mathcal{H}^{\ast}$	components. Therefore, $\mathcal{H}$, $\mathcal{H}^{\ast}$ and $B(\mathcal{H})$ have the same number of connected components.
voloshin-hypergraph_F01150	161	■ 1.000	$K_{\{4\}}^{\{1\}}$	K_4^1	In Figure 7.5, the hypergraph K_4^0 has 5 connected components, K_4^1 has 4 connected
voloshin-hypergraph_F01151	161	■ 1.000	$K_{\{4\}}^{\{2\}}, \ K_{\{4\}}^{\{3\}}$	K_4^2, K_4^3	components, K_4^2, K_4^3 and K_4^4 are connected. If we look at Figure 7.5 as one hypergraph, then
voloshin-hypergraph_F01152	161	■ 1.000	$K_{\{4\}}^{\{4\}}$	K_4^4	components, K_4^2, K_4^3 and K_4^4 are connected. If we look at Figure 7.5 as one hypergraph, then
voloshin-hypergraph_F01153	161	■ 1.000	≥ 2	≥ 2	hyperedge of cardinality ≥ 2 contains vertices from both parts. It means that there is no such
voloshin-hypergraph_F01154	161	■ 1.000	$K_{\{n_1\}, \ n_{\{2\}}, \ \ldots, \ n_{\{r\}}\}^{\{r\}}$	$K_{n_1, n_2, \dots, n_r}^r$	denoted by $K_{n_1, n_2, \dots, n_r}^r$ where n_i is the number of vertices in part X_i . Bipartite hypergraphs
voloshin-hypergraph_F01155	161	■ 1.000	$n_{\{i\}}$	n_i	denoted by $K_{n_1, n_2, \dots, n_r}^r$ where n_i is the number of vertices in part X_i . Bipartite hypergraphs
voloshin-hypergraph_F01156	161	■ 0.992	$\mathcal{H}^{\{\backslash\prime\}}=\left(X^{\{\backslash\prime\}}, \ \mathcal{D}^{\{\backslash\prime\}}\right)$	$\mathcal{H}' = (X', \mathcal{D}')$	$\mathcal{H}' = (X', \mathcal{D}')$ are called isomorphic , written $\mathcal{H} \cong \mathcal{H}'$, if there is a one-to-one correspon-
voloshin-hypergraph_F01157	161	■ 0.992	$\mathcal{H} \ \cong \ \mathcal{H}^{\{\backslash\prime\}}$	$\mathcal{H} \cong \mathcal{H}'$	$\mathcal{H}' = (X', \mathcal{D}')$ are called isomorphic , written $\mathcal{H} \cong \mathcal{H}'$, if there is a one-to-one correspon-
voloshin-hypergraph_F01158	161	■ 0.999	$\mathcal{D}^{\{\backslash\prime\}}$	$\mathcal{D}'$	$\mathcal{D}'$ such that for every vertex $x \in X$ and for every edge $D \in \mathcal{D}$ we have that $x \in D$ if and only
voloshin-hypergraph_F01159	161	■ 0.999	$x \ \text{in} \ D$	$x \in D$	$\mathcal{D}'$ such that for every vertex $x \in X$ and for every edge $D \in \mathcal{D}$ we have that $x \in D$ if and only
voloshin-hypergraph_F01160	162	■ 1.000	$x^{\{\backslash\prime\}} \ \text{in} \ X^{\{\backslash\prime\}}$	$x' \in X'$	if for the corresponding vertex $x' \in X'$ and the corresponding edge $D' \in \mathcal{D}'$ the inclusion
voloshin-hypergraph_F01161	162	■ 1.000	$D^{\{\backslash\prime\}} \ \text{in} \ \mathcal{D}^{\{\backslash\prime\}}$	$D' \in \mathcal{D}'$	if for the corresponding vertex $x' \in X'$ and the corresponding edge $D' \in \mathcal{D}'$ the inclusion
voloshin-hypergraph_F01162	162	■ 1.000	$x^{\{\backslash\prime\}} \ \text{in} \ D^{\{\backslash\prime\}}$	$x' \in D'$	$x' \in D'$ holds. Using graph terminology, one could say that two hypergraphs are isomorphic
voloshin-hypergraph_F01163	162	■ 1.000	$K_{\{4\}}^{\{3\}}, \ K_{\{5\}}^{\{3\}}, \ K_{\{5\}}^{\{1\}}, \ K_{\{6\}}^{\{5\}}$	$K_4^3, K_5^3, K_5^1, K_6^5$	3. Write down the incidence matrix of: $K_4^3, K_5^3, K_5^1, K_6^5$, and K_7^4 .
voloshin-hypergraph_F01164	162	■ 1.000	$K_{\{7\}}^{\{4\}}$	K_7^4	3. Write down the incidence matrix of: $K_4^3, K_5^3, K_5^1, K_6^5$, and K_7^4 .
voloshin-hypergraph_F01165	162	■ 1.000	$K_{\{1,2,3\}}^{\{3\}}$	$K_{1,2,3}^3$	5. Draw $K_{1,2,3}^3$ and write down the incidence matrix.
voloshin-hypergraph_F01166	162	■ 1.000	$n_{\{1\}}, \ n_{\{2\}}, \ n_{\{3\}} \ \geq 1$	$n_1, n_2, n_3 \geq 1$	2. For $\overline{n_1, n_2, n_3 \geq 1}$, generate an edge list of K_{n_1, n_2, n_3}^3 .
voloshin-hypergraph_F01167	162	■ 1.000	$K_{\{n_1\}, \ n_{\{2\}}, \ n_{\{3\}}\}^{\{3\}}$	K_{n_1, n_2, n_3}^3	2. For $n_1, n_2, n_3 \geq 1$, generate an edge list of K_{n_1, n_2, n_3}^3 .
voloshin-hypergraph_F01168	162	■ 1.000	$\mathcal{D}(x)$	$\mathcal{D}(x)$	Recall that $\mathcal{D}(x)$ denotes the set of edges containing vertex x in hypergraph $\mathcal{H}$. If
voloshin-hypergraph_F01169	162	■ 1.000	$\mathcal{D}_{\{1\}}=\mathcal{D}-\mathcal{D}(x)$	$\mathcal{D}_1 = \mathcal{D} - \mathcal{D}(x)$	$X_1 = X - \{x\}$, and $\mathcal{D}_1 = \mathcal{D} - \mathcal{D}(x)$, then strong deletion of x from $\mathcal{D}$ results in obtaining
voloshin-hypergraph_F01170	162	■ 0.996	$\mathcal{H}_{\{1\}}=\left(X_{\{1\}}, \ \mathcal{D}_{\{1\}}\right)$	$\mathcal{H}_1 = (X_1, \mathcal{D}_1)$	the hypergraph $\mathcal{H}_1 = (X_1, \mathcal{D}_1)$. We write this operation as $\mathcal{H}_1 = \mathcal{H} - x$. In $\mathcal{H}_1$, we can
voloshin-hypergraph_F01171	162	■ 0.996	$\mathcal{H}_{\{1\}}=\mathcal{H}-x$	$\mathcal{H}_1 = \mathcal{H} - x$	the hypergraph $\mathcal{H}_1 = (X_1, \mathcal{D}_1)$. We write this operation as $\mathcal{H}_1 = \mathcal{H} - x$. In $\mathcal{H}_1$, we can
voloshin-hypergraph_F01172	163	■ 1.000	$\mathcal{H} =$	$\mathcal{H} =$	Weak deletion of a vertex. As in the previous case, let us have a hypergraph $\mathcal{H} =$
voloshin-hypergraph_F01173	163	■ 1.000	$(X, \ \mathcal{D})$	$(X, \mathcal{D})$	$(X, \mathcal{D})$ and a vertex $x \in X$. A weak deletion of x from $\mathcal{H}$ is the removing of x from set X

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voloshin-hypergraph_F01174	164	■ 1.000	$\mathcal{H}_1 = \mathcal{H} - D$	$\mathcal{H}_1 = \mathcal{H} - D$	D is shown in Figure 7.8. In such case we write $\mathcal{H}_1 = \mathcal{H} - D$ and indicate that the deletion
voloshin-hypergraph_F01175	165	■ 0.974	$D^{\setminus \prime} \in \mathcal{D}$	$D' \in \mathcal{D}$	2. replace all vertices of D by one vertex belonging to each $D' \in \mathcal{D}$ such that $D \cap D' \neq \emptyset$.
voloshin-hypergraph_F01176	165	■ 0.974	$D \cap D^{\setminus \prime} \neq \emptyset$	$D \cap D' \neq \emptyset$	2. replace all vertices of D by one vertex belonging to each $D' \in \mathcal{D}$ such that $D \cap D' \neq \emptyset$.
voloshin-hypergraph_F01177	165	■ 1.000	$\mathcal{H}_1^*$	$\mathcal{H}_1^*$	example, notice that in Figure 7.10, $\mathcal{H}_1$ is isomorphic to its dual $\mathcal{H}_1^*$, while $\mathcal{H}$ is not iso-
voloshin-hypergraph_F01178	166	■ 0.997	$\{7, 11, 12\}$	$\{7, 11, 12\}$	7. Strongly delete edges $\{7, 11, 12\}$ and $\{3, 5, 9\}$.
voloshin-hypergraph_F01179	166	■ 0.997	$\{3, 5, 9\}$	$\{3, 5, 9\}$	7. Strongly delete edges $\{7, 11, 12\}$ and $\{3, 5, 9\}$.
voloshin-hypergraph_F01180	167	■ 1.000	$\mathcal{H}^{\setminus \prime} =$	$\mathcal{H}' =$	Subhypergraphs. Let us have a hypergraph $\mathcal{H} = (X, \mathcal{D})$. Any hypergraph $\mathcal{H}' =$
voloshin-hypergraph_F01181	167	■ 0.983	$(X^{\setminus \prime}, \mathcal{D}^{\setminus \prime})$	$(X', \mathcal{D}')$	$(X', \mathcal{D}')$ such that $X' \subseteq X$, and $\mathcal{D}' \subseteq \mathcal{D}$ is called a subhypergraph of $\mathcal{H}$. In such case,
voloshin-hypergraph_F01182	167	■ 0.983	$\mathcal{D}^{\setminus \prime} \subseteq \mathcal{D}$	$\mathcal{D}' \subseteq \mathcal{D}$	$(X', \mathcal{D}')$ such that $X' \subseteq X$, and $\mathcal{D}' \subseteq \mathcal{D}$ is called a subhypergraph of $\mathcal{H}$. In such case,
voloshin-hypergraph_F01183	167	■ 1.000	$\mathcal{H}^{\setminus \prime} \subseteq \mathcal{H}$	$\mathcal{H}' \subseteq \mathcal{H}$	we write $\mathcal{H}' \subseteq \mathcal{H}$. Evidently, $\mathcal{H}'$ can be obtained from $\mathcal{H}$ by strong deletion of the vertices
voloshin-hypergraph_F01184	167	■ 1.000	$\mathcal{H}^{\setminus \prime}$	$\mathcal{H}'$	we write $\mathcal{H}' \subseteq \mathcal{H}$. Evidently, $\mathcal{H}'$ can be obtained from $\mathcal{H}$ by strong deletion of the vertices
voloshin-hypergraph_F01185	167	■ 0.999	$\mathcal{D} - \mathcal{D}^{\setminus \prime}$	$\mathcal{D} - \mathcal{D}'$	maining edges from $\mathcal{D} - \mathcal{D}'$ (sequentially in any order, or at once). In Figure 7.12, both $\mathcal{H}_1$
voloshin-hypergraph_F01186	167	■ 0.978	$\mathcal{H} : \mathcal{H}_1$	$\mathcal{H} : \mathcal{H}_1$	and $\mathcal{H}_2$ are subhypergraphs of $\mathcal{H}$: $\mathcal{H}_1$ is obtained by strong deletion of vertex 5 and 1, and
voloshin-hypergraph_F01187	167	■ 0.996	$\mathcal{H}_Y$	$\mathcal{H}_Y$	a set $Y \subseteq X$ by $\mathcal{H}_Y$. In Figure 7.12, if $Y = \{2, 3, 4\}$, then $\mathcal{H}_1 = \mathcal{H}_Y$.
voloshin-hypergraph_F01188	167	■ 0.996	$Y = \{2, 3, 4\}$	$Y = \{2, 3, 4\}$	a set $Y \subseteq X$ by $\mathcal{H}_Y$. In Figure 7.12, if $Y = \{2, 3, 4\}$, then $\mathcal{H}_1 = \mathcal{H}_Y$.
voloshin-hypergraph_F01189	167	■ 0.996	$\mathcal{H}_1 = \mathcal{H}_Y$	$\mathcal{H}_1 = \mathcal{H}_Y$	a set $Y \subseteq X$ by $\mathcal{H}_Y$. In Figure 7.12, if $Y = \{2, 3, 4\}$, then $\mathcal{H}_1 = \mathcal{H}_Y$.
voloshin-hypergraph_F01190	167	■ 1.000	$\mathcal{H}^{\setminus \prime} = (X, \mathcal{D}^{\setminus \prime})$	$\mathcal{H}' = (X, \mathcal{D}')$	such that $\mathcal{H}' = (X, \mathcal{D}')$ is called a partial subhypergraph . Thus partial subhypergraphs
voloshin-hypergraph_F01191	167	■ 0.743	$\alpha(\mathcal{H})$	$\alpha(\mathcal{H})$	inclusion stable sets is called the stability (independence) number , denoted by $\alpha(\mathcal{H})$.
voloshin-hypergraph_F01192	167	■ 0.946	$1 \leq \alpha(\mathcal{H}) \leq X $	$1 \leq \alpha(\mathcal{H}) \leq X $	For any hypergraph $\mathcal{H}$ with not all vertices singletons, $1 \leq \alpha(\mathcal{H}) \leq X $. For a hypergraph
voloshin-hypergraph_F01193	167	■ 0.976	$\alpha(\mathcal{H}) = 3$	$\alpha(\mathcal{H}) = 3$	set, so $\alpha(\mathcal{H}) = 3$.
voloshin-hypergraph_F01194	168	■ 1.000	$ S \cap D \leq 1$	$ S \cap D \leq 1$	$S \subseteq X$ is called a strongly independent (stable) set if $ S \cap D \leq 1$ for every hyperedge $D \in$
voloshin-hypergraph_F01195	168	■ 1.000	$D \in$	$D \in$	$S \subseteq X$ is called a strongly independent (stable) set if $ S \cap D \leq 1$ for every hyperedge $D \in$
voloshin-hypergraph_F01196	168	■ 1.000	$\bar{\alpha}(\mathcal{H})$	$\bar{\alpha}(\mathcal{H})$	$\mathcal{D}$. The cardinality of a maximum strongly independent set is denoted by $\bar{\alpha}(\mathcal{H})$. Evidently,
voloshin-hypergraph_F01197	168	■ 1.000	$G, \alpha(G) = \bar{\alpha}(G)$	$G, \alpha(G) = \bar{\alpha}(G)$	for a graph $G, \alpha(G) = \bar{\alpha}(G)$.
voloshin-hypergraph_F01198	168	■ 1.000	$ T \cap D \geq 1$	$ T \cap D \geq 1$	$ T \cap D \geq 1$ for every edge $D \in \mathcal{D}$. The cardinality of a minimum transversal is denoted by
voloshin-hypergraph_F01199	168	■ 1.000	$\tau(\mathcal{H})$	$\tau(\mathcal{H})$	$\tau(\mathcal{H})$. Following the definition, every set $S = X - T$ is an independent set. Therefore we
voloshin-hypergraph_F01200	168	■ 1.000	$S = X - T$	$S = X - T$	$\tau(\mathcal{H})$. Following the definition, every set $S = X - T$ is an independent set. Therefore we

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voloshin-hypergraph_F01201	168	■ 1.000	$\tau=2$	$\tau = 2$	In Figure 7.12, vertices 4 and 5 form a minimum transversal, so $\tau = 2$. The comple-
voloshin-hypergraph_F01202	168	■ 0.822	$\alpha(\mathcal{H})+\tau(\mathcal{H})=3+2=5= X $	$\alpha(\mathcal{H}) + \tau(\mathcal{H}) = 3 + 2 = 5 = X $	Evidently, $\alpha(\mathcal{H}) + \tau(\mathcal{H}) = 3 + 2 = 5 = X $.
voloshin-hypergraph_F01203	168	■ 0.995	$\mathcal{V}(\mathcal{H})$	$\mathcal{V}(\mathcal{H})$	by $v(\mathcal{H})$. Any matching of $\mathcal{H}$ is a strongly independent set of vertices in the dual $\mathcal{H}^*$.
voloshin-hypergraph_F01204	168	■ 0.956	$\nu(\mathcal{H})=\bar{\alpha}(\mathcal{H}^*)$	$\nu(\mathcal{H}) = \bar{\alpha}(\mathcal{H}^*)$	Therefore, $v(\mathcal{H}) = \bar{\alpha}(\mathcal{H}^*)$. Since any matching is a set of pairwise non-intersecting edges,
voloshin-hypergraph_F01205	168	■ 0.940	$\{1,2,5\}$	$\{1,2,5\}$	ing; however, edges $\{1,2,5\}$ and $\{3,4\}$ form a perfect matching, so $v(\mathcal{H}) = 2$. Any
voloshin-hypergraph_F01206	168	■ 0.940	$\mathcal{V}(\mathcal{H})=2$	$\mathcal{V}(\mathcal{H}) = 2$	ing; however, edges $\{1,2,5\}$ and $\{3,4\}$ form a perfect matching, so $v(\mathcal{H}) = 2$. Any
voloshin-hypergraph_F01207	168	■ 0.983	$\tau(\mathcal{H}) \geq 2=v(\mathcal{H})$	$\tau(\mathcal{H}) \geq 2 = v(\mathcal{H})$	so $\tau(\mathcal{H}) \geq 2 = v(\mathcal{H})$. Since $\tau(\mathcal{H}) = v(\mathcal{H})$, $\mathcal{H}$ satisfies the König property. Notice that if a
voloshin-hypergraph_F01208	168	■ 0.983	$\tau(\mathcal{H})=v(\mathcal{H}), \mathcal{H}$	$\tau(\mathcal{H}) = v(\mathcal{H}), \mathcal{H}$	so $\tau(\mathcal{H}) \geq 2 = v(\mathcal{H})$. Since $\tau(\mathcal{H}) = v(\mathcal{H})$, $\mathcal{H}$ satisfies the König property. Notice that if a
voloshin-hypergraph_F01209	169	■ 1.000	$\rho(\mathcal{H})$	$\rho(\mathcal{H})$	number of edges in a covering is denoted by $\rho(\mathcal{H})$. One can see that each covering of $\mathcal{H}$ is
voloshin-hypergraph_F01210	169	■ 1.000	$\rho(\mathcal{H})=\bar{\alpha}(\mathcal{H})$	$\rho(\mathcal{H}) = \bar{\alpha}(\mathcal{H})$	A hypergraph $\mathcal{H}$ has the dual König property if $\rho(\mathcal{H}) = \bar{\alpha}(\mathcal{H})$.
voloshin-hypergraph_F01211	169	■ 0.522	$\rho(\mathcal{H})=2$	$\rho(\mathcal{H}) = 2$	so $\rho(\mathcal{H}) = 2$. Vertices 1 and 3 form a maximum strongly independent set, so $\bar{\alpha}(\mathcal{H}) = 2$.
voloshin-hypergraph_F01212	169	■ 0.522	$\bar{\alpha}(\mathcal{H})=2$	$\bar{\alpha}(\mathcal{H}) = 2$	so $\rho(\mathcal{H}) = 2$. Vertices 1 and 3 form a maximum strongly independent set, so $\bar{\alpha}(\mathcal{H}) = 2$.
voloshin-hypergraph_F01213	169	■ 0.991	$\tau=3>2=\nu$	$\tau = 3 > 2 = \nu$	Graph C_5 is an example of a hypergraph with $\tau = 3 > 2 = v$ and $\rho = 3 > \bar{\alpha} = \alpha = 2$.
voloshin-hypergraph_F01214	169	■ 0.991	$\rho=3>\bar{\alpha}=\alpha=2$	$\rho = 3 > \bar{\alpha} = \alpha = 2$	Graph C_5 is an example of a hypergraph with $\tau = 3 > 2 = v$ and $\bar{\rho} = 3 > \alpha = \alpha = 2$.
voloshin-hypergraph_F01215	169	■ 1.000	$X^{\prime}=\{1,2,3,4,5,7,8\}$	$X' = \{1,2,3,4,5,7,8\}$	1. Draw subhypergraph $\mathcal{H}' = (X', \mathcal{D}')$, where $X' = \{1,2,3, 4,5,7,8\}$ and $\mathcal{D}' =$
voloshin-hypergraph_F01216	169	■ 1.000	$\mathcal{D}^{\prime} =$	$\mathcal{D}' =$	1. Draw subhypergraph $\mathcal{H}' = (X', \mathcal{D}')$, where $X' = \{1,2,3, 4,5,7,8\}$ and $\mathcal{D}' =$
voloshin-hypergraph_F01217	169	■ 0.997	$v(\mathcal{H})$	$v(\mathcal{H})$	10. Find $v(\mathcal{H})$.
voloshin-hypergraph_F01218	170	■ 0.531	$\sqsubseteq(\mathcal{H})$	$\sqsubseteq(\mathcal{H})$	6. Find $v(\mathcal{H})$.
voloshin-hypergraph_F01219	170	■ 1.000	$L(\mathcal{H})$	$L(\mathcal{H})$	The line graph $L(\mathcal{H})$ of a hypergraph $\mathcal{H} = (X, \mathcal{D})$ (sometimes called the intersection
voloshin-hypergraph_F01220	170	■ 1.000	$(\mathcal{H})_2$	$(\mathcal{H})_2$	The 2-section $(\mathcal{H})_2$ of a hypergraph $\mathcal{H} = (X, \mathcal{D})$ is the graph with the same vertex set
voloshin-hypergraph_F01221	170	■ 1.000	$L(\mathcal{H}^*)$	$L(\mathcal{H}^*)$	turn, becomes the vertex set for $L(\mathcal{H}^*)$; so they have the same vertex set. Two vertices in
voloshin-hypergraph_F01222	171	■ 0.931	$\alpha((\mathcal{H})_2) . \square$	$\alpha((\mathcal{H})_2) . \square$	maximum number of pairwise non-adjacent vertices in $\mathcal{H}$ what coincides with $\alpha((\mathcal{H})_2)$. $\square$
voloshin-hypergraph_F01225	171	■ 1.000	$(\mathcal{H}^*)_2$	$(\mathcal{H}^*)_2$	One can also check that $L(\mathcal{H})$ and $(\mathcal{H}^*)_2$ are isomorphic to the graph obtained from
voloshin-hypergraph_F01226	171	■ 0.994	a, b, c, d	a, b, c, d	C_4 on vertices $\underline{a, b, c, d}$ by adding diagonal bd . Edges a and c in $\mathcal{H}$ form a matching with
voloshin-hypergraph_F01230	172	■ 1.000	$\tau(\mathcal{H})=\theta(L(\mathcal{H}))$	$\tau(\mathcal{H}) = \theta(L(\mathcal{H}))$	$L(\mathcal{H})$. Immediate conclusion is that $\tau(\mathcal{H}) = \theta(L(\mathcal{H}))$, and finding a minimum transversal
voloshin-hypergraph_F01231	172	■ 1.000	$\chi(\overline{L(\mathcal{H})})$	$\chi(\overline{L(\mathcal{H})})$	know, the last is equivalent to finding the chromatic number $\chi(\overline{L(\mathcal{H})})$ We can see that there
voloshin-hypergraph_F01232	172	■ 1.000	$\overline{L(\mathcal{H})}$	$\overline{L(\mathcal{H})}$	$\overline{L(\mathcal{H})}$. If $\mathcal{H}$ is not Helly, we do not have this relation and we are not able to use $L(\mathcal{H})$ for

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voloshin-hypergraph_ F01233	172	■ 1.000	b, c	b, c	Hypergraph $\mathcal{H}$ in Figure 7.13 is not Helly: edges b, c , and d form an intersecting family
voloshin-hypergraph_ F01239	172	■ 1.000	$Y \subseteq D$	$Y \subseteq D$	<i>a clique in $(\mathcal{H})_2$ there is an edge $D \in \mathcal{D}$ such that $Y \subseteq D$.</i>
voloshin-hypergraph_ F01240	172	■ 0.913	$G=(\mathcal{H})_2$	$G = (\mathcal{H})_2$	Proof. $\Rightarrow$ Let $\mathcal{H} = (X, \mathcal{D})$ be a conformal hypergraph, $G = (\mathcal{H})_2$, and $Y \subseteq X$ be any set
voloshin-hypergraph_ F01241	172	■ 1.000	$Z \subseteq X$	$Z \subseteq X$	clique, there exists a set $Z \subseteq X$ such that $Y \subseteq Z$ and G_Z is a clique. Then $Z = D$ for some
voloshin-hypergraph_ F01242	172	■ 1.000	$Y \subseteq Z$	$Y \subseteq Z$	clique, there exists a set $Z \subseteq X$ such that $Y \subseteq Z$ and G_Z is a clique. Then $Z = D$ for some
voloshin-hypergraph_ F01243	172	■ 1.000	$G_{\{Z\}}$	G_Z	clique, there exists a set $Z \subseteq X$ such that $Y \subseteq Z$ and G_Z is a clique. Then $Z = D$ for some
voloshin-hypergraph_ F01244	172	■ 1.000	$Z=D$	$Z = D$	clique, there exists a set $Z \subseteq X$ such that $Y \subseteq Z$ and G_Z is a clique. Then $Z = D$ for some
voloshin-hypergraph_ F01245	173	■ 1.000	$\mathcal{C}$	$\mathcal{C}$	family $\mathcal{D}'$ of maximal edges of $\mathcal{H}$ and the family $\mathcal{C}$ of maximal cliques of G coincide, i.e.
voloshin-hypergraph_ F01246	173	■ 1.000	$\mathcal{D}^{\prime}=\mathcal{C}$	$\mathcal{D}' = \mathcal{C}$	$\mathcal{D}' = \mathcal{C}$.
voloshin-hypergraph_ F01247	173	■ 1.000	$\mathcal{D}^{\prime} \subseteq \mathcal{C}$	$\mathcal{D}' \subseteq \mathcal{C}$	Case 1: Prove that $\mathcal{D}' \subseteq \mathcal{C}$. Let $D \in \mathcal{D}'$. By definition of G , all vertices of D are
voloshin-hypergraph_ F01248	173	■ 1.000	$D \in \mathcal{D}^{\prime}$	$D \in \mathcal{D}'$	Case 1: Prove that $\mathcal{D}' \subseteq \mathcal{C}$. Let $D \in \mathcal{D}'$. By definition of G , all vertices of D are
voloshin-hypergraph_ F01249	173	■ 0.615	$G_{\{D\}}$	G_D	pairwise adjacent, i.e. G_D is a clique. It is contained in a maximal clique $C \in \mathcal{C}$, i.e. $D \subseteq C$.
voloshin-hypergraph_ F01250	173	■ 0.615	$C \in \mathcal{C}$	$C \in \mathcal{C}$	pairwise adjacent, i.e. G_D is a clique. It is contained in a maximal clique $C \in \mathcal{C}$, i.e. $D \subseteq C$.
voloshin-hypergraph_ F01251	173	■ 0.615	$D \subseteq C$	$D \subseteq C$	pairwise adjacent, i.e. G_D is a clique. It is contained in a maximal clique $C \in \mathcal{C}$, i.e. $D \subseteq C$.
voloshin-hypergraph_ F01252	173	■ 1.000	$C \subseteq D^{\prime}$	$C \subseteq D'$	By the condition of Lemma, for C there exists an edge D' such that $C \subseteq D'$. So we have
voloshin-hypergraph_ F01253	173	■ 0.973	$D \subseteq C \subseteq D^{\prime}$	$D \subseteq C \subseteq D'$	$D \subseteq C \subseteq D'$. But both D and D' are maximal what implies that $D = C = D'$ and $D \in \mathcal{C}$.
voloshin-hypergraph_ F01254	173	■ 0.973	$D=C=D^{\prime}$	$D = C = D'$	$D \subseteq C \subseteq D'$. But both D and D' are maximal what implies that $D = C = D'$ and $D \in \mathcal{C}$.
voloshin-hypergraph_ F01255	173	■ 0.973	$D \in \mathcal{C}$	$D \in \mathcal{C}$	$D \subseteq C \subseteq D'$. But both D and D' are maximal what implies that $D = C = D'$ and $D \in \mathcal{C}$.
voloshin-hypergraph_ F01256	173	■ 1.000	$\mathcal{C} \subseteq \mathcal{D}^{\prime}$	$\mathcal{C} \subseteq \mathcal{D}'$	Case 2: Prove that $\mathcal{C} \subseteq \mathcal{D}'$. Let $C \in \mathcal{C}$ be an arbitrary maximal clique in G . The vertices
voloshin-hypergraph_ F01257	173	■ 1.000	$C \in \mathcal{C}$	$C \in \mathcal{C}$	Case 2: Prove that $\mathcal{C} \subseteq \mathcal{D}'$. Let $C \in \mathcal{C}$ be an arbitrary maximal clique in G . The vertices
voloshin-hypergraph_ F01258	173	■ 1.000	$C \subseteq D$	$C \subseteq D$	$C \subseteq D$. In turn, there exists an edge $D' \in \mathcal{D}'$ such that $D \subseteq D'$. Since vertices of D' are
voloshin-hypergraph_ F01259	173	■ 1.000	$D \subseteq D^{\prime}$	$D \subseteq D'$	$C \subseteq D$. In turn, there exists an edge $D' \in \mathcal{D}'$ such that $D \subseteq D'$. Since vertices of D' are
voloshin-hypergraph_ F01260	173	■ 1.000	$C^{\prime}$	C'	pairwise adjacent in G , there exists a maximal clique C' such that $D' \subseteq C'$. Since both C and
voloshin-hypergraph_ F01261	173	■ 1.000	$D^{\prime} \subseteq C^{\prime}$	$D' \subseteq C'$	pairwise adjacent in G , there exists a maximal clique C' such that $D' \subseteq C'$. Since both C and
voloshin-hypergraph_ F01262	173	■ 1.000	$C \subseteq D^{\prime} \subseteq C^{\prime}$	$C \subseteq D' \subseteq C'$	C' are maximal, and $C \subseteq D' \subseteq C'$, it follows that $C = D = C'$ and $C \in \mathcal{D}'$. Consequently,
voloshin-hypergraph_ F01263	173	■ 1.000	$C=D=C^{\prime}$	$C = D = C'$	C' are maximal, and $C \subseteq D' \subseteq C'$, it follows that $C = D = C'$ and $C \in \mathcal{D}'$. Consequently,
voloshin-hypergraph_ F01264	173	■ 1.000	$C \in \mathcal{D}^{\prime}$	$C \in \mathcal{D}'$	C' are maximal, and $C \subseteq D' \subseteq C'$, it follows that $C = D = C'$ and $C \in \mathcal{D}'$. Consequently,
voloshin-hypergraph_ F01265	173	■ 1.000	$D_{\{1\}}, D_{\{2\}}, D_{\{3\}}$	D_1, D_2, D_3	<i>any three edges D_1, D_2, D_3, there is an edge D such that</i>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F01266	173	■ 1.000	$\left(D_{\{1\}} \cap D_{\{2\}}\right) \cup \left(D_{\{1\}} \cap D_{\{3\}}\right) \cup \left(D_{\{2\}} \cap D_{\{3\}}\right)$	$(D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3)$	The vertices of $(D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3)$ are pairwise adjacent and therefore induce
voloshin-hypergraph_F01267	173	■ 1.000	$ Y $	$ Y $	fulfils. Induction on $ Y $ where Y is the set of pairwise adjacent vertices of graph $(\mathcal{H})_2$. If
voloshin-hypergraph_F01268	173	■ 1.000	$ Y =1$	$ Y = 1$	$ Y = 1$, a vertex is contained in some edge of $\mathcal{H}$ (otherwise it is isolated). If $ Y = 2$, then
voloshin-hypergraph_F01269	173	■ 1.000	$ Y =2$	$ Y = 2$	$ Y = 1$, a vertex is contained in some edge of $\mathcal{H}$ (otherwise it is isolated). If $ Y = 2$, then
voloshin-hypergraph_F01270	173	■ 1.000	$ Y < k, k \geq 3$	$ Y < k, k \geq 3$	Let the condition of Lemma 7.6.1 hold for all Y such that $ Y < k, k \geq 3$, and let Y be a
voloshin-hypergraph_F01271	173	■ 1.000	$ Y \geq 3$	$ Y \geq 3$	Since $ Y \geq 3$, choose vertices $x_1, x_2, x_3 \in Y$ and put $Y_i = Y - \{x_i\}, i = 1, 2, 3$. Notice that
voloshin-hypergraph_F01272	173	■ 1.000	$x_{\{1\}}, x_{\{2\}}, x_{\{3\}} \in Y$	$x_1, x_2, x_3 \in Y$	Since $ Y \geq 3$, choose vertices $x_1, x_2, x_3 \in Y$ and put $Y_i = Y - \{x_i\}, i = 1, 2, 3$. Notice that
voloshin-hypergraph_F01273	173	■ 1.000	$Y_{\{i\}} = Y - \{x_{\{i\}}\}, i = 1, 2, 3$	$Y_i = Y - \{x_i\}, i = 1, 2, 3$	Since $ Y \geq 3$, choose vertices $x_1, x_2, x_3 \in Y$ and put $Y_i = Y - \{x_i\}, i = 1, 2, 3$. Notice that
voloshin-hypergraph_F01274	173	■ 1.000	$Y_{\{i\}} \subseteq D_{\{i\}}, i = 1, 2, 3$	$Y_i \subseteq D_i, i = 1, 2, 3$	By the induction hypothesis, there exist edges D_1, D_2, D_3 such that $Y_i \subseteq D_i, i = 1, 2, 3$.
voloshin-hypergraph_F01275	174	■ 1.000	$x_{\{1\}}, x_{\{2\}}, \dots, x_{\{k\}}$	$x_1, x_2, \dots, x_k$	tices $x_1, x_2, \dots, x_k$. Since $\mathcal{H}$ is conformal, by Lemma 7.6.1 there is an edge D such that
voloshin-hypergraph_F01276	174	■ 0.991	$x_{\{1\}}, x_{\{2\}}, \dots, x_{\{k\}} \in D$	$x_1, x_2, \dots, x_k \in D$	$x_1, x_2, \dots, x_k \in D$. It means that in $\mathcal{H}^*$, there is a vertex d such that $d \in X_1, d \in X_2, \dots$,
voloshin-hypergraph_F01277	174	■ 0.991	$d \in X_{\{1\}}, d \in X_{\{2\}}, \dots$	$d \in X_1, d \in X_2, \dots$	$x_1, x_2, \dots, x_k \in D$. It means that in $\mathcal{H}^*$, there is a vertex d such that $d \in X_1, d \in X_2, \dots$,
voloshin-hypergraph_F01278	174	■ 1.000	$d \in X_{\{k\}}$	$d \in X_k$	$d \in X_k$, or, equivalently, $d \in X_1 \cap X_2 \cap \dots \cap X_k$. Consequently, every intersecting subfamily
voloshin-hypergraph_F01279	174	■ 1.000	$d \in X_{\{1\}} \cap X_{\{2\}} \cap \dots \cap X_{\{k\}}$	$d \in X_1 \cap X_2 \cap \dots \cap X_k$	$d \in X_k$, or, equivalently, $d \in X_1 \cap X_2 \cap \dots \cap X_k$. Consequently, every intersecting subfamily
voloshin-hypergraph_F01280	175	■ 0.996	$\mathcal{H}_2 \cong \mathcal{H}_1^*$	$\mathcal{H}_2 \cong \mathcal{H}_1^*$	hypergraph $\mathcal{H}_2$. One can easily check that $\mathcal{H}_2 \cong \mathcal{H}_1^*$. At last, if we add to $\mathcal{H}_2$ an edge
voloshin-hypergraph_F01281	175	■ 0.999	$\mathcal{H}_3$	$\mathcal{H}_3$	containing all vertices, we obtain a hypergraph $\mathcal{H}_3$ which is both Helly and conformal.
voloshin-hypergraph_F01282	175	■ 0.995	$\mathcal{H} \cong \mathcal{H}^*$	$\mathcal{H} \cong \mathcal{H}^*$	A hypergraph is called self-dual if $\mathcal{H} \cong \mathcal{H}^*$; evidently, the incidence matrix of a self-
voloshin-hypergraph_F01283	175	■ 1.000	$\mathcal{H}_3 \cong \mathcal{H}_3^*$	$\mathcal{H}_3 \cong \mathcal{H}_3^*$	Notice that hypergraph $\mathcal{H}_3$ in Figure 7.15 is bi-conformal and self-dual, i.e. $\mathcal{H}_3 \cong \mathcal{H}_3^*$.
voloshin-hypergraph_F01284	175	■ 1.000	$n, p, q, r \geq 1$	$n, p, q, r \geq 1$	5. For which $n, p, q, r \geq 1$, the graphs $P_n, C_n, W_n, E_n, K_n, K_{p,q,r}^3$ are or are not Helly ?
voloshin-hypergraph_F01285	175	■ 1.000	$P_{\{n\}}, C_{\{n\}}, W_{\{n\}}, E_{\{n\}}, K_{\{n\}}, K_{\{p, q, r\}}^3$	$P_n, C_n, W_n, E_n, K_n, K_{p,q,r}^3$	5. For which $n, p, q, r \geq 1$, the graphs $P_n, C_n, W_n, E_n, K_n, K_{p,q,r}^3$ are or are not Helly ?
voloshin-hypergraph_F01286	175	■ 1.000	$n, r \geq 1$	$n, r \geq 1$	6. For which $n, r \geq 1$ complete hypergraph K_n^r is Helly?
voloshin-hypergraph_F01287	177	■ 1.000	$E_{\{3\}}$	E_3	is an edge inducing a forest isomorphic to the empty graph E_3 in the host tree. May be
voloshin-hypergraph_F01288	178	■ 1.000	$L^{\{\prime\}}$	L'	chordal. When T is larger, let x be a pendant vertex of T . Let L' be the line graph of the
voloshin-hypergraph_F01289	178	■ 1.000	$T - x$	$T - x$	it from the family of edges. Since the host tree for $\mathcal{H}'$ is $T - x$, the hypergraph $\mathcal{H}'$ is a
voloshin-hypergraph_F01290	178	■ 1.000	$L^{\{\prime\}} = L$	$L' = L$	If no hyperedge of $\mathcal{H}$ coincides with $\{x\}$, then $L' = L$, and L is chordal. If $\{x\}$ occurs
voloshin-hypergraph_F01291	178	■ 1.000	$\mathcal{F} \subseteq \mathcal{D}$	$\mathcal{F} \subseteq \mathcal{D}$	family of $\mathcal{H}$ is a star, we prove the contrapositive. Let $\mathcal{F} \subseteq \mathcal{D}$ be an intersecting family
voloshin-hypergraph_F01292	178	■ 1.000	$D \in \mathcal{F}$	$D \in \mathcal{F}$	which is not a star. Then each $x \in X$ misses some $D \in \mathcal{F}$. Mark the edge of T that is the first

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voloshin-hypergraph_F01293	178	■ 1.000	$n(T)= X $	$n(T) = X $	on the path from x to D . Since we place $n(T) = X $ marks on $n(T) - 1$ edges, some edge
voloshin-hypergraph_F01294	178	■ 1.000	$n(T) - 1$	$n(T) - 1$	on the path from x to D . Since we place $n(T) = X $ marks on $n(T) - 1$ edges, some edge
voloshin-hypergraph_F01295	178	■ 1.000	$D_{\{j\}}$	D_j	is marked twice. This edge belongs to a path in T between some members D_i and D_j of $\mathcal{F}$,
voloshin-hypergraph_F01296	178	■ 1.000	$\mathcal{F}$	$\mathcal{F}$	is marked twice. This edge belongs to a path in T between some members D_i and D_j of $\mathcal{F}$,
voloshin-hypergraph_F01297	179	■ 1.000	$\mathcal{D}(x) \subseteq \mathcal{D}(y)$	$\mathcal{D}(x) \subseteq \mathcal{D}(y)$	In a hypergraph $\mathcal{H} = (X, \mathcal{D})$, a vertex x is pendant to a vertex y if $\mathcal{D}(x) \subseteq \mathcal{D}(y)$.
voloshin-hypergraph_F01298	179	■ 0.975	$\mathcal{H}=(Y, \mathcal{D})$	$\mathcal{H} = (Y, \mathcal{D})$	Proof. [6] $\Rightarrow$ Let $\mathcal{H} = (Y, \mathcal{D})$ be a chordal conformal hypergraph. We prove that $\mathcal{H}^*$ is
voloshin-hypergraph_F01299	179	■ 1.000	$n=n(\mathcal{H})$	$n = n(\mathcal{H})$	a hypertree by induction on $n = n(\mathcal{H})$. For $n \leq 2$, every hypergraph is a hypertree. Now
voloshin-hypergraph_F01300	179	■ 1.000	$n \leq 2$	$n \leq 2$	a hypertree by induction on $n = n(\mathcal{H})$. For $n \leq 2$, every hypergraph is a hypertree. Now
voloshin-hypergraph_F01301	179	■ 1.000	$n>2$	$n > 2$	consider $n > 2$, and suppose that the claim holds for smaller hypergraphs. The idea of the
voloshin-hypergraph_F01302	179	■ 1.000	$\mathcal{H}_{\{0\}}$	$\mathcal{H}_0$	Step 1: constructing hypergraph $\mathcal{H}_0$ and its 2-section $(\mathcal{H}_0)_2$. Since $\mathcal{H}$ is a chordal
voloshin-hypergraph_F01303	179	■ 1.000	$\left(\mathcal{H}_{\{0\}}\right)_{\{2\}}$	$(\mathcal{H}_0)_2$	Step 1: constructing hypergraph $\mathcal{H}_0$ and its 2-section $(\mathcal{H}_0)_2$. Since $\mathcal{H}$ is a chordal
voloshin-hypergraph_F01304	179	■ 0.999	$D_{\{0\}}$	D_0	$(\mathcal{H})_2$. By the conformality of $\mathcal{H}$, this set forms an edge D_0 in $\mathcal{H}$. Since x has no additional
voloshin-hypergraph_F01307	179	■ 1.000	$D_{\{0\}}-\{x\}$	$D_0 - \{x\}$	simplicial in $(\mathcal{H})_2$ and the edge $D_0 - \{x\}$ in $\mathcal{H}_0$ contains all neighbors of x in $\mathcal{H}$, we have
voloshin-hypergraph_F01309	180	■ 0.988	$\mathcal{H}_{\{0\}}^{\{ \}^{\{ * \}}}$	$\mathcal{H}_0^*$	vertices, we may apply the induction hypothesis to $\mathcal{H}_0^*$ to conclude that $\mathcal{H}_0^*$ is a hypertree.
voloshin-hypergraph_F01310	180	■ 0.622	$\mathcal{D}^{\prime}=\mathcal{V}\left(\mathcal{H}_{\{0\}}^{\{ * \}}\right)$	$\mathcal{D}' = V(\mathcal{H}_0^*)$	Thus there is a host tree T with vertex set $\mathcal{D}' = V(\mathcal{H}_0^*)$, such that for every $x' \in V(\mathcal{H}_0) =$
voloshin-hypergraph_F01311	180	■ 0.622	$x^{\prime} \in \mathcal{V}\left(\mathcal{H}_{\{0\}}\right)=$	$x' \in V(\mathcal{H}_0) =$	Thus there is a host tree T with vertex set $\mathcal{D}' = V(\mathcal{H}_0^*)$, such that for every $x' \in V(\mathcal{H}_0) =$
voloshin-hypergraph_F01312	180	■ 0.852	$\mathcal{D}\left(\mathcal{H}_{\{0\}}^{\{ * \}}\right)$	$\mathcal{D}(\mathcal{H}_0^*)$	$\mathcal{D}(\mathcal{H}_0^*)$, the elements of $\mathcal{D}'$ that contain x' comprise a set of vertices of T that induces a
voloshin-hypergraph_F01313	180	■ 0.985	$d_{\{0\}}$	d_0	$D_0 - \{x\}$ is one vertex in $\mathcal{H}_0^*$, it is also one vertex in T . Adding this vertex d_0 as a singleton
voloshin-hypergraph_F01314	180	■ 1.000	$\mathcal{H}_{\{1\}}^{\{ \}^{\{ * \}}}$	$\mathcal{H}_1^*$	X as a subtree shows that $\mathcal{H}_1^*$ is a hypertree.
voloshin-hypergraph_F01315	180	■ 1.000	$\mathcal{D}_{\{x\}}^{\prime}$	$\mathcal{D}'_x$	Good case: d is adjacent to d_0 in host tree T. Let $\mathcal{D}'_x$ be the set of edges in $\mathcal{H}_0$, other
voloshin-hypergraph_F01316	180	■ 0.707	$D \in \mathcal{D}_{\chi}^{\prime}$	$D \in \mathcal{D}'_{\chi}$	and corresponds to some $D \in \mathcal{D}'_{\chi}$, we expand the edge X of $\mathcal{H}_1^*$ to include d . This yields a
voloshin-hypergraph_F01317	180	■ 1.000	$\mathcal{H}_{\{2\}}^{\{ * \}}$	$\mathcal{H}_2^*$	hypertree $\mathcal{H}_2^*$ with host tree T , in which the edge X corresponding to x is a star with center
voloshin-hypergraph_F01322	180	■ 0.997	$D \in \mathcal{D}_{\{x\}}^{\prime}$	$D \in \mathcal{D}'_x$	from $\mathcal{H}_2$ by including x in an edge $D \in \mathcal{D}'_x$ that we have not yet expanded. The edge D of $\mathcal{H}_2$
voloshin-hypergraph_F01323	180	■ 1.000	$D \subseteq D_{\{0\}}$	$D \subset D_0$	d or d_0 are the vertices of $\mathcal{H}_2$ contained in D or D_0 , respectively, and we have $D \subset D_0$.
voloshin-hypergraph_F01324	180	■ 0.998	$\left(d, d_{\{0\}}\right)$	(d, d_0)	The vertex d in $\mathcal{H}_2^*$ occurs as a vertex of T not adjacent to d_0 . Adding the edge (d, d_0) to
voloshin-hypergraph_F01325	180	■ 0.735	$\left(d, d^{\prime}\right)$	(d, d')	T creates a unique cycle. Deleting the edge (d, d') on that cycle (with $d' \neq d_0$) yields a
voloshin-hypergraph_F01326	180	■ 0.735	$d^{\prime} \neq d_{\{0\}}$	$d' \neq d_0$	T creates a unique cycle. Deleting the edge (d, d') on that cycle (with $d' \neq d_0$) yields a

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voloshin-hypergraph_F01327	180	■ 1.000	$T_{\{3\}}$	T_3	different tree T_3 .
voloshin-hypergraph_F01328	181	■ 1.000	$\mathcal{H}_{\{3\}}^*$	$\mathcal{H}_3^*$	a star in T_3 . In order to complete the proof that $\mathcal{H}_3^*$ is a hypertree, we must show that the
voloshin-hypergraph_F01329	181	■ 1.000	$d^{\wedge\{\prime\}}$	d'	that contains d and d' . In $\mathcal{H}_2$, the vertex x' belongs to D and D' . Since d is pendant to d_0 in
voloshin-hypergraph_F01330	181	■ 1.000	$\mathcal{H}_{\{3\}}^*, \mathcal{H}_{\{4\}}^*, \ldots$	$\mathcal{H}_3^*, \mathcal{H}_4^*, \dots$	hypertrees $\mathcal{H}_3^*, \mathcal{H}_4^*, \dots$ with host trees $T_3, T_4, \dots$. At each step, the conditions are preserved
voloshin-hypergraph_F01331	181	■ 1.000	$T_{\{3\}}, T_{\{4\}}, \ldots$	$T_3, T_4, \dots$	hypertrees $\mathcal{H}_3^*, \mathcal{H}_4^*, \dots$ with host trees $T_3, T_4, \dots$. At each step, the conditions are preserved
voloshin-hypergraph_F01332	181	■ 0.998	$(\mathcal{H})_2 = L(\mathcal{H}^*)$	$(\mathcal{H})_2 = L(\mathcal{H}^*)$	$\Leftarrow$ Let $\mathcal{H}^*$ be a hypertree. By Lemma 8.1.1, $L(\mathcal{H}^*)$ is chordal. Since $(\mathcal{H})_2 = L(\mathcal{H}^*)$,
voloshin-hypergraph_F01333	181	■ 1.000	R	R	Let R be a set of pairwise adjacent vertices in $\mathcal{H}$. These become pairwise intersecting
voloshin-hypergraph_F01334	182	■ 0.995	$\left(\mathcal{H}^*\right)_2 = L(\mathcal{H})$	$(\mathcal{H}^*)_2 = L(\mathcal{H})$	$\mathcal{H}^*$ is conformal (Theorem 7.6.4), and because $(\mathcal{H}^*)_2 = L(\mathcal{H})$ (Corollary 7.6.1).
voloshin-hypergraph_F01335	183	■ 0.831	A, B, C, D	A, B, C, D	hyperedges A, B, C, D , and E in $\mathcal{H}^*$ and each induce a subtree in T . One can easily see that
voloshin-hypergraph_F01336	183	■ 0.997	$G = L(\mathcal{H}^*)$	$G = L(\mathcal{H}^*)$	$G = L(\mathcal{H}^*)$. Drawings of $\mathcal{H}$ and $\mathcal{H}^*$ are simply two different drawings of a (0,1)-matrix
voloshin-hypergraph_F01337	183	■ 0.998	α	α	Chordal conformal hypergraphs are also known as α -acyclic hypergraphs. In [2], hy-
voloshin-hypergraph_F01338	183	■ 1.000	$L(\mathcal{H}) = (\mathcal{H}^*)_2$	$L(\mathcal{H}) = (\mathcal{H}^*)_2$	7. For each of the hypergraphs in Figure 8.7, verify that $L(\mathcal{H}) = (\mathcal{H}^*)_2$.
voloshin-hypergraph_F01339	183	■ 1.000	$G = L(\mathcal{H})$	$G = L(\mathcal{H})$	4. Given a chordal graph G , construct a hypertree $\mathcal{H}$ such that $G = L(\mathcal{H})$.
voloshin-hypergraph_F01340	184	■ 1.000	$D = \{x\}$	$D = \{x\}$	an edge $D \in \mathcal{D}$ such that $D = \{x\}$ (i.e., D is a singleton). It means that x belongs to every
voloshin-hypergraph_F01341	184	■ 1.000	$y \in X, y \neq x$	$y \in X, y \neq x$	pendant) if there exist a vertex $y \in X, y \neq x$, such that $\mathcal{D}(x) \subseteq \mathcal{D}(y)$. Such a vertex y is
voloshin-hypergraph_F01342	184	■ 1.000	$\mathcal{D}(x) = \emptyset$	$\mathcal{D}(x) = \emptyset$	pendant because $\mathcal{D}(x) = \emptyset$ for any isolated vertex x . Any pendant vertex in a graph is
voloshin-hypergraph_F01343	184	■ 0.607	$\mathrm{T}(\mathcal{H}) = \mathrm{T}(\mathcal{H}_1)$	$T(\mathcal{H}) = T(\mathcal{H}_1)$	2. $\tau(\mathcal{H}) = \tau(\mathcal{H}_1)$
voloshin-hypergraph_F01344	184	■ 1.000	$\alpha(\mathcal{H}) = \alpha(\mathcal{H}_1) + 1$	$\alpha(\mathcal{H}) = \alpha(\mathcal{H}_1) + 1$	3. $\alpha(\mathcal{H}) = \alpha(\mathcal{H}_1) + 1$;
voloshin-hypergraph_F01345	184	■ 0.505	$v(\mathcal{H}) = v(\mathcal{H}_1)$	$v(\mathcal{H}) = v(\mathcal{H}_1)$	4. $v(\mathcal{H}) = v(\mathcal{H}_1)$;
voloshin-hypergraph_F01346	184	■ 1.000	$x \notin T$	$x \notin T$	Case a): y belongs to a minimum transversal $T \subseteq X$. Then $x \notin T$ because otherwise the
voloshin-hypergraph_F01347	184	■ 0.730	$T_1 = T - \{x\}$	$T_1 = T - \{x\}$	set $T_1 = T - \{x\}$ is a transversal with $ T_1 = T - 1 = \tau(\mathcal{H}) - 1$, a contradiction. Therefore,
voloshin-hypergraph_F01348	184	■ 0.730	$ T_1 = T - 1 = \tau(\mathcal{H}) - 1$	$ T_1 = T - 1 = \tau(\mathcal{H}) - 1$	set $T_1 = T - \{x\}$ is a transversal with $ T_1 = T - 1 = \tau(\mathcal{H}) - 1$, a contradiction. Therefore,
voloshin-hypergraph_F01349	184	■ 1.000	$x \in X - T = S$	$x \in X - T = S$	$x \in X - T = S$ where S is a maximum independent set.
voloshin-hypergraph_F01350	184	■ 1.000	$x \in T_1$	$x \in T_1$	sal T_1 such that $x \in T_1$, then set $T_2 = (T_1 - \{x\}) \cup \{y\}$ is a minimum transversal what contra-
voloshin-hypergraph_F01351	184	■ 1.000	$T_2 = (T_1 - \{x\}) \cup \{y\}$	$T_2 = (T_1 - \{x\}) \cup \{y\}$	sal T_1 such that $x \in T_1$, then set $T_2 = (T_1 - \{x\}) \cup \{y\}$ is a minimum transversal what contra-
voloshin-hypergraph_F01352	184	■ 0.844	$\tau(\mathcal{H}) \leq \tau(\mathcal{H}_1)$	$\tau(\mathcal{H}) \leq \tau(\mathcal{H}_1)$	transversal, so $\tau(\mathcal{H}) \leq \tau(\mathcal{H}_1)$. On the other hand, 1. implies that in $\mathcal{H}$ there is a mini-
voloshin-hypergraph_F01353	184	■ 1.000	$\tau(\mathcal{H}) \geq \tau(\mathcal{H}_1)$	$\tau(\mathcal{H}) \geq \tau(\mathcal{H}_1)$	mum transversal not containing x ; weak deletion of x cannot increase it, so $\tau(\mathcal{H}) \geq \tau(\mathcal{H}_1)$.

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voloshin-hypergraph_F01354	184	■ 0.996	$\tau(\mathcal{H})=\tau\left(\mathcal{H}_{_1}\right)$	$\tau(\mathcal{H}) = \tau(\mathcal{H}_1)$	Therefore, $\tau(\mathcal{H}) = \tau(\mathcal{H}_1)$.
voloshin-hypergraph_F01355	184	■ 0.966	$\alpha(\mathcal{H})+\tau(\mathcal{H})= X $, $\alpha\left(\mathcal{H}_{_1}\right)+\tau\left(\mathcal{H}_{_1}\right)= X -1$	$\alpha(\mathcal{H}) + \tau(\mathcal{H}) = X , \alpha(\mathcal{H}_1) + \tau(\mathcal{H}_1) = X - 1$	3. Since $\alpha(\mathcal{H}) + \tau(\mathcal{H}) = X , \alpha(\mathcal{H}_1) + \tau(\mathcal{H}_1) = X - 1$, and $\tau(\mathcal{H}) = \tau(\mathcal{H}_1)$, conclude
voloshin-hypergraph_F01356	184	■ 0.482	$\boldsymbol{\alpha}(\mathcal{H})=\boldsymbol{\alpha}\left(\mathcal{H}_{_1}\right)+1$	$\alpha(\mathcal{H}) = \alpha(\mathcal{H}_1) + 1$	that $\alpha(\mathcal{H}) = \alpha(\mathcal{H}_1) + 1$.
voloshin-hypergraph_F01357	184	■ 0.511	$\mathcal{V}(\mathcal{H}) \leq \mathcal{V}\left(\mathcal{H}_{_1}\right)$	$\mathcal{V}(\mathcal{H}) \leq \mathcal{V}(\mathcal{H}_1)$	4. Any maximum matching of $\mathcal{H}$ is a matching in $\mathcal{H}_1$, so $v(\mathcal{H}) \leq v(\mathcal{H}_1)$. On the other
voloshin-hypergraph_F01358	184	■ 0.827	$\mathcal{V}(\mathcal{H}) \geq \mathcal{V}\left(\mathcal{H}_{_1}\right)$	$\mathcal{V}(\mathcal{H}) \geq \mathcal{V}(\mathcal{H}_1)$	implies that every matching in $\mathcal{H}_1$ is a matching in $\mathcal{H}$, i.e., $v(\mathcal{H}) \geq v(\mathcal{H}_1)$.
voloshin-hypergraph_F01359	185	■ 0.998	$L(\mathcal{H})=$	$L(\mathcal{H}) =$	graph. Since x is hyper-pendant, $\mathcal{H}_1$ is a Helly hypergraph, too, and, moreover, $L(\mathcal{H}) =$
voloshin-hypergraph_F01360	185	■ 1.000	$L\left(\mathcal{H}_{_1}\right)$	$L(\mathcal{H}_1)$	$L(\mathcal{H}_1)$. Consequently, $\mathcal{H}_1$ is a hypertree.
voloshin-hypergraph_F01361	185	■ 0.927	$\tau(\mathcal{H})=\tau\left(\mathcal{H}_{_1}\right)+1$	$\tau(\mathcal{H}) = \tau(\mathcal{H}_1) + 1$	2. $\tau(\mathcal{H}) = \tau(\mathcal{H}_1) + 1$;
voloshin-hypergraph_F01362	185	■ 1.000	$\alpha(\mathcal{H})=\alpha\left(\mathcal{H}_{_1}\right)$	$\alpha(\mathcal{H}) = \alpha(\mathcal{H}_1)$	3. $\alpha(\mathcal{H}) = \alpha(\mathcal{H}_1)$;
voloshin-hypergraph_F01363	185	■ 0.804	$v(\mathcal{H})=v\left(\mathcal{H}_{_1}\right)+1$	$v(\mathcal{H}) = v(\mathcal{H}_1) + 1$	4. $v(\mathcal{H}) = v(\mathcal{H}_1) + 1$;
voloshin-hypergraph_F01364	185	■ 1.000	$\{x\} \in \mathcal{D}$	$\{x\} \in \mathcal{D}$	Proof. 1. It follows from the definition of transversal and the fact that $\{x\} \in \mathcal{D}$.
voloshin-hypergraph_F01365	185	■ 0.989	$T_{_1} \subset X_{_1}=X-\{x\}$	$T_1 \subset X_1 = X - \{x\}$	2. Let $T_1 \subset X_1 = X - \{x\}$ be a minimum transversal of $\mathcal{H}_1$. Then set $T_2 = T_1 \cup \{x\} \subset X$
voloshin-hypergraph_F01366	185	■ 0.989	$T_{_2}=T_{_1} \cup \{x\} \subset X$	$T_2 = T_1 \cup \{x\} \subset X$	2. Let $T_1 \subset X_1 = X - \{x\}$ be a minimum transversal of $\mathcal{H}_1$. Then set $T_2 = T_1 \cup \{x\} \subset X$
voloshin-hypergraph_F01367	185	■ 1.000	$\tau(\mathcal{H}) \leq \tau\left(\mathcal{H}_{_1}\right)+1$	$\tau(\mathcal{H}) \leq \tau(\mathcal{H}_1) + 1$	is a transversal of $\mathcal{H}$, what implies $\tau(\mathcal{H}) \leq \tau(\mathcal{H}_1) + 1$. On the other hand, if T is a minimum
voloshin-hypergraph_F01368	185	■ 0.999	$T_{_0}=T-\{x\}$	$T_0 = T - \{x\}$	transversal of $\mathcal{H}$, then $T_0 = T - \{x\}$ is a transversal of $\mathcal{H}_1$, what implies $\tau(\mathcal{H}) \geq \tau(\mathcal{H}_1) + 1$.
voloshin-hypergraph_F01369	185	■ 0.999	$\tau(\mathcal{H}) \geq \tau\left(\mathcal{H}_{_1}\right)+1$	$\tau(\mathcal{H}) \geq \tau(\mathcal{H}_1) + 1$	transversal of $\mathcal{H}$, then $T_0 = T - \{x\}$ is a transversal of $\mathcal{H}_1$, what implies $\tau(\mathcal{H}) \geq \tau(\mathcal{H}_1) + 1$.
voloshin-hypergraph_F01370	185	■ 0.999	$\mathcal{F} \cup \{x\}$	$\mathcal{F} \cup \{x\}$	$\mathcal{F} \cup \{x\}$ is a matching of $\mathcal{H}$, what implies $v(\mathcal{H}) \geq v(\mathcal{H}_1) + 1$.
voloshin-hypergraph_F01371	185	■ 0.999	$\mathcal{V}(\mathcal{H}) \geq \mathcal{V}\left(\mathcal{H}_{_1}\right)+1$	$\mathcal{V}(\mathcal{H}) \geq \mathcal{V}(\mathcal{H}_1) + 1$	$\mathcal{F} \cup \{x\}$ is a matching of $\mathcal{H}$, what implies $v(\mathcal{H}) \geq v(\mathcal{H}_1) + 1$.
voloshin-hypergraph_F01372	185	■ 0.940	$\mathcal{V}(\mathcal{H})-1 \leq \mathcal{V}\left(\mathcal{H}_{_1}\right)$	$\mathcal{V}(\mathcal{H}) - 1 \leq \mathcal{V}(\mathcal{H}_1)$	matching by more than 1; hence $v(\mathcal{H}) - 1 \leq v(\mathcal{H}_1)$ what implies the required equality.
voloshin-hypergraph_F01373	185	■ 1.000	$\mathcal{H}=(X, \mathcal{D}), X \geq 2$	$\mathcal{H} = (X, \mathcal{D}), X \geq 2$	Proposition 8.2.1 If $\mathcal{H} = (X, \mathcal{D}), X \geq 2$, is a hypertree, then it contains at least two ver-
voloshin-hypergraph_F01374	186	■ 0.786	$\tau=\nu=0$	$\tau = \nu = 0$	one-vertex hypergraph; either it is a hyper-pendant (isolated) vertex, then $\tau = \nu = 0$, or a
voloshin-hypergraph_F01375	186	■ 0.753	$\tau=\nu=1$	$\tau = \nu = 1$	transversal vertex, then $\tau = \nu = 1$. In all steps we have the equality, so $\tau(\mathcal{H}) = \nu(\mathcal{H})$.
voloshin-hypergraph_F01376	186	■ 0.753	$\tau(\mathcal{H})=\nu(\mathcal{H})$	$\tau(\mathcal{H}) = \nu(\mathcal{H})$	transversal vertex, then $\tau = \nu = 1$. In all steps we have the equality, so $\tau(\mathcal{H}) = \nu(\mathcal{H})$.
voloshin-hypergraph_F01377	186	■ 0.974	$\chi=\omega$	$\chi = \omega$	any chordal graph is perfect. Hence for chordal graphs $\chi = \omega$ and $\theta = \alpha$. We know that
voloshin-hypergraph_F01378	186	■ 0.974	$\theta=\alpha$	$\theta = \alpha$	any chordal graph is perfect. Hence for chordal graphs $\chi = \omega$ and $\theta = \alpha$. We know that
voloshin-hypergraph_F01379	186	■ 0.988	$\theta(L(\mathcal{H}))=\alpha(L(\mathcal{H}))$	$\theta(L(\mathcal{H})) = \alpha(L(\mathcal{H}))$	if $\mathcal{H}$ is a hypertree, then $L(\mathcal{H})$ is chordal. Therefore $\theta(L(\mathcal{H})) = \alpha(L(\mathcal{H}))$. Because $\mathcal{H}$

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voloshin-hypergraph_F01380	186	■ 0.946	$\nu(\mathcal{H})=\alpha(L(\mathcal{H}))$	$\nu(\mathcal{H}) = \alpha(L(\mathcal{H}))$	the size of maximum matching $\nu(\mathcal{H}) = \alpha(L(\mathcal{H}))$. So in total, we have the same equality
voloshin-hypergraph_F01381	186	■ 0.650	$\tau(\mathcal{H})=\mathcal{V}(\mathcal{H})$	$\tau(\mathcal{H}) = \mathcal{V}(\mathcal{H})$	$\tau(\mathcal{H}) = \nu(\mathcal{H})$. Moreover, it holds for every partial subhypergraph $\mathcal{H}' \subseteq \mathcal{H}$ and for every
voloshin-hypergraph_F01382	187	■ 0.930	$\{3\}$	$\{3\}$	of the singleton $\{3\}$. Similarly, vertex 2 in $\mathcal{H}_3$ becomes hyper-pendant with the twin 4 after
voloshin-hypergraph_F01383	187	■ 0.822	$\{2\}$	$\{2\}$	weak deletion of the singleton $\{2\}$. At last, hypergraph $\mathcal{H}_4$ has the only vertex. So, the
voloshin-hypergraph_F01384	187	■ 0.822	$\mathcal{H}_4$	$\mathcal{H}_4$	weak deletion of the singleton $\{2\}$. At last, hypergraph $\mathcal{H}_4$ has the only vertex. So, the
voloshin-hypergraph_F01385	187	■ 1.000	T_4	T_4	with T_4 . Sequentially adding vertices in inverse order 4, 2, 3, 1 and connecting them to their
voloshin-hypergraph_F01386	188	■ 1.000	$T=\emptyset, \mathcal{F}=\emptyset$	$T = \emptyset, \mathcal{F} = \emptyset$	1. Put $T = \emptyset, \mathcal{F} = \emptyset$.
voloshin-hypergraph_F01387	188	■ 0.968	$T, \mathcal{F}$	$T, \mathcal{F}$	6. Output $T, \mathcal{F}$.
voloshin-hypergraph_F01388	189	■ 1.000	$T=\{1,3,5\}$	$T = \{1, 3, 5\}$	minimum transversal $T = \{1, 3, 5\}$ and maximum matching $\mathcal{F} = \{\{1, 6\}, \{2, 3\}, \{4, 5\}\}$, so
voloshin-hypergraph_F01389	189	■ 1.000	$\mathcal{F}=\{\{1,6\},\{2,3\},\{4,5\}\}$	$\mathcal{F} = \{\{1, 6\}, \{2, 3\}, \{4, 5\}\}$	minimum transversal $T = \{1, 3, 5\}$ and maximum matching $\mathcal{F} = \{\{1, 6\}, \{2, 3\}, \{4, 5\}\}$, so
voloshin-hypergraph_F01390	189	■ 0.769	$\tau(G)=\mathcal{V}(G)$	$\tau(G) = \mathcal{V}(G)$	$\tau(G) = \nu(G)$. Maximum matching $\mathcal{F}$ is shown by double lines in the second drawing of G .
voloshin-hypergraph_F01391	189	■ 0.961	$\tau(C_5)=3>2=\nu(C_5)$	$\tau(C_5) = 3 > 2 = \nu(C_5)$	One can compare graph G with cycle C_5 where $\tau(C_5) = 3 > 2 = \nu(C_5)$.
voloshin-hypergraph_F01392	189	■ 0.987	τ, α	τ, α	3. Compute τ, α and ν and find a minimum transversal, maximum independent set and
voloshin-hypergraph_F01393	189	■ 0.772	$\tau(\mathcal{H}), \alpha(\mathcal{H})$	$\tau(\mathcal{H}), \alpha(\mathcal{H})$	and strong deletions of transversal vertices. Find $\tau(\mathcal{H}), \alpha(\mathcal{H})$ and $\nu(\mathcal{H})$.
voloshin-hypergraph_F01394	189	■ 0.772	$\nu(\mathcal{H})$	$\nu(\mathcal{H})$	and strong deletions of transversal vertices. Find $\tau(\mathcal{H}), \alpha(\mathcal{H})$ and $\nu(\mathcal{H})$.
voloshin-hypergraph_F01395	190	■ 0.977	$\Lambda(G)=$	$\Lambda(G) =$	(see Section 2.1.) that for a simple connected graph G , the cyclomatic number $\Lambda(G) =$
voloshin-hypergraph_F01396	190	■ 1.000	$m-n+1$	$m - n + 1$	$m - n + 1$. The same formula generalizes to multigraphs if we add to G any number of
voloshin-hypergraph_F01397	190	■ 1.000	$[\mathcal{H}]_2$	$[\mathcal{H}]_2$	2-section of $\mathcal{H}$, denoted by $[\mathcal{H}]_2$, if it is obtained from $\mathcal{H}$ in the following way:
voloshin-hypergraph_F01398	190	■ 1.000	≥ 3	≥ 3	2. every hyperedge of size ≥ 3 of $\mathcal{H}$ is replaced by a complete graph on the same
voloshin-hypergraph_F01399	190	■ 1.000	$(\mathcal{H})_2=[\mathcal{H}]_2$	$(\mathcal{H})_2 = [\mathcal{H}]_2$	Hence, $(\mathcal{H})_2 = [\mathcal{H}]_2$ if and only if $\mathcal{H}$ itself is a simple graph. Further, let us call any
voloshin-hypergraph_F01400	190	■ 1.000	$\bar{T}$	$\bar{T}$	a multi-forest, then $\bar{T}$ denotes a forest obtained from T by weak deletion of all loops and
voloshin-hypergraph_F01401	190	■ 1.000	$\bar{T}=(T)_2$	$\bar{T} = (T)_2$	$\bar{T} = (T)_2$. A vertex x is called pendant in a multi-forest T if it is pendant in $\bar{T}$.
voloshin-hypergraph_F01402	190	■ 1.000	$ D -1$	$ D - 1$	Proof. By definition of T , any single edge $D \in \mathcal{D}$ may provide to T maximum $ D - 1$
voloshin-hypergraph_F01403	190	■ 0.994	$l(\mathcal{H}, T)$	$l(\mathcal{H}, T)$	Let $l(\mathcal{H}, T)$ denote the number of singletons (loops) of a hypergraph $\mathcal{H}$ which <i>are not</i>
voloshin-hypergraph_F01404	191	■ 1.000	$[\mathcal{H}]_2=\mathcal{H}$	$[\mathcal{H}]_2 = \mathcal{H}$	Let $\mathcal{H}$ be a connected multigraph (with loops and multiple edges); then $[\mathcal{H}]_2 = \mathcal{H}$. If
voloshin-hypergraph_F01405	191	■ 0.998	$w(T)= X -1=n-1$	$w(T) = X - 1 = n - 1$	$w(T) = X - 1 = n - 1$. Since all loops of $\mathcal{H}$ will contribute to $l(\mathcal{H}, T)$, we obtain:
voloshin-hypergraph_F01406	191	■ 0.998	$\Lambda(\mathcal{H}, T)$	$\Lambda(\mathcal{H}, T)$	This equality shows that $\Lambda(\mathcal{H}, T)$ generalizes usual cyclomatic number of a multigraph.

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voloshin-hypergraph_ F01407	191	■ 1.000	w(T)=3	$w(T) = 3$	only one of the two loops of $\mathcal{H}$, so $w(T) = 3$. One can compute:
voloshin-hypergraph_ F01408	192	■ 0.999	E_D	E_D	i.e. a forest. Let E_D be the set of edges of size 2 which are generated by the 2-section of D
voloshin-hypergraph_ F01409	192	■ 1.000	$c(T, D)$	$c(T, D)$	and lie on T , and $c(T, D)$ be the number of connected components in such a subgraph.
voloshin-hypergraph_ F01410	192	■ 1.000	$ D \geq 2$	$ D \geq 2$	Proof. First, notice that for every edge D such that $ D \geq 2$, the following equality holds :
voloshin-hypergraph_ F01411	192	■ 1.000	$c(T, D)=1$	$c(T, D) = 1$	Further, since for any singleton D , evidently $c(T, D) = 1$, we obtain:
voloshin-hypergraph_ F01412	192	■ 0.960	$T, \Lambda(\mathcal{H}, T) \geq 0$	$T, \Lambda(\mathcal{H}, T) \geq 0$	Proposition 8.3.2 For any hypergraph $\mathcal{H}$ and forest T , $\Lambda(\mathcal{H}, T) \geq 0$.
voloshin-hypergraph_ F01413	192	■ 1.000	$l(\mathcal{H}, T) \geq 0$	$l(\mathcal{H}, T) \geq 0$	Proof. Apply inequalities (8.2) and $l(\mathcal{H}, T) \geq 0$.
voloshin-hypergraph_ F01414	192	■ 0.913	$\Lambda(\mathcal{H}, T)=0$	$\Lambda(\mathcal{H}, T) = 0$	$\Lambda(\mathcal{H}, T) = 0$ if and only if $\mathcal{H}$ is a hypertree and T is a multi-tree of maximum
voloshin-hypergraph_ F01415	193	■ 1.000	$l(\mathcal{H}, T)=0$	$l(\mathcal{H}, T) = 0$	$\mathcal{D}$, and, in addition, $l(\mathcal{H}, T) = 0$. It means that 2-section of any edge $D \in \mathcal{D}$ induces a
voloshin-hypergraph_ F01416	193	■ 0.918	$w(T)=\sum_{D \in \mathcal{D}}(D -1)$	$w(T) = \sum_{D \in \mathcal{D}} (D - 1)$	$w(T) = \sum_{D \in \mathcal{D}} (D - 1)$, and by (8.3) equality $\Lambda(\mathcal{H}, T) = 0$ holds.
voloshin-hypergraph_ F01417	193	■ 0.918	$\wedge(\mathcal{H}, T)=0$	$\wedge(\mathcal{H}, T) = 0$	$w(T) = \sum_{D \in \mathcal{D}} (D - 1)$, and by (8.3) equality $\Lambda(\mathcal{H}, T) = 0$ holds.
voloshin-hypergraph_ F01418	193	■ 1.000	$\overline{T_1}$	$\overline{T_1}$	1. x is adjacent to y and pendant in $\overline{T_1}$;
voloshin-hypergraph_ F01419	193	■ 1.000	$w(T_1)=w(T)$	$w(T_1) = w(T)$	2. $w(T_1) = w(T)$.
voloshin-hypergraph_ F01420	193	■ 1.000	$x \in X, T$	$x \in X, T$	Theorem 8.3.4 (Voloshin, 1987) Let $\mathcal{H} = (X, \mathcal{D})$ be a hypergraph, $x \in X$, T be a forest in
voloshin-hypergraph_ F01421	193	■ 1.000	$[\mathcal{H}]_2, \mathcal{D}_T(x)$	$[\mathcal{H}]_2, \mathcal{D}_T(x)$	$[\mathcal{H}]_2$, $\mathcal{D}_T(x)$ denote the set of edges of T incident to x , and m_2 be the number of edges of
voloshin-hypergraph_ F01422	193	■ 1.000	m_2	m_2	$[\mathcal{H}]_2$, $\mathcal{D}_T(x)$ denote the set of edges of T incident to x , and m_2 be the number of edges of
voloshin-hypergraph_ F01423	193	■ 1.000	$\mathcal{H} - x$	$\mathcal{H} - x$	graph $\mathcal{H} - x$ and a forest $T - x$. Again, according to the definition,
voloshin-hypergraph_ F01424	193	■ 1.000	$ \mathcal{D}(x) =3, \mathcal{D}_T(x) =1$	$ \mathcal{D}(x) = 3, \mathcal{D}_T(x) = 1$	Now notice that $ \mathcal{D}(x) = 3$, $ \mathcal{D}_T(x) = 1$, and $m_2 = 1$. Hence,
voloshin-hypergraph_ F01425	193	■ 1.000	$m_2=1$	$m_2 = 1$	Now notice that $ \mathcal{D}(x) = 3$, $ \mathcal{D}_T(x) = 1$, and $m_2 = 1$. Hence,
voloshin-hypergraph_ F01426	194	■ 1.000	$[\mathcal{H}]_2, x \in X$	$[\mathcal{H}]_2, x \in X$	$[\mathcal{H}]_2$, $x \in X$ be a vertex pendant and adjacent to y in $\overline{T}$. If hypergraph $\mathcal{H}_1$ is obtained from
voloshin-hypergraph_ F01427	194	■ 1.000	$[\mathcal{H}_1]_2$	$[\mathcal{H}_1]_2$	weight in $[\mathcal{H}_1]_2$, and moreover,
voloshin-hypergraph_ F01428	194	■ 1.000	T_1'	T_1'	Proof. Assume there is another forest, say, T_1' which is of maximum weight in $[\mathcal{H}_1]_2$.
voloshin-hypergraph_ F01429	194	■ 1.000	$w(T_1') \geq w(T_1)$	$w(T_1') \geq w(T_1)$	Evidently, $w(T_1') \geq w(T_1)$. If T_1 is not a forest of maximum weight, then $w(T_1') \geq w(T_1) + 1$.
voloshin-hypergraph_ F01430	194	■ 1.000	$w(T_1') \geq w(T_1)+1$	$w(T_1') \geq w(T_1) + 1$	Evidently, $w(T_1') \geq w(T_1)$. If T_1 is not a forest of maximum weight, then $w(T_1') \geq w(T_1) + 1$.
voloshin-hypergraph_ F01431	194	■ 1.000	$ \mathcal{D}_T(x) $	$ \mathcal{D}_T(x) $	$ \mathcal{D}_T(x) $ edges. Denote the obtained forest by T' . Then
voloshin-hypergraph_ F01432	194	■ 1.000	T'	T'	$ \mathcal{D}_T(x) $ edges. Denote the obtained forest by T' . Then
voloshin-hypergraph_ F01433	194	■ 1.000	$w(T_1')=w(T_1)$	$w(T_1') = w(T_1)$	what contradicts the maximality of T . Therefore, $w(T_1') = w(T_1)$ and the theorem follows.

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voloshin-hypergraph_F01434	195	■ 1.000	$m_2=0$	$m_2 = 0$	We are able now to apply Theorem 8.3.4 (or, equivalently, Corollary 8.3.1): $m_2 = 0$ and
voloshin-hypergraph_F01435	195	■ 1.000	$ \mathcal{D}(x) =\left \mathcal{D}_T(x)\right $	$ \mathcal{D}(x) = \mathcal{D}_T(x) $	$ \mathcal{D}(x) = \mathcal{D}_T(x) $, and therefore
voloshin-hypergraph_F01436	195	■ 0.919	$\Lambda\left(\mathcal{H}_1, T_1\right)=\Lambda\left(\mathcal{H}_1\right)$	$\Lambda\left(\mathcal{H}_1, T_1\right) = \Lambda\left(\mathcal{H}_1\right)$	$\Lambda\left(\mathcal{H}_1, T_1\right) = \Lambda\left(\mathcal{H}_1\right)$. Hence, $\Lambda(\mathcal{H}) = \Lambda(\mathcal{H}_1)$.
voloshin-hypergraph_F01437	195	■ 0.919	$\Lambda\left(\mathcal{H}\right)=\Lambda\left(\mathcal{H}_1\right)$	$\Lambda(\mathcal{H}) = \Lambda\left(\mathcal{H}_1\right)$	$\Lambda\left(\mathcal{H}_1, T_1\right) = \Lambda\left(\mathcal{H}_1\right)$. Hence, $\Lambda(\mathcal{H}) = \Lambda(\mathcal{H}_1)$.
voloshin-hypergraph_F01438	195	■ 0.876	$\wedge\left(\mathcal{H}\right)=0$	$\wedge(\mathcal{H}) = 0$	2. $\wedge(\mathcal{H}) = 0$;
voloshin-hypergraph_F01439	195	■ 1.000	$1 \rightarrow 2$	$1 \Rightarrow 2$	Proof. $1 \Rightarrow 2$: Apply Theorem 8.3.6 and evident fact that singletons do not contribute to
voloshin-hypergraph_F01440	195	■ 0.923	$\Lambda\left(\mathcal{H}\right)$	$\Lambda(\mathcal{H})$	$\Lambda(\mathcal{H})$.
voloshin-hypergraph_F01441	195	■ 0.993	$2 \rightarrow 3$	$2 \Rightarrow 3$	$2 \Rightarrow 3$: Apply Theorem 8.3.2.
voloshin-hypergraph_F01442	195	■ 1.000	$3 \rightarrow 1$	$3 \Rightarrow 1$	$3 \Rightarrow 1$: Apply Theorem 8.2.1.
voloshin-hypergraph_F01443	195	■ 0.914	$\wedge\left(\mathcal{H}\right)$	$\wedge(\mathcal{H})$	We now conclude by presenting an algorithm for computing $\Lambda(\mathcal{H})$.
voloshin-hypergraph_F01444	196	■ 1.000	$w\left(\mathcal{H}\right)$	$w(\mathcal{H})$	3. In G , find a spanning tree T of maximum weight $w(\mathcal{H})$.
voloshin-hypergraph_F01445	196	■ 0.666	$\Lambda\left(\mathcal{H}\right)=\sum_{D \in \mathcal{D}}(D -1) \cdot w\left(\mathcal{H}_D\right)$	$\Lambda(\mathcal{H}) = \sum_{D \in \mathcal{D}} (D - 1) - w(\mathcal{H})$	4. Compute $\Lambda(\mathcal{H}) = \sum_{D \in \mathcal{D}} (D - 1) - w(\mathcal{H})$.
voloshin-hypergraph_F01446	197	■ 0.999	$D_0, D_1, D_2, \ldots D_{t-1}$	$D_0, D_1, D_2, \ldots D_{t-1}$	of distinct vertices $x_0, x_1, x_2, \ldots, x_{t-1}$ and distinct edges $D_0, D_1, D_2, \ldots D_{t-1}$ satisfying
voloshin-hypergraph_F01447	198	■ 0.408	$\tau\left(\mathcal{H}'\right)=\nu\left(\mathcal{H}'\right)$	$\tau\left(\mathcal{H}'\right) = \nu\left(\mathcal{H}'\right)$	Theorem 9.1.2, $\tau(\mathcal{H}') = \nu(\mathcal{H}')$. But $\nu(\mathcal{H}') = 1$ because any matching contains maximum
voloshin-hypergraph_F01448	198	■ 0.408	$\nu\left(\mathcal{H}'\right)=1$	$\nu\left(\mathcal{H}'\right) = 1$	Theorem 9.1.2, $\tau(\mathcal{H}') = \nu(\mathcal{H}')$. But $\nu(\mathcal{H}') = 1$ because any matching contains maximum
voloshin-hypergraph_F01449	198	■ 0.999	$\tau\left(\mathcal{H}'\right)=1$	$\tau\left(\mathcal{H}'\right) = 1$	one edge from every intersecting family. Thus, $\tau(\mathcal{H}') = 1$ what means there exists a vertex
voloshin-hypergraph_F01450	198	■ 1.000	>3	> 3	Proof. Let $\mathcal{H}$ be a totally balanced hypergraph. Since every cycle of length > 3 has an
voloshin-hypergraph_F01451	199	■ 0.999	K_n, K_m, W_n	$K_n, K_{m,n}, W_n$	1. Which of the graphs $K_n, K_{m,n}, W_n$, prism, cube and Petersen graph are balanced
voloshin-hypergraph_F01452	200	■ 1.000	$C_n, n \geq 3$	$C_n, n \geq 3$	2. Explain why cycle $C_n, n \geq 3$ is not an interval hypergraph.
voloshin-hypergraph_F01453	201	■ 1.000	$\chi'\left(\mathcal{H}\right)$	$\chi'(\mathcal{H})$	color. As for graphs, we denote it by $\chi'(\mathcal{H})$. Clearly, $\chi'(\mathcal{H}) = \chi(L(\mathcal{H}))$. Now we have the
voloshin-hypergraph_F01454	201	■ 1.000	$\chi'\left(\mathcal{H}\right)=\chi\left(L\left(\mathcal{H}\right)\right)$	$\chi'(\mathcal{H}) = \chi(L(\mathcal{H}))$	color. As for graphs, we denote it by $\chi'(\mathcal{H})$. Clearly, $\chi'(\mathcal{H}) = \chi(L(\mathcal{H}))$. Now we have the
voloshin-hypergraph_F01455	201	■ 1.000	$\chi'\left(\mathcal{H}\right)=\Delta\left(\mathcal{H}\right)$	$\chi'(\mathcal{H}) = \Delta(\mathcal{H})$	We state that hypergraph $\mathcal{H}$ has the edge-coloring property if $\chi'(\mathcal{H}) = \Delta(\mathcal{H})$. The
voloshin-hypergraph_F01456	201	■ 0.996	$\chi\left(\left(\mathcal{H}^*\right)_2\right)=$	$\chi\left(\left(\mathcal{H}^*\right)_2\right) =$	edge-coloring property means that if we take dual hypergraph $\mathcal{H}^*$, then $\chi\left(\left(\mathcal{H}^*\right)_2\right) =$
voloshin-hypergraph_F01457	201	■ 1.000	$\omega\left(\left(\mathcal{H}^*\right)_2\right)$	$\omega\left(\left(\mathcal{H}^*\right)_2\right)$	$\omega\left(\left(\mathcal{H}^*\right)_2\right)$.
voloshin-hypergraph_F01458	201	■ 0.992	$\tau\left(\mathcal{H}'\right)=\nu\left(\mathcal{H}'\right)=1$	$\tau\left(\mathcal{H}'\right) = \nu\left(\mathcal{H}'\right) = 1$	$\tau(\mathcal{H}') = \nu(\mathcal{H}') = 1$ what implies that $\mathcal{H}$ is a Helly hypergraph. The last means that $\mathcal{H}^*$ is
voloshin-hypergraph_F01459	201	■ 0.998	$\chi'\left(\mathcal{H}\right)=\Delta\left(\mathcal{H}'\right)$	$\chi'(\mathcal{H}) = \Delta\left(\mathcal{H}'\right)$	conformal. Further, $\chi'(\mathcal{H}) = \Delta(\mathcal{H}')$ means $\chi\left(\left(\mathcal{H}^*\right)_2\right) = \omega\left(\left(\mathcal{H}^*\right)_2\right)$, and this equality holds
voloshin-hypergraph_F01460	201	■ 0.998	$\chi\left(\left(\mathcal{H}^*\right)_2\right)=\omega\left(\left(\mathcal{H}^*\right)_2\right)$	$\chi\left(\left(\mathcal{H}^*\right)_2\right) = \omega\left(\left(\mathcal{H}^*\right)_2\right)$	conformal. Further, $\chi'(\mathcal{H}) = \Delta(\mathcal{H}')$ means $\chi\left(\left(\mathcal{H}^*\right)_2\right) = \omega\left(\left(\mathcal{H}^*\right)_2\right)$, and this equality holds

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voloshin-hypergraph_F01461	201	0.687	$\chi^{\prime}(\mathcal{H})=$	$\chi^{\prime}(\mathcal{H}) =$	Observe that $\chi((\mathcal{H}^*)_2) = \omega((\mathcal{H}^*)_2)$ because $L(\mathcal{H})$ is perfect; this means that $\chi^{\prime}(\mathcal{H}) =$
voloshin-hypergraph_F01462	202	1.000	$\Delta(\mathcal{H})$	$\Delta(\mathcal{H})$	$\Delta(\mathcal{H})$. Since by the same reason the last equality holds for every $\mathcal{H}' \subseteq \mathcal{H}$, the hypergraph
voloshin-hypergraph_F01463	202	1.000	$K_{\{n\}}, K_{\{m, n\}}, C_{\{n\}}, W_{\{n\}}$	$K_n, K_{m,n}, C_n, W_n$	1. Which of the graphs $K_n, K_{m,n}, C_n, W_n$, cube, prism and Petersen graph are normal
voloshin-hypergraph_F01464	203	1.000	$d \in \mathcal{D}$	$d \in \mathcal{D}$	graph $B(\mathcal{H})$ with vertex set $X \cup \mathcal{D}$. The vertex $x \in X$ is adjacent to the vertex $d \in \mathcal{D}$ in
voloshin-hypergraph_F01465	204	1.000	$f_{\{2\}}$	f_2	of common vertices. In this way, the regions corresponding to the two faces f_1 and f_2
voloshin-hypergraph_F01466	204	0.725	1, 2, 3	1, 2, 3	original vertices 1, 2, 3, and 4, and the vertices corresponding to edges D_1 and D_2 .
voloshin-hypergraph_F01467	204	1.000	$\mathcal{H}=(X, \mathcal{D}), X =n, \mathcal{D} =m$	$\mathcal{H} = (X, \mathcal{D}), X = n, \mathcal{D} = m$	Theorem 9.4.1 (Euler’s formula) Let $\mathcal{H} = (X, \mathcal{D}), X = n, \mathcal{D} = m$ be a planar hyper-
voloshin-hypergraph_F01468	204	1.000	$n^{\prime}=$	$n' =$	Proof. [6] Construct the planar embedding of the bipartite graph $B(\mathcal{H})$. It contains $n' =$
voloshin-hypergraph_F01469	204	1.000	$n+m$	$n + m$	$n + m$ vertices,
voloshin-hypergraph_F01470	205	1.000	$f^{\prime}=f$	$f' = f$	edges and $f' = f$ faces. Since $B(\mathcal{H})$ is a planar graph, by Theorem 4.2.1, Euler’s formula
voloshin-hypergraph_F01471	205	1.000	$f_{\{1\}}, f_{\{2\}}, f_{\{3\}}, f_{\{4\}}, f_{\{5\}}$	f_1, f_2, f_3, f_4, f_5	f_1, f_2, f_3, f_4, f_5 and f_6 . Notice that all the faces have size 2, and face f_6 is unbounded.
voloshin-hypergraph_F01472	205	1.000	$f_{\{6\}}$	f_6	f_1, f_2, f_3, f_4, f_5 and f_6 . Notice that all the faces have size 2, and face f_6 is unbounded.
voloshin-hypergraph_F01473	205	1.000	$f_{\{1\}}, \ldots, f_{\{6\}}$	$f_1, \dots, f_6$	Faces $f_1, \dots, f_6$ can be seen as the edges of size 2 connecting the respective pairs of ver-
voloshin-hypergraph_F01474	209	1.000	≤ 1	≤ 1	the edges of size ≤ 1 , which is equivalent to the preliminary weak deletion of all such
voloshin-hypergraph_F01475	209	1.000	$\chi(\mathcal{H})$	$\chi(\mathcal{H})$	chromatic number of $\mathcal{H}$ and is denoted by $\chi(\mathcal{H})$. Since every vertex gets one color,
voloshin-hypergraph_F01476	209	1.000	$\lambda>n$	$\lambda > n$	most $n(\mathcal{H})$. If $\lambda > n$, then in every λ -coloring at least one color remains unused. Proper
voloshin-hypergraph_F01477	209	0.703	$S_{\{1\}}, S_{\{2\}}, \ldots, S_{\{k\}}$	$S_1, S_2, \dots, S_k$	tion of X into k stable sets $S_1, S_2, \dots, S_k$ called color classes . Each color class S_i represents
voloshin-hypergraph_F01478	209	0.703	$S_{\{i\}}$	S_i	tion of X into k stable sets $S_1, S_2, \dots, S_k$ called color classes . Each color class S_i represents
voloshin-hypergraph_F01479	210	1.000	$ \left S_{\{i\}}\right \leq \alpha(\mathcal{H})$	$ S_i \leq \alpha(\mathcal{H})$	oring of $\mathcal{H}$ with $\chi(\mathcal{H})$ colors, $ S_i \leq \alpha(\mathcal{H})$.
voloshin-hypergraph_F01480	210	0.999	$\tau+1$	$\tau + 1$	with a new color. Since we obtain a proper coloring of $\mathcal{H}$ and use $\tau + 1$ colors, the chro-
voloshin-hypergraph_F01481	210	0.824	$\tau+1=n-\alpha+1$	$\tau + 1 = n - \alpha + 1$	matic number χ cannot be greater than $\tau + 1 = n - \alpha + 1$.
voloshin-hypergraph_F01482	210	1.000	$\tau(\mathcal{H})=2$	$\tau(\mathcal{H}) = 2$	can easily see, $\tau(\mathcal{H}) = 2$, respectively $\alpha(\mathcal{H}) = 3$, and $\chi(\mathcal{H}) = 2$. The coloring produces a
voloshin-hypergraph_F01483	210	1.000	$\chi(\mathcal{H})=2$	$\chi(\mathcal{H}) = 2$	can easily see, $\tau(\mathcal{H}) = 2$, respectively $\alpha(\mathcal{H}) = 3$, and $\chi(\mathcal{H}) = 2$. The coloring produces a
voloshin-hypergraph_F01484	210	1.000	$\alpha(\mathcal{H}) \chi(\mathcal{H}) \geq n$	$\alpha(\mathcal{H}) \chi(\mathcal{H}) \geq n$	inequality $\alpha(\mathcal{H}) \chi(\mathcal{H}) \geq n$ turns into $3 \cdot 2 \geq 5$, and $\chi(\mathcal{H}) \leq \tau(\mathcal{H}) + 1$ into $2 \leq 2 + 1$.
voloshin-hypergraph_F01485	210	1.000	$3 \cdot 2 \geq 5$	$3 \cdot 2 \geq 5$	inequality $\alpha(\mathcal{H}) \chi(\mathcal{H}) \geq n$ turns into $3 \cdot 2 \geq 5$, and $\chi(\mathcal{H}) \leq \tau(\mathcal{H}) + 1$ into $2 \leq 2 + 1$.
voloshin-hypergraph_F01486	210	1.000	$\chi(\mathcal{H}) \leq \tau(\mathcal{H})+1$	$\chi(\mathcal{H}) \leq \tau(\mathcal{H}) + 1$	inequality $\alpha(\mathcal{H}) \chi(\mathcal{H}) \geq n$ turns into $3 \cdot 2 \geq 5$, and $\chi(\mathcal{H}) \leq \tau(\mathcal{H}) + 1$ into $2 \leq 2 + 1$.
voloshin-hypergraph_F01487	210	1.000	$2 \leq 2+1$	$2 \leq 2 + 1$	inequality $\alpha(\mathcal{H}) \chi(\mathcal{H}) \geq n$ turns into $3 \cdot 2 \geq 5$, and $\chi(\mathcal{H}) \leq \tau(\mathcal{H}) + 1$ into $2 \leq 2 + 1$.

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voloshin-hypergraph_F01488	210	■ 1.000	$\gamma(\mathcal{H})$	$\gamma(\mathcal{H})$	differently. The strong chromatic number $\gamma(\mathcal{H})$ is the smallest λ for which there exists a
voloshin-hypergraph_F01489	210	■ 0.991	$\gamma(\mathcal{H}) \geq \chi(\mathcal{H})$	$\gamma(\mathcal{H}) \geq \chi(\mathcal{H})$	strong λ -coloring of $\mathcal{H}$. It follows that $\gamma(\mathcal{H}) \geq \chi(\mathcal{H})$ because every strong coloring is also
voloshin-hypergraph_F01490	211	■ 0.998	$S_{\{i\}}, i=1,2, \ldots, \lambda$	$S_i, i = 1, 2, \dots, \lambda$	λ stable sets $S_i, i = 1, 2, \dots, \lambda$, such that for every $D \in \mathcal{D}$, and for every i the following
voloshin-hypergraph_F01491	211	■ 1.000	$\lfloor r \rfloor$	$\lfloor r \rfloor$	where $\lfloor r \rfloor$ is the largest integer not greater than r , and $\lceil r \rceil$ is the smallest integer not smaller
voloshin-hypergraph_F01492	211	■ 1.000	$\lceil r \rceil$	$\lceil r \rceil$	where $\lfloor r \rfloor$ is the largest integer not greater than r , and $\lceil r \rceil$ is the smallest integer not smaller
voloshin-hypergraph_F01493	211	■ 1.000	$i=1,2, \ldots, \lambda$	$i = 1, 2, \dots, \lambda$	$i = 1, 2, \dots, \lambda$, such that each $D \in \mathcal{D}$ has
voloshin-hypergraph_F01494	211	■ 1.000	$\lambda \geq \max_{D \in \mathcal{D}} D $	$\lambda \geq \max_{D \in \mathcal{D}} D $	then every S_i forms a transversal of $\mathcal{H}$. If $\lambda \geq \max_{D \in \mathcal{D}} D $, then a good λ -coloring is a
voloshin-hypergraph_F01495	211	■ 1.000	$(S_{\{1\}}, S_{\{2\}}, \ldots, S_{\{\lambda\}})$	$(S_1, S_2, \dots, S_\lambda)$	Uniform colorings. For a hypergraph $\mathcal{H}$ of order n , a proper λ -coloring $(S_1, S_2, \dots, S_\lambda)$
voloshin-hypergraph_F01496	211	■ 1.000	$a_{\{j\}}, b_{\{j\}}$	a_j, b_j	corresponding integers a_j, b_j such that
voloshin-hypergraph_F01497	211	■ 1.000	$a_{\{j\}}=0$	$a_j = 0$	Notice that each weak coloring is an I-regular coloring if $a_j = 0$ and $b_j = \max\{1, D_j - 1\}$;
voloshin-hypergraph_F01498	211	■ 1.000	$b_{\{j\}}=\max \left\{1, \left D_{\{j\}}\right -1\right\}$	$b_j = \max \{1, D_j - 1\}$	Notice that each weak coloring is an I-regular coloring if $a_j = 0$ and $b_j = \max\{1, D_j - 1\}$;
voloshin-hypergraph_F01499	211	■ 1.000	$b_{\{j\}}=1$	$b_j = 1$	every strong coloring is an I-regular coloring with $a_j = 0$ and $b_j = 1$; an arbitrary equitable
voloshin-hypergraph_F01500	211	■ 1.000	$a_{\{j\}}=\left\lfloor\frac{\left D_{\{j\}}\right }{\lambda}\right\rfloor$	$a_j = \left\lfloor \frac{ D_j }{\lambda} \right\rfloor$	coloring is an I-regular coloring with $a_j = \left\lfloor \frac{ D_j }{\lambda} \right\rfloor$ and $b_j = \left\lceil \frac{ D_j }{\lambda} \right\rceil$.
voloshin-hypergraph_F01501	211	■ 1.000	$b_{\{j\}}=\left\lceil\frac{\left D_{\{j\}}\right }{\lambda}\right\rceil$	$b_j = \left\lceil \frac{ D_j }{\lambda} \right\rceil$	coloring is an I-regular coloring with $a_j = \left\lfloor \frac{ D_j }{\lambda} \right\rfloor$ and $b_j = \left\lceil \frac{ D_j }{\lambda} \right\rceil$.
voloshin-hypergraph_F01502	211	■ 0.999	$b_{\{j\}}=\left D_{\{j\}}\right $	$b_j = D_j $	$b_j = D_j $. The last is equivalent to having no constraint on the coloring of D_j , i.e. it is
voloshin-hypergraph_F01503	211	■ 0.998	$\left S_{\{i\}} \cap D_{\{j\}}\right \geq 2$	$ S_i \cap D_j \geq 2$	that $ S_i \cap D_j \geq 2$. The point of this remark is that all graph generalizations of colorings
voloshin-hypergraph_F01504	212	■ 1.000	$T=$	$T =$	Proof. We use Algorithm 8.3.1 that computes the cyclomatic number of $\mathcal{H}$. Let $T =$
voloshin-hypergraph_F01505	212	■ 0.982	$\wedge(\mathcal{H})+2$	$\Lambda(\mathcal{H}) + 2$	proper coloring of $\mathcal{H}$ with $\Lambda(\mathcal{H}) + 2$ colors.
voloshin-hypergraph_F01506	212	■ 0.881	h, k, l	h, k, l	Theorem 10.1.2 (Erdős, Hajnal, 1966) For any natural numbers h, k, l , all ≥ 2 , there ex-
voloshin-hypergraph_F01507	212	■ 0.972	$\chi(\mathcal{H})=k$	$\chi(\mathcal{H}) = k$	ists an h -uniform hypergraph $\mathcal{H} = (X, \mathcal{D})$ such that $\chi(\mathcal{H}) = k$ and $\mathcal{H}$ contains no cycles
voloshin-hypergraph_F01508	212	■ 1.000	$<l$	$< l$	of length $< l$.
voloshin-hypergraph_F01509	212	■ 1.000	$\chi(\mathcal{H})=k, k \geq 3$	$\chi(\mathcal{H}) = k, k \geq 3$	A hypergraph $\mathcal{H}$ is edge-critical if it contains no isolated vertices, $\chi(\mathcal{H}) = k, k \geq 3$,
voloshin-hypergraph_F01510	212	■ 0.998	$\chi(\mathcal{H}^{\prime})=k-1$	$\chi(\mathcal{H}') = k - 1$	and weak deletion of any edge results in a hypergraph $\mathcal{H}'$ with $\chi(\mathcal{H}') = k - 1$.
voloshin-hypergraph_F01511	212	■ 1.000	$\chi(\mathcal{H}) \geq 3$	$\chi(\mathcal{H}) \geq 3$	Lemma 10.1.1 (Zykov, 1974) Any hypergraph $\mathcal{H}$ with $\chi(\mathcal{H}) \geq 3$ can be transformed into
voloshin-hypergraph_F01512	212	■ 1.000	$\chi(\mathcal{H})=3$	$\chi(\mathcal{H}) = 3$	Theorem 10.1.3 (Zykov, 1974) In each edge-critical hypergraph $\mathcal{H}$ with $\chi(\mathcal{H}) = 3$ there
voloshin-hypergraph_F01513	212	■ 0.727	$\chi(\mathcal{H})>k(\chi(\mathcal{H}) \leq k)$	$\chi(\mathcal{H}) > k(\chi(\mathcal{H}) \leq k)$	which an r -uniform hypergraph on n vertices can have if $\chi(\mathcal{H}) > k(\chi(\mathcal{H}) \leq k)$; these are

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voloshin-hypergraph_F01514	212	■ 0.999	$\alpha(\mathcal{H}), \tau(\mathcal{H})$	$\alpha(\mathcal{H}), \tau(\mathcal{H})$	2. For hypergraph $\mathcal{H}$ in Figure 10.2, determine $\alpha(\mathcal{H}), \tau(\mathcal{H})$ and verify the inequalities
voloshin-hypergraph_F01515	213	■ 1.000	$m(x, \mathcal{H})$	$m(x, \mathcal{H})$	Definition 10.2.1 The mono-degree $m(x, \mathcal{H})$ of a vertex $x \in X$ in a hypergraph $\mathcal{H} = (X, \mathcal{D})$
voloshin-hypergraph_F01516	213	■ 1.000	$\mathcal{D}_1(x) \subseteq \mathcal{D}(x)$	$\mathcal{D}_1(x) \subseteq \mathcal{D}(x)$	is the maximum cardinality of a subfamily $\mathcal{D}_1(x) \subseteq \mathcal{D}(x)$ such that:
voloshin-hypergraph_F01517	214	■ 1.000	$M(\mathcal{H})$	$M(\mathcal{H})$	If applied to graphs, $M(\mathcal{H})$ equals the Szekeres-Wilf number of a graph, see Section
voloshin-hypergraph_F01518	214	■ 0.585	ω	ω	3.3. Recall that ω denotes the maximum cardinality of a clique, and the Szekeres-Wilf
voloshin-hypergraph_F01519	214	■ 1.000	$c(x)$	$c(x)$	looks as follows. We use the notation $c(x)$ for the color of a vertex $x \in X$ and the vector
voloshin-hypergraph_F01520	214	■ 1.000	$c = (\text{c}\left(\text{c}\left(x_1\right), \text{c}\left(x_2\right), \ldots, \text{c}\left(x_n\right)\right)\right)$	$c = (c(x_1), c(x_2), \dots, c(x_n))$	$\mathfrak{c} = (c(x_1), c(x_2), \dots, c(x_n))$ for a coloring of $\mathcal{H}$; $c(x) = 0$ means that x is not colored.
voloshin-hypergraph_F01521	214	■ 1.000	$\mathcal{H}; c(x) = 0$	$\mathcal{H}; c(x) = 0$	$c = (c(x_1), c(x_2), \dots, c(x_n))$ for a coloring of $\mathcal{H}$; $c(x) = 0$ means that x is not colored.
voloshin-hypergraph_F01522	214	■ 1.000	$\mathcal{H} = (X, \mathcal{D}), X = \{1, 2, \ldots, n\}$	$\mathcal{H} = (X, \mathcal{D}), X = \{1, 2, \dots, n\}$	<i>INPUT:</i> An arbitrary hypergraph $\mathcal{H} = (X, \mathcal{D}), X = \{1, 2, \dots, n\}$.
voloshin-hypergraph_F01523	214	■ 1.000	$c = (c(1), c(2), \ldots, c(n))$	$c = (c(1), c(2), \dots, c(n))$	<i>OUTPUT:</i> A proper coloring $\mathfrak{c} = (c(1), c(2), \dots, c(n))$ of $\mathcal{H}$.
voloshin-hypergraph_F01524	214	■ 0.994	$C = (0, 0, \ldots, 0), i = n, \mathcal{H}_n = \mathcal{H}$	$C = (0, 0, \dots, 0), i = n, \mathcal{H}_n = \mathcal{H}$	1. Set $\mathfrak{C} = (0, 0, \dots, 0), i = n, \mathcal{H}_n = \mathcal{H}$. Find a vertex of minimum mono-degree in $\mathcal{H}_n$
voloshin-hypergraph_F01525	214	■ 0.994	$\mathcal{H}_n$	$\mathcal{H}_n$	1. Set $C = (0, 0, \dots, 0), i = n, \mathcal{H}_n = \mathcal{H}$. Find a vertex of minimum mono-degree in $\mathcal{H}_n$
voloshin-hypergraph_F01526	214	■ 0.998	$i := i - 1$	$i := i - 1$	2. Put $\mathfrak{i} := i - 1$; if $i = 0$, then go to step 5.
voloshin-hypergraph_F01527	214	■ 1.000	$\mathcal{H}_i = \mathcal{H}_{i+1} - x_{i+1}$	$\mathcal{H}_i = \mathcal{H}_{i+1} - x_{i+1}$	3. Strongly delete the vertex x_{i+1} and form an induced subhypergraph $\mathcal{H}_i = \mathcal{H}_{i+1} - x_{i+1}$.
voloshin-hypergraph_F01528	214	■ 1.000	$\mathcal{H}_i$	$\mathcal{H}_i$	4. Find a vertex of minimum mono-degree in $\mathcal{H}_i$ and label it x_i ; go to step 2.
voloshin-hypergraph_F01529	214	■ 1.000	$\text{c}\left(x_1\right)=1, i=1$	$c(x_1) = 1, i = 1$	5. Color x_1 with the first color: $c(x_1) = 1, i = 1$.
voloshin-hypergraph_F01530	214	■ 1.000	$i = n + 1$	$i = n + 1$	6. Put $i := i + 1$; if $\mathfrak{i} = n + 1$, then go to step 8.
voloshin-hypergraph_F01531	214	■ 0.992	$\{1, 2, \ldots, n\}$	$\{1, 2, \dots, n\}$	7. Color x_i in $\mathcal{H}_i$ with the smallest suitable color from $\mathfrak{i}, 2, \dots, n$; go to step 6.
voloshin-hypergraph_F01532	215	■ 1.000	$t \leq M(\mathcal{H})$	$t \leq M(\mathcal{H})$	the ordering generated by steps 1-4. It is clear that $t \leq M(\mathcal{H})$. We must show that $t \geq M(\mathcal{H})$
voloshin-hypergraph_F01533	215	■ 1.000	$t \geq M(\mathcal{H})$	$t \geq M(\mathcal{H})$	the ordering generated by steps 1-4. It is clear that $t \leq M(\mathcal{H})$. We must show that $t \geq M(\mathcal{H})$
voloshin-hypergraph_F01534	215	■ 0.973	$y \in Y$	$y \in Y$	also holds. There is an induced subhypergraph $\mathcal{H}_Y$ and a vertex $y \in Y$ such that
voloshin-hypergraph_F01535	215	■ 1.000	$\mathcal{H}_k$	$\mathcal{H}_k$	subhypergraph of $\mathcal{H}_k$. Therefore
voloshin-hypergraph_F01536	215	■ 1.000	$t = M(\mathcal{H})$	$t = M(\mathcal{H})$	Consequently, $t = M(\mathcal{H})$.
voloshin-hypergraph_F01537	215	■ 1.000	$1, 2, \ldots, t$	$1, 2, \dots, t$	than $M(\mathcal{H})$. Indeed, if we cannot use the colors $\mathfrak{i}, 2, \dots, t$ it means that we have the edges,
voloshin-hypergraph_F01538	215	■ 0.972	$D_1, D_2, \ldots, D_t$	$D_1, D_2, \dots, D_t$	say $\mathfrak{D}_1, \mathfrak{D}_2, \dots, \mathfrak{D}_t$ in $\mathcal{H}_i$, which are mono-colored (except the vertex x_i) with the respective
voloshin-hypergraph_F01539	215	■ 1.000	$\mathcal{H}_5 = \mathcal{H}$	$\mathcal{H}_5 = \mathcal{H}$	At the very beginning, $\mathcal{H}_5 = \mathcal{H}$ and $X = \{1, 2, 3, 4, 5\}$. In $\mathcal{H}$, the mono-degrees of vertices
voloshin-hypergraph_F01540	215	■ 1.000	$X = \{1, 2, 3, 4, 5\}$	$X = \{1, 2, 3, 4, 5\}$	At the very beginning, $\mathcal{H}_5 = \mathcal{H}$ and $\mathfrak{X} = \{1, 2, 3, 4, 5\}$. In $\mathcal{H}$, the mono-degrees of vertices

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voloshin-hypergraph_F01541	216	■ 1.000	$\min \{2,2,2,1,1\}=1$	$\min\{2,2,2,1,1\} = 1$	Since $\min\{2,2,2,1,1\} =$, either vertex 4 or vertex 5 can be deleted to obtain hypergraph
voloshin-hypergraph_F01542	216	■ 0.774	$\mathcal{H}_4=\mathcal{H}_5-5$	$\mathcal{H}_4 = \mathcal{H}_5 - 5$	$\mathcal{H}_4$. Choose vertex 5: $\mathcal{H}_4 = \mathcal{H}_5 - 5$. In $\mathcal{H}_4$, vertex 4 has the minimum mono-degree 1. Con-
voloshin-hypergraph_F01543	216	■ 1.000	$\mathcal{H}_5, \mathcal{H}_4, \mathcal{H}_3, \mathcal{H}_2, \mathcal{H}_1$	$\mathcal{H}_5, \mathcal{H}_4, \mathcal{H}_3, \mathcal{H}_2$	$\mathcal{H}_5, \mathcal{H}_4, \mathcal{H}_3, \mathcal{H}_2$ and $\mathcal{H}_1 = (\{2\}, \emptyset)$ which corresponds to the ordering of vertices
voloshin-hypergraph_F01544	216	■ 1.000	$\mathcal{H}_1=(\{2\}, \emptyset)$	$\mathcal{H}_1 = (\{2\}, \emptyset)$	$\mathcal{H}_5, \mathcal{H}_4, \mathcal{H}_3, \mathcal{H}_2$ and $\mathcal{H}_1 = (\{2\}, \emptyset)$ which corresponds to the ordering of vertices
voloshin-hypergraph_F01545	217	■ 1.000	$M(\mathcal{H})=2$	$M(\mathcal{H}) = 2$	rem 10.2.1, $M(\mathcal{H}) = 2$. At last, one can easily see that $\chi(\mathcal{H})=2 \leq M(\mathcal{H})+1=3$.
voloshin-hypergraph_F01546	217	■ 1.000	$\chi(\mathcal{H})=2 \leq M(\mathcal{H})+1=3$	$\chi(\mathcal{H}) = 2 \leq M(\mathcal{H}) + 1 = 3$	rem 10.2.1, $M(\mathcal{H}) = 2$. At last, one can easily see that $\chi(\mathcal{H})=2 \leq M(\mathcal{H})+1=3$.
voloshin-hypergraph_F01547	218	■ 1.000	$X=\left\{x_1, x_2, \ldots, x_n\right\}, n \geq 1$	$X = \{x_1, x_2, \dots, x_n\}, n \geq 1$	Let $X = \{x_1, x_2, \dots, x_n\}, n \geq 1$, be a finite set, and let $\mathcal{C} = \{C_1, C_2, \dots, C_l\}$ and $\mathcal{D} =$
voloshin-hypergraph_F01548	218	■ 1.000	$\mathcal{C}=\left\{C_1, C_2, \ldots, C_l\right\}$	$\mathcal{C} = \{C_1, C_2, \dots, C_l\}$	Let $X = \{x_1, x_2, \dots, x_n\}, n \geq 1$, be a finite set, and let $\mathcal{C} = \{C_1, C_2, \dots, C_l\}$ and $\mathcal{D} =$
voloshin-hypergraph_F01549	218	■ 1.000	$\mathcal{D} =$	$\mathcal{D} =$	Let $X = \{x_1, x_2, \dots, x_n\}, n \geq 1$, be a finite set, and let $\mathcal{C} = \{C_1, C_2, \dots, C_l\}$ and $\mathcal{D} =$
voloshin-hypergraph_F01550	218	■ 1.000	$\left\{D_1, D_2, \ldots, D_m\right\}$	$\{D_1, D_2, \dots, D_m\}$	$\{D_1, D_2, \dots, D_m\}$ be two families of subsets of X such that the <i>size of every member of</i>
voloshin-hypergraph_F01551	218	■ 1.000	$\mathcal{C} \cup \mathcal{D}$	$\mathcal{C} \cup \mathcal{D}$	$\mathcal{C} \cup \mathcal{D}$ is at least 2. Each of $\mathcal{C}, \mathcal{D}$ may be empty, and any subset of vertices may be
voloshin-hypergraph_F01552	218	■ 1.000	$\mathcal{C}, \mathcal{D}$	$\mathcal{C}, \mathcal{D}$	$\mathcal{C} \cup \mathcal{D}$ is at least 2. Each of $\mathcal{C}, \mathcal{D}$ may be empty, and any subset of vertices may be
voloshin-hypergraph_F01553	218	■ 0.224	$\mathcal{C} \neq \emptyset$	$\mathcal{C} \neq \emptyset$	contained in both $\mathcal{C}$ and $\mathcal{D}$. If $\mathcal{C} \neq \emptyset$, then denote $I = \{1, 2, \dots, l\}$, and if $\mathcal{D} \neq \emptyset$, then
voloshin-hypergraph_F01554	218	■ 0.224	$I=\{1,2, \ldots, l\}$	$I = \{1, 2, \dots, l\}$	contained in both $\mathcal{C}$ and $\mathcal{D}$. If $\mathcal{C} \neq \emptyset$, then denote $I = \{1, 2, \dots, l\}$, and if $\mathcal{D} \neq \emptyset$, then
voloshin-hypergraph_F01555	218	■ 0.224	$\mathcal{D} \neq \emptyset$	$\mathcal{D} \neq \emptyset$	contained in both $\mathcal{C}$ and $\mathcal{D}$. If $\mathcal{C} \neq \emptyset$, then denote $I = \{1, 2, \dots, l\}$, and if $\mathcal{D} \neq \emptyset$, then
voloshin-hypergraph_F01556	218	■ 1.000	$J=\{1,2, \ldots, m\}$	$J = \{1, 2, \dots, m\}$	denote $J = \{1, 2, \dots, m\}$.
voloshin-hypergraph_F01557	218	■ 0.993	$\mathcal{H}=(X, \mathcal{C}, \mathcal{D})$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$	Definition 10.3.1 A mixed hypergraph is a triple $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ where X is called the
voloshin-hypergraph_F01558	218	■ 1.000	$V(\mathcal{H}), \mathcal{C}$	$V(\mathcal{H}), \mathcal{C}$	vertex set, also denoted by $V(\mathcal{H})$, $\mathcal{C}$ is the family of subsets called $\mathcal{C}$ -edges, also denoted
voloshin-hypergraph_F01559	218	■ 1.000	$\mathcal{C}(\mathcal{H})$	$\mathcal{C}(\mathcal{H})$	by $\mathcal{C}(\mathcal{H})$, and $\mathcal{D}$ is the family of subsets called $\mathcal{D}$ -edges, also denoted by $\mathcal{D}(\mathcal{H})$.
voloshin-hypergraph_F01560	218	■ 1.000	$c(x), x \in X$	$c(x), x \in X$	In such a mapping, the number $c(x), x \in X$, is called the color of vertex x .
voloshin-hypergraph_F01561	218	■ 0.833	$\mathcal{H}=(X, \mathcal{C}, \mathcal{D})$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$	Definition 10.3.2 A proper λ -coloring of a mixed hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ is a mapping
voloshin-hypergraph_F01562	218	■ 1.000	$c: X \rightarrow \{1,2, \ldots, \lambda\}$	$c: X \rightarrow \{1, 2, \dots, \lambda\}$	$c: X \rightarrow \{1, 2, \dots, \lambda\}$ such that the following two conditions hold:
voloshin-hypergraph_F01563	218	■ 0.952	$\mathcal{C}=\emptyset$	$\mathcal{C} = \emptyset$	10.1. and 10.2. that correspond to the case $\mathcal{C} = \emptyset$. In contrast to graphs and hypergraphs,
voloshin-hypergraph_F01564	218	■ 1.000	$X=\{1,2\}, \mathcal{C}=\{\{1,2\}\}$	$X = \{1, 2\}, \mathcal{C} = \{\{1, 2\}\}$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ where $X = \{1, 2\}$, $\mathcal{C} = \{\{1, 2\}\}$, and $\mathcal{D} = \{\{1, 2\}\}$. Indeed, according to
voloshin-hypergraph_F01565	218	■ 1.000	$\mathcal{D}=\{\{1,2\}\}$	$\mathcal{D} = \{\{1, 2\}\}$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ where $X = \{1, 2\}$, $\mathcal{C} = \{\{1, 2\}\}$, and $\mathcal{D} = \{\{1, 2\}\}$. Indeed, according to
voloshin-hypergraph_F01566	218	■ 0.848	$\mathcal{C}=\{1,2\}$	$\mathcal{C} = \{1, 2\}$	Definition 10.3.2, $\mathcal{C}$ -edge $\mathcal{C} = \{1, 2\}$ requires the vertices 1 and 2 to be of a common color;
voloshin-hypergraph_F01567	218	■ 1.000	$\mathcal{D}=\{1,2\}$	$\mathcal{D} = \{1, 2\}$	however, $\mathcal{D}$ -edge $\mathcal{D} = \{1, 2\}$ requires the vertices 1 and 2 to be of different colors. This

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voloshin-hypergraph_F01568	219	■ 0.999	$c_{\{1\}}, c_{\{2\}}$	c_1, c_2	Two proper colorings c_1, c_2 of a mixed hypergraph $\mathcal{H}$ are said to be different , if there
voloshin-hypergraph_F01569	219	■ 1.000	$c_{\{1\}}(x) \neq c_{\{2\}}(x)$	$c_1(x) \neq c_2(x)$	exists at least one vertex $x \in X$ that changes the color, i.e. $c_1(x) \neq c_2(x)$. Hence any permu-
voloshin-hypergraph_F01570	219	■ 1.000	$P(\mathcal{H}, \lambda)$	$P(\mathcal{H}, \lambda)$	$P(\mathcal{H}, \lambda)$ be the number of different proper λ-colorings of a mixed hypergraph $\mathcal{H}$. As in
voloshin-hypergraph_F01571	219	■ 1.000	$1 \leq i \leq n$	$1 \leq i \leq n$	Strict i -colorings exist only for $1 \leq i \leq n$. Note that a proper $\chi(\mathcal{H})$ -coloring is necessar-
voloshin-hypergraph_F01572	219	■ 1.000	$\bar{\chi}(\mathcal{H})$	$\bar{\chi}(\mathcal{H})$	<i>pergraph $\mathcal{H}$ is called the upper chromatic number of $\mathcal{H}$, denoted by $\bar{\chi}(\mathcal{H})$.</i>
voloshin-hypergraph_F01573	219	■ 1.000	$c(y), y \in Y$	$c(y), y \in Y$	called monochromatic if all the colors $c(y), y \in Y$, have the same value, and it is called
voloshin-hypergraph_F01574	219	■ 0.991	$\dots, X_i$	$\dots, X_i$	$\dots, X_i$ is called a feasible partition of $\mathcal{H}$ if every C -edge has at least two vertices in a
voloshin-hypergraph_F01575	219	■ 1.000	$c = X_1 \cup X_2 \cup \dots \cup X_i$	$c = X_1 \cup X_2 \cup \dots \cup X_i$	Sometimes we will write the feasible partition or strict i -coloring c as $c = X_1 \cup X_2 \cup \dots \cup X_i$.
voloshin-hypergraph_F01576	219	■ 1.000	$r_{\{i\}}(\mathcal{H}) = r_i, 1 \leq i \leq n$	$r_i(\mathcal{H}) = r_i, 1 \leq i \leq n$	Let $r_i(\mathcal{H}) = r_i, 1 \leq i \leq n$, denote the number of feasible partitions of a mixed hypergraph
voloshin-hypergraph_F01577	220	■ 0.693	$\mathcal{X}$	$\mathcal{X}$	hypergraph $\mathcal{H}$, there are no feasible partitions into less than χ color classes, and there no
voloshin-hypergraph_F01578	220	■ 0.998	$\bar{\chi}$	$\bar{\chi}$	feasible partitions into more than $\bar{\chi}$ color classes. Therefore, chromatic spectrum generally
voloshin-hypergraph_F01579	220	■ 0.924	$r_i i!$	$r_i i!$	equals $r_i i!$. In general, if we have $\lambda \geq i$ colors, then to count proper λ -colorings we have
voloshin-hypergraph_F01580	220	■ 0.924	$\lambda \geq i$	$\lambda \geq i$	equals $r_i i!$. In general, if we have $\lambda \geq i$ colors, then to count proper λ -colorings we have
voloshin-hypergraph_F01581	220	■ 1.000	$\binom{\lambda}{i}$	$\binom{\lambda}{i}$	$\binom{\lambda}{i}$ ways to choose the subset of i colors. Consequently the number of proper λ -colorings
voloshin-hypergraph_F01582	220	■ 0.993	$\binom{\lambda}{i} r_i i! = r_i \lambda(\lambda - 1) \dots (\lambda - i + 1) =$	$\binom{\lambda}{i} r_i i! = r_i \lambda(\lambda - 1) \dots (\lambda - i + 1) =$	generated by all feasible partitions into i subsets is $\binom{\lambda}{i} r_i i! = r_i \lambda(\lambda - 1) \dots (\lambda - i + 1) =$
voloshin-hypergraph_F01583	220	■ 0.998	$r_i \lambda^{(i)}$	$r_i \lambda^{(i)}$	$r_i \lambda^{(i)}$. To obtain the number of all proper λ -coloring it is sufficient then to add all these
voloshin-hypergraph_F01584	220	■ 0.907	$\chi(\mathcal{H}) \leq i \leq \bar{\chi}(\mathcal{H})$	$\chi(\mathcal{H}) \leq i \leq \bar{\chi}(\mathcal{H})$	numbers for every $\chi(\mathcal{H}) \leq i \leq \bar{\chi}(\mathcal{H})$. Thus we arrive to the following conclusion:
voloshin-hypergraph_F01585	220	■ 0.968	$r_{\bar{\chi}}$	$r_{\bar{\chi}}$	<i>polynomial equals $\bar{\chi}$, and the leading coefficient equals $r_{\bar{\chi}}$.</i>
voloshin-hypergraph_F01586	220	■ 1.000	$\mathcal{H} = (X, \mathcal{C}, \emptyset)$	$\mathcal{H} = (X, \mathcal{C}, \emptyset)$	For a mixed hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$, the partial mixed subhypergraph $\mathcal{H} = (X, \mathcal{C}, \emptyset)$,
voloshin-hypergraph_F01587	220	■ 0.996	$\mathcal{H}_{\mathcal{C}} = (X, \mathcal{C})$	$\mathcal{H}_{\mathcal{C}} = (X, \mathcal{C})$	also denoted by $\mathcal{H}_{\mathcal{C}} = (X, \mathcal{C})$, is called a C-hypergraph , and the partial mixed subhyper-
voloshin-hypergraph_F01588	220	■ 1.000	$\mathcal{H} = (X, \emptyset, \mathcal{D})$	$\mathcal{H} = (X, \emptyset, \mathcal{D})$	graph $\mathcal{H} = (X, \emptyset, \mathcal{D})$, also denoted by $\mathcal{H}_{\mathcal{D}} = (X, \mathcal{D})$, is called a $\mathcal{D}$-hypergraph .
voloshin-hypergraph_F01589	220	■ 1.000	$\mathcal{H}_{\mathcal{D}} = (X, \mathcal{D})$	$\mathcal{H}_{\mathcal{D}} = (X, \mathcal{D})$	graph $\mathcal{H} = (X, \emptyset, \mathcal{D})$, also denoted by $\mathcal{H}_{\mathcal{D}} = (X, \mathcal{D})$, is called a $\mathcal{D}$-hypergraph .
voloshin-hypergraph_F01590	220	■ 0.699	$\mathcal{H}_{\mathcal{C}}$	$\mathcal{H}_{\mathcal{C}}$	and structural properties of $\mathcal{H}_{\mathcal{C}}$ and $\mathcal{H}_{\mathcal{D}}$ respectively. For example, a $\mathcal{D}$-graph is a $\mathcal{D}$ -
voloshin-hypergraph_F01591	220	■ 0.699	$\mathcal{H}_{\mathcal{D}}$	$\mathcal{H}_{\mathcal{D}}$	and structural properties of $\mathcal{H}_{\mathcal{C}}$ and $\mathcal{H}_{\mathcal{D}}$ respectively. For example, a $\mathcal{D}$-graph is a $\mathcal{D}$ -
voloshin-hypergraph_F01592	221	■ 0.942	$\bar{\chi}(\mathcal{H}_{\mathcal{C}})$	$\bar{\chi}(\mathcal{H}_{\mathcal{C}})$	$\bar{\chi}(\mathcal{H}_{\mathcal{C}})$. Notice that, for any mixed hypergraph $\mathcal{H}$ (colorable or not), the partial hypergraphs
voloshin-hypergraph_F01593	221	■ 0.997	$\mathcal{H}_{\mathcal{C}}$	$\mathcal{H}_{\mathcal{C}}$	$\mathcal{H}_{\mathcal{C}}$ and $\mathcal{H}_{\mathcal{D}}$ are both colorable, trivially with one color and $n(\mathcal{H})$ colors respectively. Obvi-
voloshin-hypergraph_F01594	221	■ 0.742	$\chi(\mathcal{H}_{\mathcal{C}}) = 1, r_1(\mathcal{H}_{\mathcal{C}}) = 1$	$\chi(\mathcal{H}_{\mathcal{C}}) = 1, r_1(\mathcal{H}_{\mathcal{C}}) = 1$	ously, $\chi(\mathcal{H}_{\mathcal{C}}) = 1, r_1(\mathcal{H}_{\mathcal{C}}) = 1$, and $\bar{\chi}(\mathcal{H}_{\mathcal{D}}) = n(\mathcal{H}), r_n(\mathcal{H}_{\mathcal{D}}) = 1$. Generally, for a colorable

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voloshin-hypergraph_F01595	221	■ 0.742	$\bar{\chi}(\mathcal{H}_{\mathcal{D}}) = n(\mathcal{H}), r_n(\mathcal{H}_{\mathcal{D}}) = 1$	$\bar{\chi}(\mathcal{H}_{\mathcal{D}}) = n(\mathcal{H}), r_n(\mathcal{H}_{\mathcal{D}}) = 1$	ously, $\chi(\mathcal{H}_{\mathcal{C}}) = 1$, $r_1(\mathcal{H}_{\mathcal{C}}) = 1$, and $\bar{\chi}(\mathcal{H}_{\mathcal{D}}) = n(\mathcal{H})$, $r_n(\mathcal{H}_{\mathcal{D}}) = 1$. Generally, for a colorable
voloshin-hypergraph_F01602	222	■ 0.808	$\bar{\chi}(\mathcal{H}_{\mathcal{C}}) < \chi(\mathcal{H}_{\mathcal{D}})$	$\bar{\chi}(\mathcal{H}_{\mathcal{C}}) < \chi(\mathcal{H}_{\mathcal{D}})$	One more direct observation is that any mixed hypergraph with $\bar{\chi}(\mathcal{H}_{\mathcal{C}}) < \chi(\mathcal{H}_{\mathcal{D}})$ is
voloshin-hypergraph_F01603	222	■ 1.000	$\{1, 3, 4\}$	$\{1, 3, 4\}$	then it is a bi-edge . For example, in Figure 10.6, the subset of vertices $\{1, 3, 4\}$ is a C -edge
voloshin-hypergraph_F01604	222	■ 1.000	$\mathcal{C} = \mathcal{D}$	$\mathcal{C} = \mathcal{D}$	bihypergraph if $\mathcal{C} = \mathcal{D}$.
voloshin-hypergraph_F01605	222	■ 0.806	$\mathcal{E} = \mathcal{C} \cup \mathcal{D}$	$\mathcal{E} = \mathcal{C} \cup \mathcal{D}$	Let us denote the underlying families of $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ by $\mathcal{E} = \mathcal{C} \cup \mathcal{D}$. We say that
voloshin-hypergraph_F01606	222	■ 0.994	$\mathcal{H}' = (X, \mathcal{E})$	$\mathcal{H}' = (X, \mathcal{E})$	$\mathcal{H}' = (X, \mathcal{E})$ is the underlying hypergraph of $\mathcal{H}$. Using underlying hypergraphs, many
voloshin-hypergraph_F01607	222	■ 0.907	$\mathcal{H}_{\mathcal{C}}(\mathcal{H}_{\mathcal{D}})$	$\mathcal{H}_{\mathcal{C}}(\mathcal{H}_{\mathcal{D}})$	if it is a path in $\mathcal{H}_{\mathcal{C}}(\mathcal{H}_{\mathcal{D}})$, respectively. $\mathcal{H}$ is an r-uniform mixed hypergraph if $\mathcal{H}'$ is an
voloshin-hypergraph_F01608	222	■ 1.000	$\mathcal{H}_Y = (Y, \mathcal{C}', \mathcal{D}')$	$\mathcal{H}_Y = (Y, \mathcal{C}', \mathcal{D}')$	For any subset $Y \subseteq X$, a mixed hypergraph $\mathcal{H}_Y = (Y, \mathcal{C}', \mathcal{D}')$ is the induced subhypergraph
voloshin-hypergraph_F01609	222	■ 1.000	$\mathcal{C}'$	$\mathcal{C}'$	that $\mathcal{C}'$ and $\mathcal{D}'$ consist of all those members of $\mathcal{C}$ and of $\mathcal{D}$, respectively, which are entirely
voloshin-hypergraph_F01610	222	■ 1.000	$R(\mathcal{H})$	$R(\mathcal{H})$	feasible partitions; estimate the chromatic spectrum $R(\mathcal{H})$.
voloshin-hypergraph_F01611	223	■ 1.000	$\mathcal{H} = (X, \mathcal{C})$	$\mathcal{H} = (X, \mathcal{C})$	Let $\mathcal{H} = (X, \mathcal{C}, \emptyset)$ be a C -hypergraph, abbreviated simply by $\mathcal{H} = (X, \mathcal{C})$ and let $\mathcal{C}(x)$
voloshin-hypergraph_F01612	223	■ 1.000	$\mathcal{C}(x)$	$\mathcal{C}(x)$	Let $\mathcal{H} = (X, \mathcal{C}, \emptyset)$ be a C -hypergraph, abbreviated simply by $\mathcal{H} = (X, \mathcal{C})$ and let $\mathcal{C}(x)$
voloshin-hypergraph_F01613	223	■ 0.639	$\mathcal{C}(x) \bigcap \mathcal{C}(y) \neq \emptyset$	$\mathcal{C}(x) \bigcap \mathcal{C}(y) \neq \emptyset$	only if $\mathcal{C}(x) \bigcap \mathcal{C}(y) \neq \emptyset$. Call the set $\mathcal{C}(x) \bigcap \mathcal{C}(y)$ a bistar of the vertex $x \in X$ with respect
voloshin-hypergraph_F01614	223	■ 0.639	$\mathcal{C}(x) \cap \mathcal{C}(y)$	$\mathcal{C}(x) \cap \mathcal{C}(y)$	only if $\mathcal{C}(x) \bigcap \mathcal{C}(y) \neq \emptyset$. Call the set $\mathcal{C}(x) \bigcap \mathcal{C}(y)$ a bistar of the vertex $x \in X$ with respect
voloshin-hypergraph_F01615	224	■ 1.000	$o(x, \mathcal{H}) = 0$	$o(x, \mathcal{H}) = 0$	Thus $o(x, \mathcal{H}) = 0$ implies that there is another vertex $y \in X$ which is contained in all
voloshin-hypergraph_F01616	224	■ 0.997	$ \mathcal{C}(x) - 1$	$ \mathcal{C}(x) - 1$	vertex is to its neighbors. The upper limit here is $ \mathcal{C}(x) - 1$ if x is the center of a monostar.
voloshin-hypergraph_F01617	224	■ 1.000	$O(\mathcal{H})$	$O(\mathcal{H})$	We will see that $O(\mathcal{H})$ plays the role which is similar to the role of $M(\mathcal{H})$.
voloshin-hypergraph_F01618	224	■ 0.988	$M \subseteq X$	$M \subseteq X$	Definition 10.4.2 In a coloring of a mixed hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$, a set $M \subseteq X$ of ver-
voloshin-hypergraph_F01619	224	■ 1.000	$x \in M$	$x \in M$	tices is called a monochromatic component of a vertex $x \in M$, and is denoted by $MC(x)$,
voloshin-hypergraph_F01620	224	■ 1.000	$MC(x)$	$MC(x)$	tices is called a monochromatic component of a vertex $x \in M$, and is denoted by $MC(x)$,
voloshin-hypergraph_F01621	224	■ 0.783	M	M	if all the vertices of M have the same color as x and are reachable from x by C -paths.
voloshin-hypergraph_F01622	224	■ 1.000	$\mathcal{H} = (X, \mathcal{C}), X = n$	$\mathcal{H} = (X, \mathcal{C}), X = n$	INPUT: An arbitrary C -hypergraph $\mathcal{H} = (X, \mathcal{C}), X = n$.
voloshin-hypergraph_F01623	224	■ 1.000	$i = n, \mathcal{H}_n = \mathcal{H}$	$i = n, \mathcal{H}_n = \mathcal{H}$	1. Set $i = n$, $\mathcal{H}_n = \mathcal{H}$; find a vertex of minimum originality and label it x_n .
voloshin-hypergraph_F01624	225	■ 0.706	$U = \{1\}$	$U = \{1\}$	5. Set the list of used colors $U = \{1\}$, color the vertex x_1 with the color 1; set $i =$
voloshin-hypergraph_F01625	225	■ 0.706	$i =$	$i =$	5. Set the list of used colors $U = \{1\}$, color the vertex x_1 with the color 1; set $i =$
voloshin-hypergraph_F01626	225	■ 0.999	$i = n$	$i = n$	6. Put $i := i + 1$; if $i = n$, then go to step 10; color the vertex x_i of $\mathcal{H}_i$ with color new;
voloshin-hypergraph_F01627	225	■ 1.000	$U := U \cup \{$	$U := U \cup \{$	put $U := U \cup \{new\}$, $new := new + 1$.

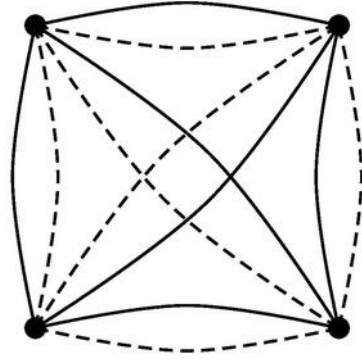
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voloshin-hypergraph_F01628	225	■ 1.000	$\backslash\}$	$\}$	$put\ U := U \cup \{new\},\ new := new + 1.$
voloshin-hypergraph_F01629	225	■ 1.000	$:=$	$:=$	$put\ U := U \cup \{new\},\ new := new + 1.$
voloshin-hypergraph_F01630	225	■ 0.758	$U:=U-\backslash\{$	$U := U - \{$	$the\ color\ of\ y, put\ U := U - \{new\},\ new := new - 1, and\ go\ to\ step\ 7.$
voloshin-hypergraph_F01631	225	■ 0.971	$M\ C(y)$	$MC(y)$	9. Re-color all the vertices from the monochromatic component $MC(y)$ with the color
voloshin-hypergraph_F01632	225	■ 1.000	U	U	10. Renumber the colors of U in increasing order; end.
voloshin-hypergraph_F01633	225	■ 1.000	$ X =n, \mathcal{C} =k$	$ X = n, \mathcal{C} = k$	Complexity. Let us suppose that the hypergraph $\mathcal{H} = (X, C)$, with $ X = n, C = k$, is
voloshin-hypergraph_F01634	225	■ 1.000	$n\ \times\ k$	$n \times k$	represented by its incidence matrix $I(\mathcal{H})$ of the size $n \times k$. Since finding the originality of a
voloshin-hypergraph_F01635	225	■ 1.000	$O(n\ k)$	$O(nk)$	vertex requires $O(nk)$ steps, finding the minimum originality requires $O(n^2k)$ steps. Hence,
voloshin-hypergraph_F01636	225	■ 1.000	$O\left(n^{\{2\}}\ k\right)$	$O\left(n^2k\right)$	vertex requires $O(nk)$ steps, finding the minimum originality requires $O(n^2k)$ steps. Hence,
voloshin-hypergraph_F01637	225	■ 1.000	$O\left(n^{\{3\}}\ k\right)$	$O\left(n^3k\right)$	steps 1-4 may be implemented, in the worst case, in $O(n^3k)$ steps.
voloshin-hypergraph_F01638	225	■ 1.000	$X=$	$X =$	Example 10.4.1 Consider the C -hypergraph $\mathcal{H} = (X, C)$ (see Figure 10.9) such that $X =$
voloshin-hypergraph_F01639	225	■ 0.852	$\backslash\{1,2,3,4,5\},\ C=\backslash\left\{C_{\{1\}},\ C_{\{2\}},\ C_{\{3\}},\ C_{\{4\}},\ C_{\{5\}}\right\}\backslash\right\},\ C_{\{1\}}=\backslash\{1,2,3\}\backslash,\ C_{\{2\}}=\backslash\{2,3,4\}\backslash,\ C_{\{3\}}=\backslash\{3,4,5\}\backslash\}$	$\{1,2,3,4,5\}, C = \{C_1, C_2, C_3, C_4, C_5\}, C_1 = \{1,2,3\}, C_2 = \{2,3,4\}, C_3 = \{3,4,5\}$	$\{1, \underline{2, 3, 4, 5}\},\ C = \{C_1, C_2, C_3, C_4, C_5\}, C_1 = \{1,2,3\},\ C_2 = \{2, 3, 4\}, C_3 = \underline{\{3,4,5\}},$
voloshin-hypergraph_F01640	225	■ 1.000	$C_{\{4\}}=\backslash\{4,5,1\}$	$C_4 = \{4,5,1\}$	$C_4 = \{4,5,1\}$, and $C_5 = \{5,1,2\}$. Set $\mathcal{H}_5 = \mathcal{H}$. All the vertices have the same originality
voloshin-hypergraph_F01641	225	■ 1.000	$C_{\{5\}}=\backslash\{5,1,2\}$	$C_5 = \{5,1,2\}$	$C_4 = \{4,5,1\}$, and $C_5 = \{5,1,2\}$. Set $\mathcal{H}_5 = \mathcal{H}$. All the vertices have the same originality
voloshin-hypergraph_F01642	225	■ 0.669	$x_{\{5\}}=1$	$x_5 = 1$	1, therefore, let us start with the first vertex: $x_5 = 1$.
voloshin-hypergraph_F01643	225	■ 0.967	$\mathcal{H}_{\{4\}}=\backslash\left(X_{\{4\}},\ \mathcal{C}_{\{4\}}\right)\backslash\right)$	$\mathcal{H}_4 = (X_4, C_4)$	Form the induced C -hypergraph $\mathcal{H}_4 = (X_4, C_4)$ with $X_4 = \{2,3,4,5\}$, $C_4 = \{C_2, C_3\}$.
voloshin-hypergraph_F01644	225	■ 0.967	$X_{\{4\}}=\backslash\{2,3,4,5\}\backslash,\ \mathcal{C}_{\{4\}}=\backslash\left\{C_{\{2\}},\ C_{\{3\}}\right\}\backslash\right)\backslash\}$	$X_4 = \{2,3,4,5\}, C_4 = \{C_2, C_3\}$	Form the induced C -hypergraph $\mathcal{H}_4 = (X_4, C_4)$ with $X_4 = \{2,3,4,5\}$, $C_4 = \{C_2, C_3\}$.
voloshin-hypergraph_F01645	225	■ 1.000	$x_{\{4\}}=2$	$x_4 = 2$	The first vertex with minimum originality is $x_4 = 2$.
voloshin-hypergraph_F01646	225	■ 1.000	$\mathcal{H}_{\{3\}}=\backslash\left(X_{\{3\}},\ \mathcal{C}_{\{3\}}\right)\backslash\right)$	$\mathcal{H}_3 = (X_3, C_3)$	Form the induced C -hypergraph $\mathcal{H}_3 = (X_3, C_3)$ with $X_3 = \{3,4,5\}$, $C_3 = \{C_3\}$. The
voloshin-hypergraph_F01647	225	■ 1.000	$X_{\{3\}}=\backslash\{3,4,5\}\backslash,\ \mathcal{C}_{\{3\}}=\backslash\left\{C_{\{3\}}\right\}\backslash\right)\backslash\}$	$X_3 = \{3,4,5\}, C_3 = \{C_3\}$	Form the induced C -hypergraph $\mathcal{H}_3 = (X_3, C_3)$ with $X_3 = \{3,4,5\}$, $C_3 = \{C_3\}$. The
voloshin-hypergraph_F01648	225	■ 1.000	$x_{\{3\}}=3$	$x_3 = 3$	first vertex with minimum originality is $x_3 = 3$.
voloshin-hypergraph_F01649	225	■ 1.000	$\mathcal{H}_{\{2\}}=\backslash\left(X_{\{2\}},\ \mathcal{C}_{\{2\}}\right)\backslash\right)$	$\mathcal{H}_2 = (X_2, C_2)$	Form the induced C -hypergraph $\mathcal{H}_2 = (X_2, C_2)$ with $X_2 = \{4,5\}$, $C_2 = \{\emptyset\}$. The first
voloshin-hypergraph_F01650	225	■ 1.000	$X_{\{2\}}=\backslash\{4,5\}\backslash,\ \mathcal{C}_{\{2\}}=\backslash\{\emptyset\}\backslash\}$	$X_2 = \{4,5\}, C_2 = \{\emptyset\}$	Form the induced C -hypergraph $\mathcal{H}_2 = (X_2, C_2)$ with $X_2 = \{4,5\}$, $C_2 = \{\emptyset\}$. The first
voloshin-hypergraph_F01651	225	■ 1.000	$x_{\{2\}}=4$	$x_2 = 4$	vertex with minimum originality is $x_2 = 4$.
voloshin-hypergraph_F01652	226	■ 0.984	$\mathcal{H}_{\{1\}}=\backslash\left(X_{\{1\}},\ \mathcal{C}_{\{1\}}\right)\backslash\right)$	$\mathcal{H}_1 = (X_1, C_1)$	Form the induced C -hypergraph $\mathcal{H}_1 = (X_1, C_1)$ with $X_1 = \{5\}$, $C_1 = \{\emptyset\}$. The last
voloshin-hypergraph_F01653	226	■ 0.984	$X_{\{1\}}=\backslash\{5\}\backslash,\ \mathcal{C}_{\{1\}}=\backslash\{\emptyset\}\backslash\}$	$X_1 = \{5\}, C_1 = \{\emptyset\}$	Form the induced C -hypergraph $\mathcal{H}_1 = (X_1, C_1)$ with $X_1 = \{5\}$, $C_1 = \{\emptyset\}$. The last

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voloshin-hypergraph_F01654	226	■ 1.000	$x_{\{1\}}=5$	$x_1 = 5$	vertex is $x_1 = 5$.
voloshin-hypergraph_F01655	226	■ 0.976	$c(i)$	$c(i)$	These are the results of steps 1-4. As usual, let $c(i)$ be the color of vertex i , $c(i) = 0$
voloshin-hypergraph_F01656	226	■ 0.976	$i, c(i)=0$	$i, c(i) = 0$	These are the results of steps 1-4. As usual, let $c(i)$ be the color of vertex i , $c(i) = 0$
voloshin-hypergraph_F01657	226	■ 1.000	$i=1, \ldots, 5$	$i = 1, \dots, 5$	means the vertex is not colored, $\overline{t} = 1, \dots, 5$.
voloshin-hypergraph_F01658	226	■ 1.000	$c=(c(1), c(2), c(3), c(4), c(5))=(0,0,0,0,0)$	$c = (c(1), c(2), c(3), c(4), c(5)) = (0, 0, 0, 0, 0)$	Set $c \neq (c(1), c(2), c(3), c(4), c(5)) = (0, 0, 0, 0, 0)$. Start coloring. Step 5: $c =$
voloshin-hypergraph_F01659	226	■ 1.000	$c=$	$c =$	Set $c = (c(1), c(2), c(3), c(4), c(5)) = (0, 0, 0, 0, 0)$. Start coloring. Step 5: $c =$
voloshin-hypergraph_F01660	226	■ 0.921	$c=(0,0,0,2,1)$	$c = (0, 0, 0, 2, 1)$	$(0, 0, 0, 0, 1)$. Step 6: $\overline{c} = (0, 0, 0, 2, 1)$. Step 7: there are no polychromatic C -edges in
voloshin-hypergraph_F01661	226	■ 1.000	$c=(0,0,3,2,1)$	$c = (0, 0, 3, 2, 1)$	Step 6: $\overline{c} = (0, 0, 3, 2, 1)$. Step 7: C -edge C_3 is polychromatic.
voloshin-hypergraph_F01662	226	■ 0.991	$c=(0,0,2,2,1)$	$c = (0, 0, 2, 2, 1)$	Step 8: re-coloring: $\overline{c} = (0, 0, 2, 2, 1)$.
voloshin-hypergraph_F01663	226	■ 0.980	$c=(0,3,2,2,1)$	$c = (0, 3, 2, 2, 1)$	Step 6: $\overline{c} = (0, 3, 2, 2, 1)$.
voloshin-hypergraph_F01664	226	■ 0.964	$c=(4,3,2,2,1)$	$c = (4, 3, 2, 2, 1)$	Step 6: $\overline{c} = (4, 3, 2, 2, 1)$.
voloshin-hypergraph_F01665	226	■ 1.000	$C_{\{1\}}, C_{\{4\}}, C_{\{5\}}$	C_1, C_4, C_5	Step 7: the C -edges C_1, C_4, C_5 are each polychromatic in the C -hypergraph $\mathcal{H}_5$.
voloshin-hypergraph_F01666	226	■ 1.000	$\mathcal{H}_{\{5\}}$	$\mathcal{H}_5$	Step 7: the C -edges C_1, C_4, C_5 are each polychromatic in the C -hypergraph $\mathcal{H}_5$.
voloshin-hypergraph_F01667	226	■ 0.910	$c(5)=1: c=(1,3,2,2,1)$	$c(5) = 1 : c = (1, 3, 2, 2, 1)$	re-color it with the color $\overline{c}(5) = 1 : c = (1, 3, 2, 2, 1)$.
voloshin-hypergraph_F01668	226	■ 1.000	$M C(3)=\{3,4\}: c=(1,3,1,1,1)$	$MC(3) = \{3, 4\} : c = (1, 3, 1, 1, 1)$	Step 9: re-color the monochromatic component $MC(3)=\{3,4\}: c=(1,3,1,1,1)$.
voloshin-hypergraph_F01669	226	■ 0.948	$\mathcal{H}_{\{1\}}=\mathcal{H}$	$\mathcal{H}_1 = \mathcal{H}$	Step 7: $\mathcal{H}_1 = \mathcal{H}$ is colored properly.
voloshin-hypergraph_F01670	227	■ 1.000	$c=(1,2,1,1,1)$	$c = (1, 2, 1, 1, 1)$	Step 10: renumber the colors in increasing order: $\overline{c} = (1, 2, 1, 1, 1)$. End.
voloshin-hypergraph_F01671	227	■ 1.000	$C=(4,3,2,2,1)$	$C = (4, 3, 2, 2, 1)$	the coloring $\overline{C} = (4, 3, 2, 2, 1)$ is obtained, for vertex $x_5 = 1$ none from the available colors
voloshin-hypergraph_F01672	227	■ 0.982	$1,2,3,4,5 \ldots$	$1, 2, 3, 4, 5 \dots$	$\overline{1,2,3,4,5\dots}$ is appropriate. At this point we reach a deadlock , i.e. the situation when none
voloshin-hypergraph_F01673	227	■ 0.999	$t \leq 0(\mathcal{H})$	$t \leq O(\mathcal{H})$	order generated by steps 1-4. It is clear that $\overline{t} \leq O(\mathcal{H})$.
voloshin-hypergraph_F01674	227	■ 1.000	$t \leq 0(\mathcal{H})-1$	$t \leq O(\mathcal{H}) - 1$	Suppose that $t \leq O(\mathcal{H}) - 1$. Hence in an induced subhypergraph $\mathcal{H}' \subseteq \mathcal{H}$ there is a
voloshin-hypergraph_F01675	227	■ 1.000	$\geq t+1$	$\geq t + 1$	by the algorithm had originality $\geq t + 1$, a contradiction. Consequently, $t = O(\mathcal{H})$.
voloshin-hypergraph_F01676	227	■ 1.000	$t=0(\mathcal{H})$	$t = O(\mathcal{H})$	by the algorithm had originality $\geq t + 1$, a contradiction. Consequently, $t = O(\mathcal{H})$.
voloshin-hypergraph_F01677	227	■ 1.000	$0(\mathcal{H})+1$	$O(\mathcal{H}) + 1$	<i>does not exceed the value $O(\mathcal{H}) + 1$.</i>
voloshin-hypergraph_F01678	227	■ 1.000	$c(y)$	$c(y)$	neighbor y forming the largest bistar of x_i . If we re-color vertex x_i with $c(y)$, then we are
voloshin-hypergraph_F01679	227	■ 0.971	$b\left(x_{\{i\}}, \mathcal{H}_{\{i\}}\right) \mathcal{C}$	$b(x_i, \mathcal{H}_i) \mathcal{C}$	losing one color (<i>new</i>) and at the same time are properly coloring $b(x_i, \mathcal{H}_i)$ C -edges. Hence,

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voloshin-hypergraph_F01680	227	■ 0.990	$o\left(x_{\{i\}}, \mathcal{H}_{\{i\}}\right)=\left \mathcal{C}\left(x_{\{i\}}\right)-b\left(x_{\{i\}}, \mathcal{H}_{\{i\}}\right)\right \mathcal{C}$	$o\left(x_i, \mathcal{H}_i\right)=\left \mathcal{C}\left(x_i\right)\right -b\left(x_i, \mathcal{H}_i\right) \mathcal{C}$	in the worst case, there remain at most $o\left(x_i, \mathcal{H}_i\right)=\left \mathcal{C}\left(x_i\right)\right -b\left(x_i, \mathcal{H}_i\right)$ $\mathcal{C}$ -edges that are still
voloshin-hypergraph_F01681	227	■ 1.000	$o\left(x_{\{i\}}, \mathcal{H}_{\{i\}}\right)$	$o\left(x_i, \mathcal{H}_i\right)$	at most $o\left(x_i, \mathcal{H}_i\right)$ colors. Consequently, the total number of colors lost is not greater than
voloshin-hypergraph_F01682	227	■ 1.000	$o\left(x_{\{i\}}, \mathcal{H}_{\{i\}}\right)+1$	$o\left(x_i, \mathcal{H}_i\right)+1$	$o\left(x_i, \mathcal{H}_i\right)+1$. Since for any i , $1 \leq i \leq n$,
voloshin-hypergraph_F01683	227	■ 1.000	$i, 1 \leq i \leq n$	$i, 1 \leq i \leq n$	$o\left(x_i, \mathcal{H}_i\right)+1$. Since for any i , $1 \leq i \leq n$,
voloshin-hypergraph_F01684	227	■ 1.000	$O\left(\mathcal{H}\right)=0$	$O\left(\mathcal{H}\right)=0$	Corollary 10.4.1 <i>If $O\left(\mathcal{H}\right)=0$, then there exists an ordering of the vertex set X such that</i>
voloshin-hypergraph_F01685	227	■ 1.000	$\left U\right =p$	$\left U\right =p$	Corollary 10.4.2 <i>In Algorithm 10.4.1, if $\left U\right =p$, then $\bar{\chi}\left(\mathcal{H}\right) \geq p$.</i>
voloshin-hypergraph_F01686	227	■ 1.000	$\bar{\chi}\left(\mathcal{H}\right) \geq p$	$\bar{\chi}\left(\mathcal{H}\right) \geq p$	Corollary 10.4.2 <i>In Algorithm 10.4.1, if $\left U\right =p$, then $\bar{\chi}\left(\mathcal{H}\right) \geq p$.</i>
voloshin-hypergraph_F01687	228	■ 1.000	$n=2,3$	$n=2,3$	Proof. Induction on $\left X\right =n$. For $n=2,3$ the assertion is trivial. Assume it holds for all
voloshin-hypergraph_F01688	228	■ 1.000	$O\left(\mathcal{H}_Y\right)=0$	$O\left(\mathcal{H}_Y\right)=0$	$\mathcal{C}$ -edge of $\mathcal{H}$ has cardinality at least 2, $o\left(x, \mathcal{H}\right)=0$. From this and $O\left(\mathcal{H}_Y\right)=0$ for any $Y \subset X$
voloshin-hypergraph_F01689	228	■ 1.000	$Y \subset X$	$Y \subset X$	$\mathcal{C}$ -edge of $\mathcal{H}$ has cardinality at least 2, $o\left(x, \mathcal{H}\right)=0$. From this and $O\left(\mathcal{H}_Y\right)=0$ for any $Y \subset X$
voloshin-hypergraph_F01690	228	■ 0.902	$\overline{\chi}\left(\mathcal{H}\right)$	$\bar{\chi}\left(\mathcal{H}\right)$	4. For $\mathcal{C}$ -hypergraph $\mathcal{H}$ in Figure 10.10, find the upper chromatic number $\bar{\chi}\left(\mathcal{H}\right)$, the
voloshin-hypergraph_F01691	229	■ 0.934	$\mathcal{C} \neq \emptyset$	$\mathcal{C} \neq \emptyset$	$\left\{D_1, D_2, \ldots, D_m\right\}$; if $\mathcal{C} \neq \emptyset$, then $I=\left\{1,2, \ldots, l\right\}$, and if $\mathcal{D} \neq \emptyset$, then $J=\left\{1,2, \ldots, m\right\}$. An
voloshin-hypergraph_F01692	229	■ 0.938	$\left(X, \mathcal{C}, \mathcal{D}\right)$	$\left(X, \mathcal{C}, \mathcal{D}\right)$	$\left(X, \mathcal{C}, \mathcal{D}\right)$, we provide the following 5 rules.
voloshin-hypergraph_F01693	229	■ 1.000	$C_{\{i\}} \subseteq C_{\{j\}}$	$C_i \subseteq C_j$	2. If $C_i \subseteq C_j$, then $P\left(\mathcal{H}, \lambda\right)=P\left(\mathcal{H}-C_j, \lambda\right), R\left(\mathcal{H}\right)=R\left(\mathcal{H}-C_j\right), i, j \in I$; it means that
voloshin-hypergraph_F01694	229	■ 1.000	$P\left(\mathcal{H}, \lambda\right)=P\left(\mathcal{H}-C_{\{j\}}, \lambda\right), R\left(\mathcal{H}\right)=R\left(\mathcal{H}-C_{\{j\}}\right), i, j \in I$	$P\left(\mathcal{H}, \lambda\right)=P\left(\mathcal{H}-C_j, \lambda\right), R\left(\mathcal{H}\right)=R\left(\mathcal{H}-C_j\right), i, j \in I$	2. If $C_i \subseteq C_j$, then $P\left(\mathcal{H}, \lambda\right)=P\left(\mathcal{H}-C_j, \lambda\right), R\left(\mathcal{H}\right)=R\left(\mathcal{H}-C_j\right), i, j \in I$; it means that
voloshin-hypergraph_F01695	229	■ 1.000	$C_{\{j\}}$	C_j	including edge C_j can be weakly deleted because the respective coloring constraints
voloshin-hypergraph_F01696	229	■ 1.000	$D_{\{i\}} \subseteq D_{\{j\}}$	$D_i \subseteq D_j$	3. If $D_i \subseteq D_j$, then $P\left(\mathcal{H}, \lambda\right)=P\left(\mathcal{H}-D_j, \lambda\right), R\left(\mathcal{H}\right)=R\left(\mathcal{H}-D_j\right), i, j \in J$; it means that
voloshin-hypergraph_F01697	229	■ 1.000	$P\left(\mathcal{H}, \lambda\right)=P\left(\mathcal{H}-D_{\{j\}}, \lambda\right), R\left(\mathcal{H}\right)=R\left(\mathcal{H}-D_{\{j\}}\right), i, j \in J$	$P\left(\mathcal{H}, \lambda\right)=P\left(\mathcal{H}-D_j, \lambda\right), R\left(\mathcal{H}\right)=R\left(\mathcal{H}-D_j\right), i, j \in J$	3. If $D_i \subseteq D_j$, then $P\left(\mathcal{H}, \lambda\right)=P\left(\mathcal{H}-D_j, \lambda\right), R\left(\mathcal{H}\right)=R\left(\mathcal{H}-D_j\right), i, j \in J$; it means that
voloshin-hypergraph_F01698	230	■ 1.000	$\left\{x_{\{k\}}, x_{\{l\}}\right\} \notin \mathcal{D}$	$\left\{x_k, x_l\right\} \notin \mathcal{D}$	4. If $\left\{x_k, x_l\right\} \notin \mathcal{D}$ and $\left\{x_k, x_l\right\} \notin \mathcal{C}$, then
voloshin-hypergraph_F01699	230	■ 1.000	$\left\{x_{\{k\}}, x_{\{l\}}\right\} \notin \mathcal{C}$	$\left\{x_k, x_l\right\} \notin \mathcal{C}$	4. If $\left\{x_k, x_l\right\} \notin \mathcal{D}$ and $\left\{x_k, x_l\right\} \notin \mathcal{C}$, then
voloshin-hypergraph_F01700	230	■ 0.931	$x_{\{l\}}$	x_l	colorings of $\mathcal{H}_2$; in $\mathcal{H}_1$ vertices x_k and x_l have different colors, and in $\mathcal{H}_2$ vertices x_k
voloshin-hypergraph_F01701	230	■ 1.000	$C_{\{t\}}=\left\{x_{\{k\}}, x_{\{l\}}\right\}$	$C_t=\left\{x_k, x_l\right\}$	5. If $C_t=\left\{x_k, x_l\right\}$, for some $t \in I$ and $x_k, x_l \in X$, such that $C_t \neq D_s$ for any $s \in J$, then
voloshin-hypergraph_F01702	230	■ 1.000	$t \in I$	$t \in I$	5. If $C_t=\left\{x_k, x_l\right\}$, for some $t \in I$ and $x_k, x_l \in X$, such that $C_t \neq D_s$ for any $s \in J$, then
voloshin-hypergraph_F01703	230	■ 1.000	$x_{\{k\}}, x_{\{l\}} \in X$	$x_k, x_l \in X$	5. If $C_t=\left\{x_k, x_l\right\}$, for some $t \in I$ and $x_k, x_l \in X$, such that $C_t \neq D_s$ for any $s \in J$, then
voloshin-hypergraph_F01704	230	■ 1.000	$C_{\{t\}} \neq D_{\{s\}}$	$C_t \neq D_s$	5. If $C_t=\left\{x_k, x_l\right\}$, for some $t \in I$ and $x_k, x_l \in X$, such that $C_t \neq D_s$ for any $s \in J$, then

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voloshin-hypergraph_F01705	230	■ 1.000	$s \in J$	$s \in J$	5. If $C_t = \{x_k, x_l\}$, for some $t \in I$ and $x_k, x_l \in X$, such that $C_t \neq D_s$ for any $s \in J$, then
voloshin-hypergraph_F01706	230	■ 0.619	$\mathcal{H}_1 = \left(X_1, \mathcal{C}^1, \mathcal{D}^1 \right), X_1 = (X \setminus \{x_k, x_l\}) \cup \{y\},$ $\mathcal{D}_j^1 = (D_j \setminus \{x_k, x_l\}) \cup \{y\}$	$\mathcal{H}_1 = (X_1, \mathcal{C}^1, \mathcal{D}^1), X_1 = (X \setminus \{x_k, x_l\}) \cup \{y\},$ $\mathcal{D}_j^1 = (D_j \setminus \{x_k, x_l\}) \cup \{y\}$	$\mathcal{H}_1 = (X_1, \mathcal{C}^1, \mathcal{D}^1), X_1 = (X \setminus \{x_k, x_l\}) \cup \{y\},$ y is a new vertex;
voloshin-hypergraph_F01707	230	■ 1.000	$x_k \in D_j$	$x_k \in D_j$	if $x_k \in D_j$, or $x_l \in D_j, j \in J$, then $D_j^1 = (D_j \setminus \{x_k, x_l\}) \cup \{y\}$, otherwise $D_j^1 = D_j$;
voloshin-hypergraph_F01708	230	■ 1.000	$x_l \in D_j, j \in J$	$x_l \in D_j, j \in J$	if $x_k \in D_j$, or $x_l \in D_j, j \in J$, then $D_j^1 = (D_j \setminus \{x_k, x_l\}) \cup \{y\}$, otherwise $D_j^1 = D_j$;
voloshin-hypergraph_F01709	230	■ 1.000	$D_j^1 = (D_j \setminus \{x_k, x_l\}) \cup \{y\}$	$D_j^1 = (D_j \setminus \{x_k, x_l\}) \cup \{y\}$	if $x_k \in D_j$, or $x_l \in D_j, j \in J$, then $D_j^1 = (D_j \setminus \{x_k, x_l\}) \cup \{y\}$, otherwise $D_j^1 = D_j$;
voloshin-hypergraph_F01710	230	■ 1.000	$D_j^1 = D_j$	$D_j^1 = D_j$	if $x_k \in D_j$, or $x_l \in D_j, j \in J$, then $D_j^1 = (D_j \setminus \{x_k, x_l\}) \cup \{y\}$, otherwise $D_j^1 = D_j$;
voloshin-hypergraph_F01711	230	■ 1.000	$x_k \in C_i$	$x_k \in C_i$	if $x_k \in C_i$, or $x_l \in C_i, i \in I, i \neq t$, then $C_i^1 = (C_i \setminus \{x_k, x_l\}) \cup \{y\}$, otherwise $C_i^1 = C_i$;
voloshin-hypergraph_F01712	230	■ 1.000	$x_l \in C_i, i \in I, i \neq t$	$x_l \in C_i, i \in I, i \neq t$	if $x_k \in C_i$, or $x_l \in C_i, i \in I, i \neq t$, then $C_i^1 = (C_i \setminus \{x_k, x_l\}) \cup \{y\}$, otherwise $C_i^1 = C_i$;
voloshin-hypergraph_F01713	230	■ 1.000	$C_i^1 = (C_i \setminus \{x_k, x_l\}) \cup \{y\}$	$C_i^1 = (C_i \setminus \{x_k, x_l\}) \cup \{y\}$	if $x_k \in C_i$, or $x_l \in C_i, i \in I, i \neq t$, then $C_i^1 = (C_i \setminus \{x_k, x_l\}) \cup \{y\}$, otherwise $C_i^1 = C_i$;
voloshin-hypergraph_F01714	230	■ 1.000	$C_i^1 = C_i$	$C_i^1 = C_i$	if $x_k \in C_i$, or $x_l \in C_i, i \in I, i \neq t$, then $C_i^1 = (C_i \setminus \{x_k, x_l\}) \cup \{y\}$, otherwise $C_i^1 = C_i$;
voloshin-hypergraph_F01715	230	■ 1.000	$n(\mathcal{H}) = n(\mathcal{H}_1) + 1$	$n(\mathcal{H}) = n(\mathcal{H}_1) + 1$	Remark. In 5., $n(\mathcal{H}) = n(\mathcal{H}_1) + 1$ and the equality $R(\mathcal{H}) = R(\mathcal{H}_1)$ means that
voloshin-hypergraph_F01716	230	■ 1.000	$R(\mathcal{H}) = R(\mathcal{H}_1)$	$R(\mathcal{H}) = R(\mathcal{H}_1)$	Remark. In 5., $n(\mathcal{H}) = n(\mathcal{H}_1) + 1$ and the equality $R(\mathcal{H}) = R(\mathcal{H}_1)$ means that
voloshin-hypergraph_F01717	230	■ 1.000	$r_i(\mathcal{H}) = r_i(\mathcal{H}_1)$	$r_i(\mathcal{H}) = r_i(\mathcal{H}_1)$	$r_i(\mathcal{H}) = r_i(\mathcal{H}_1)$ for $i = 1, 2, \dots, n-1$ and $r_n(\mathcal{H}) = 0$ since $\mathcal{H}$ contains one C -edge C_t of size
voloshin-hypergraph_F01718	230	■ 1.000	$i = 1, 2, \dots, n-1$	$i = 1, 2, \dots, n-1$	$r_i(\mathcal{H}) = r_i(\mathcal{H}_1)$ for $i = 1, 2, \dots, n-1$ and $r_n(\mathcal{H}) = 0$ since $\mathcal{H}$ contains one C -edge C_t of size
voloshin-hypergraph_F01719	230	■ 1.000	$r_n(\mathcal{H}) = 0$	$r_n(\mathcal{H}) = 0$	$r_i(\mathcal{H}) = r_i(\mathcal{H}_1)$ for $i = 1, 2, \dots, n-1$ and $r_n(\mathcal{H}) = 0$ since $\mathcal{H}$ contains one C -edge C_t of size
voloshin-hypergraph_F01720	230	■ 1.000	C_t	C_t	$r_i(\mathcal{H}) = r_i(\mathcal{H}_1)$ for $i = 1, 2, \dots, n-1$ and $r_n(\mathcal{H}) = 0$ since $\mathcal{H}$ contains one C -edge C_t of size
voloshin-hypergraph_F01721	230	■ 0.982	$1, 2, \dots, n$	$1, 2, \dots, n$	INPUT: An arbitrary mixed hypergraph $\mathcal{H} = (X, C, \mathcal{D})$ with X labeled $1, 2, \dots, n$.
voloshin-hypergraph_F01722	231	■ 0.998	$L = Z = Y = \emptyset, R(\mathcal{H}) = (0, 0, \dots, 0), P(\mathcal{H}, \lambda) = 0, \chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 0$	$L = Z = Y = \emptyset, R(\mathcal{H}) = (0, 0, \dots, 0), P(\mathcal{H}, \lambda) = 0, \chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 0$	1. Set lists $L = Z = Y = \emptyset, R(\mathcal{H}) = (0, 0, \dots, 0), P(\mathcal{H}, \lambda) = 0, \chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 0$. Add
voloshin-hypergraph_F01723	231	■ 0.588	$i = 1, 2, \dots, n$	$i = 1, 2, \dots, n$	graphs in Z having exactly i vertices, $i = 1, 2, \dots, n$.
voloshin-hypergraph_F01724	231	■ 1.000	$\chi(\mathcal{H}), \bar{\chi}(\mathcal{H})$	$\chi(\mathcal{H}), \bar{\chi}(\mathcal{H})$	9. Determine $\chi(\mathcal{H}), \bar{\chi}(\mathcal{H})$ using $R(\mathcal{H})$.
voloshin-hypergraph_F01725	231	■ 0.988	$X = \{1, 2\}$	$X = \{1, 2\}$	ellipse and lines. We have here a mixed hypergraph $\mathcal{H} = (X, C, \mathcal{D})$, where $X = \{1, 2,$
voloshin-hypergraph_F01726	231	■ 0.698	$3, 4\}, \mathcal{C} = \{C\} = \{\{1, 2, 3, 4\}\}, \mathcal{D} = \{D_1, D_2, D_3, D_4\} = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}\}$	$3, 4\}, \mathcal{C} = \{C\} = \{\{1, 2, 3, 4\}\}, \mathcal{D} = \{D_1, D_2, D_3, D_4\} = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}\}$	$\{3, 4\}, \mathcal{C} = \{C\} = \{\{1, 2, 3, 4\}\}, \mathcal{D} = \{D_1, D_2, D_3, D_4\} = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}\}$.
voloshin-hypergraph_F01727	231	■ 1.000	$\chi = 2, \bar{\chi} = 3$	$\chi = 2, \bar{\chi} = 3$	$\chi = 2, \bar{\chi} = 3$, and the corresponding list L of three strict colorings in the form
voloshin-hypergraph_F01728	231	■ 0.999	$(c(1), c(2), c(3), c(4))$	$(c(1), c(2), c(3), c(4))$	$(c(1), c(2), c(3), c(4))$ is the following (now the numbers are colors):
voloshin-hypergraph_F01729	233	■ 1.000	$X = \{1, 2, 3, 4, 5\}, \mathcal{C} = \{\{1, 2, 3\}, \{1, 3, 4\}\}$	$X = \{1, 2, 3, 4, 5\}, \mathcal{C} = \{\{1, 2, 3\}, \{1, 3, 4\}\}$	Example 10.5.2 Let $\mathcal{H} = (X, C, \mathcal{D})$, where $X = \{1, 2, 3, 4, 5\}, \mathcal{C} = \{\{1, 2, 3\}, \{1, 3, 4\}\},$

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voloshin-hypergraph_F01730	233	■ 0.997	$\{1,4,5\}, \{1,5,2\}$, $\quad \mathcal{D} = \{\{3,5\}\}$	$\{1,4,5\}, \{1,5,2\}, \quad \mathcal{D} = \{\{3,5\}\}$	$\{1,4,5\}, \{1,5,2\}, \quad \mathcal{D} = \{\{3,5\}\}$; we have that $\bar{\chi}(\mathcal{H}) = 3$ and, after adding the $\mathcal{D}$ -edge
voloshin-hypergraph_F01731	233	■ 0.997	$\bar{\chi}(\mathcal{H})=3$	$\bar{\chi}(\mathcal{H}) = 3$	$\{1,4,5\}, \{1,5,2\}, \quad \mathcal{D} = \{\{3,5\}\}$; we have that $\bar{\chi}(\mathcal{H}) = 3$ and, after adding the $\mathcal{D}$ -edge
voloshin-hypergraph_F01732	233	■ 0.967	$\{2,4\}$	$\{2,4\}$	$\{2,4\}$, we obtain the new mixed hypergraph $\mathcal{H}_1$ for which $\bar{\chi}(\mathcal{H}_1) = 2$. It is easy to see
voloshin-hypergraph_F01733	233	■ 0.967	$\bar{\chi}(\mathcal{H}_1)=2$	$\bar{\chi}(\mathcal{H}_1) = 2$	$\{2,4\}$, we obtain the new mixed hypergraph $\mathcal{H}_1$ for which $\bar{\chi}(\mathcal{H}_1) = 2$. It is easy to see
voloshin-hypergraph_F01734	233	■ 0.989	$C \in \mathcal{C}(\mathcal{D})$	$C \in \mathcal{C}(\mathcal{D})$	Definition 10.5.1 In a mixed hypergraph $\mathcal{H} = (X, C, \mathcal{D})$, a C -edge $C \in \mathcal{C}$ ($\mathcal{D}$ -edge $D \in$
voloshin-hypergraph_F01735	233	■ 1.000	$X=\{1,2,3,4\}$	$X = \{1,2,3,4\}$	Example 10.5.3 Let $\mathcal{H} = (X, C, \emptyset)$, where $X = \{1,2,3,4\}$ and $C = \{\{1,2,3\},$
voloshin-hypergraph_F01736	233	■ 1.000	$\mathcal{C}=\{\{1,2,3\}$	$\mathcal{C} = \{\{1,2,3\}$	Example 10.5.3 Let $\mathcal{H} = (X, C, \emptyset)$, where $X = \{1,2,3,4\}$ and $C = \{\{1,2,3\},$
voloshin-hypergraph_F01737	233	■ 0.987	$\{1,3,4\}, \{1,2,4\}, \{2,3,4\}$	$\{1,3,4\}, \{1,2,4\}, \{2,3,4\}$	$\{1,3,4\}, \{1,2,4\}, \{2,3,4\}$, see Figure 10.12. One can see that any C -edge in $\mathcal{H}$ is re-
voloshin-hypergraph_F01738	233	■ 1.000	$\mathcal{H}'=(X, \mathcal{C}, \mathcal{D})$	$\mathcal{H}' = (X, \mathcal{C}, \mathcal{D})$	forming a simple cycle, then in the mixed hypergraph $\mathcal{H}' = (X, \mathcal{C}, \mathcal{D})$, any $\mathcal{D}$ -edge is also
voloshin-hypergraph_F01739	233	■ 1.000	$\mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_k, k \geq 2$	$\mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_k, k \geq 2$	Proposition 10.5.2 If $\mathcal{H}$ is a disconnected mixed hypergraph and $\mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_k, k \geq 2$,
voloshin-hypergraph_F01740	235	■ 1.000	$\mathcal{H}_1=\left(\{1,2,3,4\}, K_4^3, \emptyset\right)$	$\mathcal{H}_1 = (\{1,2,3,4\}, K_4^3, \emptyset)$	2. Apply the splitting-contraction algorithm to $\mathcal{H}_1 = (\{1,2,3,4\}, K_4^3, \emptyset)$ and $\mathcal{H}_2 =$
voloshin-hypergraph_F01741	235	■ 1.000	$\mathcal{H}_2=$	$\mathcal{H}_2 =$	2. Apply the splitting-contraction algorithm to $\mathcal{H}_1 = (\{1,2,3,4\}, K_4^3, \emptyset)$ and $\mathcal{H}_2 =$
voloshin-hypergraph_F01742	235	■ 0.957	$\left(\{1,2,3,4\}, \emptyset, K_4^3\right)$	$(\{1,2,3,4\}, \emptyset, K_4^3)$	$(\{1,2,3,4\}, \emptyset, K_4^3)$ to find the feasible partitions, chromatic polynomials, and chro-
voloshin-hypergraph_F01743	235	■ 1.000	$\mathcal{D}=\left\{D_1, D_2\right\}$.	$\mathcal{D} = \{D_1, D_2$	Let $\mathcal{H} = (X, C, \mathcal{D})$ be a mixed hypergraph and $C = \{C_1, C_2, \dots, C_l\}$ and $\mathcal{D} = \{D_1, D_2,$
voloshin-hypergraph_F01744	235	■ 0.999	$\left.\left.\dots, D_m\right\}\right\}$	$\dots, D_m\}$	$\dots, D_m\}$ be two nonempty families of edges. As usual, denote $I = \{1, 2, \dots, l\}$ and $J =$
voloshin-hypergraph_F01745	235	■ 0.999	$J=$	$J =$	$\dots, D_m\}$ be two nonempty families of edges. As usual, denote $I = \{1, 2, \dots, l\}$ and $J =$
voloshin-hypergraph_F01746	235	■ 1.000	$\{1,2, \dots, m\}$	$\{1,2, \dots, m\}$	$\{1,2, \dots, m\}$. In this section we assume that mixed hypergraph $\mathcal{H}$ is simple, i.e. no edge is
voloshin-hypergraph_F01747	236	■ 1.000	K_n^{k+1}	K_n^{k+1}	add C -edges of the complete hypergraph K_n^{k+1} . Then a $\mathcal{D}$ -hypergraph $\mathcal{H} = (X, \mathcal{D})$, is k -
voloshin-hypergraph_F01748	236	■ 1.000	$\mathcal{H}'=\left(X, \binom{X}{k+1}, \mathcal{D}\right)$	$\mathcal{H}' = \left(X, \binom{X}{k+1}, \mathcal{D}\right)$	colorable if and only if the mixed hypergraph $\mathcal{H}' = (X, \binom{X}{k+1}, \mathcal{D})$ is colorable.
voloshin-hypergraph_F01749	236	■ 1.000	$\left\{x_1, x_2\right\}$	$\{x_1, x_2\}$	straints on $\{x_1, x_2\}$. Thus, minimal uncolorability in this situation immediately implies that
voloshin-hypergraph_F01750	236	■ 1.000	$v(k)$	$v(k)$	To show that, let us consider the following problem: <i>For every integer $k \geq 0$, let $v(k)$</i>
voloshin-hypergraph_F01751	236	■ 1.000	$\mathcal{H}=(X, \mathcal{C}, \mathcal{D}), X =n$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D}), X = n$	colorable mixed hypergraph $\mathcal{H} = (X, C, \mathcal{D}), X = n$, for which $\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_D) = k$. What
voloshin-hypergraph_F01752	236	■ 1.000	$\bar{\chi}(\mathcal{H}_C)-\chi(\mathcal{H}_D)=k$	$\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_D) = k$	colorable mixed hypergraph $\mathcal{H} = (X, C, \mathcal{D}), X = n$, for which $\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_D) = k$. What
voloshin-hypergraph_F01753	236	■ 0.816	$k=0,1,2, \dots$	$k = 0, 1, 2, \dots$	are the values of $v(k)$ for $k=0,1,2, \dots$..?
voloshin-hypergraph_F01754	236	■ 0.554	$\bar{\chi}(\mathcal{H}_C)-\chi(\mathcal{H}_D)$	$\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_D)$	orable mixed hypergraph $\mathcal{H}$ with large $\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_D)$ by taking just one bi-edge of size
voloshin-hypergraph_F01755	236	■ 0.996	$n-2=k+1$	$n - 2 = k + 1$	2 together with $n-2=k+1$ isolated vertices ; then the lower chromatic number of $\mathcal{H}_D$

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voloshin-hypergraph_ F01756	237	—	$\mathcal{U}_{\{4\}}$	$\mathcal{U}_4$	
voloshin-hypergraph_ F01757	237	■ 1.000	$n= X \geq 2$	$n = X \geq 2$	mixed hypergraph is evidently uncolorable. For every $n = X \geq 2$, the “most uncolorable”
voloshin-hypergraph_ F01758	237	■ 0.836	$\mathcal{U}_{\{n\}} = \left(X, \binom{X}{2}, \binom{X}{2}\right)$	$\mathcal{U}_n = \left(X, \binom{X}{2}, \binom{X}{2}\right)$	example (if minimality is not required) is the mixed hypergraph $\mathcal{U}_n = (X, \binom{X}{2}, \binom{X}{2})$ which
voloshin-hypergraph_ F01759	237	■ 0.910	$\mathcal{U}_{\{n\}}$	$\mathcal{U}_n$	is uncolorable together with each of its induced subhypergraphs of order ≥ 2 . In $\mathcal{U}_n$, each
voloshin-hypergraph_ F01760	237	■ 0.706	$\mathcal{U}_{\{ \}_{4\}}$	$\mathcal{U}_4$	uncolorable mixed hypergraph on four vertices $\mathcal{U}_4$.
voloshin-hypergraph_ F01761	237	■ 0.906	$ X =n=v(k)$	$ X = n = v(k)$	such that $ X = n = v(k)$, and $\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_D) = k$. We have to prove that $n = v(k) = k + 4$.
voloshin-hypergraph_ F01762	237	■ 0.906	$n=v(k)=k+4$	$n = v(k) = k + 4$	such that $ X = n = v(k)$, and $\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_D) = k$. We have to prove that $n = v(k) = k + 4$.
voloshin-hypergraph_ F01763	237	■ 1.000	$n \geq k+4$	$n \geq k + 4$	We show first that $n \geq k + 4$. For a contradiction, suppose that $n < k + 4$. Since $\mathcal{H}$
voloshin-hypergraph_ F01764	237	■ 1.000	$n < k+4$	$n < k + 4$	We show first that $n \geq k + 4$. For a contradiction, suppose that $n < k + 4$. Since $\mathcal{H}$
voloshin-hypergraph_ F01765	237	■ 0.959	$\chi(\mathcal{H}_D) \geq 2$	$\chi(\mathcal{H}_D) \geq 2$	is uncolorable, $\chi(\mathcal{H}_D) \geq 2$. If $\chi(\mathcal{H}_D) \geq 3$, then $\bar{\chi}(\mathcal{H}_C) \geq k + 3$, which implies $n = k + 3$,
voloshin-hypergraph_ F01766	237	■ 0.959	$\chi(\mathcal{H}_D) \geq 3$	$\chi(\mathcal{H}_D) \geq 3$	is uncolorable, $\chi(\mathcal{H}_D) \geq 2$. If $\chi(\mathcal{H}_D) \geq 3$, then $\bar{\chi}(\mathcal{H}_C) \geq k + 3$, which implies $n = k + 3$,
voloshin-hypergraph_ F01767	237	■ 0.959	$\bar{\chi}(\mathcal{H}_C) \geq k+3$	$\bar{\chi}(\mathcal{H}_C) \geq k + 3$	is uncolorable, $\chi(\mathcal{H}_D) \geq 2$. If $\chi(\mathcal{H}_D) \geq 3$, then $\bar{\chi}(\mathcal{H}_C) \geq k + 3$, which implies $n = k + 3$,
voloshin-hypergraph_ F01768	237	■ 0.959	$n=k+3$	$n = k + 3$	is uncolorable, $\chi(\mathcal{H}_D) \geq 2$. If $\chi(\mathcal{H}_D) \geq 3$, then $\bar{\chi}(\mathcal{H}_C) \geq k + 3$, which implies $n = k + 3$,
voloshin-hypergraph_ F01769	237	■ 0.984	$\chi(\mathcal{H}_C) = n$	$\chi(\mathcal{H}_C) = n$	$\chi(\mathcal{H}_C) = n$; therefore, $\mathcal{H}$ contains no C -edges and thus it is colorable, a contradiction.
voloshin-hypergraph_ F01770	237	■ 0.999	$\chi(\mathcal{H}_D) = 2$	$\chi(\mathcal{H}_D) = 2$	Hence, $\chi(\mathcal{H}_D) = 2$. Then we have only two possibilities for the number of vertices:
voloshin-hypergraph_ F01771	237	■ 1.000	$n=k+2$	$n = k + 2$	$n = k + 2$ or $n = k + 3$.
voloshin-hypergraph_ F01772	237	■ 0.983	$\bar{\chi}(\mathcal{H}_C) = k+2$	$\bar{\chi}(\mathcal{H}_C) = k + 2$	Similarly to the previous case, for $n = k + 2$ and $\bar{\chi}(\mathcal{H}_C) = k + 2$ it follows that the mixed
voloshin-hypergraph_ F01773	237	■ 0.715	$\bar{\chi}(\mathcal{H}_C) = k+2=n-1$	$\bar{\chi}(\mathcal{H}_C) = k + 2 = n - 1$	the last case $n = k + 3$. Since $\bar{\chi}(\mathcal{H}_C) = k + 2 = n - 1$, the partial C -hypergraph $\mathcal{H}_C =$
voloshin-hypergraph_ F01774	237	■ 0.715	$\mathcal{H}_C =$	$\mathcal{H}_C =$	the last case $n = k + 3$. Since $\bar{\chi}(\mathcal{H}_C) = k + 2 = n - 1$, the partial C -hypergraph $\mathcal{H}_C =$
voloshin-hypergraph_ F01775	237	■ 0.994	$(X, \mathcal{C})$	$(X, \mathcal{C})$	$(X, \mathcal{C})$ is a bistar , i.e., a C -hypergraph having two vertices, say x_1 and x_2 , that belong to

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voloshin-hypergraph_ F01776	238	—	$k=3$	$k = 3$	
voloshin-hypergraph_ F01777	238	■ 1.000	$v(k)=$	$v(k) =$	vertices. Thus, again, we obtain that $\mathcal{H}$ is colorable. This contradiction implies that $v(k) =$
voloshin-hypergraph_ F01778	238	■ 0.999	$v(k) \leq k+4$	$v(k) \leq k + 4$	Now, in order to prove the converse inequality $v(k) \leq k + 4$, we construct a series of
voloshin-hypergraph_ F01779	238	■ 0.373	$\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_D) = k$	$\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_D) = k$	examples of minimal uncolorable mixed hypergraphs with $\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_D) = k$ and $n =$
voloshin-hypergraph_ F01780	238	■ 0.373	$n=$	$n =$	examples of minimal uncolorable mixed hypergraphs with $\bar{\chi}(\mathcal{H}_C) - \chi(\mathcal{H}_D) = k$ and $n =$
voloshin-hypergraph_ F01781	238	■ 1.000	$k+4, k=0,1,2, \dots$	$k + 4, k = 0, 1, 2, \dots$	$k+4, k=0,1,2,\dots$. The construction will depend on the parity of k ; we first describe the
voloshin-hypergraph_ F01782	238	■ 1.000	$k=0,1$	$k = 0, 1$	particular cases $k=0,1$ that can be verified directly.
voloshin-hypergraph_ F01783	238	■ 1.000	$k=0$	$k = 0$	$k = 0$.
voloshin-hypergraph_ F01784	238	■ 1.000	$X=\{1,2,3,4\}, \mathcal{C}=\{\{1,2,3\},\{1,2,4\}\}, \mathcal{D}=\{\{1,2\}$	$X = \{1, 2, 3, 4\}, \mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}\}, \mathcal{D} = \{\{1, 2\}$	Let $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$, where $X = \{1, 2, 3, 4\}$, $\mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}\}$, $\mathcal{D} = \{\{1, 2\}$
voloshin-hypergraph_ F01785	238	■ 0.665	$\{2,3\},\{2,4\},\{3,4\}\}$	$\{2, 3\}, \{2, 4\}, \{3, 4\}$	$\{2, 3\}, \{2, 4\}, \{3, 4\}$.
voloshin-hypergraph_ F01786	238	■ 1.000	$X=\{1,2,3,4,5\}, \mathcal{C}=\{\{1,2,3\},\{1,2,4\}, \{1,2,5\}\}$	$X = \{1, 2, 3, 4, 5\}, \mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}, \{1, 2, 5\}\}$	Consider $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$, where $X = \{1, 2, 3, 4, 5\}$, $\mathcal{C} = \{\{1, 2, 3\}, \{1, 2, 4\}, \{1, 2, 5\}\}$,
voloshin-hypergraph_ F01787	238	■ 0.950	$\mathcal{D}=\{\{1,2\},\{3,4\},\{4,5\},\{3,5\}\}$	$\mathcal{D} = \{\{1, 2\}, \{3, 4\}, \{4, 5\}, \{3, 5\}\}$	$\mathcal{D} = \{\{1, 2\}, \{3, 4\}, \{4, 5\}, \{3, 5\}\}$.
voloshin-hypergraph_ F01788	238	■ 1.000	$k=2l, l \geq 1$	$k = 2l, l \geq 1$	$k = 2l, l \geq 1$.
voloshin-hypergraph_ F01789	238	■ 0.999	$X=\{1,2,3, \dots, k+4\}, \mathcal{C}=$	$X = \{1, 2, 3, \dots, k + 4\}, \mathcal{C} =$	Construct the mixed hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$, where $X = \{1, 2, 3, \dots, k + 4\}$, $\mathcal{C} =$
voloshin-hypergraph_ F01790	238	■ 0.990	$\{\{1,2, i\}: 3 \leq i \leq k+4\}$	$\{\{1, 2, i\} : 3 \leq i \leq k + 4\}$	$\{\{1, 2, i\} : 3 \leq i \leq k + 4\}$, and $\mathcal{D} = \{\{i, i + 1\} : 1 \leq i \leq k + 3\} \cup \{k + 4, 2\}$.
voloshin-hypergraph_ F01791	238	■ 0.990	$\mathcal{D}=\{\{i, i+1\}: 1 \leq i \leq k+3\} \cup \{k+4, 2\}$	$\mathcal{D} = \{\{i, i + 1\} : 1 \leq i \leq k + 3\} \cup \{k + 4, 2\}$	$\{\{1, 2, i\} : 3 \leq i \leq k + 4\}$, and $\mathcal{D} = \{\{i, i + 1\} : 1 \leq i \leq k + 3\} \cup \{k + 4, 2\}$.
voloshin-hypergraph_ F01792	238	■ 0.860	$\bar{\chi}(\mathcal{H}_C) = n - 1 = k + 3$	$\bar{\chi}(\mathcal{H}_C) = n - 1 = k + 3$	to all C -edges, and therefore $\bar{\chi}(\mathcal{H}_C) = n - 1 = k + 3$. Moreover, $\mathcal{H}_D = (X, \mathcal{D})$ is the odd
voloshin-hypergraph_ F01793	238	■ 0.990	$(2, 3, 4, \dots, k + 4, 2)$	$(2, 3, 4, \dots, k + 4, 2)$	cycle $(2, 3, 4, \dots, k + 4, 2)$ with the pendant $\mathcal{D}$ -edge $\{1, 2\}$, so that $\chi(\mathcal{H}_D) = 3$.

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voloshin-hypergraph_ F01794	238	■ 0.990	$\chi(\text{left}(\mathcal{H}_{\mathcal{D}})\text{right})=3$	$\chi(\mathcal{H}_{\mathcal{D}}) = 3$	cycle $(2, 3, 4, \dots, k+4, 2)$ with the pendant $\mathcal{D}$ -edge $\{1, 2\}$, so that $\chi(\mathcal{H}_{\mathcal{D}}) = 3$.
voloshin-hypergraph_ F01795	239	■ 0.999	$i, i=1, 2, \dots, n$	$i, i = 1, 2, \dots, n$	vertex $i, i=1, 2, \dots, n$. In any possible coloring of $\mathcal{H}$, vertices 1 and 2 have to be colored
voloshin-hypergraph_ F01796	239	■ 0.998	$c(1)=1, c(2)=2$	$c(1) = 1, c(2) = 2$	differently, say $c(1) = 1, c(2) = 2$. Since $\{2, 3\}$ is a $\mathcal{D}$ -edge, we have $c(3) \neq c(2)$ and,
voloshin-hypergraph_ F01797	239	■ 0.998	$c(3) \neq c(2)$	$c(3) \neq c(2)$	differently, say $c(1) = 1, c(2) = 2$. Since $\{2, 3\}$ is a $\mathcal{D}$ -edge, we have $c(3) \neq c(2)$ and,
voloshin-hypergraph_ F01798	239	■ 0.600	$c(3)=$	$c(3) =$	because of the C -edge $\{1, 2, 3\}$, the unique possibility for vertex 3 to get colored is $c(3) =$
voloshin-hypergraph_ F01799	239	■ 0.993	$c(1)=1$	$c(1) = 1$	$c(1) = 1$. In the same way, $c(4) \neq c(3)$ and, because of the C -edge $\{1, 2, 4\}$, the unique
voloshin-hypergraph_ F01800	239	■ 0.993	$c(4) \neq c(3)$	$c(4) \neq c(3)$	$c(1) = 1$. In the same way, $c(4) \neq c(3)$ and, because of the C -edge $\{1, 2, 4\}$, the unique
voloshin-hypergraph_ F01801	239	■ 0.993	$\{1, 2, 4\}$	$\{1, 2, 4\}$	$c(1) = 1$. In the same way, $c(4) \neq c(3)$ and, because of the C -edge $\{1, 2, 4\}$, the unique
voloshin-hypergraph_ F01802	239	■ 1.000	$c(4)=c(2)=2$	$c(4) = c(2) = 2$	possibility for vertex 4 to be colored is $c(4) = c(2) = 2$.
voloshin-hypergraph_ F01803	239	■ 0.996	$(2, 3, 4, \dots, k+4)$	$(2, 3, 4, \dots, k+4)$	It is now clear that the colors have to alternate on the cycle $(2, 3, 4, \dots, k+4)$. Since
voloshin-hypergraph_ F01804	239	■ 1.000	$c(k+3)=1$	$c(k+3) = 1$	$c(k+3) = 1$ and $c(2) = 2$, we can color vertex $k+4$ neither with color 1 nor with color
voloshin-hypergraph_ F01805	239	■ 1.000	$c(2)=2$	$c(2) = 2$	$c(k+3) = 1$ and $c(2) = 2$, we can color vertex $k+4$ neither with color 1 nor with color
voloshin-hypergraph_ F01806	239	■ 1.000	$k+4$	$k+4$	$c(k+3) = 1$ and $c(2) = 2$, we can color vertex $k+4$ neither with color 1 nor with color
voloshin-hypergraph_ F01807	239	■ 0.994	$c(k+4)$	$c(k+4)$	2. However, any other color $c(k+4)$ is infeasible because of the C -edge $\{1, 2, k+4\}$.
voloshin-hypergraph_ F01808	239	■ 0.994	$\{1, 2, k+4\}$	$\{1, 2, k+4\}$	2. However, any other color $c(k+4)$ is infeasible because of the C -edge $\{1, 2, k+4\}$.
voloshin-hypergraph_ F01809	239	■ 1.000	$k=2l+1, l \geq 1$	$k = 2l+1, l \geq 1$	$k=2l+1, l \geq 1$.
voloshin-hypergraph_ F01810	239	■ 0.936	$\mathcal{D}=\{1, 2\} \cup \{i, i+1\} : 3 \leq i \leq k+3\} \cup \{k+4, 3\}$	$\mathcal{D} = \{1, 2\} \cup \{i, i+1\} : 3 \leq i \leq k+3\} \cup \{k+4, 3\}$	$\{\{1, 2, i\} : 3 \leq i \leq k+4\}$, and $\mathcal{D} = \{1, 2\} \cup \{i, i+1\} : 3 \leq i \leq k+3\} \cup \{k+4, 3\}$.
voloshin-hypergraph_ F01811	239	■ 0.929	$(3, 4, \dots, k+4, 3)$	$(3, 4, \dots, k+4, 3)$	graph having $\mathcal{D}$ -edge $\{1, 2\}$ as the first component and the odd cycle $(3, 4, \dots, k+4, 3)$ as
voloshin-hypergraph_ F01812	239	■ 1.000	$c(3)$	$c(3)$	Let $c(1) = 1, c(2) = 2$. For $c(3)$ there are only two possibilities: $c(3) = 1$, or $c(3) = 2$.
voloshin-hypergraph_ F01813	239	■ 1.000	$c(3)=1$	$c(3) = 1$	Let $c(1) = 1, c(2) = 2$. For $c(3)$ there are only two possibilities: $c(3) = 1$, or $c(3) = 2$.
voloshin-hypergraph_ F01814	239	■ 1.000	$c(3)=2$	$c(3) = 2$	Let $c(1) = 1, c(2) = 2$. For $c(3)$ there are only two possibilities: $c(3) = 1$, or $c(3) = 2$.
voloshin-hypergraph_ F01815	239	■ 1.000	$c(4)=2, c(5)=1, c(6)=2$	$c(4) = 2, c(5) = 1, c(6) = 2$	$c(4) = 2, c(5) = 1, c(6) = 2$, and so on; i.e., the colors have to alternate along the odd cycle.
voloshin-hypergraph_ F01816	239	■ 1.000	$\{k+3, k+4\}$	$\{k+3, k+4\}$	the $\mathcal{D}$ -edges $\{k+3, k+4\}$ and $\{k+4, 3\}$), we conclude that $\mathcal{H}$ is uncolorable. Minimality
voloshin-hypergraph_ F01817	239	■ 1.000	$\{k+4, 3\}$	$\{k+4, 3\}$	the $\mathcal{D}$ -edges $\{k+3, k+4\}$ and $\{k+4, 3\}$), we conclude that $\mathcal{H}$ is uncolorable. Minimality
voloshin-hypergraph_ F01818	239	■ 1.000	$2 \leq l, m \leq n= X $	$2 \leq l, m \leq n = X $	Complete uncolorable mixed hypergraphs. For $2 \leq l, m \leq n = X $, let
voloshin-hypergraph_ F01819	239	■ 0.996	$ \mathcal{C} = \binom{n}{l}$	$ \mathcal{C} = \binom{n}{l}$	Hence, $ \mathcal{C} = \binom{n}{l}$ and $ \mathcal{D} = \binom{n}{m}$. Call $\mathcal{K}(n, l, m)$ the complete (l, m)-uniform mixed hy-
voloshin-hypergraph_ F01820	239	■ 0.996	$ \mathcal{D} = \binom{n}{m}$	$ \mathcal{D} = \binom{n}{m}$	Hence, $ \mathcal{C} = \binom{n}{l}$ and $ \mathcal{D} = \binom{n}{m}$. Call $\mathcal{K}(n, l, m)$ the complete (l, m)-uniform mixed hy-

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voloshin-hypergraph_ F01821	239	■ 0.996	$\mathcal{K}(n, l, m)$	$\mathcal{K}(n, l, m)$	Hence, $ C = \binom{n}{l}$ and $ \mathcal{D} = \binom{n}{m}$. Call $\mathcal{K}(n, l, m)$ the complete (l, m)-uniform mixed hy-
voloshin-hypergraph_ F01822	239	■ 0.996	(l, m)	(l, m)	Hence, $ C = \binom{n}{l}$ and $ \mathcal{D} = \binom{n}{m}$. Call $\mathcal{K}(n, l, m)$ the complete (l, m)-uniform mixed hy-
voloshin-hypergraph_ F01823	239	■ 1.000	n, l, m	n, l, m	every m vertices form a $\mathcal{D}$ -edge. Evidently, for given n, l, m there exists exactly one (up to
voloshin-hypergraph_ F01824	239	■ 0.740	$\mathcal{K}(4, 4, 2)$	$\mathcal{K}(4, 4, 2)$	isomorphism) $\mathcal{K}(n, l, m)$. Complete $(4, 2)$ -uniform mixed hypergraph $\mathcal{K}(4, 4, 2)$ is shown in
voloshin-hypergraph_ F01825	240	■ 1.000	$n \leq (l-1)(m-1)$	$n \leq (l-1)(m-1)$	Proof. $\Rightarrow$ Let $n \leq (l-1)(m-1)$. We color $m-1$ vertices with the first color, the next
voloshin-hypergraph_ F01826	240	■ 1.000	$m-1$	$m-1$	Proof. $\Rightarrow$ Let $n \leq (l-1)(m-1)$. We color $m-1$ vertices with the first color, the next
voloshin-hypergraph_ F01827	240	■ 0.994	$l-1$	$l-1$	at most $l-1$ colors, and a proper coloring of $\mathcal{K}(n, l, m)$ is obtained; i.e., the hypergraph is
voloshin-hypergraph_ F01828	240	■ 0.998	$n \geq (l-1)(m-1)+1$	$n \geq (l-1)(m-1)+1$	$\Leftarrow$ Let $n \geq (l-1)(m-1)+1$. Suppose there exists a proper coloring of $\mathcal{K}(n, l, m)$. Since
voloshin-hypergraph_ F01829	240	■ 0.998	$(l-1)(m-1)$	$(l-1)(m-1)$	$(l-1)(m-1)$ of colorable mixed hypergraphs $\mathcal{K}(n, l, m)$ and any number of uncolorable
voloshin-hypergraph_ F01830	240	■ 1.000	$n+1$	$n+1$	$n+1$ instead of n . Since $l=1$ and $m=1$ are excluded by definition, we have n^2 possibilities
voloshin-hypergraph_ F01831	240	■ 1.000	$l=1$	$l=1$	$n+1$ instead of n . Since $l=1$ and $m=1$ are excluded by definition, we have n^2 possibilities
voloshin-hypergraph_ F01832	240	■ 1.000	$m=1$	$m=1$	$n+1$ instead of n . Since $l=1$ and $m=1$ are excluded by definition, we have n^2 possibilities
voloshin-hypergraph_ F01833	240	■ 1.000	n^2	n^2	$n+1$ instead of n . Since $l=1$ and $m=1$ are excluded by definition, we have n^2 possibilities
voloshin-hypergraph_ F01834	240	■ 1.000	$2 \leq l \leq n+1, 2 \leq m \leq n+1$	$2 \leq l \leq n+1, 2 \leq m \leq n+1$	to choose the pair (l, m) in the range $2 \leq l \leq n+1, 2 \leq m \leq n+1$. Applying Theorem 10.6.3,
voloshin-hypergraph_ F01835	240	■ 1.000	$\mathcal{K}(n, l, m+1)$	$\mathcal{K}(n, l, m+1)$	we obtain that $\mathcal{K}(n, l, m+1)$ is uncolorable if and only if
voloshin-hypergraph_ F01836	240	■ 0.997	$l \geq 2$	$l \geq 2$	Here the smallest possible value of $m-1$ is 1. Thus, for each $l \geq 2$, there are precisely $\lfloor \frac{n}{l-1} \rfloor$
voloshin-hypergraph_ F01837	240	■ 0.997	$\left\lfloor \frac{n}{l-1} \right\rfloor$	$\left\lfloor \frac{n}{l-1} \right\rfloor$	Here the smallest possible value of $m-1$ is 1. Thus, for each $l \geq 2$, there are precisely $\lfloor \frac{n}{l-1} \rfloor$
voloshin-hypergraph_ F01838	240	■ 1.000	N_n	N_n	uncolorable complete mixed hypergraphs of order $n+1$. Consequently, the total number N_n
voloshin-hypergraph_ F01840	241	■ 1.000	$\mathcal{K}(8, l, m)$	$\mathcal{K}(8, l, m)$	Example 10.6.1 The behavior of the uncolorability of $\mathcal{K}(8, l, m)$ is shown in Figure 10.17.
voloshin-hypergraph_ F01841	241	■ 0.926	l, m	l, m	The signs “+” and “-” mean that for given l, m , the mixed hypergraph $\mathcal{K}(8, l, m)$ is colorable
voloshin-hypergraph_ F01842	241	■ 0.999	$x, y \in D$	$x, y \in D$	uncolorable if each pair of vertices $x, y \in D$ is connected by a C -path of $\mathcal{H}$ consisting of
voloshin-hypergraph_ F01843	242	■ 1.000	$\chi(\mathcal{H}) \leq 2$	$\chi(\mathcal{H}) \leq 2$	Corollary 10.6.3 If $\mathcal{H}$ is a colorable mixed hypertree, then $\chi(\mathcal{H}) \leq 2$.
voloshin-hypergraph_ F01844	243	■ 0.996	$\mathcal{UC}$	$\mathcal{UC}$	class of uniquely colorable mixed hypergraphs is denoted by $\mathcal{UC}$.
voloshin-hypergraph_ F01845	243	■ 1.000	$r=2$	$r=2$	uc mixed hypergraphs. This reflects the fact that K_n^r is uc if and only if $r=2$.
voloshin-hypergraph_ F01846	244	■ 0.712	$x =$	$x =$	Definition 10.7.2 In a mixed hypergraph $\mathcal{H} = (X, C, \mathcal{D})$, a sequence of vertices $x =$
voloshin-hypergraph_ F01847	244	■ 0.977	$x_{\{0\}}, x_{\{1\}}, \dots, x_{\{k\}} = y, k \geq 1$	$x_0, x_1, \dots, x_k = y, k \geq 1$	$x_0, x_1, \dots, x_k = y, k \geq 1$, is called an (x, y) -invertor if and only if $x_i \neq x_{i+1}$ and $(x_i, x_{i+1}) \in \mathcal{D}$
voloshin-hypergraph_ F01848	244	■ 0.977	$x_{\{i\}} \neq x_{\{i+1\}}$	$x_i \neq x_{i+1}$	$x_0, x_1, \dots, x_k = y, k \geq 1$, is called an (x, y) -invertor if and only if $x_i \neq x_{i+1}$ and $(x_i, x_{i+1}) \in \mathcal{D}$

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voloshin-hypergraph_ F01849	244	■ 0.977	$\left(x_{\{i\}}, x_{\{i+1\}}\right) \in \mathcal{D}$	$(x_i, x_{i+1}) \in \mathcal{D}$	$x_0, x_1, \dots, x_k = y, k \geq 1$, is called an (x, y) - invertor if and only if $x_i \neq x_{i+1}$ and $(x_i, x_{i+1}) \in \mathcal{D}$
voloshin-hypergraph_ F01850	244	■ 1.000	$x_{\{1\}}, \ldots, x_{\{k-1\}}$	$x_1, \dots, x_{k-1}$	if $x = y$; and if $k \geq 2$, then $x_1, \dots, x_{k-1}$ are termed internal vertices.
voloshin-hypergraph_ F01851	244	■ 0.999	$x_{\{0\}}, x_{\{1\}}$	x_0, x_1	Two examples of invertors are shown in Figure 10.20. $\mathcal{H}_1$ is an odd invertor x_0, x_1 ,
voloshin-hypergraph_ F01852	244	■ 1.000	$x_{\{2\}}, x_{\{3\}}$	x_2, x_3	x_2, x_3 , while $\mathcal{H}_2$ is an even invertor x_0, x_1, x_2, x_3, x_4 . The colors are shown by numbers 1
voloshin-hypergraph_ F01853	244	■ 1.000	$x_{\{0\}}, x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, x_{\{4\}}$	x_0, x_1, x_2, x_3, x_4	x_2, x_3 , while $\mathcal{H}_2$ is an even invertor x_0, x_1, x_2, x_3, x_4 . The colors are shown by numbers 1
voloshin-hypergraph_ F01854	244	■ 1.000	$\left(x_{\{0\}}, x_{\{4\}}\right)$	(x_0, x_4)	(x_0, x_4) -path uses edge $\{x_1, x_2\}$ twice, i.e. it is not a simple path. The important feature
voloshin-hypergraph_ F01855	244	■ 0.979	$\chi=\bar{\chi}=2$	$\chi = \bar{\chi} = 2$	vertex x_4 . One can easily see that for both invertors $\chi = \bar{\chi} = 2$, and, moreover
voloshin-hypergraph_ F01856	245	■ 0.835	$c=X_{\{1\}} \cup \dots \cup X_{\{t\}}$	$c = X_1 \cup \dots \cup X_t$	hypergraph and $c = X_1 \cup \dots \cup X_t$ be a strict t -coloring. Then there exists a uc mixed hyper-
voloshin-hypergraph_ F01857	245	■ 0.999	$\mathcal{H}^{\prime}=\left(X^{\prime}, \mathcal{C}^{\prime}, \mathcal{D}^{\prime}\right)$	$\mathcal{H}' = (X', \mathcal{C}', \mathcal{D}')$	graph $\mathcal{H}' = (X', \mathcal{C}', \mathcal{D}')$ with the following properties:
voloshin-hypergraph_ F01858	245	■ 0.999	$\chi\left(\mathcal{H}^{\prime}\right)=\bar{\chi}\left(\mathcal{H}^{\prime}\right)=t$	$\chi(\mathcal{H}') = \bar{\chi}(\mathcal{H}') = t$	2. $\chi(\mathcal{H}') = \bar{\chi}(\mathcal{H}') = t$ and an extension of c is the unique strict coloring of $\mathcal{H}'$.
voloshin-hypergraph_ F01859	245	■ 1.000	$T_{\{i\}}$	T_i	ning tree T_i inside each partition class X_i of c . For every edge xy of T_i we put an even
voloshin-hypergraph_ F01860	245	■ 0.737	$\left\{x=x_{\{1\}}, x_{\{2\}}, z\right\}$	$\{x = x_1, x_2, z\}$	$x = x_1, x_2, z$, for each even (x, y) -invertor, where z is any vertex such that x and z belong to
voloshin-hypergraph_ F01861	245	■ 1.000	$X_{\{j\}}$	X_j	Next, for each pair (i, j) such that X_i and X_j are two distinct classes of c , we choose
voloshin-hypergraph_ F01862	245	■ 0.591	$x_{\{i\}} \in X_{\{i\}}$	$x_i \in X_i$	two vertices $x_i \in X_i$ and $x_j \in X_j$, and build an odd (x_i, x_j) -invertor on them. Then any two
voloshin-hypergraph_ F01863	245	■ 0.591	$x_{\{j\}} \in X_{\{j\}}$	$x_j \in X_j$	two vertices $x_i \in X_i$ and $x_j \in X_j$, and build an odd (x_i, x_j) -invertor on them. Then any two
voloshin-hypergraph_ F01864	246	■ 1.000	$\mu=\left(z_{\{0\}}, z_{\{1\}}, \ldots, z_{\{k\}}=z_{\{0\}}\right), k \geq 6$	$\mu = (z_0, z_1, \dots, z_k = z_0), k \geq 6$	Let $\mu = (z_0, z_1, \dots, z_k = z_0)$, $k \geq 6$ be a simple cyclic invertor in a mixed hypertree.
voloshin-hypergraph_ F01865	246	■ 0.983	$z_{\{0\}} \neq z_{\{1\}} \neq z_{\{2\}} \neq z_{\{0\}}$	$z_0 \neq z_1 \neq z_2 \neq z_0$	Without loss of generality, assume that $z_0 \neq z_1 \neq z_2 \neq z_0$. From the definition of a simple
voloshin-hypergraph_ F01866	246	■ 1.000	$z_{\{0\}} \neq z_{\{2\}} \neq \ldots \neq z_{\{k-2\}}$	$z_0 \neq z_2 \neq \dots \neq z_{k-2}$	cyclic invertor, it follows that $z_0 \neq z_2 \neq \dots \neq z_{k-2}$ and $z_1 = z_3 = \dots = z_{k-1} = y$, where y is
voloshin-hypergraph_ F01867	246	■ 1.000	$z_{\{1\}}=z_{\{3\}}=\ldots=z_{\{k-1\}}=y$	$z_1 = z_3 = \dots = z_{k-1} = y$	cyclic invertor, it follows that $z_0 \neq z_2 \neq \dots \neq z_{k-2}$ and $z_1 = z_3 = \dots = z_{k-1} = y$, where y is
voloshin-hypergraph_ F01868	246	■ 0.907	$ \mathcal{D} \leq n-2$	$ \mathcal{D} \leq n - 2$	Theorem 10.7.2 If $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ is a reduced mixed hypertree with $ \mathcal{D} \leq n - 2$, then
voloshin-hypergraph_ F01869	246	■ 1.000	$r_{\{2\}}(\mathcal{H}) \geq 2$	$r_2(\mathcal{H}) \geq 2$	$r_2(\mathcal{H}) \geq 2$.
voloshin-hypergraph_ F01870	246	■ 1.000	$T=(X, \mathcal{E})$	$T = (X, \mathcal{E})$	Proof. Let $T = (X, \mathcal{E})$ be a host tree of the mixed hypertree $\mathcal{H}$. Since $ \mathcal{D} \leq n - 2$,
voloshin-hypergraph_ F01871	246	■ 1.000	$e=\{x, y\} \notin \mathcal{D}$	$e = \{x, y\} \notin \mathcal{D}$	in T there exists an edge $e = \{x, y\} \notin \mathcal{D}$. Starting with vertices x, y , we can construct
voloshin-hypergraph_ F01872	246	■ 1.000	$c(x)=c(y)=1$	$c(x) = c(y) = 1$	$c(x) = c(y) = 1$ and color all the other vertices alternatively along the tree T with the colors
voloshin-hypergraph_ F01873	246	■ 0.941	$2, 1, 2, \ldots$	$2, 1, 2, \dots$	$2, 1, 2, \dots$. Second, apply the same procedure starting with $c(x) = 1$ and $c(y) = 2$.
voloshin-hypergraph_ F01874	246	■ 0.941	$c(x)=1$	$c(x) = 1$	$2, 1, 2, \dots$. Second, apply the same procedure starting with $c(x) = 1$ and $c(y) = 2$.
voloshin-hypergraph_ F01875	246	■ 0.941	$c(y)=2$	$c(y) = 2$	$2, 1, 2, \dots$. Second, apply the same procedure starting with $c(x) = 1$ and $c(y) = 2$.

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voloshin-hypergraph_F01903	247	■ 0.999	$x_{\{1\}}=z_{\{0\}}, z_{\{1\}}, \ldots, z_{\{k\}}=x_{\{3\}}=\left(x_{\{1\}}, x_{\{3\}}\right)^{\prime}$	$x_1 = z_0, z_1, \dots, z_k = x_3 = (x_1, x_3)'$	(x_1, x_3) -invertors: $\{x_1, x_2, x_3\} = C$ and $\boxed{x_1 = z_0, z_1, \dots, z_k = x_3 = (x_1, x_3)'$ -inverter. In each
voloshin-hypergraph_F01904	247	■ 0.999	$\left(x_{\{1\}}, x_{\{3\}}\right)^{\prime}$	$(x_1, x_3)'$	(x, y) -inverter containing C , replace this C -edge by the $(x_1, x_3)'$ -inverter. Thus, $\mathcal{H}' = (X, C \setminus$
voloshin-hypergraph_F01905	247	■ 0.999	$\mathcal{H}^{\prime}=(X, \mathcal{C} \backslash$	$\mathcal{H}' = (X, C \setminus$	(x, y) -inverter containing C , replace this C -edge by the $(x_1, x_3)'$ -inverter. Thus, $\mathcal{H}' = (X, C \setminus$
voloshin-hypergraph_F01906	247	■ 0.886	$\{C\}, \mathcal{D})$	$\{C\}, \mathcal{D})$	$\{C\}, \mathcal{D})$ is uniquely colorable; i.e., the C -edge C is redundant.
voloshin-hypergraph_F01907	247	■ 1.000	$X=X_{\{1\}} \cup X_{\{2\}} \cup \ldots \cup X_{\{i\}}$	$X = X_1 \cup X_2 \cup \dots \cup X_i$	Consider a mixed hypergraph $\mathcal{H} = (X, C, \mathcal{D})$. Let $\boxed{X=X_1 \cup X_2 \cup \dots \cup X_i}$ be a proper
voloshin-hypergraph_F01908	247	■ 0.913	$\mathcal{H}, \chi(\mathcal{H}) \leq i \leq \bar{\chi}(\mathcal{H})$	$\mathcal{H}, \chi(\mathcal{H}) \leq i \leq \bar{\chi}(\mathcal{H})$	i -coloring of $\mathcal{H}$, $\boxed{\chi(\mathcal{H}) \leq i \leq \bar{\chi}(\mathcal{H})}$ and choose any X_j .
voloshin-hypergraph_F01909	247	■ 1.000	$L_{\{j\}}=\left(X_{\{j\}}, E_{\{j\}}\right)$	$L_j = (X_j, E_j)$	Definition 10.7.4 The touching graph of a color class X_j is the graph $L_j = (X_j, E_j)$ where
voloshin-hypergraph_F01910	247	■ 0.699	$E_{\{j\}}$	E_j	the edge set E_j is defined in the following way: $\{x, y\} \in E_j$ if and only if some $C \in \mathcal{C}$ satisfies
voloshin-hypergraph_F01911	247	■ 0.699	$\{x, y\} \in E_{\{j\}}$	$\{x, y\} \in E_j$	the edge set E_j is defined in the following way: $\{x, y\} \in E_j$ if and only if some $C \in \mathcal{C}$ satisfies
voloshin-hypergraph_F01912	247	■ 1.000	$C \cap X_{\{j\}}=\{x, y\}$	$C \cap X_j = \{x, y\}$	$\boxed{C \cap X_j = \{x, y\}}$, and $ C \cap X_k \leq 1$ for any $k \neq j$.
voloshin-hypergraph_F01913	247	■ 1.000	$\left C \cap X_{\{k\}}\right \leq 1$	$ C \cap X_k \leq 1$	$C \cap X_j = \{x, y\}$, and $ C \cap X_k \leq 1$ for any $k \neq j$.
voloshin-hypergraph_F01914	247	■ 1.000	$k \neq j$	$k \neq j$	$C \cap X_j = \{x, y\}$, and $ C \cap X_k \leq 1$ for any $k \neq j$.
voloshin-hypergraph_F01915	247	■ 0.998	$\bar{\chi}+1$	$\bar{\chi} + 1$	coloring of $\mathcal{H}$ with $\bar{\chi} + 1$ colors by assigning a new color to the vertices of one component.
voloshin-hypergraph_F01916	247	■ 1.000	$L_{\{1\}}$	L_1	in its 2-coloring, the touching graphs L_1 and L_2 are connected.
voloshin-hypergraph_F01917	247	■ 1.000	$L_{\{2\}}$	L_2	in its 2-coloring, the touching graphs L_1 and L_2 are connected.
voloshin-hypergraph_F01918	248	■ 1.000	$\chi(\mathcal{H})=\chi$	$\chi(\mathcal{H}) = \chi$	Proof. Let $\mathcal{H}$ be a uniquely colorable mixed hypergraph, $\chi(\mathcal{H}) = \chi$. Consider a unique
voloshin-hypergraph_F01919	248	■ 1.000	$X=X_{\{1\}} \cup X_{\{2\}} \cup \ldots \cup X_{\{\chi\}}$	$X = X_1 \cup X_2 \cup \dots \cup X_\chi$	χ -coloring $\boxed{X = X_1 \cup X_2 \cup \dots \cup X_\chi}$ and construct the touching graphs
voloshin-hypergraph_F01920	248	■ 1.000	$L_{\{i\}}$	L_i	The minimum number of edges in L_i required for L_i to be connected is $ X_i - 1$, and, in this
voloshin-hypergraph_F01921	248	■ 1.000	$\left X_{\{i\}}\right -1$	$ X_i - 1$	The minimum number of edges in L_i required for L_i to be connected is $ X_i - 1$, and, in this
voloshin-hypergraph_F01922	248	■ 1.000	$i=1,2, \ldots, \chi$	$i = 1, 2, \dots, \chi$	case, each L_i is a tree, $\boxed{i=1,2,\dots,\chi}$. Since every edge in L_i corresponds to some C -edge of
voloshin-hypergraph_F01923	248	■ 0.562	$ \mathcal{C} <n-\chi(\mathcal{H})$	$ \mathcal{C} < n - \chi(\mathcal{H})$	Corollary 10.7.6 A mixed hypergraph $\mathcal{H}$ with $ \mathcal{C} < n - \chi(\mathcal{H})$ is not uc.
voloshin-hypergraph_F01924	248	■ 1.000	$ \mathcal{C} -n+2$	$ \mathcal{C} - n + 2$	number of redundant C -edges is $ \mathcal{C} - n + 2$.
voloshin-hypergraph_F01925	248	■ 1.000	$T_{\{i\}}, i=1,2$	$T_i, i = 1, 2$	Proof. Indeed, for each touching graph L_i , construct a spanning tree T_i , $i = 1, 2$. Each
voloshin-hypergraph_F01926	248	■ 0.997	$ \mathcal{D} =n-1, \quad n-2 \leq \mathcal{C} \leq M$	$ \mathcal{D} = n - 1, \quad n - 2 \leq \mathcal{C} \leq M$	one can construct a mixed hypertree $\mathcal{H} = (X, C, \mathcal{D})$ with $ \mathcal{D} = n - 1, \quad n - 2 \leq \mathcal{C} \leq M$ and
voloshin-hypergraph_F01927	248	■ 0.996	$\bar{\chi}(\mathcal{H})=k$	$\bar{\chi}(\mathcal{H}) = k$	$\bar{\chi}(\mathcal{H}) = k$. Therefore, these bounds on $ \mathcal{D} $ and $ \mathcal{C} $ are not sufficient for a mixed hypertree
voloshin-hypergraph_F01928	248	■ 0.996	$ \mathcal{D} $	$ \mathcal{D} $	$\bar{\chi}(\mathcal{H}) = k$. Therefore, these bounds on $ \mathcal{D} $ and $ \mathcal{C} $ are not sufficient for a mixed hypertree
voloshin-hypergraph_F01929	248	■ 0.996	$ \mathcal{C} $	$ \mathcal{C} $	$\bar{\chi}(\mathcal{H}) = k$. Therefore, these bounds on $ \mathcal{D} $ and $ \mathcal{C} $ are not sufficient for a mixed hypertree

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F01930	249	■ 1.000	$C=\{x, y, z\}$	$C = \{x, y, z\}$	is pendant in T and in, say T_1 . Then x belongs to precisely one C -edge $C = \{x, y, z\}$.
voloshin-hypergraph_F01931	249	■ 1.000	$\{x, y\}, \{y, z\} \in \mathcal{D}$	$\{x, y\}, \{y, z\} \in \mathcal{D}$	Evidently, $\{x, y\}, \{y, z\} \in \mathcal{D}$. Consequently, $c(x) = c(z)$ in the unique partition c of $\mathcal{H}$.
voloshin-hypergraph_F01932	249	■ 1.000	$c(x)=c(z)$	$c(x) = c(z)$	Evidently, $\{x, y\}, \{y, z\} \in \mathcal{D}$. Consequently, $c(x) = c(z)$ in the unique partition c of $\mathcal{H}$.
voloshin-hypergraph_F01933	249	■ 1.000	$ \mathcal{C}(\mathcal{H}-x) = \mathcal{C}(\mathcal{H}) -1$	$ \mathcal{C}(\mathcal{H}-x) = \mathcal{C}(\mathcal{H}) - 1$	$ \mathcal{C}(\mathcal{H}-x) = \mathcal{C}(\mathcal{H}) - 1$. If it is not uc, then it admits at least two strict colorings. In
voloshin-hypergraph_F01934	249	■ 1.000	$\left(x_n, x_{n-1}, \dots, x_1\right)$	$(x_n, x_{n-1}, \dots, x_1)$	decompose $\mathcal{H}$ by elimination of vertices in order inverse to σ , i.e. $\boxed{x_n, x_{n-1}, \dots, x_1}$, and at
voloshin-hypergraph_F01935	249	■ 0.637	$\mathcal{H}=(X, \mathcal{C}, \mathcal{D}), \sigma$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D}), \sigma$	• <i>INPUT: A reduced mixed hypertree $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$, σ (n-dimensional empty vector).</i>
voloshin-hypergraph_F01936	249	■ 1.000	L_1, L_2	L_1, L_2	<i>graphs L_1, L_2, respectively, by pendant vertices.</i>
voloshin-hypergraph_F01937	250	■ 1.000	$L_i, i=1, 2$	$L_i, i = 1, 2$	<i>4. If $L_i, i = 1, 2$, is not connected, then return NON uc.</i>
voloshin-hypergraph_F01938	250	■ 1.000	$i:=1$	$i := 1$	<i>6. $\boxed{i:=1}$.</i>
voloshin-hypergraph_F01939	250	■ 0.965	$i:=3-i$	$i := 3 - i$	<i>8. If at least one of T_1 and T_2 is not empty, assign $\boxed{i:=3-i}$ and go to 7; otherwise,</i>
voloshin-hypergraph_F01940	250	■ 0.565	$\sigma=$	$\sigma =$	<i>return uc, σ=uc-ordering.</i>
voloshin-hypergraph_F01941	250	■ 0.999	$T_1=L_1$	$T_1 = L_1$	shown in Figure 10.21. Graph L_1 is a tree, so $T_1 = L_1$. Graph L_2 is not a tree; it means
voloshin-hypergraph_F01942	250	■ 1.000	$\left\{x_1, x_3, x_4\right\}, \left\{x_4, x_3, x_5\right\}, \left\{x_2, x_3, x_5\right\}$	$\{x_1, x_3, x_4\}, \{x_4, x_3, x_5\}, \{x_2, x_3, x_5\}$	that $\mathcal{H}$ has a redundant C -edge. Any of three C -edges $\boxed{\{x_1, x_3, x_4\}, \{x_4, x_3, x_5\}, \{x_2, x_3, x_5\}}$
voloshin-hypergraph_F01943	250	■ 1.000	$\left\{x_2, x_5\right\}$	$\{x_2, x_5\}$	is redundant. In graph L_2 , choose edge $\{x_2, x_5\}$ to be a chord of spanning tree T_2 . At the
voloshin-hypergraph_F01944	250	■ 1.000	x_1, x_2, x_4, x_3	x_1, x_2, x_4, x_3	and so on. Thus the algorithm decomposes $\mathcal{H}$ in the ordering x_1, x_2, x_4, x_3 , and x_5 . Hence
voloshin-hypergraph_F01945	251	■ 1.000	$R(\mathcal{H})=(0,1,0, \dots, 0)$	$R(\mathcal{H}) = (0, 1, 0, \dots, 0)$	(1) $\boxed{R(\mathcal{H}) = (0, 1, 0, \dots, 0)}$;
voloshin-hypergraph_F01946	251	■ 0.994	$P(\mathcal{H}, \lambda)=\lambda(\lambda-1)$	$P(\mathcal{H}, \lambda) = \lambda(\lambda - 1)$	(2) $\boxed{P(\mathcal{H}, \lambda) = \lambda(\lambda - 1)}$;
voloshin-hypergraph_F01947	252	■ 0.995	$\mathcal{K}_n^r, r \geq 3$	$\mathcal{K}_n^r, r \geq 3$	graphs (cliques). For example, the $\mathcal{D}$ -hypergraph $\mathcal{K}_n^r, r \geq 3$, being a hypergraph general-
voloshin-hypergraph_F01948	252	■ 0.998	$\mathcal{H}=\left(X, \emptyset, \binom{X}{r}\right)$	$\mathcal{H} = \left(X, \emptyset, \binom{X}{r}\right)$	ization of the clique K_n and considered as a mixed hypergraph $\mathcal{H} = (X, \emptyset, \binom{X}{r})$, has many
voloshin-hypergraph_F01949	252	■ 0.972	$\mathcal{H}=\left(X, \emptyset, \binom{X}{2}\right)$	$\mathcal{H} = \left(X, \emptyset, \binom{X}{2}\right)$	proper colorings and its properties are not so nice as those of the clique $\mathcal{H} = (X, \emptyset, \binom{X}{2})$.
voloshin-hypergraph_F01950	252	■ 0.986	$\alpha_c(\mathcal{H})$	$\alpha_c(\mathcal{H})$	(C-independence) number $\alpha_c(\mathcal{H})$.
voloshin-hypergraph_F01951	252	■ 0.976	$\bar{\chi}(\mathcal{H})=\alpha_c(\mathcal{H})$	$\bar{\chi}(\mathcal{H}) = \alpha_c(\mathcal{H})$	Mixed hypergraphs with $\bar{\chi}(\mathcal{H}) = \alpha_c(\mathcal{H})$ may be constructed easily, and it is now seen
voloshin-hypergraph_F01952	252	■ 1.000	$\chi(G) \geq \omega(G)$	$\chi(G) \geq \omega(G)$	to the inequality $\boxed{\chi(G) \geq \omega(G)}$, see Section 5.7. The significant difference, however, is that
voloshin-hypergraph_F01953	253	■ 1.000	$\bar{\chi} \geq k+1$	$\bar{\chi} \geq k + 1$	Theorem 10.8.1 <i>For any $k \geq 0$ and $\bar{\chi} \geq k + 1$, there exists a 3-uniform C-hypergraph $\mathcal{H}$</i>
voloshin-hypergraph_F01954	253	■ 1.000	$X=\{1,2, \dots, 2 k+5\}$	$X = \{1, 2, \dots, 2k + 5\}$	Proof. Let $\mathcal{H} = (X, \mathcal{C})$ be a 3-uniform C -hypergraph with $\boxed{X = \{1, 2, \dots, 2k + 5\}}$, and
voloshin-hypergraph_F01955	253	■ 1.000	$C_1=\{1,2,3\}, C_2=\{1,4,5\}, \dots, C_{k+2}=\{1,2 k+4,2 k+5\}$	$C_1 = \{1, 2, 3\}, C_2 = \{1, 4, 5\}, \dots, C_{k+2} = \{1, 2k + 4, 2k + 5\}$	$\boxed{C_1 = \{1, 2, 3\}, C_2 = \{1, 4, 5\}, \dots, C_{k+2} = \{1, 2k + 4, 2k + 5\}}$ such that
voloshin-hypergraph_F01962	253	■ 0.957	$\mathcal{H}=\left(X, \binom{X}{r}, \mathcal{D}\right), r \geq 2$	$\mathcal{H} = \left(X, \binom{X}{r}, \mathcal{D}\right), r \geq 2$	Example 10.8.1 Every colorable mixed hypergraph $\mathcal{H} = (X, \binom{X}{r}, \mathcal{D}), r \geq 2$, is perfect. In-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F01963	253	■ 0.998	$\bar{\chi}(\mathcal{H}_Y) = r - 1 = \alpha_c(\mathcal{H}_Y)$	$\bar{\chi}(\mathcal{H}_Y) = r - 1 = \alpha_c(\mathcal{H}_Y)$	deed, $\bar{\chi}(\mathcal{H}_Y) = r - 1 = \alpha_c(\mathcal{H}_Y)$ for each $ Y \geq r$, and $\bar{\chi}(\mathcal{H}_Y) = Y = \alpha_c(\mathcal{H}_Y)$ for each $ Y < r$.
voloshin-hypergraph_F01964	253	■ 0.998	$ Y \geq r$	$ Y \geq r$	deed, $\bar{\chi}(\mathcal{H}_Y) = r - 1 = \alpha_c(\mathcal{H}_Y)$ for each $ Y \geq r$, and $\bar{\chi}(\mathcal{H}_Y) = Y = \alpha_c(\mathcal{H}_Y)$ for each $ Y < r$.
voloshin-hypergraph_F01965	253	■ 0.998	$\bar{\chi}(\mathcal{H}_Y) = Y = \alpha_c(\mathcal{H}_Y)$	$\bar{\chi}(\mathcal{H}_Y) = Y = \alpha_c(\mathcal{H}_Y)$	deed, $\bar{\chi}(\mathcal{H}_Y) = r - 1 = \alpha_c(\mathcal{H}_Y)$ for each $ Y \geq r$, and $\bar{\chi}(\mathcal{H}_Y) = Y = \alpha_c(\mathcal{H}_Y)$ for each $ Y < r$.
voloshin-hypergraph_F01966	253	■ 0.998	$ Y < r$	$ Y < r$	deed, $\bar{\chi}(\mathcal{H}_Y) = r - 1 = \alpha_c(\mathcal{H}_Y)$ for each $ Y \geq r$, and $\bar{\chi}(\mathcal{H}_Y) = Y = \alpha_c(\mathcal{H}_Y)$ for each $ Y < r$.
voloshin-hypergraph_F01967	254	■ 0.570	$\bar{\chi}(\mathcal{H}) = n - 1 = \alpha_c(\mathcal{H})$	$\bar{\chi}(\mathcal{H}) = n - 1 = \alpha_c(\mathcal{H})$	all different colors, so $\bar{\chi}(\mathcal{H}) = n - 1 = \alpha_c(\mathcal{H})$. Consider any $Y \subseteq X$. If $\mathcal{H}_Y$ contains at least
voloshin-hypergraph_F01968	254	■ 0.904	$\bar{\chi}(\mathcal{H}_Y) = Y - 1 = \alpha_c(\mathcal{H}_Y)$	$\bar{\chi}(\mathcal{H}_Y) = Y - 1 = \alpha_c(\mathcal{H}_Y)$	one C -edge, then $\bar{\chi}(\mathcal{H}_Y) = Y - 1 = \alpha_c(\mathcal{H}_Y)$. Otherwise, $\bar{\chi}(\mathcal{H}_Y) = Y = \alpha_c(\mathcal{H}_Y)$.
voloshin-hypergraph_F01969	254	■ 0.904	$\bar{\chi}(\mathcal{H}_Y) = Y = \alpha_c(\mathcal{H}_Y)$	$\bar{\chi}(\mathcal{H}_Y) = Y = \alpha_c(\mathcal{H}_Y)$	one C -edge, then $\bar{\chi}(\mathcal{H}_Y) = Y - 1 = \alpha_c(\mathcal{H}_Y)$. Otherwise, $\bar{\chi}(\mathcal{H}_Y) = Y = \alpha_c(\mathcal{H}_Y)$.
voloshin-hypergraph_F01970	254	■ 0.793	$\tau(\mathcal{H}_C)$	$\tau(\mathcal{H}_C)$	Minimal non-perfect mixed hypergraphs. Recall that we denote by $\tau(\mathcal{H}_C)$ the
voloshin-hypergraph_F01971	254	■ 0.998	$\tau_2(\mathcal{H}_C)$	$\tau_2(\mathcal{H}_C)$	at least one vertex from every C -edge of a mixed hypergraph $\mathcal{H} = (X, C, \mathcal{D})$. Let $\tau_2(\mathcal{H}_C)$
voloshin-hypergraph_F01972	254	■ 0.997	$\tau(\mathcal{H}_C) = 1$	$\tau(\mathcal{H}_C) = 1$	monostar, then $\tau(\mathcal{H}_C) = 1$, and $\tau_2(\mathcal{H}_C) \geq 3$. However, if $\mathcal{H} = (X, C, \mathcal{D})$ is a C -bistar, then it
voloshin-hypergraph_F01973	254	■ 0.997	$\tau_2(\mathcal{H}_C) \geq 3$	$\tau_2(\mathcal{H}_C) \geq 3$	monostar, then $\tau(\mathcal{H}_C) = 1$, and $\tau_2(\mathcal{H}_C) \geq 3$. However, if $\mathcal{H} = (X, C, \mathcal{D})$ is a C -bistar, then it
voloshin-hypergraph_F01974	254	■ 0.977	$\tau_2(\mathcal{H}_C) = 2$	$\tau_2(\mathcal{H}_C) = 2$	has two vertices in common with all the C -edges, what implies $\tau(\mathcal{H}_C) = 1$, and $\tau_2(\mathcal{H}_C) = 2$.
voloshin-hypergraph_F01975	254	■ 0.982	$\alpha_c(\mathcal{H}) = X - 1$	$\alpha_c(\mathcal{H}) = X - 1$	Proof. It follows from the definition of C -monostar that $\alpha_c(\mathcal{H}) = X - 1$. Since also by
voloshin-hypergraph_F01976	254	■ 1.000	$ X - 2$	$ X - 2$	number of colors is not greater than $ X - 2$. This implies that $\bar{\chi}(\mathcal{H}) \leq X - 2$, so $\bar{\chi}(\mathcal{H}) \neq$
voloshin-hypergraph_F01977	254	■ 1.000	$\bar{\chi}(\mathcal{H}) \leq X - 2$	$\bar{\chi}(\mathcal{H}) \leq X - 2$	number of colors is not greater than $ X - 2$. This implies that $\bar{\chi}(\mathcal{H}) \leq X - 2$, so $\bar{\chi}(\mathcal{H}) \neq$
voloshin-hypergraph_F01978	254	■ 1.000	$\bar{\chi}(\mathcal{H}) \neq$	$\bar{\chi}(\mathcal{H}) \neq$	number of colors is not greater than $ X - 2$. This implies that $\bar{\chi}(\mathcal{H}) \leq X - 2$, so $\bar{\chi}(\mathcal{H}) \neq$
voloshin-hypergraph_F01979	254	■ 0.899	$\alpha_c(\mathcal{H})$	$\alpha_c(\mathcal{H})$	$\alpha_c(\mathcal{H})$.
voloshin-hypergraph_F01980	254	■ 0.997	$\mathcal{H} = (X, C), X = n \geq 3, r \geq 2$	$\mathcal{H} = (X, C), X = n \geq 3, r \geq 2$	Definition 10.8.3 An r -uniform C -hypergraph $\mathcal{H} = (X, C)$, $ X = n \geq 3$, $r \geq 2$, is called
voloshin-hypergraph_F01981	254	■ 0.844	C_n^r	C_n^r	a cycloid and denoted by C_n^r if $X = \{0, 1, \dots, n - 1\}$ and $C = \{\{i, i + 1 \pmod n\}, \dots, i + r -$
voloshin-hypergraph_F01982	254	■ 0.844	$X = \{0, 1, \dots, n - 1\}$	$X = \{0, 1, \dots, n - 1\}$	a cycloid and denoted by C_n^r if $X = \{0, 1, \dots, n - 1\}$ and $C = \{\{i, i + 1 \pmod n\}, \dots, i + r -$
voloshin-hypergraph_F01983	254	■ 0.844	$C = \{\{i, i + 1 \pmod n\}, \dots, i + r -$	$C = \{\{i, i + 1 \pmod n\}, \dots, i + r -$	a cycloid and denoted by C_n^r if $X = \{0, 1, \dots, n - 1\}$ and $C = \{\{i, i + 1 \pmod n\}, \dots, i + r -$
voloshin-hypergraph_F01984	254	■ 1.000	$1 \pmod n\} : i = 0, 1, \dots, n - 1\}$	$1 \pmod n\} : i = 0, 1, \dots, n - 1\}$	$1 \pmod n\} : i = 0, 1, \dots, n - 1\}$.
voloshin-hypergraph_F01985	254	■ 1.000	$C_n = (X, E)$	$C_n = (X, E)$	In other words, one can say that for a cycloid there exists a host graph $C_n = (X, E)$
voloshin-hypergraph_F01986	254	■ 1.000	$r - 1$	$r - 1$	paths of length $r - 1$ on C_n . Thus, the usual cycle C_n (considered as a C -hypergraph) is
voloshin-hypergraph_F01987	254	■ 1.000	C_n^2	C_n^2	C_n^2 for any $n \geq 3$. Note that the example given for Algorithm 10.4.1, see Figure 10.9, is the
voloshin-hypergraph_F01988	254	■ 0.614	C_5^3	C_5^3	cycloid C_5^3 considered as the mixed hypergraph $(X, C_5^3, \emptyset)$.
voloshin-hypergraph_F01989	254	■ 0.614	$(X, C_5^3, \emptyset)$	$(X, C_5^3, \emptyset)$	cycloid C_5^3 considered as the mixed hypergraph $(X, C_5^3, \emptyset)$.

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voloshin-hypergraph_ F01990	254	■ 0.995	$C_{\{n\}^{\{r\}}}=(X, \mathcal{C}), 3 \leq r \leq n$	$C_n^r = (X, \mathcal{C}), 3 \leq r \leq n$	Theorem 10.8.4 A cycloid $C_n^r = (X, \mathcal{C}), 3 \leq r \leq n$, is perfect if and only if $2r \geq n + 2$.
voloshin-hypergraph_ F01991	254	■ 0.995	$2r \geq n + 2$	$2r \geq n + 2$	Theorem 10.8.4 A cycloid $C_n^r = (X, \mathcal{C}), 3 \leq r \leq n$, is perfect if and only if $2r \geq n + 2$.
voloshin-hypergraph_ F01992	255	■ 1.000	$r=n$	$r = n$	Case 1: $2r \geq n + 2$. Since for $r = n$ the theorem is evident, let $r \leq n - 1$. Hence,
voloshin-hypergraph_ F01993	255	■ 1.000	$r \leq n-1$	$r \leq n - 1$	Case 1: $2r \geq n + 2$. Since for $r = n$ the theorem is evident, let $r \leq n - 1$. Hence,
voloshin-hypergraph_ F01994	255	■ 1.000	$ \left C_{\{i\}} \cap C_{\{j\}}\right \right \geq 2, i, j \in I$	$ C_i \cap C_j \geq 2, i, j \in I$	$ C_i \cap C_j \geq 2, i, j \in I$. Let x_1 be a neighbor of x_2 , x_3 be a neighbor of x_4 , and the pair $\{x_1, x_2\}$
voloshin-hypergraph_ F01995	255	■ 1.000	$\left\{x_{\{3\}}, x_{\{4\}}\right\}$	$\{x_3, x_4\}$	be opposite to the pair $\{x_3, x_4\}$ on the host cycle of C_n^r . The inequality $2r \geq n + 2$ implies
voloshin-hypergraph_ F01996	255	■ 1.000	$C_{\{i\}} \in \mathcal{C}$	$C_i \in \mathcal{C}$	that for any $C_i \in \mathcal{C}$ either $ C_i \cap \{x_1, x_2\} \geq 2$ or $ C_i \cap \{x_3, x_4\} \geq 2$. Thus, C_n^r is the union
voloshin-hypergraph_ F01997	255	■ 1.000	$\left \left C_{\{i\}} \cap \left\{x_{\{1\}}, x_{\{2\}}\right\}\right \right \geq 2$	$ C_i \cap \{x_1, x_2\} \geq 2$	that for any $C_i \in \mathcal{C}$ either $ C_i \cap \{x_1, x_2\} \geq 2$ or $ C_i \cap \{x_3, x_4\} \geq 2$. Thus, C_n^r is the union
voloshin-hypergraph_ F01998	255	■ 1.000	$\left \left C_{\{i\}} \cap \left\{x_{\{3\}}, x_{\{4\}}\right\}\right \right \geq 2$	$ C_i \cap \{x_3, x_4\} \geq 2$	that for any $C_i \in \mathcal{C}$ either $ C_i \cap \{x_1, x_2\} \geq 2$ or $ C_i \cap \{x_3, x_4\} \geq 2$. Thus, C_n^r is the union
voloshin-hypergraph_ F01999	255	■ 1.000	$\bar{\chi}\left(C_{\{n\}^{\{r\}}}\right)<n-1$	$\bar{\chi}\left(C_n^r\right) < n - 1$	vertices belonging to all $\mathcal{C}$ -edges, so $\bar{\chi}\left(C_n^r\right) < n - 1$. On the other hand, we can color x_1, x_2
voloshin-hypergraph_ F02000	255	■ 1.000	$x_{\{3\}}, x_{\{4\}}$	x_3, x_4	with the first color, x_3, x_4 with the second color and all the remaining $n - 4$ vertices with all
voloshin-hypergraph_ F02001	255	■ 1.000	$n-4$	$n - 4$	with the first color, x_3, x_4 with the second color and all the remaining $n - 4$ vertices with all
voloshin-hypergraph_ F02002	255	■ 1.000	$\bar{\chi}\left(C_{\{n\}^{\{r\}}}\right)=n-2$	$\bar{\chi}\left(C_n^r\right) = n - 2$	different colors. Therefore, $\bar{\chi}\left(C_n^r\right) = n - 2$. Since C_n^r is not a $\mathcal{C}$ -monostar, and vertices x_1
voloshin-hypergraph_ F02003	255	■ 1.000	$\tau\left(C_{\{n\}^{\{r\}}}\right)=2$	$\tau\left(C_n^r\right) = 2$	and x_3 form a minimum transversal, $\tau\left(C_n^r\right) = 2$. It implies $\alpha\left(C_n^r\right) = n - 2 = \bar{\chi}\left(C_n^r\right)$. At last,
voloshin-hypergraph_ F02004	255	■ 1.000	$\alpha\left(C_{\{n\}^{\{r\}}}\right)=n-2=\bar{\chi}\left(C_{\{n\}^{\{r\}}}\right)$	$\alpha\left(C_n^r\right) = n - 2 = \bar{\chi}\left(C_n^r\right)$	and x_3 form a minimum transversal, $\tau\left(C_n^r\right) = 2$. It implies $\alpha\left(C_n^r\right) = n - 2 = \bar{\chi}\left(C_n^r\right)$. At last,
voloshin-hypergraph_ F02005	255	■ 1.000	$2r=n+1$	$2r = n + 1$	Case 2: $2r = n + 1$. Observe that C_n^r does not contain any $\mathcal{C}$ -monostar as an induced
voloshin-hypergraph_ F02006	255	■ 1.000	$r=3,4$	$r = 3, 4$	subhypergraph. For $r = 3, 4$, it can be verified directly that C_n^r is not $\mathcal{C}$ -perfect. Hence, let
voloshin-hypergraph_ F02007	255	■ 1.000	$r \geq 5$	$r \geq 5$	$r \geq 5$. Since $\tau\left(C_n^r\right) = 2$, it follows that $\alpha\left(C_n^r\right) = n - 2$. We show that $\bar{\chi}\left(C_n^r\right) < n - 2$. For a
voloshin-hypergraph_ F02008	255	■ 1.000	$\alpha\left(C_{\{n\}^{\{r\}}}\right)=n-2$	$\alpha\left(C_n^r\right) = n - 2$	$r \geq 5$. Since $\tau\left(C_n^r\right) = 2$, it follows that $\alpha\left(C_n^r\right) = n - 2$. We show that $\bar{\chi}\left(C_n^r\right) < n - 2$. For a
voloshin-hypergraph_ F02009	255	■ 1.000	$\bar{\chi}\left(C_{\{n\}^{\{r\}}}\right)<n-2$	$\bar{\chi}\left(C_n^r\right) < n - 2$	$r \geq 5$. Since $\tau\left(C_n^r\right) = 2$, it follows that $\alpha\left(C_n^r\right) = n - 2$. We show that $\bar{\chi}\left(C_n^r\right) < n - 2$. For a
voloshin-hypergraph_ F02010	255	■ 0.988	$x_{\{1\}}, x_{\{2\}} \in X$	$x_1, x_2 \in X$	that $x_1, x_2 \in X$ represent a center of the first $\mathcal{C}$ -bistar, and x_3, x_4 a center of the second $\mathcal{C}$ -
voloshin-hypergraph_ F02011	255	■ 1.000	$n_{\{i\ j\}}$	n_{ij}	bistar, and, moreover, x_1, x_2, x_3, x_4 are placed on C_n^r clockwise in this order. Let n_{ij} be the
voloshin-hypergraph_ F02012	255	■ 1.000	$x_{\{j\}}, i, j=1,2,3,4$	$x_j, i, j = 1, 2, 3, 4$	number of vertices between x_i and $x_j, i, j = 1, 2, 3, 4$. If $n_{12} = 0$ and $n_{34} = 0$, then a $\mathcal{C}$ -edge
voloshin-hypergraph_ F02013	255	■ 1.000	$n_{\{12\}}=0$	$n_{12} = 0$	number of vertices between x_i and $x_j, i, j = 1, 2, 3, 4$. If $n_{12} = 0$ and $n_{34} = 0$, then a $\mathcal{C}$ -edge
voloshin-hypergraph_ F02014	255	■ 1.000	$n_{\{34\}}=0$	$n_{34} = 0$	number of vertices between x_i and $x_j, i, j = 1, 2, 3, 4$. If $n_{12} = 0$ and $n_{34} = 0$, then a $\mathcal{C}$ -edge
voloshin-hypergraph_ F02015	255	■ 0.942	$ \left C \cap \left\{x_{\{1\}}, x_{\{2\}}\right\}\right \right \leq 1$	$ C \cap \{x_1, x_2\} \leq 1$	$C \in \mathcal{C}$ may be found easily such that $ C \cap \{x_1, x_2\} \leq 1$ and $ C \cap \{x_3, x_4\} \leq 1$; so, assume
voloshin-hypergraph_ F02016	255	■ 0.942	$ \left C \cap \left\{x_{\{3\}}, x_{\{4\}}\right\}\right \right \leq 1$	$ C \cap \{x_3, x_4\} \leq 1$	$C \in \mathcal{C}$ may be found easily such that $ C \cap \{x_1, x_2\} \leq 1$ and $ C \cap \{x_3, x_4\} \leq 1$; so, assume

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F02017	255	■ 1.000	$n_{12}+n_{34} \geq 1$	$n_{12} + n_{34} \geq 1$	that $n_{12} + n_{34} \geq 1$. We have $n_{12} + n_{23} + n_{34} + n_{41} + 4 = n = 2r - 1$. Since any C -edge must
voloshin-hypergraph_F02018	255	■ 1.000	$n_{12}+n_{23}+n_{34}+n_{41}+4=n=2r-1$	$n_{12} + n_{23} + n_{34} + n_{41} + 4 = n = 2r - 1$	that $n_{12} + n_{34} \geq 1$. We have $n_{12} + n_{23} + n_{34} + n_{41} + 4 = n = 2r - 1$. Since any C -edge must
voloshin-hypergraph_F02019	255	■ 1.000	$2r-1+n_{12}+n_{34}<2r$	$2r - 1 + n_{12} + n_{34} < 2r$	By summing the above two inequalities, we have $2r - 1 + n_{12} + n_{34} < 2r$ which gives the
voloshin-hypergraph_F02020	255	■ 1.000	$r < r$	$r < r$	contradiction $r < r$.
voloshin-hypergraph_F02021	255	■ 1.000	C_{2r-1}^r	C_{2r-1}^r	may easily be found such that $ C \cap \{x_1, x_2\} \leq 1$ and $ C \cap \{x_3, x_4\} \leq 1$. Consequently, C_{2r-1}^r
voloshin-hypergraph_F02022	255	■ 1.000	$r \geq 3$	$r \geq 3$	cannot be the union of two C -bistars with disjoint centers for any $r \geq 3$.
voloshin-hypergraph_F02023	255	■ 1.000	$x_{\{1\}}, x_{\{2\}}, x_{\{3\}}$	x_1, x_2, x_3	clockwise in this order. Then x_1, x_2, x_3 is a minimum C -bitransversal. Let n_{ij} be the number
voloshin-hypergraph_F02024	255	■ 1.000	$x_{\{j\}}, i, j=1,2,3$	$x_j, i, j = 1, 2, 3$	of vertices between x_i and x_j , $i, j = 1, 2, 3$. We have $n_{12} + n_{23} + n_{31} + 3 = n = 2r - 1$. Since
voloshin-hypergraph_F02025	255	■ 1.000	$n_{12}+n_{23}+n_{31}+3=n=2r-1$	$n_{12} + n_{23} + n_{31} + 3 = n = 2r - 1$	of vertices between x_i and x_j , $i, j = 1, 2, 3$. We have $n_{12} + n_{23} + n_{31} + 3 = n = 2r - 1$. Since
voloshin-hypergraph_F02026	256	■ 1.000	$r < 5$	$r < 5$	Summing these inequalities implies $r < 5$, a contradiction. Consequently, C_{2r-1}^r cannot be
voloshin-hypergraph_F02027	256	■ 1.000	$2r \leq n$	$2r \leq n$	Case 3: $2r \leq n$. In this case, all C -edges containing a fixed vertex form an induced
voloshin-hypergraph_F02028	256	■ 0.995	$\mathcal{H}^r = (X, \mathcal{C})$	$\mathcal{H}^r = (X, \mathcal{C})$	$\mathcal{H}^r = (X, \mathcal{C})$ in the following way. Let
voloshin-hypergraph_F02029	256	■ 1.000	$2r+2$	$2r + 2$	and the vertices be drawn in the plane in cyclic clockwise order. The edge family has $2r + 2$
voloshin-hypergraph_F02030	256	■ 1.000	$2r$	$2r$	Thus C_1 contains consecutive vertices 1 and 2, skips vertex 3 and contains all (except $2r$)
voloshin-hypergraph_F02031	256	■ 0.978	C_{2r}	C_{2r}	in the same direction around the cycle, and so on, $\dots$, edge C_{2r} is obtained when shifting
voloshin-hypergraph_F02032	256	■ 1.000	C_{2r-1}	C_{2r-1}	C_{2r-1} by 1 around the cycle. In other words, edges $C_2, C_3, \dots, C_{2r}$ are all copies of C_1
voloshin-hypergraph_F02033	256	■ 1.000	$C_{\{2\}}, C_{\{3\}}, \dots, C_{\{2r\}}$	$C_2, C_3, \dots, C_{2r}$	C_{2r-1} by 1 around the cycle. In other words, edges $C_2, C_3, \dots, C_{2r}$ are all copies of C_1
voloshin-hypergraph_F02034	256	■ 1.000	$C_{\{o\}}$	C_o	obtained by shifting around the cycle. At last, edge C_o contains all odd vertices, and edge
voloshin-hypergraph_F02035	256	■ 0.950	$C_{\{e\}}$	C_e	C_e contains all even vertices. Clearly, $\mathcal{H}^r$ is an r -uniform $(r+1)$ -regular C -hypergraph on
voloshin-hypergraph_F02036	256	■ 0.950	$\mathcal{H}^r$	$\mathcal{H}^r$	C_e contains all even vertices. Clearly, $\mathcal{H}^r$ is an r -uniform $(r+1)$ -regular C -hypergraph on
voloshin-hypergraph_F02037	256	■ 0.950	$r+1$	$r + 1$	C_e contains all even vertices. Clearly, $\mathcal{H}^r$ is an r -uniform $(r+1)$ -regular C -hypergraph on
voloshin-hypergraph_F02038	256	■ 0.997	$r=3$	$r = 3$	For $r = 3$ the example is shown in Figure 10.24. To avoid confusion, edges $C_2, \dots, C_6$
voloshin-hypergraph_F02039	256	■ 0.997	$C_{\{2\}}, \dots, C_{\{6\}}$	$C_2, \dots, C_6$	For $r = 3$ the example is shown in Figure 10.24. To avoid confusion, edges $C_2, \dots, C_6$
voloshin-hypergraph_F02040	256	■ 0.997	$2r-1$	$2r - 1$	Theorem 10.8.5 (Kráľ', 2003) <i>The C-hypergraph $\mathcal{H}^r$ contains no cycloid on $2r - 1$ ver-</i>
voloshin-hypergraph_F02041	257	■ 0.901	$\alpha(\mathcal{H}_C) = n-1$	$\alpha(\mathcal{H}_C) = n - 1$	$\alpha(\mathcal{H}_C) = n - 1$. If $\mathcal{H}$ contains only one C -edge, then it is uncolorable and $\bar{\chi}(\mathcal{H}) =$
voloshin-hypergraph_F02042	257	■ 0.901	$\bar{\chi}(\mathcal{H}) =$	$\bar{\chi}(\mathcal{H}) =$	$\alpha(\mathcal{H}_C) = n - 1$. If $\mathcal{H}$ contains only one C -edge, then it is uncolorable and $\bar{\chi}(\mathcal{H}) =$
voloshin-hypergraph_F02043	257	■ 0.960	$\alpha_C(\mathcal{H})$	$\alpha_C(\mathcal{H})$	show that we can color $\mathcal{H}$ with $\alpha_C(\mathcal{H})$ colors.
voloshin-hypergraph_F02044	257	■ 0.998	$\mathcal{H}_C = (X, \mathcal{C}, \emptyset)$	$\mathcal{H}_C = (X, \mathcal{C}, \emptyset)$	By Theorem 8.1.2 the hypertree $\mathcal{H}_C = (X, \mathcal{C}, \emptyset)$ satisfies the Helly property; i.e., for any

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F02045	257	■ 1.000	$\mathcal{C}_1 \subseteq \mathcal{C}$	$C_1 \subseteq \mathcal{C}$	subfamily $C_1 \subseteq \mathcal{C}$ with $C \cap C' \neq \emptyset$ for each pair $C, C' \in C_1$, it follows that
voloshin-hypergraph_F02046	257	■ 1.000	$C \cap C^{\prime} \neq \emptyset$	$C \cap C' \neq \emptyset$	subfamily $C_1 \subseteq \mathcal{C}$ with $C \cap C' \neq \emptyset$ for each pair $C, C' \in C_1$, it follows that
voloshin-hypergraph_F02047	257	■ 1.000	$C, C^{\prime} \in \mathcal{C}_1$	$C, C' \in C_1$	subfamily $C_1 \subseteq \mathcal{C}$ with $C \cap C' \neq \emptyset$ for each pair $C, C' \in C_1$, it follows that
voloshin-hypergraph_F02048	257	■ 1.000	$ \bigcap_{C \in \mathcal{C}_1} C \geq 2$	$\left \bigcap_{C \in C_1} C \right \geq 2$	In fact, for every such subfamily $ \bigcap_{C \in C_1} C \geq 2$ because otherwise C_1 forms an eclipse. Thus
voloshin-hypergraph_F02049	257	■ 1.000	$\mathcal{C}_1$	C_1	In fact, for every such subfamily $ \bigcap_{C \in C_1} C \geq 2$ because otherwise C_1 forms an eclipse. Thus
voloshin-hypergraph_F02050	257	■ 0.989	$ C \cap C^{\prime} \geq 2$	$ C \cap C' \geq 2$	$\mathcal{H}_C$ is a bi-Helly hypergraph, and moreover, $C \cap C' \neq \emptyset$ implies $ C \cap C' \geq 2$ for any $C, C' \in \mathcal{C}$.
voloshin-hypergraph_F02051	257	■ 0.989	$C, C^{\prime} \in \mathcal{C}$	$C, C' \in \mathcal{C}$	$\mathcal{H}_C$ is a bi-Helly hypergraph, and moreover, $C \cap C' \neq \emptyset$ implies $ C \cap C' \geq 2$ for any $C, C' \in \mathcal{C}$.
voloshin-hypergraph_F02052	257	■ 0.815	$L(\mathcal{H}_C) = (\mathcal{C}, \mathcal{E})$	$L(\mathcal{H}_C) = (\mathcal{C}, \mathcal{E})$	line graph of $\mathcal{H}_C$, that is the graph $L(\mathcal{H}_C) = (\mathcal{C}, \mathcal{E})$, where $(C, C') \in \mathcal{E} \Leftrightarrow C \cap C' \neq \emptyset$ in $\mathcal{H}$.
voloshin-hypergraph_F02053	257	■ 0.815	$(C, C^{\prime}) \in \mathcal{E} \Leftrightarrow C \cap C^{\prime} \neq \emptyset$	$(C, C') \in \mathcal{E} \Leftrightarrow C \cap C' \neq \emptyset$	line graph of $\mathcal{H}_C$, that is the graph $L(\mathcal{H}_C) = (\mathcal{C}, \mathcal{E})$, where $(C, C') \in \mathcal{E} \Leftrightarrow C \cap C' \neq \emptyset$ in $\mathcal{H}$.
voloshin-hypergraph_F02054	257	■ 0.718	$K_1, K_2, \dots, K_t$	$K_1, K_2, \dots, K_t$	mum transversal of $\mathcal{H}_C$. Let $K_1, K_2, \dots, K_t$ be the cliques of such a minimum covering of L .
voloshin-hypergraph_F02055	257	■ 0.930	$\tau(\mathcal{H}_C) = t$	$\tau(\mathcal{H}_C) = t$	Thus $\tau(\mathcal{H}_C) = t$ and $\alpha_C(\mathcal{H}) = X - t$.
voloshin-hypergraph_F02056	257	■ 0.930	$\alpha_C(\mathcal{H}) = X - t$	$\alpha_C(\mathcal{H}) = X - t$	Thus $\tau(\mathcal{H}_C) = t$ and $\alpha_C(\mathcal{H}) = X - t$.
voloshin-hypergraph_F02057	258	■ 1.000	$\{x_i, y_i\}$	$\{x_i, y_i\}$	Let the pair $\{x_i, y_i\}$ be the C -bitransversal corresponding to the clique K_i , $i = 1, 2, \dots, t$
voloshin-hypergraph_F02058	258	■ 1.000	$K_i, i=1,2, \dots, t$	$K_i, i = 1, 2, \dots, t$	Let the pair $\{x_i, y_i\}$ be the C -bitransversal corresponding to the clique K_i , $i = 1, 2, \dots, t$
voloshin-hypergraph_F02059	258	■ 1.000	$\{x_i, y_i\} \notin \mathcal{D}$	$\{x_i, y_i\} \notin \mathcal{D}$	such that $\{x_i, y_i\} \notin \mathcal{D}$. They all are different because $K_i \neq K_j$ implies that $\{x_i, y_i\} \cap \{x_j, y_j\} =$
voloshin-hypergraph_F02060	258	■ 1.000	$K_i \neq K_j$	$K_i \neq K_j$	such that $\{x_i, y_i\} \notin \mathcal{D}$. They all are different because $K_i \neq K_j$ implies that $\{x_i, y_i\} \cap \{x_j, y_j\} =$
voloshin-hypergraph_F02061	258	■ 1.000	$\{x_i, y_i\} \cap \{x_j, y_j\} =$	$\{x_i, y_i\} \cap \{x_j, y_j\} =$	such that $\{x_i, y_i\} \notin \mathcal{D}$. They all are different because $K_i \neq K_j$ implies that $\{x_i, y_i\} \cap \{x_j, y_j\} =$
voloshin-hypergraph_F02062	258	■ 0.961	x_1, y_1	x_1, y_1	$\emptyset$. Color the vertices x_1, y_1 with the first color, x_2, y_2 with the second color, $\dots, x_t, y_t$ with
voloshin-hypergraph_F02063	258	■ 0.961	x_2, y_2	x_2, y_2	$\emptyset$. Color the vertices x_1, y_1 with the first color, x_2, y_2 with the second color, $\dots, x_t, y_t$ with
voloshin-hypergraph_F02064	258	■ 0.961	$\dots, x_t, y_t$	$\dots, x_t, y_t$	$\emptyset$. Color the vertices x_1, y_1 with the first color, x_2, y_2 with the second color, $\dots, x_t, y_t$ with
voloshin-hypergraph_F02065	258	■ 0.536	$t+1, t+2, \dots, X - t$	$t + 1, t + 2, \dots, X - t$	$t + 1, t + 2, \dots, X - t$. Thus, we obtain a proper coloring of $\mathcal{H}$ with $ X - t = \alpha_C(\mathcal{H})$
voloshin-hypergraph_F02066	258	■ 0.536	$ X - t = \alpha_C(\mathcal{H})$	$ X - t = \alpha_C(\mathcal{H})$	$t + 1, t + 2, \dots, X - t$. Thus, we obtain a proper coloring of $\mathcal{H}$ with $ X - t = \alpha_C(\mathcal{H})$
voloshin-hypergraph_F02067	258	■ 0.447	$v(\mathcal{H}_C)$	$v(\mathcal{H}_C)$	Corollary 10.8.1 If $\mathcal{H} = (X, C, \mathcal{D})$ is a perfect mixed hypertree, $v(\mathcal{H}_C)$ and $\tau(\mathcal{H}_C)$ are the
voloshin-hypergraph_F02068	258	■ 0.999	$X = \{1, 2, 3, 4, 5\}, C = \{C_1, C_2, C_3\}, C_1 = \{1, 2, 3\}, C_2 = \{2, 3, 4, 5\}, C_3 = \{1, 4, 5\}$	$X = \{1, 2, 3, 4, 5\}, C = \{C_1, C_2, C_3\}, C_1 = \{1, 2, 3\}, C_2 = \{2, 3, 4, 5\}, C_3 = \{1, 4, 5\}$	$X = \{1, 2, 3, 4, 5\}, C = \{C_1, C_2, C_3\}, C_1 = \{1, 2, 3\}, C_2 = \{2, 3, 4, 5\}, C_3 = \{1, 4, 5\}$. We
voloshin-hypergraph_F02069	258	■ 1.000	$C_1 \cap C_3 = \{1\}$	$C_1 \cap C_3 = \{1\}$	see that $C_1 \cap C_3 = \{1\}$ and we have a monostar as a partial subhypergraph. However, $\mathcal{H}$ is
voloshin-hypergraph_F02070	258	■ 0.934	$\bar{\chi}(\mathcal{H}) = \alpha_C(\mathcal{H}) = 3$	$\bar{\chi}(\mathcal{H}) = \alpha_C(\mathcal{H}) = 3$	perfect with $\bar{\chi}(\mathcal{H}) = \alpha_C(\mathcal{H}) = 3$. On the other hand, the cycloid C_3^3 contains monostars as
voloshin-hypergraph_F02071	259	■ 1.000	$\tau_2(\mathcal{H}_1)$	$\tau_2(\mathcal{H}_1)$	$\tau_2(\mathcal{H}_1)$ and $\tau_2(\mathcal{H}_2)$.

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voloshin-hypergraph_F02072	259	■ 1.000	$\tau_2(\mathcal{H}_2)$	$\tau_2(\mathcal{H}_2)$	$\tau_2(\mathcal{H}_1)$ and $\tau_2(\mathcal{H}_2)$.
voloshin-hypergraph_F02073	260	■ 1.000	$S(\mathcal{H})$	$S(\mathcal{H})$	denoted by $S(\mathcal{H})$. In other words, $S(\mathcal{H})$ is the set of indices k such that $r_k(\mathcal{H}) > 0$.
voloshin-hypergraph_F02074	260	■ 1.000	$r_k(\mathcal{H}) > 0$	$r_k(\mathcal{H}) > 0$	denoted by $S(\mathcal{H})$. In other words, $S(\mathcal{H})$ is the set of indices k such that $r_k(\mathcal{H}) > 0$.
voloshin-hypergraph_F02075	260	■ 1.000	$\{a, b\}$	$\{a, b\}$	“inflation” of a $\mathcal{D}$ -edge. Let us have a $\mathcal{D}$ -edge $\{a, b\}$, see Figure 10.27. We double the
voloshin-hypergraph_F02076	260	■ 1.000	a, a'	a, a'	vertices, i.e. replace vertices a, b with the pairs a, a' and b, b' respectively. Thus, $\mathcal{D}$ -edge
voloshin-hypergraph_F02077	260	■ 1.000	b, b'	b, b'	vertices, i.e. replace vertices a, b with the pairs a, a' and b, b' respectively. Thus, $\mathcal{D}$ -edge
voloshin-hypergraph_F02078	260	■ 0.997	$\{a, a', b, b'\}$	$\{a, a', b, b'\}$	$\{a, b\}$ becomes a quadruple, i.e. $\mathcal{D}$ -edge $\{a, a', b, b'\}$.
voloshin-hypergraph_F02079	260	■ 1.000	a, a', b, b'	a, a', b, b'	Next, we add all triples on vertices a, a', b, b' as $\mathcal{C}$ -edges. If $\mathcal{H}$ denotes the mixed
voloshin-hypergraph_F02080	260	■ 1.000	$\mathcal{H} = (X, K_4^3, K_4^4)$	$\mathcal{H} = (X, K_4^3, K_4^4)$	hypergraph obtained, then simply $\mathcal{H} = (X, K_4^3, K_4^4)$ where $X = \{a, a', b, b'\}$. The inflation
voloshin-hypergraph_F02081	260	■ 1.000	$X = \{a, a', b, b'\}$	$X = \{a, a', b, b'\}$	hypergraph obtained, then simply $\mathcal{H} = (X, K_4^3, K_4^4)$ where $X = \{a, a', b, b'\}$. The inflation
voloshin-hypergraph_F02082	261	■ 0.949	a'	a'	when vertex a has the same color as a' (color 1), and vertex b has the same color as b' (color
voloshin-hypergraph_F02083	261	■ 0.949	b'	b'	when vertex a has the same color as a' (color 1), and vertex b has the same color as b' (color
voloshin-hypergraph_F02084	261	■ 0.999	$\{a, a'\}$	$\{a, a'\}$	Consider now the proper colorings of $\mathcal{H}$. It has four pairs of vertices, namely, $\{a, a'\}$,
voloshin-hypergraph_F02085	261	■ 0.998	$\{b, b'\}, \{c, c'\}$	$\{b, b'\}, \{c, c'\}$	$\{b, b'\}$, $\{c, c'\}$, and $\{d, d'\}$. In any proper coloring of $\mathcal{H}$, if there is a pair, say $\{a, a'\}$,
voloshin-hypergraph_F02086	261	■ 0.998	$\{d, d'\}$	$\{d, d'\}$	$\{b, b'\}$, $\{c, c'\}$, and $\{d, d'\}$. In any proper coloring of $\mathcal{H}$, if there is a pair, say $\{a, a'\}$,
voloshin-hypergraph_F02087	261	■ 1.000	$\{a, a', c\}$	$\{a, a', c\}$	say c , immediately results in a polychromatic triple $\{a, a', c\}$ which is a $\mathcal{C}$ -edge. So, it
voloshin-hypergraph_F02088	261	■ 0.982	$r_2 \geq 2$	$r_2 \geq 2$	strict 2-colorings ($r_2 \geq 2$), a strict 4-coloring ($r_4 = 1$) and no strict 3-coloring ($r_3 = 0$), i.e.
voloshin-hypergraph_F02089	261	■ 0.982	$r_4 = 1$	$r_4 = 1$	strict 2-colorings ($r_2 \geq 2$), a strict 4-coloring ($r_4 = 1$) and no strict 3-coloring ($r_3 = 0$), i.e.
voloshin-hypergraph_F02090	261	■ 0.982	$r_3 = 0$	$r_3 = 0$	strict 2-colorings ($r_2 \geq 2$), a strict 4-coloring ($r_4 = 1$) and no strict 3-coloring ($r_3 = 0$), i.e.
voloshin-hypergraph_F02091	262	■ 0.991	$\{a, c, c'\}, \{a, d, d'\}, \{b, c, c'\}, \{b, d, d'\}$	$\{a, c, c'\}, \{a, d, d'\}, \{b, c, c'\}, \{b, d, d'\}$	inflating $\mathcal{D}$ -edge $\{c, d\}$ and adding triples $\{a, c, c'\}, \{a, d, d'\}, \{b, c, c'\}, \{b, d, d'\}$ as $\mathcal{C}$ -
voloshin-hypergraph_F02092	262	■ 1.000	$\mathcal{H}_{2,4}$	$\mathcal{H}_{2,4}$	denoted by $\mathcal{H}_{2,4}$ where index means $S(\mathcal{H}) = \{2, 4\}$.
voloshin-hypergraph_F02093	262	■ 1.000	$S(\mathcal{H}) = \{2, 4\}$	$S(\mathcal{H}) = \{2, 4\}$	denoted by $\mathcal{H}_{2,4}$ where index means $S(\mathcal{H}) = \{2, 4\}$.
voloshin-hypergraph_F02094	262	■ 1.000	s, t	s, t	The smallest mixed hypergraphs with gaps. Let s, t be two integer numbers such
voloshin-hypergraph_F02095	262	■ 1.000	$2 \leq s \leq t - 2$	$2 \leq s \leq t - 2$	that $2 \leq s \leq t - 2$. In this subsection we construct a mixed hypergraph $\mathcal{H}_{s,t}$ with feasible
voloshin-hypergraph_F02096	262	■ 1.000	$\mathcal{H}_{s,t}$	$\mathcal{H}_{s,t}$	that $2 \leq s \leq t - 2$. In this subsection we construct a mixed hypergraph $\mathcal{H}_{s,t}$ with feasible
voloshin-hypergraph_F02097	262	■ 0.998	$S(\mathcal{H}_{s,t}) = \{s, t\}$	$S(\mathcal{H}_{s,t}) = \{s, t\}$	set $S(\mathcal{H}_{s,t}) = \{s, t\}$ and prove that $\mathcal{H}_{s,t}$ has the fewest vertices among all s -colorable mixed
voloshin-hypergraph_F02098	262	■ 0.998	s	s	set $S(\mathcal{H}_{s,t}) = \{s, t\}$ and prove that $\mathcal{H}_{s,t}$ has the fewest vertices among all s -colorable mixed

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voloshin-hypergraph_ F02099	262	■ 1.000	t-1	$t - 1$	hypergraphs with a gap at $t - 1$; this minimum number of vertices is $2t - s$.
voloshin-hypergraph_ F02100	262	■ 1.000	2 t-s	$2t - s$	hypergraphs with a gap at $t - 1$; this minimum number of vertices is $2t - s$.
voloshin-hypergraph_ F02101	262	■ 1.000	2 t-2	$2t - 2$	We begin with an explicit construction of a mixed hypergraph with $2t - 2$ vertices and
voloshin-hypergraph_ F02102	262	■ 0.723	$\{2, t\}$	$\{2, t\}$	feasible set $\{2, t\}$. As usual, K_n is viewed as the mixed hypergraph $(X, \emptyset, \binom{X}{2})$; trivially,
voloshin-hypergraph_ F02103	262	■ 0.723	$\left(X, \emptyset, \binom{X}{2}\right)$	$\left(X, \emptyset, \binom{X}{2}\right)$	feasible set $\{2, t\}$. As usual, K_n is viewed as the mixed hypergraph $(X, \emptyset, \binom{X}{2})$; trivially,
voloshin-hypergraph_ F02104	262	■ 0.945	$S(K_n) = \{n\}$	$S(K_n) = \{n\}$	$S(K_n) = \{n\}$. First, we describe the construction informally. Beginning with K_t , expand $t -$
voloshin-hypergraph_ F02105	262	■ 0.945	K_t	K_t	$S(K_n) = \{n\}$. First, we describe the construction informally. Beginning with K_t , expand $t -$
voloshin-hypergraph_ F02106	262	■ 0.945	t-	$t -$	$S(K_n) = \{n\}$. First, we describe the construction informally. Beginning with K_t , expand $t -$
voloshin-hypergraph_ F02107	263	■ 0.924	t=4	$t = 4$	the proofs. The smallest instance is for $t = 4$; from K_4 , a 6-vertex mixed hypergraph with
voloshin-hypergraph_ F02108	263	■ 0.999	$[m] = \{1, \dots, m\}$	$[m] = \{1, \dots, m\}$	spectrum $\{2, 4\}$ is produced. Let $[m] = \{1, \dots, m\}$.
voloshin-hypergraph_ F02109	263	■ 1.000	$\mathcal{H}_{2,t}$	$\mathcal{H}_{2,t}$	Construction 1. Define a hypergraph $\mathcal{H}_{2,t}$ with vertex set
voloshin-hypergraph_ F02110	263	■ 1.000	$x_r a_i b_i$	$x_r a_i b_i$	Let T be the set of triples of the form $x_r a_i b_i$, for $r \in \{1, 2\}$ and $i \in [t - 2]$. Let U be the set
voloshin-hypergraph_ F02111	263	■ 1.000	$r \in \{1, 2\}$	$r \in \{1, 2\}$	Let T be the set of triples of the form $x_r a_i b_i$, for $r \in \{1, 2\}$ and $i \in [t - 2]$. Let U be the set
voloshin-hypergraph_ F02112	263	■ 1.000	$i \in [t - 2]$	$i \in [t - 2]$	Let T be the set of triples of the form $x_r a_i b_i$, for $r \in \{1, 2\}$ and $i \in [t - 2]$. Let U be the set
voloshin-hypergraph_ F02113	263	■ 1.000	$a_i b_i a_j b_j$	$a_i b_i a_j b_j$	of quadruples of the form $a_i b_i a_j b_j$ for $i, j \in [t - 2]$. Let W be the union, over $i, j \in [t - 2]$,
voloshin-hypergraph_ F02114	263	■ 1.000	$i, j \in [t - 2]$	$i, j \in [t - 2]$	of quadruples of the form $a_i b_i a_j b_j$ for $i, j \in [t - 2]$. Let W be the union, over $i, j \in [t - 2]$,
voloshin-hypergraph_ F02115	263	■ 1.000	W	W	of quadruples of the form $a_i b_i a_j b_j$ for $i, j \in [t - 2]$. Let W be the union, over $i, j \in [t - 2]$,
voloshin-hypergraph_ F02116	263	■ 1.000	$\{a_i, b_i, a_j, b_j\}$	$\{a_i, b_i, a_j, b_j\}$	of the sets of four triples contained in $\{a_i, b_i, a_j, b_j\}$. The C -edges in $\mathcal{H}_{2,t}$ are $T \cup W$. The
voloshin-hypergraph_ F02117	263	■ 1.000	$T \cup W$	$T \cup W$	of the sets of four triples contained in $\{a_i, b_i, a_j, b_j\}$. The C -edges in $\mathcal{H}_{2,t}$ are $T \cup W$. The
voloshin-hypergraph_ F02118	263	■ 1.000	$T \cup U \cup \{x_1 x_2\}$	$T \cup U \cup \{x_1 x_2\}$	$\mathcal{D}$ -edges are $T \cup U \cup \{x_1 x_2\}$.
voloshin-hypergraph_ F02119	263	■ 1.000	$S(\mathcal{H}_{2,t}) = \{2, t\}$	$S(\mathcal{H}_{2,t}) = \{2, t\}$	Lemma 10.9.1 $S(\mathcal{H}_{2,t}) = \{2, t\}$.
voloshin-hypergraph_ F02120	263	■ 1.000	$c(a_i) \neq c(b_i)$	$c(a_i) \neq c(b_i)$	Proof. Let c be an arbitrary coloring of $\mathcal{H}_{2,t}$. If $c(a_i) \neq c(b_i)$ for some $i \in [t - 2]$, then the
voloshin-hypergraph_ F02121	263	■ 1.000	a_i	a_i	C -edges in T and W that contain a_i and b_i force all other vertices to have the same color as
voloshin-hypergraph_ F02122	263	■ 1.000	b_i	b_i	C -edges in T and W that contain a_i and b_i force all other vertices to have the same color as
voloshin-hypergraph_ F02123	263	■ 0.987	$c(a_i) = c(x_1) = 1$	$c(a_i) = c(x_1) = 1$	a proper 1-coloring, and setting all $c(a_i) = c(x_1) = 1$ and $c(b_i) = c(x_2) = 2$ completes a
voloshin-hypergraph_ F02124	263	■ 0.987	$c(b_i) = c(x_2) = 2$	$c(b_i) = c(x_2) = 2$	a proper 1-coloring, and setting all $c(a_i) = c(x_1) = 1$ and $c(b_i) = c(x_2) = 2$ completes a
voloshin-hypergraph_ F02125	263	■ 1.000	$c(a_i) = c(b_i)$	$c(a_i) = c(b_i)$	Hence, we may assume that $c(a_i) = c(b_i)$ for all $i \in [t - 2]$. Now the $\mathcal{D}$ -edges in U

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voloshin-hypergraph_F02126	263	■ 1.000	$\left\{x_{\{1\}} x_{\{2\}}\right\}$	$\{x_1 x_2\}$	force these colors to be distinct for all i , and the $\mathcal{D}$ -edges in T along with $\{x_1 x_2\}$ require
voloshin-hypergraph_F02127	263	■ 1.000	$\left(X_{\{1\}}, \mathcal{C}_{\{1\}}, \mathcal{D}_{\{1\}}\right)$	$(X_1, \mathcal{C}_1, \mathcal{D}_1)$	lemma about combining feasible sets. The join of two mixed hypergraphs $(X_1, \mathcal{C}_1, \mathcal{D}_1)$
voloshin-hypergraph_F02128	263	■ 0.739	$\left(X_{\{2\}}, \mathcal{C}_{\{2\}}, \mathcal{D}_{\{2\}}\right)$	$(X_2, \mathcal{C}_2, \mathcal{D}_2)$	and $(X_2, \mathcal{C}_2, \mathcal{D}_2)$ with disjoint vertex sets is the mixed hypergraph $(X, \mathcal{C}, \mathcal{D})$ defined by
voloshin-hypergraph_F02129	263	■ 0.739	$(X, \mathcal{C}, \mathcal{D})$	$(X, \mathcal{C}, \mathcal{D})$	and $(X_2, \mathcal{C}_2, \mathcal{D}_2)$ with disjoint vertex sets is the mixed hypergraph $(X, \mathcal{C}, \mathcal{D})$ defined by
voloshin-hypergraph_F02130	263	■ 0.998	$X=X_{\{1\}} \cup X_{\{2\}}, \mathcal{C}=\mathcal{C}_{\{1\}} \cup \mathcal{C}_{\{2\}}$	$X=X_1 \cup X_2, \mathcal{C}=\mathcal{C}_1 \cup \mathcal{C}_2$	$X=X_1 \cup X_2$, $\mathcal{C}=\mathcal{C}_1 \cup \mathcal{C}_2$, and $\mathcal{D}=\mathcal{D}_1 \cup \mathcal{D}_2 \cup R$, where R is the set of all pairs consisting
voloshin-hypergraph_F02131	263	■ 0.998	$\mathcal{D}=\mathcal{D}_{\{1\}} \cup \mathcal{D}_{\{2\}} \cup R$	$\mathcal{D}=\mathcal{D}_1 \cup \mathcal{D}_2 \cup R$	$X=X_1 \cup X_2$, $\mathcal{C}=\mathcal{C}_1 \cup \mathcal{C}_2$, and $\mathcal{D}=\mathcal{D}_1 \cup \mathcal{D}_2 \cup R$, where R is the set of all pairs consisting
voloshin-hypergraph_F02132	263	■ 0.991	$\left\{i+j: i \in S \left(\mathcal{H}_{\{1\}}\right), j \in S \left(\mathcal{H}_{\{2\}}\right)\right\}$	$\{i+j: i \in S(\mathcal{H}_1), j \in S(\mathcal{H}_2)\}$	and $\mathcal{H}_2$ is $\{i+j: i \in S(\mathcal{H}_1), j \in S(\mathcal{H}_2)\}$.
voloshin-hypergraph_F02133	263	■ 1.000	$z \notin X$	$z \notin X$	$z \notin X$ an elementary shifting of the chromatic spectrum of $\mathcal{H}$. Evidently, elementary
voloshin-hypergraph_F02134	263	■ 1.000	$\left(r_{\{1\}}, r_{\{2\}}, \ldots, r_{\{n\}}\right)$	$(r_1, r_2, \ldots, r_n)$	$(r_1, r_2, \ldots, r_n)$, then $R(\mathcal{H}') = (0, r_1, r_2, \ldots, r_n)$. Joining with the complete graph K_i results
voloshin-hypergraph_F02135	263	■ 1.000	$R \left(\mathcal{H}^{\prime}\right)=\left(0, r_{\{1\}}, r_{\{2\}}, \ldots, r_{\{n\}}\right)$	$R(\mathcal{H}') = (0, r_1, r_2, \ldots, r_n)$	$(r_1, r_2, \ldots, r_n)$, then $R(\mathcal{H}') = (0, r_1, r_2, \ldots, r_n)$. Joining with the complete graph K_i results
voloshin-hypergraph_F02136	263	■ 1.000	$n \geq$	$n \geq$	Theorem 10.9.1 If $\mathcal{H}$ is an s -colorable mixed hypergraph with a gap at $t-1$, then $n \geq$
voloshin-hypergraph_F02137	264	■ 1.000	$n < 2t - s$	$n < 2t - s$	the feasible set larger than $t-1$. If $n < 2t - s$, then using at least t colors requires having
voloshin-hypergraph_F02138	264	■ 0.996	$s+1$	$s+1$	at least $s+1$ color classes of size 1. Two such color classes can be combined to obtain a
voloshin-hypergraph_F02139	264	■ 0.999	K_{s+1}	K_{s+1}	in the feasible set, $\mathcal{H}$ contains K_{s+1} . Now $\mathcal{H}$ is not s -colorable; the contradiction yields
voloshin-hypergraph_F02140	264	■ 1.000	$n \geq 2t - s$	$n \geq 2t - s$	$n > 2t - s$.
voloshin-hypergraph_F02141	264	■ 1.000	$s=2$	$s=2$	For $s=2$, Lemma 10.9.1 shows that Construction 10.9. achieves the bound. For $s > 2$,
voloshin-hypergraph_F02142	264	■ 1.000	$s>2$	$s > 2$	For $s=2$, Lemma 10.9.1 shows that Construction 10.9. achieves the bound. For $s > 2$,
voloshin-hypergraph_F02143	264	■ 0.996	K_{s-2}	K_{s-2}	define $\mathcal{H}_{s,t}$ to be the join of K_{s-2} and $\mathcal{H}_{2,t-s+2}$. By Lemma 10.9.2, the feasible set of $\mathcal{H}_{s,t}$ is
voloshin-hypergraph_F02144	264	■ 0.996	$\mathcal{H}_{\{2\}, t-s+2}$	$\mathcal{H}_{2,t-s+2}$	define $\mathcal{H}_{s,t}$ to be the join of K_{s-2} and $\mathcal{H}_{2,t-s+2}$. By Lemma 10.9.2, the feasible set of $\mathcal{H}_{s,t}$ is
voloshin-hypergraph_F02145	264	■ 1.000	$\{s-2\}+\{2, t-s+2\}=\{s, t\}$	$\{s-2\}+\{2, t-s+2\}=\{s, t\}$	$\{s-2\}+\{2, t-s+2\}=\{s, t\}$. The number of vertices in $\mathcal{H}_{s,t}$ is $s-2+2(t-s+2)-2=$
voloshin-hypergraph_F02146	264	■ 1.000	$s-2+2(t-s+2)-2=$	$s-2+2(t-s+2)-2=$	$\{s-2\}+\{2, t-s+2\}=\{s, t\}$. The number of vertices in $\mathcal{H}_{s,t}$ is $s-2+2(t-s+2)-2=$
voloshin-hypergraph_F02147	264	■ 1.000	$t-1>s \geq 2$	$t-1>s \geq 2$	$t-1$, for some s, t with $t-1>s \geq 2$ (notice that $s \geq 2$, otherwise there are no gaps). Thus
voloshin-hypergraph_F02148	264	■ 1.000	$s \geq 2$	$s \geq 2$	$t-1$, for some s, t with $t-1>s \geq 2$ (notice that $s \geq 2$, otherwise there are no gaps). Thus
voloshin-hypergraph_F02149	264	■ 1.000	$t \geq 4$	$t \geq 4$	$t \geq 4$ and $t-s \geq 2$. By Theorem 10.9.1, $n \geq t+(t-s) \geq 6$.
voloshin-hypergraph_F02150	264	■ 1.000	$t-s \geq 2$	$t-s \geq 2$	$t \geq 4$ and $t-s \geq 2$. By Theorem 10.9.1, $n \geq t+(t-s) \geq 6$.
voloshin-hypergraph_F02151	264	■ 1.000	$n \geq t+(t-s) \geq 6$	$n \geq t+(t-s) \geq 6$	$t \geq 4$ and $t-s \geq 2$. By Theorem 10.9.1, $n \geq t+(t-s) \geq 6$.
voloshin-hypergraph_F02152	264	■ 0.999	$7 \mathcal{C}$	$7\mathcal{C}$	be dropped without changing the spectrum. Thus, 7 $\mathcal{C}$ -edges and 6 $\mathcal{D}$ -edges suffice. The
voloshin-hypergraph_F02153	264	■ 0.999	$6 \mathcal{D}$	$6\mathcal{D}$	be dropped without changing the spectrum. Thus, 7 $\mathcal{C}$ -edges and 6 $\mathcal{D}$ -edges suffice. The

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voloshin-hypergraph_ F02154	264	■ 0.615	$X=\{1,2,3,4,5,6\}, C=\{\{1,2,3\}\}$	$X = \{1, 2, 3, 4, 5, 6\}, C = \{\{1, 2, 3\}\}$	mixed hypergraph $\mathcal{H}_{2,4}$ has the following structure: $X = \{1, 2, 3, 4, 5, 6\}, C = \{\{1, 2, 3\}\},$
voloshin-hypergraph_ F02155	264	■ 0.783	$\{1,4,5\},\{6,2,3\},\{6,4,5\},\{2,3,4\},\{3,4,5\},\{2,4,5\}\}, \mathcal{D}=\{\{1,6\},\{1,2,3\},\{1,4,5\}\},$	$\{1, 4, 5\}, \{6, 2, 3\}, \{6, 4, 5\}, \{2, 3, 4\}, \{3, 4, 5\}, \{2, 4, 5\}\}, \mathcal{D} = \{\{1, 6\}, \{1, 2, 3\}, \{1, 4, 5\}\},$	$\{1, 4, 5\}, \{6, 2, 3\}, \{6, 4, 5\}, \{2, 3, 4\}, \{3, 4, 5\}, \{2, 4, 5\}\}, \mathcal{D} = \{\{1, 6\}, \{1, 2, 3\}, \{1, 4, 5\}\},$
voloshin-hypergraph_ F02156	264	■ 0.906	$X, \emptyset, \emptyset$	$X, \emptyset, \emptyset$	The n -vertex trivial mixed hypergraph $(X, \emptyset, \emptyset)$ has, evidently, feasible set
voloshin-hypergraph_ F02157	264	■ 1.000	$\{1, \ldots, n\}$	$\{1, \ldots, n\}$	$\{1, \ldots, n\}$. We construct mixed hypergraphs realizing all other feasible sets using the trivial
voloshin-hypergraph_ F02158	265	■ 0.990	$X^{\prime}=\bigcup_{v \in X}\left\{v^{-}, v^{+}\right\}$	$X' = \bigcup_{v \in X} \{v^-, v^+\}$	hypergraph $\mathcal{H}' = (X', C', \mathcal{D}')$ with $X' = \bigcup_{v \in X} \{v^-, v^+\}$. For each edge $D \in \mathcal{D}$, we add
voloshin-hypergraph_ F02159	265	■ 0.957	$D^{\prime}=\bigcup_{v \in D}\left\{v^{-}, v^{+}\right\}$	$D' = \bigcup_{v \in D} \{v^-, v^+\}$	$D' = \bigcup_{v \in D} \{v^-, v^+\}$ to $\mathcal{D}'$. For each edge $C \in \mathcal{C}$, we add $C' = \{v^- : v \in C\}$ to $\mathcal{C}'$. Finally,
voloshin-hypergraph_ F02160	265	■ 0.957	$C^{\prime}=\left\{v^{-}: v \in C\right\}$	$C' = \{v^- : v \in C\}$	$D' = \bigcup_{v \in D} \{v^-, v^+\}$ to $\mathcal{D}'$. For each edge $C \in \mathcal{C}$, we add $C' = \{v^- : v \in C\}$ to $\mathcal{C}'$. Finally,
voloshin-hypergraph_ F02161	265	■ 0.999	$\left\{v^{-}, v^{+}, u^{-}\right\}$	$\{v^-, v^+, u^-\}$	for each ordered pair $u, v \in X$, we add the triples $\{v^-, v^+, u^-\}$ and $\{v^-, v^+, u^+\}$ to $\mathcal{C}'$.
voloshin-hypergraph_ F02162	265	■ 0.999	$\left\{v^{-}, v^{+}, u^{+}\right\}$	$\{v^-, v^+, u^+\}$	for each ordered pair $u, v \in X$, we add the triples $\{v^-, v^+, u^-\}$ and $\{v^-, v^+, u^+\}$ to $\mathcal{C}'$.
voloshin-hypergraph_ F02163	265	■ 1.000	$\chi(\mathcal{H}) \geq 2$	$\chi(\mathcal{H}) \geq 2$	Lemma 10.9.3 Let $\mathcal{H}$ be a mixed hypergraph with feasible set S . If $\chi(\mathcal{H}) \geq 2$, then the
voloshin-hypergraph_ F02164	265	■ 1.000	$S \cup\{2\}$	$S \cup \{2\}$	mixed hypergraph $\mathcal{H}'$ obtained from $\mathcal{H}$ via Construction 2 has feasible set $S \cup \{2\}$.
voloshin-hypergraph_ F02165	265	■ 0.986	$c\left(v^{-}\right) \neq c\left(v^{+}\right)$	$c\left(v^{-}\right) \neq c\left(v^{+}\right)$	Proof. Let c be a proper coloring of $\mathcal{H}'$. If $c(v^-) \neq c(v^+)$ for a vertex v of $\mathcal{H}$, then the
voloshin-hypergraph_ F02166	265	■ 1.000	v^{-}, v^{+}	v^-, v^+	$\mathcal{C}$ -edges that are triples containing v^-, v^+ force all other vertices to have color $c(v^-)$ or
voloshin-hypergraph_ F02167	265	■ 1.000	$c\left(v^{-}\right)$	$c\left(v^{-}\right)$	$\mathcal{C}$ -edges that are triples containing v^-, v^+ force all other vertices to have color $c(v^-)$ or
voloshin-hypergraph_ F02168	265	■ 1.000	$c\left(v^{+}\right)$	$c\left(v^{+}\right)$	$c(v^+)$. Thus, such a coloring uses exactly two colors. We obtain a strict 2-coloring by
voloshin-hypergraph_ F02169	265	■ 1.000	$c\left(u^{-}\right)=c\left(v^{-}\right)$	$c\left(u^{-}\right)=c\left(v^{-}\right)$	setting $c(u^-) = c(v^-)$ and $c(u^+) = c(v^+)$ for all u . Since each member of $\mathcal{D}'$ consists of
voloshin-hypergraph_ F02170	265	■ 1.000	$c\left(u^{+}\right)=c\left(v^{+}\right)$	$c\left(u^{+}\right)=c\left(v^{+}\right)$	setting $c(u^-) = c(v^-)$ and $c(u^+) = c(v^+)$ for all u . Since each member of $\mathcal{D}'$ consists of
voloshin-hypergraph_ F02171	265	■ 1.000	$c\left(v^{-}\right)=c\left(v^{+}\right)$	$c\left(v^{-}\right)=c\left(v^{+}\right)$	It remains to consider proper colorings with $c(v^-) = c(v^+)$ for each vertex v of $\mathcal{H}$. Let
voloshin-hypergraph_ F02172	265	■ 1.000	$\tilde{c}$	$\tilde{c}$	$\tilde{c}$ be the coloring of $\mathcal{H}$ defined by $\tilde{c}(v) = c(v^-)$. For each member of $\mathcal{D}'$, the coloring
voloshin-hypergraph_ F02173	265	■ 1.000	$\tilde{c}(v)=c\left(v^{-}\right)$	$\tilde{c}(v)=c\left(v^{-}\right)$	$\tilde{c}$ be the coloring of $\mathcal{H}$ defined by $\tilde{c}(v) = c(v^-)$. For each member of $\mathcal{D}'$, the coloring
voloshin-hypergraph_ F02174	266	■ 0.813	$R\left(\mathcal{H}^{\prime}\right)$	$R\left(\mathcal{H}^{\prime}\right)$	number of 0s on the right part of the chromatic spectrum of $R(\mathcal{H}')$ is greater by $n(\mathcal{H})$ since
voloshin-hypergraph_ F02175	266	■ 1.000	$n\left(\mathcal{H}^{\prime}\right)=2 n(\mathcal{H})$	$n\left(\mathcal{H}^{\prime}\right)=2 n(\mathcal{H})$	$n\left(\mathcal{H}^{\prime}\right)=2 n(\mathcal{H})$.
voloshin-hypergraph_ F02176	266	■ 1.000	$\mathcal{H}(T)$	$\mathcal{H}(T)$	graph $\mathcal{H}(T)$ with feasible set T . We use induction on the size of set T , and within each
voloshin-hypergraph_ F02177	266	■ 1.000	$T=\{t\}$	$T=\{t\}$	size, we use induction on the smallest element t of T . For $T=\{t\}$, we set $\mathcal{H}(T)=K_t$.
voloshin-hypergraph_ F02178	266	■ 1.000	$\mathcal{H}(T)=K_t$	$\mathcal{H}(T)=K_t$	size, we use induction on the smallest element t of T . For $T=\{t\}$, we set $\mathcal{H}(T)=K_t$.
voloshin-hypergraph_ F02179	266	■ 1.000	$ T >1$	$ T >1$	For $ T >1$ and $t=2$, we let $\mathcal{H}(T)$ be the mixed hypergraph obtained by applying
voloshin-hypergraph_ F02180	266	■ 1.000	$t=2$	$t=2$	For $ T >1$ and $t=2$, we let $\mathcal{H}(T)$ be the mixed hypergraph obtained by applying

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voloshin-hypergraph_ F02181	266	■ 1.000	$\mathcal{H}(T - \{2\})$	$\mathcal{H}(T - \{2\})$	Construction 2 to $\mathcal{H}(T - \{2\})$. Lemma 10.9.3 implies that this works.
voloshin-hypergraph_ F02182	266	■ 1.000	$t > 2$	$t > 2$	For $ T > 1$ and $t > 2$, we let $\mathcal{H}(T)$ be the join of K_{t-2} with the mixed hypergraph
voloshin-hypergraph_ F02183	266	■ 1.000	K_{t-2}	K_{t-2}	For $ T > 1$ and $t > 2$, we let $\mathcal{H}(T)$ be the join of K_{t-2} with the mixed hypergraph
voloshin-hypergraph_ F02184	266	■ 0.999	$\mathcal{H}(\mathcal{T}^{\prime})$	$\mathcal{H}(T')$	$\mathcal{H}(T')$, where T' is obtained from T by subtracting $t - 2$ from each element. Lemma 10.9.2
voloshin-hypergraph_ F02185	266	■ 0.999	$t - 2$	$t - 2$	$\mathcal{H}(T')$, where T' is obtained from T by subtracting $t - 2$ from each element. Lemma 10.9.2
voloshin-hypergraph_ F02186	266	■ 0.861	$S = \{n_1, n_2, \dots, n_p\}$	$S = \{n_1, n_2, \dots, n_p\}$	<i>INPUT:</i> A set $S = \{n_1, n_2, \dots, n_p\}$ of increasing integer numbers, $n_1 \geq 2$.
voloshin-hypergraph_ F02187	266	■ 0.861	$n_1 \geq 2$	$n_1 \geq 2$	<i>INPUT:</i> A set $S = \{n_1, n_2, \dots, n_p\}$ of increasing integer numbers, $n_1 \geq 2$.
voloshin-hypergraph_ F02188	266	■ 1.000	$i = n_p - 1; \mathcal{H} = K_3$	$i = n_p - 1; \mathcal{H} = K_3$	Initialization: $i = n_p - 1; \mathcal{H} = K_3$.
voloshin-hypergraph_ F02189	266	■ 0.982	$i \neq 1$	$i \neq 1$	Iteration: while $i \neq 1$, do:
voloshin-hypergraph_ F02190	266	■ 0.999	$i \in S$	$i \in S$	1. If $i \in S$, do doubling.
voloshin-hypergraph_ F02191	266	■ 0.999	$i \notin S$	$i \notin S$	2. If $i \notin S$, do elementary shifting.
voloshin-hypergraph_ F02192	266	■ 1.000	$i = i - 1$	$i = i - 1$	3. $i = i - 1$.
voloshin-hypergraph_ F02193	267	■ 1.000	$\mathcal{H} = (X, \mathcal{E})$	$\mathcal{H} = (X, \mathcal{E})$	Let $\mathcal{H} = (X, \mathcal{E})$ be a colorable 3-uniform bihypergraph with $X = \{1, 2, \dots, 6\}$, with
voloshin-hypergraph_ F02194	267	■ 1.000	$X = \{1, 2, \dots, 6\}$	$X = \{1, 2, \dots, 6\}$	Let $\mathcal{H} = (X, \mathcal{E})$ be a colorable 3-uniform bihypergraph with $X = \{1, 2, \dots, 6\}$, with
voloshin-hypergraph_ F02195	267	■ 0.640	$\mathcal{E}$	$\mathcal{E}$	all the elements of $\mathcal{E}$ being triples (bi-edges of size 3), and $R(\mathcal{H}) = (r_1, r_2, \dots, r_6)$. For a
voloshin-hypergraph_ F02196	267	■ 0.640	$R(\mathcal{H}) = (r_1, r_2, \dots, r_6)$	$R(\mathcal{H}) = (r_1, r_2, \dots, r_6)$	all the elements of $\mathcal{E}$ being triples (bi-edges of size 3), and $R(\mathcal{H}) = (r_1, r_2, \dots, r_6)$. For a
voloshin-hypergraph_ F02197	267	■ 0.998	$r_1 = 0, r_6 = 0$	$r_1 = 0, r_6 = 0$	Evidently, $r_1 = 0, r_6 = 0$. If $r_5 \neq 0$, then in a strict 5-coloring of $\mathcal{H}$, all bi-edges contain a
voloshin-hypergraph_ F02198	267	■ 0.998	$r_5 \neq 0$	$r_5 \neq 0$	Evidently, $r_1 = 0, r_6 = 0$. If $r_5 \neq 0$, then in a strict 5-coloring of $\mathcal{H}$, all bi-edges contain a
voloshin-hypergraph_ F02199	267	■ 1.000	$r_4 \neq 0, r_3 \neq 0, r_2 \neq 0$	$r_4 \neq 0, r_3 \neq 0, r_2 \neq 0$	These color classes can easily be combined, so $r_4 \neq 0, r_3 \neq 0, r_2 \neq 0$, a contradiction.
voloshin-hypergraph_ F02200	267	■ 1.000	$r_5 = 0$	$r_5 = 0$	Hence $r_5 = 0$. If $r_4 \neq 0$, then in a strict 4-coloring of $\mathcal{H}$, there are at least two singleton
voloshin-hypergraph_ F02201	267	■ 1.000	$r_4 \neq 0$	$r_4 \neq 0$	Hence $r_5 = 0$. If $r_4 \neq 0$, then in a strict 4-coloring of $\mathcal{H}$, there are at least two singleton
voloshin-hypergraph_ F02202	267	■ 1.000	$r_3 \neq 0$	$r_3 \neq 0$	color classes which implies $r_3 \neq 0$, a contradiction. Hence, there are no gaps in $R(\mathcal{H})$.
voloshin-hypergraph_ F02210	268	■ 0.995	$\{A, B\}$	$\{A, B\}$	vertices are colored with the colors from $\{A, B\}$; i.e., all the colorings are 2-colorings. If
voloshin-hypergraph_ F02211	268	■ 1.000	$c(1) = c(2) = A$	$c(1) = c(2) = A$	$c(1) = c(2) = A$, then $c(3) = c(4) = B$, and consequently, $c(5), c(6), c(7) \in \{C, D\}$; i.e., we
voloshin-hypergraph_ F02212	268	■ 1.000	$c(3) = c(4) = B$	$c(3) = c(4) = B$	$c(1) = c(2) = A$, then $c(3) = c(4) = B$, and consequently, $c(5), c(6), c(7) \in \{C, D\}$; i.e., we
voloshin-hypergraph_ F02213	268	■ 1.000	$c(5), c(6), c(7) \in \{C, D\}$	$c(5), c(6), c(7) \in \{C, D\}$	$c(1) = c(2) = A$, then $c(3) = c(4) = B$, and consequently, $c(5), c(6), c(7) \in \{C, D\}$ i.e., we
voloshin-hypergraph_ F02214	268	■ 0.999	$R(\mathcal{H}) = (0, 12, 0, 3, 0, 0, 0), P(\mathcal{H}, \lambda) = 3\lambda(\lambda - 1)(\lambda^2 - 5\lambda + 10)$	$R(\mathcal{H}) = (0, 12, 0, 3, 0, 0, 0), P(\mathcal{H}, \lambda) = 3\lambda(\lambda - 1)(\lambda^2 - 5\lambda + 10)$	analysis shows that $R(\mathcal{H}) = (0, 12, 0, 3, 0, 0, 0), P(\mathcal{H}, \lambda) = 3\lambda(\lambda - 1)(\lambda^2 - 5\lambda + 10)$.

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voloshin-hypergraph_F02215	268	■ 0.999	$X=\{1,2, \ldots, 7\}$	$X = \{1, 2, \dots, 7\}$	$\mathcal{H} = (X, \mathcal{E})$ be a 3-uniform bihypergraph with $X = \{1, 2, \dots, 7\}$ and $ \mathcal{E} \leq 8$. For a con-
voloshin-hypergraph_F02216	268	■ 0.999	$ \mathcal{H} \leq 8$	$ \mathcal{E} \leq 8$	$\mathcal{H} = (X, \mathcal{E})$ be a 3-uniform bihypergraph with $X = \{1, 2, \dots, 7\}$ and $ \mathcal{E} \leq 8$. For a con-
voloshin-hypergraph_F02217	268	■ 1.000	$R(\mathcal{H})=\left(r_{\{1\}}, r_{\{2\}}, \ldots, r_{\{7\}}\right)$	$R(\mathcal{H}) = (r_1, r_2, \dots, r_7)$	tradiction, suppose that $R(\mathcal{H}) = (r_1, r_2, \dots, r_7)$ has at least one gap. We immediately have
voloshin-hypergraph_F02218	268	■ 0.995	$r_{\{1\}}=r_{\{7\}}=0$	$r_1 = r_7 = 0$	$r_1 = r_7 = 0$. If $r_6 \neq 0$ (respectively $r_6 = 0, r_5 \neq 0$), then at least 3 color classes are singletons
voloshin-hypergraph_F02219	268	■ 0.995	$r_{\{6\}} \neq 0$	$r_6 \neq 0$	$r_1 = r_7 = 0$. If $r_6 \neq 0$ (respectively $r_6 = 0, r_5 \neq 0$), then at least 3 color classes are singletons
voloshin-hypergraph_F02220	268	■ 0.995	$r_{\{6\}}=0, r_{\{5\}} \neq 0$	$r_6 = 0, r_5 \neq 0$	$r_1 = r_7 = 0$. If $r_6 \neq 0$ (respectively $r_6 = 0, r_5 \neq 0$), then at least 3 color classes are singletons
voloshin-hypergraph_F02221	268	■ 0.960	$r_{\{7\}}=r_{\{6\}}=r_{\{5\}}=0, r_{\{4\}} \neq 0$	$r_7 = r_6 = r_5 = 0, r_4 \neq 0$	Therefore, suppose $r_7 = r_6 = r_5 = 0, r_4 \neq 0$. In a strict 4-coloring, if the number of
voloshin-hypergraph_F02222	268	■ 0.531	$\{1,2,7\} \notin \mathcal{H}$	$\{1, 2, 7\} \notin \mathcal{E}$	If $\{1, 2, 7\} \notin \mathcal{E}$ (respectively $\{3, 4, 7\} \notin \mathcal{E}, \{5, 6, 7\} \notin \mathcal{E}$), then there exists a 3-coloring
voloshin-hypergraph_F02223	268	■ 0.531	$\{3,4,7\} \notin \mathcal{H}, \{5,6,7\} \notin \mathcal{H}$	$\{3, 4, 7\} \notin \mathcal{E}, \{5, 6, 7\} \notin \mathcal{E}$	If $\{1, 2, 7\} \notin \mathcal{E}$ (respectively $\{3, 4, 7\} \notin \mathcal{E}, \{5, 6, 7\} \notin \mathcal{E}$), then there exists a 3-coloring
voloshin-hypergraph_F02224	268	■ 0.998	(A, A, B, B, C, C, A)	(A, A, B, B, C, C, A)	(A, A, B, B, C, C, A) (respectively $(A, A, B, B, C, C, B), (A, A, B, B, C, C, C)$) and thus $r_3 \neq 0$,
voloshin-hypergraph_F02225	268	■ 0.998	$(A, A, B, B, C, C, B), (A, A, B, B, C, C, C)$	$(A, A, B, B, C, C, B), (A, A, B, B, C, C, C)$	(A, A, B, B, C, C, A) (respectively $(A, A, B, B, C, C, B), (A, A, B, B, C, C, C)$) and thus $r_3 \neq 0$,
voloshin-hypergraph_F02226	268	■ 1.000	$\{1,2,3\} \notin \mathcal{H}, \{1,2,4\} \notin \mathcal{H}, \{1,2,5\} \notin \mathcal{H}$	$\{1, 2, 3\} \notin \mathcal{E}, \{1, 2, 4\} \notin \mathcal{E}, \{1, 2, 5\} \notin \mathcal{E}$	of $\mathcal{H}$. Choose the first column and let $\{1, 2, 3\} \notin \mathcal{E}, \{1, 2, 4\} \notin \mathcal{E}, \{1, 2, 5\} \notin \mathcal{E}$. If
voloshin-hypergraph_F02227	268	■ 1.000	$\{3,4,1\}, \{3,4,2\} \notin \mathcal{H}$	$\{3, 4, 1\}, \{3, 4, 2\} \notin \mathcal{E}$	$\{3, 4, 1\}, \{3, 4, 2\} \notin \mathcal{E}$, then the following 3-coloring exists: (A, A, A, B, B, C) , a contra-
voloshin-hypergraph_F02228	268	■ 0.995	$\{3,4,1\} \in \mathcal{H}$	$\{3, 4, 1\} \in \mathcal{E}$	diction. Therefore $\{3, 4, 1\} \in \mathcal{E}$, or $\{3, 4, 2\} \in \mathcal{E}$. If $\{3, 4, 5\}, \{3, 4, 6\} \notin \mathcal{E}$ then the
voloshin-hypergraph_F02229	268	■ 0.995	$\{3,4,2\} \in \mathcal{H}$	$\{3, 4, 2\} \in \mathcal{E}$	diction. Therefore $\{3, 4, 1\} \in \mathcal{E}$, or $\{3, 4, 2\} \in \mathcal{E}$. If $\{3, 4, 5\}, \{3, 4, 6\} \notin \mathcal{E}$ then the
voloshin-hypergraph_F02230	268	■ 0.995	$\{3,4,5\}, \{3,4,6\} \notin \mathcal{H}$	$\{3, 4, 5\}, \{3, 4, 6\} \notin \mathcal{E}$	diction. Therefore $\{3, 4, 1\} \in \mathcal{E}$, or $\{3, 4, 2\} \in \mathcal{E}$. If $\{3, 4, 5\}, \{3, 4, 6\} \notin \mathcal{E}$ then the
voloshin-hypergraph_F02231	268	■ 0.745	$\{3,4,5\} \in \mathcal{H}$	$\{3, 4, 5\} \in \mathcal{E}$	following 3-coloring exists: (A, A, A, B, C, C, B) , a contradiction. Hence $\{3, 4, 5\} \in \mathcal{E}$ or
voloshin-hypergraph_F02232	268	■ 0.996	$\{3,4,6\} \in \mathcal{H}$	$\{3, 4, 6\} \in \mathcal{E}$	$\{3, 4, 6\} \in \mathcal{E}$. If $\{5, 6, 1\} \notin \mathcal{E}$ (respectively, $\{5, 6, 2\} \notin \mathcal{E}$), then the following 3-coloring ex-
voloshin-hypergraph_F02233	268	■ 0.996	$\{5,6,1\} \notin \mathcal{H}$	$\{5, 6, 1\} \notin \mathcal{E}$	$\{3, 4, 6\} \in \mathcal{E}$. If $\{5, 6, 1\} \notin \mathcal{E}$ (respectively, $\{5, 6, 2\} \notin \mathcal{E}$), then the following 3-coloring ex-
voloshin-hypergraph_F02234	268	■ 0.996	$\{5,6,2\} \notin \mathcal{H}$	$\{5, 6, 2\} \notin \mathcal{E}$	$\{3, 4, 6\} \in \mathcal{E}$. If $\{5, 6, 1\} \notin \mathcal{E}$ (respectively, $\{5, 6, 2\} \notin \mathcal{E}$), then the following 3-coloring ex-
voloshin-hypergraph_F02235	268	■ 0.999	$\{5,6,2\} \in \mathcal{H}$	$\{5, 6, 2\} \in \mathcal{E}$	$\{5, 6, 2\} \in \mathcal{E}$. If $\{1, 2, 6\} \in \mathcal{D}$ then all 8 bi-edges of $\mathcal{H}$ are exhausted and the 3-coloring
voloshin-hypergraph_F02236	268	■ 0.999	$\{1,2,6\} \in \mathcal{H}$	$\{1, 2, 6\} \in \mathcal{D}$	$\{5, 6, 2\} \in \mathcal{E}$. If $\{1, 2, 6\} \in \mathcal{D}$ then all 8 bi-edges of $\mathcal{H}$ are exhausted and the 3-coloring
voloshin-hypergraph_F02237	268	■ 1.000	$\{1,2,6\} \notin \mathcal{H}$	$\{1, 2, 6\} \notin \mathcal{E}$	Suppose $\{1, 2, 6\} \notin \mathcal{E}$. If $\{3, 4, 1\} \notin \mathcal{E}$ (respectively $\{3, 4, 2\} \notin \mathcal{E}$), then the follow-
voloshin-hypergraph_F02238	268	■ 1.000	$\{3,4,1\} \notin \mathcal{H}$	$\{3, 4, 1\} \notin \mathcal{E}$	Suppose $\{1, 2, 6\} \notin \mathcal{E}$. If $\{3, 4, 1\} \notin \mathcal{E}$ (respectively $\{3, 4, 2\} \notin \mathcal{E}$), then the follow-
voloshin-hypergraph_F02239	268	■ 1.000	$\{3,4,2\} \notin \mathcal{H}$	$\{3, 4, 2\} \notin \mathcal{E}$	Suppose $\{1, 2, 6\} \notin \mathcal{E}$. If $\{3, 4, 1\} \notin \mathcal{E}$ (respectively $\{3, 4, 2\} \notin \mathcal{E}$), then the follow-
voloshin-hypergraph_F02240	268	■ 0.999	$\{3,4,1\}, \{3,4,2\} \in \mathcal{H}$	$\{3, 4, 1\}, \{3, 4, 2\} \in \mathcal{E}$	Therefore both $\{3, 4, 1\}, \{3, 4, 2\} \in \mathcal{E}$. Hence all 8 bi-edges of $\mathcal{H}$ are exhausted, and the
voloshin-hypergraph_F02241	269	■ 0.523	$S\left(\mathcal{H}_{\mathcal{C}}\right)$	$S\left(\mathcal{H}_{\mathcal{C}}\right)$	1. For mixed hypergraph $\mathcal{H}$ in Figure 10.31, find feasible sets $S(\mathcal{H}_{\mathcal{C}})$ and $S(\mathcal{H}_{\mathcal{D}})$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_F02242	269	■ 0.523	$S(\text{\mathcal{H}}_{\text{\mathcal{D}}})$	$S(\mathcal{H}_{\mathcal{D}})$	1. For mixed hypergraph $\mathcal{H}$ in Figure 10.31, find feasible sets $S(\mathcal{H}_C)$ and $S(\mathcal{H}_{\mathcal{D}})$.
voloshin-hypergraph_F02243	269	■ 1.000	$S=\{2,3,5\}$	$S = \{2, 3, 5\}$	3. Given set of integers $S = \{2, 3, 5\}$, apply Algorithm 10.9.1 to construct a mixed hy-
voloshin-hypergraph_F02244	269	■ 1.000	$S(\text{\mathcal{H}})=S$	$S(\mathcal{H}) = S$	pergraph $\mathcal{H}$ with $S(\mathcal{H}) = S$.
voloshin-hypergraph_F02245	270	■ 1.000	$B(\text{\mathcal{H}}^{\prime})$	$B(\mathcal{H}')$	from Section 9.4., a hypergraph $\mathcal{H}'$ is planar if and only if bipartite representation $B(\mathcal{H}')$ is
voloshin-hypergraph_F02246	270	■ 0.991	B, C	B, C	with B, C or D appropriately, according to whether they are bi-edges, C -edges or $\mathcal{D}$ -edges.
voloshin-hypergraph_F02247	270	■ 0.981	$\text{\mathcal{H}}=(X, \text{\mathcal{C}}, \text{\mathcal{D}})=\left(X, \text{\binom{X}{3}}, \text{\binom{X}{2}}\right)$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D}) = \left(X, \binom{X}{3}, \binom{X}{2}\right)$	non-trivial (reduced) uncolorable planar mixed hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D}) = (X, \binom{X}{3}, \binom{X}{2})$
voloshin-hypergraph_F02248	270	■ 1.000	$X=\{1,2,3\}, \text{\mathcal{C}}=\{C\}=\{\{1,2,3\}\}, \text{\mathcal{D}}=\left\{D_1, D_2, D_3\right\}=$	$X = \{1, 2, 3\}, \mathcal{C} = \{C\} = \{\{1, 2, 3\}\}, \mathcal{D} = \{D_1, D_2, D_3\} =$	is shown in Figure 10.32: $X = \{1, 2, 3\}, \mathcal{C} = \{C\} = \{\{1, 2, 3\}\}, \mathcal{D} = \{D_1, D_2, D_3\} =$
voloshin-hypergraph_F02249	270	■ 0.991	$\left\{\{1,2\}, \{2,3\}, \{1,3\}\right\}$	$\{\{1, 2\}, \{2, 3\}, \{1, 3\}\}$	$\{\{1, 2\}, \{2, 3\}, \{1, 3\}\}$, the four faces are f_1, f_2, f_3, f_4 . It is not difficult to extend this
voloshin-hypergraph_F02250	270	■ 0.991	f_1, f_2, f_3, f_4	f_1, f_2, f_3, f_4	$\{\{1, 2\}, \{2, 3\}, \{1, 3\}\}$, the four faces are f_1, f_2, f_3, f_4 . It is not difficult to extend this
voloshin-hypergraph_F02251	270	■ 1.000	$G(\text{\mathcal{H}})$	$G(\mathcal{H})$	$G(\mathcal{H})$ on the same vertex set with $\mathcal{H}$: replace every face in $\mathcal{H}$ by an edge in G , so that every
voloshin-hypergraph_F02252	272	■ 1.000	$S(n, k)$	$S(n, k)$	As in Section 5.4., let $S(n, k)$ denote the Stirling numbers of the second kind, i.e. the
voloshin-hypergraph_F02253	272	■ 1.000	$f_k(\text{\mathcal{G}}^*)$	$f_k(G^*)$	number of ways of partitioning a set of n elements into exactly k sets. Also define $f_k(G^*)$ to
voloshin-hypergraph_F02254	272	■ 1.000	$f(\text{\mathcal{G}}^*)=\sum_{i \geq 1} f_i(\text{\mathcal{G}}^*)$	$f(G^*) = \sum_{i \geq 1} f_i(G^*)$	cycles), and let $f(G^*) = \sum_{i \geq 1} f_i(G^*)$ be the total number of 2-factors of G^* .
voloshin-hypergraph_F02255	272	■ 1.000	$f_{k-1}(\text{\mathcal{G}}^*)$	$f_{k-1}(G^*)$	<i>mal coloring and there are exactly $f_{k-1}(G^*)$ maximal k-colorings of G.</i>
voloshin-hypergraph_F02256	273	■ 0.995	$f(\text{\mathcal{G}}^*)$	$f(G^*)$	Corollary 10.10.2 <i>Every bitriangulation G has exactly $f(G^*)$ strict 2-colorings. In gen-</i>
voloshin-hypergraph_F02257	273	■ 1.000	$k=2$	$k = 2$	Proof. The first statement follows from both summation formulas, by setting $k = 2$ or
voloshin-hypergraph_F02258	273	■ 1.000	$\lambda=2$	$\lambda = 2$	$\lambda = 2$ respectively. For the first formula, it suffices, by Theorem 10.10.3, to show that every
voloshin-hypergraph_F02259	273	■ 0.994	$S(i, k-1)$	$S(i, k - 1)$	2-factor consisting of i cycles can be k -face-colored in exactly $S(i, k - 1)$ ways. To see this,
voloshin-hypergraph_F02260	273	■ 1.000	$r_k(T)$	$r_k(T)$	and must therefore form a tree T . Let $r_k(T)$ be the number of proper k -colorings of T .
voloshin-hypergraph_F02261	273	■ 1.000	$r_k(T)=S(e(T), k-1)$	$r_k(T) = S(e(T), k - 1)$	To see that $r_k(T) = S(e(T), k - 1)$, observe that $r_1(K_1) = 1$ and $r_k(K_1) = 0$ for $k \geq 2$. By
voloshin-hypergraph_F02262	273	■ 1.000	$r_1(K_1)=1$	$r_1(K_1) = 1$	To see that $r_k(T) = S(e(T), k - 1)$, observe that $r_1(K_1) = 1$ and $r_k(K_1) = 0$ for $k \geq 2$. By
voloshin-hypergraph_F02263	273	■ 1.000	$r_k(K_1)=0$	$r_k(K_1) = 0$	To see that $r_k(T) = S(e(T), k - 1)$, observe that $r_1(K_1) = 1$ and $r_k(K_1) = 0$ for $k \geq 2$. By
voloshin-hypergraph_F02264	273	■ 1.000	$r_k(T)=(k-1) r_k(T-x)+r_{k-1}(T-x)$	$r_k(T) = (k - 1)r_k(T - x) + r_{k-1}(T - x)$	removing a pendant vertex x , we can see that $r_k(T) = (k - 1)r_k(T - x) + r_{k-1}(T - x)$, the
voloshin-hypergraph_F02265	273	■ 1.000	$\lambda(\lambda-1)^i$	$\lambda(\lambda - 1)^i$	$\lambda(\lambda - 1)^i$ (Theorem 5.5.1).
voloshin-hypergraph_F02266	273	■ 1.000	$\bar{\chi}(G)=1+\max \left\{k: f_k(\text{\mathcal{G}}^*) \geq 1\right\}$	$\bar{\chi}(G) = 1 + \max \{k : f_k(G^*) \geq 1\}$	$\chi(G) = 2$ and $\bar{\chi}(G) = 1 + \max \{k : f_k(G^*) \geq 1\}$.
voloshin-hypergraph_F02267	273	■ 1.000	$\bar{\chi}(G)$	$\bar{\chi}(G)$	the upper chromatic number must be maximal, so that the value of $\bar{\chi}(G)$ follows from
voloshin-hypergraph_F02268	273	■ 1.000	$k=\bar{\chi}(G)$	$k = \bar{\chi}(G)$	Corollary 10.10.1. If $k = \bar{\chi}(G)$, then $f_{k-1}(G^*) \geq 1$; so, since $S(k - 1, i - 1) \geq 1$ for every

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voloshin-hypergraph_F02269	273	■ 1.000	$f_{\{k-1\}}(\left(G^*\right)) \geq 1$	$f_{k-1}(G^*) \geq 1$	Corollary 10.10.1. If $k = \bar{\chi}(G)$, then $f_{k-1}(G^*) \geq 1$; so, since $S(k-1, i-1) \geq 1$ for every
voloshin-hypergraph_F02270	273	■ 1.000	$S(k-1, i-1) \geq 1$	$S(k-1, i-1) \geq 1$	Corollary 10.10.1. If $k = \bar{\chi}(G)$, then $f_{k-1}(G^*) \geq 1$; so, since $S(k-1, i-1) \geq 1$ for every
voloshin-hypergraph_F02271	273	■ 1.000	$2 \leq i \leq k$	$2 \leq i \leq k$	$2 \leq i \leq k$, we get that $r_i(G) \geq 1$ in this range and that the chromatic spectrum is continuous.
voloshin-hypergraph_F02272	273	■ 1.000	$r_{\{i\}}(G) \geq 1$	$r_i(G) \geq 1$	$2 \leq i \leq k$, we get that $r_i(G) \geq 1$ in this range and that the chromatic spectrum is continuous.
voloshin-hypergraph_F02273	274	■ 1.000	$\bar{\chi}(G)=2$	$\bar{\chi}(G) = 2$	may assume that G is a bitriangulation. By Corollary 10.10.3, $\bar{\chi}(G) = 2$; then, by Corol-
voloshin-hypergraph_F02274	274	■ 1.000	$u \ v$	uv	Proof. We sketch the elegant proof of Thomason. Let uv be the given edge. Consider
voloshin-hypergraph_F02275	274	■ 1.000	$a, \ b, \ c$	a, b, c	embedding of K_4 there are four faces denoted by a, b, c , and d . The dual “vertices - faces”
voloshin-hypergraph_F02282	275	■ 0.978	$\mathcal{H}_{2,4}^{\prime}=(X, \mathcal{C}, \mathcal{D})$	$\mathcal{H}_{2,4}' = (X, \mathcal{C}, \mathcal{D})$	Lemma 10.10.2 (Kobler, Kündgen, 2001) Let $\mathcal{H}_{2,4}' = (X, \mathcal{C}, \mathcal{D})$ be the mixed hypergraph
voloshin-hypergraph_F02283	275	■ 0.991	$X=\{1,2, \ldots, 6\}, \mathcal{C}=\left\{C_{\{1\}}, \ldots, C_{\{4\}}\right\}=\left\{\{1,2,3\},\{2,3,4\},\{2,4,5\},\{4,5,6\}\right\}$	$X = \{1, 2, \dots, 6\}, \mathcal{C} = \{C_1, \dots, C_4\} = \{\{1, 2, 3\}, \{2, 3, 4\}, \{2, 4, 5\}, \{4, 5, 6\}\}$	with $X = \{1, 2, \dots, 6\}$, $\mathcal{C} = \{C_1, \dots, C_4\} = \{\{1, 2, 3\}, \{2, 3, 4\}, \{2, 4, 5\}, \{4, 5, 6\}\}$, and
voloshin-hypergraph_F02284	275	■ 0.990	$\mathcal{D}=\left\{D_{\{1\}}, \ldots, D_{\{6\}}\right\}=\left\{\{1,2\},\{1,5\},\{1,6\},\{2,4\},\{3,6\},\{4,6\}\right\}$	$\mathcal{D} = \{D_1, \dots, D_6\} = \{\{1, 2\}, \{1, 5\}, \{1, 6\}, \{2, 4\}, \{3, 6\}, \{4, 6\}\}$	$\mathcal{D} = \{D_1, \dots, D_6\} = \{\{1, 2\}, \{1, 5\}, \{1, 6\}, \{2, 4\}, \{3, 6\}, \{4, 6\}\}$. $\mathcal{H}_{2,4}'$ is planar and has
voloshin-hypergraph_F02285	275	■ 0.990	$\mathcal{H}_{2,4}^{\prime}$	$\mathcal{H}_{2,4}'$	$\mathcal{D} = \{D_1, \dots, D_6\} = \{\{1, 2\}, \{1, 5\}, \{1, 6\}, \{2, 4\}, \{3, 6\}, \{4, 6\}\}$. $\mathcal{H}_{2,4}'$ is planar and has
voloshin-hypergraph_F02286	276	■ 1.000	$S\left(\mathcal{H}_{2,4}^{\prime}\right)=\{2,4\}$	$S\left(\mathcal{H}_{2,4}'\right) = \{2, 4\}$	feasible set $S(\mathcal{H}_{2,4}') = \{2, 4\}$.
voloshin-hypergraph_F02287	276	■ 1.000	$c(2) \neq c(3)$	$c(2) \neq c(3)$	indicated. Let c be a strict coloring of $\mathcal{H}_{2,4}'$. If $c(2) \neq c(3)$, then the remaining colors are
voloshin-hypergraph_F02288	276	■ 1.000	$\{1,3,4\} \cup\{2,5,6\}$	$\{1, 3, 4\} \cup \{2, 5, 6\}$	forced and we have the 2-coloring $\{1, 3, 4\} \cup \{2, 5, 6\}$. If $c(2) = c(3)$, then $c(4) = c(5)$ and
voloshin-hypergraph_F02289	276	■ 1.000	$c(2)=c(3)$	$c(2) = c(3)$	forced and we have the 2-coloring $\{1, 3, 4\} \cup \{2, 5, 6\}$. If $c(2) = c(3)$, then $c(4) = c(5)$ and
voloshin-hypergraph_F02290	276	■ 1.000	$c(4)=c(5)$	$c(4) = c(5)$	forced and we have the 2-coloring $\{1, 3, 4\} \cup \{2, 5, 6\}$. If $c(2) = c(3)$, then $c(4) = c(5)$ and
voloshin-hypergraph_F02291	276	■ 1.000	$c(2) \neq c(4)$	$c(2) \neq c(4)$	$c(2) \neq c(4)$, which results in the strict 4-coloring $\{1\} \cup \{2, 3\} \cup \{4, 5\} \cup \{6\}$. One can see
voloshin-hypergraph_F02292	276	■ 1.000	$\{1\} \cup\{2,3\} \cup\{4,5\} \cup\{6\}$	$\{1\} \cup \{2, 3\} \cup \{4, 5\} \cup \{6\}$	$c(2) \neq c(4)$, which results in the strict 4-coloring $\{1\} \cup \{2, 3\} \cup \{4, 5\} \cup \{6\}$. One can see
voloshin-hypergraph_F02293	276	■ 1.000	$\{s, \ s+1, \ldots, \ t\}$	$\{s, s+1, \dots, t\}$	feasible set of some planar mixed hypergraph if and only if S is an interval $[s, s+1, \dots, t]$
voloshin-hypergraph_F02294	276	■ 0.995	$1 \leq s \leq 4$	$1 \leq s \leq 4$	with $1 \leq s \leq 4$ or of the form $\{2, 4, 5, \dots, t\}$.
voloshin-hypergraph_F02295	276	■ 0.995	$\{2,4,5, \ldots, \ t\}$	$\{2, 4, 5, \dots, t\}$	with $1 \leq s \leq 4$ or of the form $[2, 4, 5, \dots, t]$.
voloshin-hypergraph_F02296	276	■ 0.999	$S(G)=\{\chi(G)$	$S(G) = \{\chi(G)$	Proof. Let $G = (X, \mathcal{D})$ be a t -vertex planar $\mathcal{D}$ -graph. The feasible set $S(G) = \{\chi(G)$,
voloshin-hypergraph_F02297	276	■ 1.000	$\chi(G)+1, \ldots, \ t\}$	$\chi(G) + 1, \dots, t\}$	$\chi(G) + 1, \dots, t\}$, where $\chi(G) \leq 4$. This shows the sufficiency of the condition when S is
voloshin-hypergraph_F02298	277	■ 1.000	$\{7, \ldots, \ t+2\}$	$\{7, \dots, t+2\}$	taking the vertices $[7, \dots, t+2]$ and placing them in the region containing vertices 2 and 3.
voloshin-hypergraph_F02299	277	■ 1.000	$\{\{2,3,7\},\{2,3,8\}, \ldots,\{2,3, \ t+2\}\}$	$\{\{2, 3, 7\}, \{2, 3, 8\}, \dots, \{2, 3, t+2\}\}$	Then include the $\mathcal{C}$ -edges $\{\{2, 3, 7\}, \{2, 3, 8\}, \dots, \{2, 3, t+2\}\}$. If we have a 2-coloring on
voloshin-hypergraph_F02300	277	■ 1.000	$S \neq \emptyset$	$S \neq \emptyset$	graph $\mathcal{H}$, with feasible set $S \neq \emptyset$. If $1 \in S$, then S trivially forms an interval. Let c be a
voloshin-hypergraph_F02301	277	■ 1.000	$1 \in S$	$1 \in S$	graph $\mathcal{H}$, with feasible set $S \neq \emptyset$. If $1 \in S$, then S trivially forms an interval. Let c be a

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voloshin-hypergraph_F02302	277	■ 0.999	$\{4,5,\ldots,t\}\subset S(\mathcal{H}^{\prime})\subset S$	$\{4,5,\ldots,t\}\subset S(\mathcal{H}^{\prime})\subset S$	hypergraph $\mathcal{H}^{\prime}$ with $\{4,5,\ldots,t\}\subset S(\mathcal{H}^{\prime})\subset S$. So, by the choice of t , it follows that S is of
voloshin-hypergraph_F02303	277	■ 1.000	u, v	u, v	≥ 3 vertices. If the corresponding region is a C -edge, then it contains vertices u, v with
voloshin-hypergraph_F02304	277	■ 1.000	$c(u)=c(v)$	$c(u)=c(v)$	$c(u)=c(v)$. We replace the region by a C -edge $\{u, v\}$. If the region is a $\mathcal{D}$ -edge, then it
voloshin-hypergraph_F02305	277	■ 1.000	$\{u, v\}$	$\{u, v\}$	$c(u)=c(v)$. We replace the region by a C -edge $\{u, v\}$. If the region is a $\mathcal{D}$ -edge, then it
voloshin-hypergraph_F02306	277	■ 1.000	$c(u)\neq c(v)$	$c(u)\neq c(v)$	contains vertices u, v with $c(u)\neq c(v)$. We replace the region by a $\mathcal{D}$ -edge $\{u, v\}$. If the
voloshin-hypergraph_F02307	277	■ 1.000	u, v, w	u, v, w	region is a bi-edge, then it contains vertices u, v, w with $c(u)=c(v)\neq c(w)$. We replace
voloshin-hypergraph_F02308	277	■ 1.000	$c(u)=c(v)\neq c(w)$	$c(u)=c(v)\neq c(w)$	region is a bi-edge, then it contains vertices u, v, w with $c(u)=c(v)\neq c(w)$. We replace
voloshin-hypergraph_F02309	277	■ 1.000	$\{v, w\}$	$\{v, w\}$	the region by a C -edge $\{u, v\}$ and a $\mathcal{D}$ -edge $\{v, w\}$. The mixed hypergraph $\mathcal{H}^{\prime}$ we obtain
voloshin-hypergraph_F02310	277	■ 1.000	$S(G)=$	$S(G)=$	between strict colorings of G and strict colorings of $\mathcal{H}^{\prime}$. Since the feasible set $S(G)=$
voloshin-hypergraph_F02311	277	■ 0.990	$\{\chi(G), \chi(G)+1, \ldots, t\}$	$\{\chi(G), \chi(G)+1, \ldots, t\}$	$\{\chi(G), \chi(G)+1, \ldots, t\}$, we see that $\{4, 5, \ldots, t\}\subset S(\mathcal{H}^{\prime})$.
voloshin-hypergraph_F02312	277	■ 0.990	$\{4,5,\ldots,t\}\subset S(\mathcal{H}^{\prime})$	$\{4,5,\ldots,t\}\subset S(\mathcal{H}^{\prime})$	$\{\chi(G), \chi(G)+1, \ldots, t\}$, we see that $\{4, 5, \ldots, t\}\subset S(\mathcal{H}^{\prime})$.
voloshin-hypergraph_F02313	279	■ 1.000	$Z_{\{1\}}, Z_{\{2\}}$	Z_1, Z_2	frequency at any time. Suppose we have three zones in a region, say Z_1, Z_2 and Z_3 . Zone
voloshin-hypergraph_F02314	279	■ 1.000	$Z_{\{3\}}$	Z_3	frequency at any time. Suppose we have three zones in a region, say Z_1, Z_2 and Z_3 . Zone
voloshin-hypergraph_F02315	279	■ 1.000	$Z_{\{1\}}$	Z_1	Z_1 must use frequencies a or c , zone Z_2 must use frequencies a or b , and zone Z_3 must use
voloshin-hypergraph_F02316	279	■ 1.000	$Z_{\{2\}}$	Z_2	Z_1 must use frequencies a or c , zone Z_2 must use frequencies a or b , and zone Z_3 must use
voloshin-hypergraph_F02317	279	■ 0.793	$b, c\}$	$b, c\}$	$\{b, c\}$ is assigned to vertex Z_3 . Now the original problem can be formulated as follows: in
voloshin-hypergraph_F02318	279	■ 1.000	$\{Z_{\{1\}}, Z_{\{2\}}, Z_{\{3\}}, a, b, c\}$	$\{Z_1, Z_2, Z_3, a, b, c\}$	$\{Z_1, Z_2, Z_3, a, b, c\}$. Set $\mathcal{D}$ to include all edges of G plus all edges forming complete graph
voloshin-hypergraph_F02319	279	■ 1.000	$\mathcal{C}=\{\{Z_{\{1\}}, a, c\}, \{Z_{\{2\}}, a, b\}, \{Z_{\{3\}}, b, c\}\}$	$\mathcal{C}=\{\{Z_1, a, c\}, \{Z_2, a, b\}, \{Z_3, b, c\}\}$	on vertices a, b and c . At last, set $\mathcal{C}=\{\{Z_1, a, c\}, \{Z_2, a, b\}, \{Z_3, b, c\}\}$.
voloshin-hypergraph_F02320	279	■ 1.000	$\{Z_{\{1\}}, a, c\}$	$\{Z_1, a, c\}$	mixed hypergraph $\mathcal{H}$, and vice versa. Indeed, since the first C -edge $\{Z_1, a, c\}$ must have at
voloshin-hypergraph_F02321	279	■ 0.998	$\chi(\mathcal{H})=\bar{\chi}(\mathcal{H})=$	$\chi(\mathcal{H})=\bar{\chi}(\mathcal{H})=$	For our simple example in Figure 11.1, one can manually find that $\chi(\mathcal{H})=\bar{\chi}(\mathcal{H})=$
voloshin-hypergraph_F02322	280	■ 1.000	$R(\mathcal{H})=(0,0,3,0,0,0)$	$R(\mathcal{H})=(0,0,3,0,0,0)$	3, chromatic spectrum $R(\mathcal{H})=(0,0,3,0,0,0)$, chromatic polynomial $P(\mathcal{H},\lambda)=3\lambda(\lambda-$
voloshin-hypergraph_F02323	280	■ 1.000	$P(\mathcal{H},\lambda)=3\lambda(\lambda-$	$P(\mathcal{H},\lambda)=3\lambda(\lambda-$	3, chromatic spectrum $R(\mathcal{H})=(0,0,3,0,0,0)$, chromatic polynomial $P(\mathcal{H},\lambda)=3\lambda(\lambda-$
voloshin-hypergraph_F02324	280	■ 0.780	$1)(\lambda-2)$	$1)(\lambda-2)$	$1)(\lambda-2)$, and moreover, the three list colorings of G in order (Z_1, Z_2, Z_3) are as follows:
voloshin-hypergraph_F02325	280	■ 0.780	$(Z_{\{1\}}, Z_{\{2\}}, Z_{\{3\}})$	(Z_1, Z_2, Z_3)	$1)(\lambda-2)$, and moreover, the three list colorings of G in order (Z_1, Z_2, Z_3) are as follows:
voloshin-hypergraph_F02326	280	■ 1.000	$(a, b, c), (c, b, c), (c, a, c)$	$(a, b, c), (c, b, c), (c, a, c)$	$(a, b, c), (c, b, c), (c, a, c)$.
voloshin-hypergraph_F02327	280	■ 0.995	$n=4$	$n=4$	Consider the following example. Let us have $n=4$ elementary jobs $X=\{x_1, x_2, x_3,$
voloshin-hypergraph_F02328	280	■ 0.995	$X=\{x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\}$	$X=\{x_1, x_2, x_3$	Consider the following example. Let us have $n=4$ elementary jobs $X=\{x_1, x_2, x_3,$
voloshin-hypergraph_F02329	280	■ 0.987	$x_{\{4\}}\}$	$x_4\}$	$x_4\}$ which are to be executed by allocating $m=5$ available resources $Y=\{y_1, y_2, y_3,$

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voloshin-hypergraph_F02330	280	■ 0.987	m=5	$m = 5$	x_4 which are to be executed by allocating $m = 5$ available resources $Y = \{y_1, y_2, y_3,$
voloshin-hypergraph_F02331	280	■ 0.987	$Y = \left\{ y_1, y_2, y_3 \right\}$.	$Y = \{y_1, y_2, y_3$	x_4 which are to be executed by allocating $m = 5$ available resources $Y = \{y_1, y_2, y_3,$
voloshin-hypergraph_F02332	280	■ 1.000	$\left. y_4, y_5 \right\}$	$y_4, y_5 \}$	$y_4, y_5 \}$. Suppose that time is discrete and each of these elementary jobs can be executed
voloshin-hypergraph_F02333	280	■ 1.000	$S \left(x_i \right) \subseteq Y$	$S \left(x_i \right) \subseteq Y$	during one unit of time. A set of resources $S(x_i) \subseteq Y$ must be available for job $x_i \in X$
voloshin-hypergraph_F02334	280	■ 1.000	$S_{\{1\}} = \left\{ y_1 \right\}$	$S_1 = \{y_1\}$	to be performed. For execution, job x_1 requires resource $S_1 = \{y_1\}$, job x_2 requires re-
voloshin-hypergraph_F02335	280	■ 1.000	$S_{\{2\}} = \left\{ y_1, y_2, y_3 \right\}$	$S_2 = \{y_1, y_2, y_3\}$	sources $S_2 = \{y_1, y_2, y_3\}$, job x_3 requires resources $S_3 = \{y_3, y_4\}$, job x_4 requires resources
voloshin-hypergraph_F02336	280	■ 1.000	$S_{\{3\}} = \left\{ y_3, y_4 \right\}$	$S_3 = \{y_3, y_4\}$	sources $S_2 = \{y_1, y_2, y_3\}$, job x_3 requires resources $S_3 = \{y_3, y_4\}$, job x_4 requires resources
voloshin-hypergraph_F02337	280	■ 1.000	$S_{\{4\}} = \left\{ y_2, y_4, y_5 \right\}$	$S_4 = \{y_2, y_4, y_5\}$	$S_4 = \{y_2, y_4, y_5\}$. For any resource used by at least two jobs simultaneously, a penalty must
voloshin-hypergraph_F02338	280	■ 1.000	$x_{\{1\}}, x_{\{2\}}, x_{\{4\}}$	x_1, x_2, x_4	of time, and jobs x_1, x_2, x_4 must also be performed for a maximum two units of time without
voloshin-hypergraph_F02339	281	■ 1.000	$S_{\{1\}} \cap S_{\{2\}} \neq \emptyset$	$S_1 \cap S_2 \neq \emptyset$	a mixed hypergraph $\mathcal{H}$ as follows. Since $S_1 \cap S_2 \neq \emptyset$, construct the first $\mathcal{D}$ -edge $D_1 =$
voloshin-hypergraph_F02340	281	■ 1.000	$D_{\{1\}} =$	$D_1 =$	a mixed hypergraph $\mathcal{H}$ as follows. Since $S_1 \cap S_2 \neq \emptyset$, construct the first $\mathcal{D}$ -edge $D_1 =$
voloshin-hypergraph_F02341	281	■ 0.999	$S_{\{2\}} \cap S_{\{3\}} \neq \emptyset$	$S_2 \cap S_3 \neq \emptyset$	$\{x_1, x_2\}$. Since $S_2 \cap S_3 \neq \emptyset$, construct the second $\mathcal{D}$ -edge $D_2 = \{x_2, x_3\}$. Since $S_2 \cap S_4 \neq \emptyset$,
voloshin-hypergraph_F02342	281	■ 0.999	$D_{\{2\}} = \left\{ x_2, x_3 \right\}$	$D_2 = \{x_2, x_3\}$	$\{x_1, x_2\}$. Since $S_2 \cap S_3 \neq \emptyset$, construct the second $\mathcal{D}$ -edge $D_2 = \{x_2, x_3\}$. Since $S_2 \cap S_4 \neq \emptyset$,
voloshin-hypergraph_F02343	281	■ 0.999	$S_{\{2\}} \cap S_{\{4\}} \neq \emptyset$	$S_2 \cap S_4 \neq \emptyset$	$\{x_1, x_2\}$. Since $S_2 \cap S_3 \neq \emptyset$, construct the second $\mathcal{D}$ -edge $D_2 = \{x_2, x_3\}$. Since $S_2 \cap S_4 \neq \emptyset$,
voloshin-hypergraph_F02344	281	■ 1.000	$D_{\{3\}} = \left\{ x_2, x_4 \right\}$	$D_3 = \{x_2, x_4\}$	construct the third $\mathcal{D}$ -edge $D_3 = \{x_2, x_4\}$. Since $S_3 \cap S_4 \neq \emptyset$, construct the fourth $\mathcal{D}$ -edge
voloshin-hypergraph_F02345	281	■ 1.000	$S_{\{3\}} \cap S_{\{4\}} \neq \emptyset$	$S_3 \cap S_4 \neq \emptyset$	construct the third $\mathcal{D}$ -edge $D_3 = \{x_2, x_4\}$. Since $S_3 \cap S_4 \neq \emptyset$, construct the fourth $\mathcal{D}$ -edge
voloshin-hypergraph_F02346	281	■ 0.993	$D_{\{4\}} = \left\{ x_3, x_4 \right\}$	$D_4 = \{x_3, x_4\}$	$D_4 = \{x_3, x_4\}$. Since $S_1 \cap S_3 = \emptyset$ and $S_1 \cap S_4 = \emptyset$, no other $\mathcal{D}$ -edge can be added to $\mathcal{H}$.
voloshin-hypergraph_F02347	281	■ 0.993	$S_{\{1\}} \cap S_{\{3\}} = \emptyset$	$S_1 \cap S_3 = \emptyset$	$D_4 = \{x_3, x_4\}$. Since $S_1 \cap S_3 = \emptyset$ and $S_1 \cap S_4 = \emptyset$, no other $\mathcal{D}$ -edge can be added to $\mathcal{H}$.
voloshin-hypergraph_F02348	281	■ 0.993	$S_{\{1\}} \cap S_{\{4\}} = \emptyset$	$S_1 \cap S_4 = \emptyset$	$D_4 = \{x_3, x_4\}$. Since $S_1 \cap S_3 = \emptyset$ and $S_1 \cap S_4 = \emptyset$, no other $\mathcal{D}$ -edge can be added to $\mathcal{H}$.
voloshin-hypergraph_F02349	281	■ 1.000	$C_{\{1\}} = \left\{ x_{\{1\}}, x_{\{2\}}, x_{\{3\}} \right\}, C_{\{2\}} =$	$C_1 = \{x_1, x_2, x_3\}, C_2 =$	Management requirements give us the following two $\mathcal{C}$ -edges: $C_1 = \{x_1, x_2, x_3\}, C_2 =$
voloshin-hypergraph_F02350	281	■ 1.000	$\left\{ x_{\{1\}}, x_{\{2\}}, x_{\{4\}} \right\}$	$\{x_1, x_2, x_4\}$	$\{x_1, x_2, x_4\}$. Put $\mathcal{C} = \{C_1, C_2\}, \mathcal{D} = \{D_1, D_2, D_3, D_4\}$. Thus, the mixed hypergraph $\mathcal{H} =$
voloshin-hypergraph_F02351	281	■ 1.000	$\mathcal{C} = \left\{ C_{\{1\}}, C_{\{2\}} \right\}, \mathcal{D} = \left\{ D_{\{1\}}, D_{\{2\}}, D_{\{3\}}, D_{\{4\}} \right\}$	$\mathcal{C} = \{C_1, C_2\}, \mathcal{D} = \{D_1, D_2, D_3, D_4\}$	$\{x_1, x_2, x_4\}$. Put $\mathcal{C} = \{C_1, C_2\}, \mathcal{D} = \{D_1, D_2, D_3, D_4\}$. Thus, the mixed hypergraph $\mathcal{H} =$
voloshin-hypergraph_F02352	283	■ 0.998	$n = 1, 2, 3, \ldots$	$n = 1, 2, 3, \dots$	where $n = 1, 2, 3, \dots$. This is equivalent to prove an infinite list of statements $P_1, P_2, \dots$ to
voloshin-hypergraph_F02353	283	■ 0.998	$P_{\{1\}}, P_{\{2\}}, \ldots$	$P_1, P_2, \dots$	where $n = 1, 2, 3, \dots$. This is equivalent to prove an infinite list of statements $P_1, P_2, \dots$ to
voloshin-hypergraph_F02354	283	■ 1.000	$P_{\{1\}}$	P_1	1. Prove P_1 .
voloshin-hypergraph_F02355	283	■ 1.000	$P_{\{k\}}$	P_k	2. For any $k \geq 1$, prove that if P_k is true, then P_{k+1} is true.
voloshin-hypergraph_F02356	283	■ 1.000	$P_{\{k+1\}}$	P_{k+1}	2. For any $k \geq 1$, prove that if P_k is true, then P_{k+1} is true.
voloshin-hypergraph_F02357	283	■ 1.000	$n = 1, 2, \ldots$	$n = 1, 2, \dots$	It is easy to see that this is sufficient to conclude that P_n is true for all $n = 1, 2, \dots$, or,

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voloshin-hypergraph_F02358	283	■ 1.000	$P_{\{k\}} \rightarrow P_{\{k+1\}}$	$P_k \rightarrow P_{k+1}$	step. It can be abbreviated as $P_k \rightarrow P_{k+1}$ and called an implication . In step 2., the statement
voloshin-hypergraph_F02359	283	■ 1.000	$P_{\{n\}}, n$	P_n, n	P_k is called the induction hypothesis . In notation P_n , n is called the induction parameter .
voloshin-hypergraph_F02360	283	■ 1.000	$P_{\{1\}} \rightarrow P_{\{2\}}$	$P_1 \rightarrow P_2$	instead of proving $P_1 \rightarrow P_2$, then $P_2 \rightarrow P_3$, and so on (up to infinity!), one just prove the
voloshin-hypergraph_F02361	283	■ 1.000	$P_{\{2\}} \rightarrow P_{\{3\}}$	$P_2 \rightarrow P_3$	instead of proving $P_1 \rightarrow P_2$, then $P_2 \rightarrow P_3$, and so on (up to infinity!), one just prove the
voloshin-hypergraph_F02362	283	■ 1.000	$P_{\{n_{\{0\}}\}}$	P_{n_0}	2. Induction basis: write down and prove the statement P_{n_0} , where property P_{n_0} is
voloshin-hypergraph_F02363	283	■ 1.000	$n_{\{0\}}=3$	$n_0 = 3$	directly observed; usually, n is the number of vertices, and $n_0 = 3$, or $n_0 = 4$.
voloshin-hypergraph_F02364	283	■ 1.000	$n_{\{0\}}=4$	$n_0 = 4$	directly observed; usually, n is the number of vertices, and $n_0 = 3$, or $n_0 = 4$.
voloshin-hypergraph_F02365	284	■ 1.000	$k+1$	$k + 1$	that P_k is true, must be used . To prove P_{k+1} , one consider a graph on $k + 1$ vertices.
voloshin-hypergraph_F02366	284	■ 1.000	$n \geq n_{\{0\}}$	$n \geq n_0$	6. Conclude that P_n is true for all $n \geq n_0$.
voloshin-hypergraph_F02367	285	■ 1.000	$P_{\{3\}}$	P_3	true, and if P_3 is true, and so on, and if P_k is true, then P_{k+1} is true. This can be abbreviated
voloshin-hypergraph_F02368	285	■ 1.000	$P_{\{1\}}, P_{\{2\}}, \ldots, P_{\{k\}} \rightarrow P_{\{k+1\}}$	$P_1, P_2, \dots, P_k \rightarrow P_{k+1}$	as $P_1, P_2, \dots, P_k \rightarrow P_{k+1}$. Implicitly it is clear that if we started at P_1 and arrived to P_k , then
voloshin-hypergraph_F02369	285	■ 1.000	$P_{\{2\}}, P_{\{3\}}, \ldots, P_{\{k-1\}}$	$P_2, P_3, \dots, P_{k-1}$	all intermediate statements $P_2, P_3, \dots, P_{k-1}$ must be true. In some complicated cases we
voloshin-hypergraph_F02370	285	■ 1.000	$f(n)$	$f(n)$	function $f(n)$. If we have another algorithm for the same problem, then the running time
voloshin-hypergraph_F02371	285	■ 1.000	$g(n)$	$g(n)$	is another function of n , say, $g(n)$. How to compare the algorithms? The first algorithm is
voloshin-hypergraph_F02372	285	■ 1.000	$f(n) \leq g(n)$	$f(n) \leq g(n)$	better if $f(n) \leq g(n)$ beginning with some fixed number of vertices, say $n \geq N_1$. If the first
voloshin-hypergraph_F02373	285	■ 1.000	$n \geq N_{\{1\}}$	$n \geq N_1$	better if $f(n) \leq g(n)$ beginning with some fixed number of vertices, say $n \geq N_1$. If the first
voloshin-hypergraph_F02374	285	■ 1.000	$f(n) / c_{\{1\}} \leq g(n) / c_{\{2\}}$	$f(n)/c_1 \leq g(n)/c_2$	a computer which is c_2 times faster, then the first is better when $f(n)/c_1 \leq g(n)/c_2$ holds
voloshin-hypergraph_F02375	285	■ 1.000	$n \geq N_{\{2\}}$	$n \geq N_2$	for all $n \geq N_2$ where N_2 is some other natural number. This is equivalent to $f(n) \leq Cg(n)$
voloshin-hypergraph_F02376	285	■ 1.000	$f(n) \leq C g(n)$	$f(n) \leq Cg(n)$	for all $n \geq N_2$ where N_2 is some other natural number. This is equivalent to $f(n) \leq Cg(n)$
voloshin-hypergraph_F02377	285	■ 1.000	$C=c_{\{1\}} / c_{\{2\}}$	$C = c_1/c_2$	for some constant $C = c_1/c_2$. Evidently, it holds for any other constant $C' \geq C$ and any
voloshin-hypergraph_F02378	285	■ 1.000	$C^{\{\prime\}}$ $\geq C$	$C' \geq C$	for some constant $C = c_1/c_2$. Evidently, it holds for any other constant $C' \geq C$ and any
voloshin-hypergraph_F02379	285	■ 1.000	$N \geq N_{\{2\}}$	$N \geq N_2$	other $N \geq N_2$. In other words, no matter how fast the new computers would be, if the
voloshin-hypergraph_F02380	285	■ 1.000	$n \geq N$	$n \geq N$	inequality $f(n) \leq Cg(n)$ holds for some sufficiently large constant C and all $n \geq N$, then the
voloshin-hypergraph_F02381	285	■ 1.000	$0(g(n))$	$O(g(n))$	We say that $f(n)$ is $O(g(n))$ (read “ $f(n)$ is big-oh of $g(n)$ ”, sometimes denoted by $f(n)$
voloshin-hypergraph_F02382	285	■ 0.999	$=0(g(n))$	$= O(g(n))$	$= O(g(n))$) if there are constant C and a number N such that
voloshin-hypergraph_F02383	285	■ 1.000	$0(g(n))$	$O(g(n))$	the complexity of the algorithm is $O(g(n))$ where $g(n)$ is that very same upper bound.
voloshin-hypergraph_F02384	285	■ 1.000	$g(n)=n$	$g(n) = n$	If $g(n) = n$, then the algorithm is called linear-time and its complexity is denoted by
voloshin-hypergraph_F02385	285	■ 1.000	$0(n)$	$O(n)$	$O(n)$. Generally, if $g(n)$ is a polynomial of degree k on n , then the algorithm is called

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voloshin-hypergraph_F02386	286	■ 1.000	$0\left(n^k\right)$	$O\left(n^k\right)$	polynomial-time and complexity is denoted by $O(n^k)$. If $g(n)$ is an exponential function,
voloshin-hypergraph_F02387	286	■ 1.000	$0\left(a^n\right)$	$O\left(a^n\right)$	then the algorithm is called exponential-time and complexity is denoted by $O(a^n)$ where
voloshin-hypergraph_F02388	286	■ 1.000	$a>1$	$a>1$	$a>1$. There is also a constant complexity denoted by $O(1)$, logarithmic complexity
voloshin-hypergraph_F02389	286	■ 1.000	$0(1)$	$O(1)$	$a>1$. There is also a constant complexity denoted by $O(1)$, logarithmic complexity
voloshin-hypergraph_F02390	286	■ 1.000	$0(\log n)$	$O(\log n)$	denoted by $O(\log n)$ and even factorial complexity denoted by $O(n!)$.
voloshin-hypergraph_F02391	286	■ 1.000	$0(n!)$	$O(n!)$	denoted by $O(\log n)$ and even factorial complexity denoted by $O(n!)$.
voloshin-hypergraph_F02392	286	■ 0.798	$\left(n^2-n\right) / 2$	$\left(n^2-n\right) / 2$	every entry of the lower triangle if it is 0 or 1. The number of such checks is $(n^2-n)/2$
voloshin-hypergraph_F02393	286	■ 1.000	$0\left(n^2\right)$	$O\left(n^2\right)$	which is the polynomial of n . Therefore the complexity of such procedure is $O(n^2)$.
voloshin-hypergraph_F02394	286	■ 1.000	$m n$	$m n$	n vertices and m edges given by its incidence matrix, then we need to check every of $m n$
voloshin-hypergraph_F02395	286	■ 0.974	$0(m n)$	$O(m n)$	entries if it is 0 or 1. So the total time is $O(m n)$. However, if the hypergraph is given by its
voloshin-hypergraph_F02396	286	■ 1.000	0	O	the total time is O (the sum of all edge cardinalities). As one can see, the complexity of
voloshin-hypergraph_F02397	286	■ 1.000	$n\left(G_1\right)=4, m\left(G_1\right)=3, n\left(G_2\right)=5, m\left(G_2\right)=7, n\left(G_3\right)=8, m\left(G_3\right)=12$	$n\left(G_1\right)=4, m\left(G_1\right)=3, n\left(G_2\right)=5, m\left(G_2\right)=7, n\left(G_3\right)=8, m\left(G_3\right)=12$	1. $n\left(G_1\right)=4, m\left(G_1\right)=3, n\left(G_2\right)=5, m\left(G_2\right)=7, n\left(G_3\right)=8, m\left(G_3\right)=12$. 5. Degree se-
voloshin-hypergraph_F02398	286	■ 1.000	$G_1:(1,1,1,3), G_2:(2,2,2,4,4), G_3:(2,2,3,3,3,3,4,4)$	$G_1:(1,1,1,3), G_2:(2,2,2,4,4), G_3:(2,2,3,3,3,3,4,4)$	quences: $G_1:(1,1,1,3), G_2:(2,2,2,4,4), G_3:(2,2,3,3,3,3,4,4)$.
voloshin-hypergraph_F02399	286	■ 1.000	$L\left(G_1\right)=\left\{\left\{2\right\},\left\{1,3,4\right\},\left\{2,4\right\},\left\{2,3\right\}\right\}$	$L\left(G_1\right)=\left\{\left\{2\right\},\left\{1,3,4\right\},\left\{2,4\right\},\left\{2,3\right\}\right\}$	1. $L\left(G_1\right)=\left\{\left\{2\right\},\left\{1,3,4\right\},\left\{2,4\right\},\left\{2,3\right\}\right\}$.
voloshin-hypergraph_F02400	287	■ 0.998	$J\left(G_1\right)=\left\{\left\{1,2\right\},\left\{2,3\right\},\left\{3,4\right\},\left\{2,4\right\}\right\}$	$J\left(G_1\right)=\left\{\left\{1,2\right\},\left\{2,3\right\},\left\{3,4\right\},\left\{2,4\right\}\right\}$	4. $J\left(G_1\right)=\left\{\left\{1,2\right\},\left\{2,3\right\},\left\{3,4\right\},\left\{2,4\right\}\right\}$.
voloshin-hypergraph_F02401	287	■ 1.000	G_5, G_6, G_7, G_8, G_9	G_5, G_6, G_7, G_8, G_9	5. G_5, G_6, G_7, G_8, G_9 . 7. Because $K_{s,r}$ is obtained from $K_{r,s}$ by just interchanging the parts.
voloshin-hypergraph_F02402	287	■ 0.999	$G_1 \cong G_2, G_3 \cong G_4, G_7 \cong G_8$	$G_1 \cong G_2, G_3 \cong G_4, G_7 \cong G_8 \cong G_9$	8. $G_1 \cong G_2, G_3 \cong G_4, G_7 \cong G_8 \cong G_9$. 9. When the names and order of vertices correspond
voloshin-hypergraph_F02403	287	■ 1.000	$\eta(T)=2, \eta\left(C_5\right)=3, \eta\left(W_5\right)=4, \eta\left(K_{2,3}\right)=3$	$\eta(T)=2, \eta\left(C_5\right)=3, \eta\left(W_5\right)=4, \eta\left(K_{2,3}\right)=3$	6. $\eta(T)=2, \eta\left(C_5\right)=3, \eta\left(W_5\right)=4, \eta\left(K_{2,3}\right)=3$. 10. If we reverse the order in degree
voloshin-hypergraph_F02404	287	■ 1.000	$K_{4,4}: C_4, C_8$	$K_{4,4}: C_4, C_8$	11. In $K_{4,4}$: C_4, C_8 , in cube: C_4, C_8 , in Petersen graph: C_5, C_9 . 13. For graphs in Figure 1.21:
voloshin-hypergraph_F02405	287	■ 1.000	C_4, C_8	C_4, C_8	11. In $K_{4,4}$: C_4, C_8 , in cube: C_4, C_8 , in Petersen graph: C_5, C_9 . 13. For graphs in Figure 1.21:
voloshin-hypergraph_F02406	287	■ 1.000	C_5, C_9	C_5, C_9	11. In $K_{4,4}$: C_4, C_8 , in cube: C_4, C_8 , in Petersen graph: C_5, C_9 . 13. For graphs in Figure 1.21:
voloshin-hypergraph_F02407	287	■ 0.890	$\omega\left(G_1\right)=\omega\left(G_2\right)=\omega\left(G_3\right)=\omega\left(G_4\right)=3, \omega\left(G_5\right)=\omega\left(G_6\right)=\omega\left(G_7\right)=\omega\left(G_8\right)=\omega\left(G_9\right)=2$	$\omega\left(G_1\right)=\omega\left(G_2\right)=\omega\left(G_3\right)=\omega\left(G_4\right)=3, \omega\left(G_5\right)=\omega\left(G_6\right)=\omega\left(G_7\right)=\omega\left(G_8\right)=\omega\left(G_9\right)=2$	$\omega\left(G_1\right)=\omega\left(G_2\right)=\omega\left(G_3\right)=\omega\left(G_4\right)=3, \omega\left(G_5\right)=\omega\left(G_6\right)=\omega\left(G_7\right)=\omega\left(G_8\right)=\omega\left(G_9\right)=2$.
voloshin-hypergraph_F02408	287	■ 1.000	$\alpha\left(G_1\right)=\alpha\left(G_2\right)=\alpha\left(G_3\right)=\alpha\left(G_4\right)=2, \alpha\left(G_5\right)=\alpha\left(G_6\right)=4$	$\alpha\left(G_1\right)=\alpha\left(G_2\right)=\alpha\left(G_3\right)=\alpha\left(G_4\right)=2, \alpha\left(G_5\right)=\alpha\left(G_6\right)=4$	14. For graphs in Figure 1.21: $\alpha\left(G_1\right)=\alpha\left(G_2\right)=\alpha\left(G_3\right)=\alpha\left(G_4\right)=2, \alpha\left(G_5\right)=\alpha\left(G_6\right)=4$,
voloshin-hypergraph_F02409	287	■ 0.803	$\alpha\left(G_7\right)=\alpha\left(G_8\right)=\alpha\left(G_9\right)=3$	$\alpha\left(G_7\right)=\alpha\left(G_8\right)=\alpha\left(G_9\right)=3$	$\alpha\left(G_7\right)=\alpha\left(G_8\right)=\alpha\left(G_9\right)=3$. 15. For graphs in Figure 1.21: $\tau\left(G_1\right)=\tau\left(G_2\right)=2, \tau\left(G_3\right)=$

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voloshin-hypergraph_F02410	287	■ 0.803	$\boldsymbol{\tau}(G_1)=\boldsymbol{\tau}(G_2)=2, \boldsymbol{\tau}(G_3)=$	$\tau(G_1) = \tau(G_2) = 2, \tau(G_3) =$	$\alpha(G_7) = \alpha(G_8) = \alpha(G_9) = 3$. 15. For graphs in Figure 1.21: $\tau(G_1) = \tau(G_2) = 2, \tau(G_3) =$
voloshin-hypergraph_F02411	287	■ 0.997	$\tau(G_4)=3, \tau(G_5)=\tau(G_6)=4, \tau(G_7)=\tau(G_8)=\tau(G_9)=2$	$\tau(G_4) = 3, \tau(G_5) = \tau(G_6) = 4, \tau(G_7) = \tau(G_8) = \tau(G_9) = 2$	$\tau(G_4) = 3, \tau(G_5) = \tau(G_6) = 4, \tau(G_7) = \tau(G_8) = \tau(G_9) = 2$. 16. For graphs in Figure 1.21:
voloshin-hypergraph_F02412	287	■ 0.995	$\nu(G_1)=\nu(G_2)=\nu(G_3)=\nu(G_4)=2, \nu(G_5)=\nu(G_6)=4, \nu(G_7)=\nu(G_8)=\nu(G_9)=2$	$\nu(G_1) = \nu(G_2) = \nu(G_3) = \nu(G_4) = 2, \nu(G_5) = \nu(G_6) = 4, \nu(G_7) = \nu(G_8) = \nu(G_9) = 2$	$\nu(G_1) = \nu(G_2) = \nu(G_3) = \nu(G_4) = 2, \nu(G_5) = \nu(G_6) = 4, \nu(G_7) = \nu(G_8) = \nu(G_9) = 2$.
voloshin-hypergraph_F02413	287	■ 1.000	$\alpha(C_n)=\tau(C_n)=\nu(C_n)=n/2$	$\alpha(C_n) = \tau(C_n) = \nu(C_n) = n/2$	17. If n is even: $\alpha(C_n) = \tau(C_n) = \nu(C_n) = n/2$, if n is odd: $\alpha(C_n) = \nu(C_n) = (n-1)/2$,
voloshin-hypergraph_F02414	287	■ 1.000	$\alpha(C_n)=\nu(C_n)=(n-1)/2$	$\alpha(C_n) = \nu(C_n) = (n-1)/2$	17. If n is even: $\alpha(C_n) = \tau(C_n) = \nu(C_n) = n/2$, if n is odd: $\alpha(C_n) = \nu(C_n) = (n-1)/2$,
voloshin-hypergraph_F02415	287	■ 1.000	$\tau(C_n)=(n+1)/2$	$\tau(C_n) = (n+1)/2$	$\tau(C_n) = (n+1)/2$, for all n : $\omega(C_n) = 2$.
voloshin-hypergraph_F02416	287	■ 1.000	$n: \omega(C_n)=2$	$n: \omega(C_n) = 2$	$\tau(C_n) = (n+1)/2$, for all n : $\omega(C_n) = 2$.
voloshin-hypergraph_F02417	287	■ 0.905	$\Lambda(E_n)=0, \Lambda(C_n)=1, \Lambda(K_n)=(n-1)(n-2)/2, \Lambda(W_n)=n-1$	$\Lambda(E_n) = 0, \Lambda(C_n) = 1, \Lambda(K_n) = (n-1)(n-2)/2, \Lambda(W_n) = n-1$	1. $\Lambda(E_n) = 0, \Lambda(C_n) = 1, \Lambda(K_n) = (n-1)(n-2)/2, \Lambda(W_n) = n-1$.
voloshin-hypergraph_F02418	287	■ 0.782	$\operatorname{diam}(T)=9$	$\operatorname{diam}(T) = 9$	1. $\operatorname{diam}(T) = 9, \operatorname{radius} = 5$.
voloshin-hypergraph_F02419	287	■ 0.782	$=5$	$= 5$	1. $\operatorname{diam}(T) = 9, \operatorname{radius} = 5$.
voloshin-hypergraph_F02420	287	■ 0.604	$=28$	$= 28$	2. Minimum weight = 28. 3. Maximum weight = 49. 4. n^{n-2} .
voloshin-hypergraph_F02421	287	■ 0.604	$=49$	$= 49$	2. Minimum weight = 28. 3. Maximum weight = 49. 4. n^{n-2} .
voloshin-hypergraph_F02422	287	■ 0.745	$m=n$	$m = n$	2. $m = n$. 4. $\tau(K_{m,n}) = \nu(K_{m,n}) = \min\{m, n\}$. 6. $\tau = \nu = 13$.
voloshin-hypergraph_F02423	287	■ 0.745	$\tau(K_{m,n})=\nu(K_{m,n})=\min\{m, n\}$	$\tau(K_{m,n}) = \nu(K_{m,n}) = \min\{m, n\}$	2. $m = n$. 4. $\tau(K_{m,n}) = \nu(K_{m,n}) = \min\{m, n\}$. 6. $\tau = \nu = 13$.
voloshin-hypergraph_F02424	287	■ 0.745	$\tau=\nu=13$	$\tau = \nu = 13$	2. $m = n$. 4. $\tau(K_{m,n}) = \nu(K_{m,n}) = \min\{m, n\}$. 6. $\tau = \nu = 13$.
voloshin-hypergraph_F02425	288	■ 0.646	$\Theta(G_1)=\alpha(G_1)=3, \Theta(G_2)=$	$\Theta(G_1) = \alpha(G_1) = 3, \Theta(G_2) =$	2. G_1 and G_3 are chordal; G_2 is not chordal. 3. G_2 . 7. $\Theta(G_1) = \alpha(G_1) = 3, \Theta(G_2) =$
voloshin-hypergraph_F02426	288	■ 0.544	$\alpha(G_2)=5, \theta(G_3)=\alpha(G_3)=3$	$\alpha(G_2) = 5, \theta(G_3) = \alpha(G_3) = 3$	$\alpha(G_2) = 5, \theta(G_3) = \alpha(G_3) = 3$. 8. For cube: 6. 9. From cube: 5.
voloshin-hypergraph_F02427	288	■ 0.987	$M(G_1)=3, \omega(G_1)-1=2; M(G_2)=4, \omega(G_2)-1=2; M(G_3)=2, \omega(G_3)-1=2.2$	$M(G_1) = 3, \omega(G_1) - 1 = 2; M(G_2) = 4, \omega(G_2) - 1 = 2; M(G_3) = 2, \omega(G_3) - 1 = 2.2$	1. $M(G_1) = 3, \omega(G_1) - 1 = 2; M(G_2) = 4, \omega(G_2) - 1 = 2; M(G_3) = 2, \omega(G_3) - 1 = 2$. 2.
voloshin-hypergraph_F02428	288	■ 1.000	$M(C_n)=2, M(K_n)=n-1, M(W_n)=3$	$M(C_n) = 2, M(K_n) = n-1, M(W_n) = 3$	G_1, G_2 are not chordal; G_3 is chordal. 3. $M(C_n) = 2, M(K_n) = n-1, M(W_n) = 3$; for cube,
voloshin-hypergraph_F02429	288	■ 1.000	$M(G)=3$	$M(G) = 3$	prism and Petersen graph: $M(G) = 3$. 4. Take empty graph E_n with sufficient large n ; add
voloshin-hypergraph_F02430	288	■ 1.000	$\leq k-1$	$\leq k-1$	k -regular, at any step except the first, there is a vertex of degree $\leq k-1$.
voloshin-hypergraph_F02431	288	■ 0.934	$\operatorname{diam}(G)=6$	$\operatorname{diam}(G) = 6$	2. $\operatorname{diam}(G) = 6$, radius = 3.
voloshin-hypergraph_F02432	288	■ 1.000	$n=3, 4, K_n$	$n = 3, 4, K_n$	C_n for $n = 3, 4, K_n$ for $n \geq 3, W_n$ for $n = 4, 5$.

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voloshin-hypergraph_ F02433	288	■ 1.000	$n \geq 3, W_{\{n\}}$	$n \geq 3, W_n$	C_n for $n = 3, 4$, K_n for $n \geq 3$, W_n for $n = 4, 5$.
voloshin-hypergraph_ F02434	288	■ 1.000	$n=4, 5$	$n = 4, 5$	C_n for $n = 3, 4$, K_n for $n \geq 3$, W_n for $n = 4, 5$.
voloshin-hypergraph_ F02435	288	■ 0.999	$G_{\{1\}}: 5 ; G_{\{2\}}: 7$	$G_1 : 5; G_2 : 7$	1. $G_1: 5; G_2: 7$. 3. Yes.
voloshin-hypergraph_ F02436	288	■ 0.427	$K_{\{3,3\}}: 0, 1, 2, 3$	$K_{3,3} : 0, 1, 2, 3$	1. For $K_{3,3}$: 0,1,2,3; for K_5 : 0,1,2,3,4. 6. 1.
voloshin-hypergraph_ F02437	288	■ 0.427	$K_{\{5\}}: 0, 1, 2, 3, 4$	$K_5 : 0, 1, 2, 3, 4$	1. For $K_{3,3}$: 0,1,2,3; for K_5 : 0,1,2,3,4. 6. 1.
voloshin-hypergraph_ F02438	288	■ 1.000	$K_{\{n\}}: n=3, 4 ; K_{\{m, n\}}: m=1, n \geq 3$	$K_n : n = 3, 4; K_{m,n} : m = 1, n \geq 3$	1. K_n : $n = 3, 4$; $K_{m,n}$: $m = 1, n \geq 3$, or $m = 2, n \geq 3$; W_n : $n \geq 4$. 3. G_1 is not planar; G_2 is
voloshin-hypergraph_ F02439	288	■ 1.000	$m=2, n \geq 3 ; W_{\{n\}}: n \geq 4$	$m = 2, n \geq 3; W_n : n \geq 4$	1. K_n : $n = 3, 4$; $K_{m,n}$: $m = 1, n \geq 3$, or $m = 2, n \geq 3$; W_n : $n \geq 4$. 3. G_1 is not planar; G_2 is
voloshin-hypergraph_ F02440	288	■ 1.000	$\chi(E_{\{n\}})=1, \chi(K_{\{n\}})=n, \chi(K_{\{m, n\}})=2, \chi(T_{\{n\}})=2, \chi(C_{\{2 n\}})=2, \chi(C_{\{2 n+1\}})=3, \chi(W_{\{2 n\}})=4$	$\chi(E_n) = 1, \chi(K_n) = n, \chi(K_{m,n}) = 2, \chi(T_n) = 2, \chi(C_{2n}) = 2, \chi(C_{2n+1}) = 3, \chi(W_{2n}) = 4$	1. $\chi(E_n) = 1, \chi(K_n) = n, \chi(K_{m,n}) = 2, \chi(T_n) = 2, \chi(C_{2n}) = 2, \chi(C_{2n+1}) = 3, \chi(W_{2n}) = 4$,
voloshin-hypergraph_ F02441	288	■ 0.928	$\chi(W_{\{2 n+1\}})=3$	$\chi(W_{2n+1}) = 3$	$\chi(W_{2n+1}) = 3$. 4. $\lambda \geq 136$. 6. $\chi(G) \leq 6$. 7. $\chi(G) = 5$.
voloshin-hypergraph_ F02442	288	■ 0.928	$\lambda \geq 136$	$\lambda \geq 136$	$\chi(W_{2n+1}) = 3$. 4. $\lambda \geq 136$. 6. $\chi(G) \leq 6$. 7. $\chi(G) = 5$.
voloshin-hypergraph_ F02443	288	■ 0.928	$\chi(G)=5$	$\chi(G) = 5$	$\chi(W_{2n+1}) = 3$. 4. $\lambda \geq 136$. 6. $\chi(G) \leq 6$. 7. $\chi(G) = 5$.
voloshin-hypergraph_ F02444	288	■ 1.000	$P(E_{\{4\}}, \lambda)=\lambda^4 ; P(C_{\{5\}}, \lambda)=(\lambda-1)^5-(\lambda-1) ; P(W_{\{4\}}, \lambda)=\lambda^4 ; P(P_{\{n\}}, \lambda)=\lambda(\lambda-1)^{n-1}$	$P(E_4, \lambda) = \lambda^4; P(C_5, \lambda) = (\lambda - 1)^5 - (\lambda - 1); P(W_4, \lambda) = \lambda^4; P(P_n, \lambda) = \lambda(\lambda - 1)^{n-1}$	1. $P(E_4, \lambda) = \lambda^4; P(C_5, \lambda) = (\lambda - 1)^5 - (\lambda - 1); P(W_4, \lambda) = \lambda^4; P(P_n, \lambda) = \lambda(\lambda - 1)^{n-1}$.
voloshin-hypergraph_ F02445	288	■ 1.000	$S(7,1)=1, S(7,2)=63, S(7,3)=301, S(7,4)=350, S(7,5)=140, S(7,6)=21$	$S(7,1) = 1, S(7,2) = 63, S(7,3) = 301, S(7,4) = 350, S(7,5) = 140, S(7,6) = 21$	2. $S(7,1) = 1, S(7,2) = 63, S(7,3) = 301, S(7,4) = 350, S(7,5) = 140, S(7,6) = 21$,
voloshin-hypergraph_ F02446	288	■ 0.999	$S(7,7)=1 ; s(3,1)=2, s(3,2)=-3, s(3,3)=1$	$S(7,7) = 1; s(3,1) = 2, s(3,2) = -3, s(3,3) = 1$	$S(7,7) = 1; s(3,1) = 2, s(3,2) = -3, s(3,3) = 1$. 3. $P(G_1, \lambda) = \lambda(\lambda - 1)^3(\lambda - 2)^2(\lambda^2 -$
voloshin-hypergraph_ F02447	288	■ 0.999	$P(G_{\{1\}}, \lambda)=\lambda(\lambda-1)^3(\lambda-2)^2$	$P(G_1, \lambda) = \lambda(\lambda - 1)^3(\lambda - 2)^2(\lambda^2 -$	$S(7,7) = 1; s(3,1) = 2, s(3,2) = -3, s(3,3) = 1$. 3. $P(G_1, \lambda) = \lambda(\lambda - 1)^3(\lambda - 2)^2(\lambda^2 -$
voloshin-hypergraph_ F02448	288	■ 1.000	$3 \lambda+3)^2 ; P(G_{\{2\}}, \lambda)=\lambda(\lambda-1)(\lambda-2)^4 ; P(G_{\{3\}}, \lambda)=\lambda(\lambda-1)(\lambda-2)^4(\lambda^2-3 \lambda+3)$	$3 \lambda+3)^2; P(G_2, \lambda) = \lambda(\lambda - 1)(\lambda - 2)^4; P(G_3, \lambda) = \lambda(\lambda - 1)(\lambda - 2)^4(\lambda^2 - 3 \lambda + 3)$	$3 \lambda+3)^2; P(G_2, \lambda) = \lambda(\lambda - 1)(\lambda - 2)^4; P(G_3, \lambda) = \lambda(\lambda - 1)(\lambda - 2)^4(\lambda^2 - 3 \lambda + 3)$.
voloshin-hypergraph_ F02449	289	■ 1.000	$L(G)=\{\{2,4\}\}$	$L(G) = \{\{2, 4\}\}$	3. Graph $G = (X, E)$ with $X = \{1, 2, 3, 4, 5\}$ and adjacency list $L(G) = \{\{2, 4\},$
voloshin-hypergraph_ F02450	289	■ 0.970	$\{1,3,4,5\},\{2,5\},\{1,2,5\},\{2,3,4\}\}$	$\{1, 3, 4, 5\}, \{2, 5\}, \{1, 2, 5\}, \{2, 3, 4\}\}$	$\{1, 3, 4, 5\}, \{2, 5\}, \{1, 2, 5\}, \{2, 3, 4\}\}$; the order of online coloring: 1,2,3,4,5.
voloshin-hypergraph_ F02451	289	■ 0.970	$1, 2, 3, 4, 5$	$1, 2, 3, 4, 5$	$\{1, 3, 4, 5\}, \{2, 5\}, \{1, 2, 5\}, \{2, 3, 4\}\}$; the order of online coloring: 1,2,3,4,5.
voloshin-hypergraph_ F02452	289	■ 1.000	$M=3$	$M = 3$	1. For Petersen graph: $M = 3$, therefore $\chi \leq 4$. 2. Idea: switch the colors in the regions on
voloshin-hypergraph_ F02453	289	■ 1.000	$\chi \leq 4$	$\chi \leq 4$	1. For Petersen graph: $M = 3$, therefore $\chi \leq 4$. 2. Idea: switch the colors in the regions on
voloshin-hypergraph_ F02454	289	■ 1.000	$W_{\{2 k-1\}}$	W_{2k-1}	1. None of them is perfect because each contains induced C_5 . 2. W_{2k-1} are perfect, W_{2k} are
voloshin-hypergraph_ F02455	289	■ 1.000	$W_{\{2 k\}}$	W_{2k}	1. None of them is perfect because each contains induced C_5 . 2. W_{2k-1} are perfect, W_{2k} are
voloshin-hypergraph_ F02456	289	■ 1.000	$\chi^{\prime}\left(K_{\{n\}}\right)=n$	$\chi^{\prime}\left(K_n\right)=n$	1. $\chi^{\prime}\left(K_n\right)=n$ if $n \geq 3$ is odd, $\chi^{\prime}\left(K_n\right)=n-1$ if $n \geq 2$ is even. $\chi^{\prime}\left(C_n\right)=3$ if $n \geq 3$ is odd,

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voloshin-hypergraph_F02457	289	■ 1.000	$\chi^{\prime}(\text{prime})\left(K_n\right)=n-1$	$\chi^{\prime}\left(K_n\right)=n-1$	1. $\chi^{\prime}\left(K_n\right)=n$ if $n \geq 3$ is odd, $\chi^{\prime}\left(K_n\right)=n-1$ if $n \geq 2$ is even. $\chi^{\prime}\left(C_n\right)=3$ if $n \geq 3$ is odd,
voloshin-hypergraph_F02458	289	■ 1.000	$\chi^{\prime}(\text{prime})\left(C_n\right)=3$	$\chi^{\prime}\left(C_n\right)=3$	1. $\chi^{\prime}\left(K_n\right)=n$ if $n \geq 3$ is odd, $\chi^{\prime}\left(K_n\right)=n-1$ if $n \geq 2$ is even. $\chi^{\prime}\left(C_n\right)=3$ if $n \geq 3$ is odd,
voloshin-hypergraph_F02459	289	■ 0.997	$\chi^{\prime}(\text{prime})\left(C_n\right)=2$	$\chi^{\prime}\left(C_n\right)=2$	$\chi^{\prime}\left(C_n\right)=2$ if $n \geq 2$ is even. $\chi^{\prime}\left(W_n\right)=n-1$ for all $n \geq 4$. $\chi^{\prime}\left(K_{m,n}\right)=\max \{m, n\}$.
voloshin-hypergraph_F02460	289	■ 0.997	$\chi^{\prime}(\text{prime})\left(W_n\right)=n-1$	$\chi^{\prime}\left(W_n\right)=n-1$	$\chi^{\prime}\left(C_n\right)=2$ if $n \geq 2$ is even. $\chi^{\prime}\left(W_n\right)=n-1$ for all $n \geq 4$. $\chi^{\prime}\left(K_{m,n}\right)=\max \{m, n\}$.
voloshin-hypergraph_F02461	289	■ 0.997	$n \geq 4 . \chi^{\prime}(\text{prime})\left(K_{\{m, n\}}\right)=\max \{m, n\}$	$n \geq 4 . \chi^{\prime}\left(K_{m,n}\right)=\max \{m, n\}$	$\chi^{\prime}\left(C_n\right)=2$ if $n \geq 2$ is even. $\chi^{\prime}\left(W_n\right)=n-1$ for all $n \geq 4$. $\chi^{\prime}\left(K_{m,n}\right)=\max \{m, n\}$.
voloshin-hypergraph_F02462	289	■ 0.973	$\bar{\chi}^{\prime}(\text{prime})(G)=8$	$\bar{\chi}^{\prime}(G)=8$	2. $\bar{\chi}^{\prime}(G)=8$. 5. $\bar{\chi}^{\prime}\left(C_n\right)=1$, $\bar{\chi}^{\prime}\left(W_n\right)=n-1$. 6. For Petersen graph $\bar{\chi}^{\prime}=7$, for cube $\bar{\chi}^{\prime}=6$,
voloshin-hypergraph_F02463	289	■ 0.973	$\bar{\chi}^{\prime}(\text{prime})\left(C_n\right)=1, \bar{\chi}^{\prime}(\text{prime})\left(W_n\right)=n-1$	$\bar{\chi}^{\prime}\left(C_n\right)=1, \bar{\chi}^{\prime}\left(W_n\right)=n-1$	2. $\bar{\chi}^{\prime}(G)=8$. 5. $\bar{\chi}^{\prime}\left(C_n\right)=1$, $\bar{\chi}^{\prime}\left(W_n\right)=n-1$. 6. For Petersen graph $\bar{\chi}^{\prime}=7$, for cube $\bar{\chi}^{\prime}=6$,
voloshin-hypergraph_F02464	289	■ 0.973	$\bar{\chi}^{\prime}(\text{prime})=7$	$\bar{\chi}^{\prime}=7$	2. $\bar{\chi}^{\prime}(G)=8$. 5. $\bar{\chi}^{\prime}\left(C_n\right)=1$, $\bar{\chi}^{\prime}\left(W_n\right)=n-1$. 6. For Petersen graph $\bar{\chi}^{\prime}=7$, for cube $\bar{\chi}^{\prime}=6$,
voloshin-hypergraph_F02465	289	■ 0.973	$\bar{\chi}^{\prime}(\text{prime})=6$	$\bar{\chi}^{\prime}=6$	2. $\bar{\chi}^{\prime}(G)=8$. 5. $\bar{\chi}^{\prime}\left(C_n\right)=1$, $\bar{\chi}^{\prime}\left(W_n\right)=n-1$. 6. For Petersen graph $\bar{\chi}^{\prime}=7$, for cube $\bar{\chi}^{\prime}=6$,
voloshin-hypergraph_F02466	289	■ 1.000	$\bar{\chi}^{\prime}(\text{prime})=5$	$\bar{\chi}^{\prime}=5$	and for prism $\bar{\chi}^{\prime}=5$.
voloshin-hypergraph_F02467	289	■ 1.000	$K_n: n \geq 3 ; K_{\{m, n\}}: m=n ; W_n: n \geq 4$	$K_n: n \geq 3 ; K_{m,n}: m=n ; W_n: n \geq 4$	1. $K_n: n \geq 3 ; K_{m,n}: m=n ; W_n: n \geq 4$. 2. Prism and cube - yes; Petersen graph - not. 3.
voloshin-hypergraph_F02468	289	■ 0.998	$n\left(\mathcal{H}_1\right)=6, m\left(\mathcal{H}_1\right)=5 ; n\left(\mathcal{H}_2\right)=6, m\left(\mathcal{H}_2\right)=11$	$n\left(\mathcal{H}_1\right)=6, m\left(\mathcal{H}_1\right)=5 ; n\left(\mathcal{H}_2\right)=6, m\left(\mathcal{H}_2\right)=11$	1. $n\left(\mathcal{H}_1\right)=6, m\left(\mathcal{H}_1\right)=5 ; n\left(\mathcal{H}_2\right)=6, m\left(\mathcal{H}_2\right)=11$. 2. Two in $\mathcal{H}_1$, seven in $\mathcal{H}_2$. 3. None. 4.
voloshin-hypergraph_F02469	289	■ 1.000	$\Delta\left(\mathcal{H}_1\right)=3, \Delta\left(\mathcal{H}_2\right)=5$	$\Delta\left(\mathcal{H}_1\right)=3, \Delta\left(\mathcal{H}_2\right)=5$	None. 9. $\Delta\left(\mathcal{H}_1\right)=3, \Delta\left(\mathcal{H}_2\right)=5$. 10. None. 11. None. 12. $r\left(\mathcal{H}_1\right)=3 ; r\left(\mathcal{H}_2\right)=4$. 13. $\mathcal{H}_1$:
voloshin-hypergraph_F02470	289	■ 1.000	$r\left(\mathcal{H}_1\right)=3 ; r\left(\mathcal{H}_2\right)=4$	$r\left(\mathcal{H}_1\right)=3 ; r\left(\mathcal{H}_2\right)=4$	None. 9. $\Delta\left(\mathcal{H}_1\right)=3, \Delta\left(\mathcal{H}_2\right)=5$. 10. None. 11. None. 12. $r\left(\mathcal{H}_1\right)=3 ; r\left(\mathcal{H}_2\right)=4$. 13. $\mathcal{H}_1$:
voloshin-hypergraph_F02471	289	■ 1.000	$\mathcal{H}: N(1)=\{2,4\}, N(2)=\{1,3,4,5\}, N(3)=\{2,4,5\}, N(4)=\{1,2,3,5\}$	$\mathcal{H}: N(1)=\{2,4\}, N(2)=\{1,3,4,5\}, N(3)=\{2,4,5\}, N(4)=\{1,2,3,5\}$	1. In $\mathcal{H}$: $N(1)=\{2,4\}, N(2)=\{1,3,4,5\}, N(3)=\{2,4,5\}, N(4)=\{1,2,3,5\}$,
voloshin-hypergraph_F02472	289	■ 0.773	$N(5)=\{2,3,4\}, N(6)=\emptyset$	$N(5)=\{2,3,4\}, N(6)=\emptyset$	$N(5)=\{2,3,4\}, N(6)=\emptyset$; in $\mathcal{H}^*$: $N\left(d_1\right)=\left\{d_2, d_3\right\}, N\left(d_2\right)=\left\{d_1, d_3, d_4\right\}, N\left(d_3\right)=$
voloshin-hypergraph_F02473	289	■ 0.773	$\mathcal{H}^*: N\left(d_1\right)=\left\{d_2, d_3\right\}, N\left(d_2\right)=\left\{d_1, d_3, d_4\right\}, N\left(d_3\right)=$	$\mathcal{H}^*: N\left(d_1\right)=\left\{d_2, d_3\right\}, N\left(d_2\right)=\left\{d_1, d_3, d_4\right\}, N\left(d_3\right)=$	$N(5)=\{2,3,4\}, N(6)=\emptyset$; in $\mathcal{H}^*$: $N\left(d_1\right)=\left\{d_2, d_3\right\}, N\left(d_2\right)=\left\{d_1, d_3, d_4\right\}, N\left(d_3\right)=$
voloshin-hypergraph_F02474	289	■ 0.493	$\left\{d_1, d_2, d_4, d_5\right\}, N\left(d_4\right)=\left\{d_2, d_3, d_5\right\}, N\left(d_5\right)=\left\{d_3, d_4\right\}$	$\left\{d_1, d_2, d_4, d_5\right\}, N\left(d_4\right)=\left\{d_2, d_3, d_5\right\}, N\left(d_5\right)=\left\{d_3, d_4\right\}$	$\left\{d_1, d_2, d_4, d_5\right\}, N\left(d_4\right)=\left\{d_2, d_3, d_5\right\}, N\left(d_5\right)=\left\{d_3, d_4\right\}$. 3. $r(\mathcal{H})=r\left(\mathcal{H}^*\right)=3$. 7. $L(G)=$
voloshin-hypergraph_F02475	289	■ 0.493	$r(\mathcal{H})=r\left(\mathcal{H}^*\right)=3.7 . L(G)=$	$r(\mathcal{H})=r\left(\mathcal{H}^*\right)=3.7 . L(G)=$	$\left\{d_1, d_2, d_4, d_5\right\}, N\left(d_4\right)=\left\{d_2, d_3, d_5\right\}, N\left(d_5\right)=\left\{d_3, d_4\right\}$. 3. $r(\mathcal{H})=r\left(\mathcal{H}^*\right)=3$. 7. $L(G)=$
voloshin-hypergraph_F02476	289	■ 0.398	$\left\{\{1,5\},\{1,2\},\{2,3\},\{2,4\},\{4,5\}\right\} ; L\left(G^*\right)=\left\{\{a, b\},\{b, c, d\},\{c\},\{d, e\},\{a, e\}\right\} . 9 . E_n^*$	$\left\{\{1,5\},\{1,2\},\{2,3\},\{2,4\},\{4,5\}\right\} ; L\left(G^*\right)=\left\{\{a, b\},\{b, c, d\},\{c\},\{d, e\},\{a, e\}\right\} . 9 . E_n^*$	$\left\{\{1,5\},\{1,2\},\{2,3\},\{2,4\},\{4,5\}\right\} ; L\left(G^*\right)=\left\{\{a, b\},\{b, c, d\},\{c\},\{d, e\},\{a, e\}\right\} . 9 . E_n^*$
voloshin-hypergraph_F02477	290	■ 1.000	$L\left(K_4^3\right)=\left\{\{1,2,3\},\{1,3,4\},\{2,3,4\},\{1,2,4\}\right\}$	$L\left(K_4^3\right)=\left\{\{1,2,3\},\{1,3,4\},\{2,3,4\},\{1,2,4\}\right\}$	4. $L\left(K_4^3\right)=\left\{\{1,2,3\},\{1,3,4\},\{2,3,4\},\{1,2,4\}\right\}$. 6. When order of the vertices is the
voloshin-hypergraph_F02478	290	■ 1.000	$1 \rightarrow 7 \rightarrow 11 \rightarrow 6$	$1 \rightarrow 7 \rightarrow 11 \rightarrow 6$	2. $1 \rightarrow 7 \rightarrow 11 \rightarrow 6$. 10. $\mathcal{H}^*$ is the Petersen graph.
voloshin-hypergraph_F02479	290	■ 0.990	$S=\{3,4,5,6,7,8,10,11,13,14\}$	$S=\{3,4,5,6,7,8,10,11,13,14\}$	4. For example, $S=\{3,4,5,6,7,8,10,11,13,14\}$. 5. $\alpha(\mathcal{H})=10$. 7. For ex-
voloshin-hypergraph_F02480	290	■ 0.990	$\alpha(\mathcal{H})=10$	$\alpha(\mathcal{H})=10$	4. For example, $S=\{3,4,5,6,7,8,10,11,13,14\}$. 5. $\alpha(\mathcal{H})=10$. 7. For ex-
voloshin-hypergraph_F02481	290	■ 0.927	$T=\{1,2,9,12,15\}$	$T=\{1,2,9,12,15\}$	ample, $T=\{1,2,9,12,15\}$. 8. $\tau(\mathcal{H})=5$. 9. For example, $\mathcal{D}'=\left\{\{6,11,15\},\right.$

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voloshin-hypergraph_F02482	290	■ 0.927	$\tau(\mathcal{H})=5$	$\tau(\mathcal{H}) = 5$	ample, $T = \{1, 2, 9, 12, 15\}$. 8. $\tau(\mathcal{H}) = 5$. 9. For example, $\mathcal{D}' = \{\{6, 11, 15\},$
voloshin-hypergraph_F02483	290	■ 0.927	$\mathcal{D}^{\prime}=\{\{6, 11, 15\}$	$\mathcal{D}' = \{\{6, 11, 15\}$	ample, $T = \{1, 2, 9, 12, 15\}$. 8. $\tau(\mathcal{H}) = 5$. 9. For example, $\mathcal{D}' = \{\{6, 11, 15\},$
voloshin-hypergraph_F02484	290	■ 0.279	$\{9, 13, 14\}, \{1, 3, 7\}, \{2, 8, 4\}\}$. $\mathbf{10.} \nu(\mathcal{H}) = 4$	$\{9, 13, 14\}, \{1, 3, 7\}, \{2, 8, 4\}\}$. 10. $\nu(\mathcal{H}) = 4$	$\{9, 13, 14\}, \{1, 3, 7\}, \{2, 8, 4\}\}$. 10. $\nu(\mathcal{H}) = 4$. 11. $\rho(\mathcal{H}) = 6$.
voloshin-hypergraph_F02485	290	■ 0.279	$\rho(\mathcal{H})=6$	$\rho(\mathcal{H}) = 6$	$\{9, 13, 14\}, \{1, 3, 7\}, \{2, 8, 4\}\}$. 10. $\nu(\mathcal{H}) = 4$. 11. $\rho(\mathcal{H}) = 6$.
voloshin-hypergraph_F02486	290	■ 0.971	$\tau\left(\mathcal{H}_1\right)=2, \alpha\left(\mathcal{H}_1\right)=4, \nu\left(\mathcal{H}_1\right)=2 ; \tau\left(\mathcal{H}_2\right)=3, \alpha\left(\mathcal{H}_2\right)=5, \nu\left(\mathcal{H}_2\right)=3$	$\tau\left(\mathcal{H}_1\right)=2, \alpha\left(\mathcal{H}_1\right)=4, \nu\left(\mathcal{H}_1\right)=2 ; \tau\left(\mathcal{H}_2\right)=3, \alpha\left(\mathcal{H}_2\right)=5, \nu\left(\mathcal{H}_2\right)=3$	3. $\tau\left(\mathcal{H}_1\right)=2, \alpha\left(\mathcal{H}_1\right)=4, \nu\left(\mathcal{H}_1\right)=2 ; \tau\left(\mathcal{H}_2\right)=3, \alpha\left(\mathcal{H}_2\right)=5, \nu\left(\mathcal{H}_2\right)=3$.
voloshin-hypergraph_F02487	290	■ 1.000	$\wedge\left(\mathcal{H}_1\right)=1, \wedge\left(\mathcal{H}_2\right)=0$	$\wedge\left(\mathcal{H}_1\right)=1, \wedge\left(\mathcal{H}_2\right)=0$	5. $\wedge\left(\mathcal{H}_1\right)=1, \wedge\left(\mathcal{H}_2\right)=0$.
voloshin-hypergraph_F02488	290	■ 1.000	$K_n: n=1, 2 ; K_{m, n}: m, n \geq 1 ; W_n$	$K_n: n=1, 2 ; K_{m, n}: m, n \geq 1 ; W_n$	1. $K_n: n=1, 2 ; K_{m, n}: m, n \geq 1 ; W_n$: none; prism: no; cube: yes; Petersen graph:
voloshin-hypergraph_F02489	290	■ 0.326	$\mathcal{D}=\{\{1, 2, 3\}, \{1, 2, 4\}, \{2, 3, 4\}\}$	$\mathcal{D} = \{\{1, 2, 3\}, \{1, 2, 4\}, \{2, 3, 4\}\}$	$\mathcal{D}=\{\{1, 2, 3\}, \{1, 2, 4\}, \{2, 3, 4\}\}$. 4. $\mathcal{H}=(X, \mathcal{D})$ with $X=\{1, 2, 3\}$ and $\mathcal{D}=\{\{1, 2, 3\},$
voloshin-hypergraph_F02490	290	■ 0.326	$X=\{1, 2, 3\}$	$X = \{1, 2, 3\}$	$\mathcal{D}=\{\{1, 2, 3\}, \{1, 2, 4\}, \{2, 3, 4\}\}$. 4. $\mathcal{H}=(X, \mathcal{D})$ with $X=\{1, 2, 3\}$ and $\mathcal{D}=\{\{1, 2, 3\},$
voloshin-hypergraph_F02491	290	■ 0.326	$\mathcal{D}=\{\{1, 2, 3\}$	$\mathcal{D} = \{\{1, 2, 3\}$	$\mathcal{D}=\{\{1, 2, 3\}, \{1, 2, 4\}, \{2, 3, 4\}\}$. 4. $\mathcal{H}=(X, \mathcal{D})$ with $X=\{1, 2, 3\}$ and $\mathcal{D}=\{\{1, 2, 3\},$
voloshin-hypergraph_F02492	290	■ 1.000	$\{1, 2\}, \{2, 3\}, \{1, 3\}\}$	$\{1, 2\}, \{2, 3\}, \{1, 3\}\}$	$\{1, 2\}, \{2, 3\}, \{1, 3\}\}$.
voloshin-hypergraph_F02493	290	■ 1.000	$P_n, n \geq 1$	$P_n, n \geq 1$	1. Paths $P_n, n \geq 1$. 2. In any linear ordering of the vertices of $C_n, n \geq 3$, there is always
voloshin-hypergraph_F02494	290	■ 1.000	$X=\{1, 2, 3, 4\}, \mathcal{D} =$	$X = \{1, 2, 3, 4\}, \mathcal{D} =$	an edge which is not an interval in the ordering. 3. $\mathcal{H}=(X, \mathcal{D})$ with $X=\{1, 2, 3, 4\}, \mathcal{D} =$
voloshin-hypergraph_F02495	290	■ 0.513	$\{\{1, 4\}, \{2, 4\}, \{3, 4\}\}$	$\{\{1, 4\}, \{2, 4\}, \{3, 4\}\}$	$\{\{1, 4\}, \{2, 4\}, \{3, 4\}\}$. 4. $\mathcal{H}=(X, \mathcal{D})$ with $X=\{1, 2, 3, 4\}, \mathcal{D}=\{\{1, 4\}, \{2, 4\}, \{3, 4\}\}$.
voloshin-hypergraph_F02496	290	■ 0.513	$X=\{1, 2, 3, 4\}, \mathcal{D}=\{\{1, 4\}, \{2, 4\}, \{3, 4\}\}$	$X = \{1, 2, 3, 4\}, \mathcal{D} = \{\{1, 4\}, \{2, 4\}, \{3, 4\}\}$	$\{\{1, 4\}, \{2, 4\}, \{3, 4\}\}$. 4. $\mathcal{H}=(X, \mathcal{D})$ with $X=\{1, 2, 3, 4\}, \mathcal{D}=\{\{1, 4\}, \{2, 4\}, \{3, 4\}\}$.
voloshin-hypergraph_F02497	290	■ 0.644	$L\left(\mathcal{H}_2\right)=C_5$	$L\left(\mathcal{H}_2\right)=C_5$	2. $\mathcal{H}_1$: not because it is not Helly; $\mathcal{H}_2$: not because $L\left(\mathcal{H}_2\right)=C_5$ which is not perfect. 3.
voloshin-hypergraph_F02498	290	■ 1.000	$\chi^{\prime}\left(\mathcal{H}_1\right)=3 ; \chi^{\prime}\left(\mathcal{H}_2\right)=3$	$\chi^{\prime}\left(\mathcal{H}_1\right)=3 ; \chi^{\prime}\left(\mathcal{H}_2\right)=3$	None. 4. $\chi^{\prime}\left(\mathcal{H}_1\right)=3 ; \chi^{\prime}\left(\mathcal{H}_2\right)=3$. 5. None.
voloshin-hypergraph_F02499	291	■ 0.259	$\alpha(\mathcal{H})=6 ; \tau(\mathcal{H})=2$	$\alpha(\mathcal{H}) = 6 ; \tau(\mathcal{H}) = 2$	1. $\chi(\mathcal{H}) = 2$. 2. $\alpha(\mathcal{H}) = 6 ; \tau(\mathcal{H}) = 2$. 3. $\gamma(\mathcal{H}) = 4$. 5. $\Lambda(\mathcal{H}) = 0$. 6. No.
voloshin-hypergraph_F02500	291	■ 0.259	$\gamma(\mathcal{H})=4.5$	$\gamma(\mathcal{H}) = 4.5$	1. $\chi(\mathcal{H}) = 2$. 2. $\alpha(\mathcal{H}) = 6 ; \tau(\mathcal{H}) = 2$. 3. $\gamma(\mathcal{H}) = 4$. 5. $\Lambda(\mathcal{H}) = 0$. 6. No.
voloshin-hypergraph_F02501	291	■ 0.817	$M(\mathcal{H})=2, \chi(\mathcal{H}) \geq 3$	$M(\mathcal{H}) = 2, \chi(\mathcal{H}) \geq 3$	2. $M(\mathcal{H}) = 2, \chi(\mathcal{H}) \geq 3$. 2. $\chi(\mathcal{H}) = 3$.
voloshin-hypergraph_F02502	291	■ 0.713	$\chi(\mathcal{H})=2, \bar{\chi}(\mathcal{H})=3$	$\chi(\mathcal{H}) = 2, \bar{\chi}(\mathcal{H}) = 3$	2. $\chi(\mathcal{H}) = 2, \bar{\chi}(\mathcal{H}) = 3$. 3. 3. 4. $\chi\left(\mathcal{H}_C\right)=1, \bar{\chi}\left(\mathcal{H}_C\right)=3 ; \chi\left(\mathcal{H}_D\right)=2, \bar{\chi}\left(\mathcal{H}_D\right)=5$.
voloshin-hypergraph_F02503	291	■ 0.713	$\chi\left(\mathcal{H}_C\right)=1, \bar{\chi}\left(\mathcal{H}_C\right)=3 ; \chi\left(\mathcal{H}_D\right)=2, \bar{\chi}\left(\mathcal{H}_D\right)=5$	$\chi\left(\mathcal{H}_C\right)=1, \bar{\chi}\left(\mathcal{H}_C\right)=3 ; \chi\left(\mathcal{H}_D\right)=2, \bar{\chi}\left(\mathcal{H}_D\right)=5$	2. $\chi(\mathcal{H}) = 2, \bar{\chi}(\mathcal{H}) = 3$. 3. 3. 4. $\chi\left(\mathcal{H}_C\right)=1, \bar{\chi}\left(\mathcal{H}_C\right)=3 ; \chi\left(\mathcal{H}_D\right)=2, \bar{\chi}\left(\mathcal{H}_D\right)=5$.
voloshin-hypergraph_F02504	291	■ 0.693	$\bar{\chi}(\mathcal{H}) \geq 3$	$\bar{\chi}(\mathcal{H}) \geq 3$	2. 0. 3. $\bar{\chi}(\mathcal{H}) \geq 3$. 4. $\bar{\chi}(\mathcal{H}) = 3$. 5. Because the resistance $O(\mathcal{H}) = 0$.

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voloshin-hypergraph_F02505	291	■ 1.000	$P\left(\mathcal{H}_{\{1\}}, \lambda\right)=3 \lambda^{\{3\}}+6 \lambda^{\{2\}}, R\left(\mathcal{H}_{\{1\}}\right)=(0,6,3,0) ; P\left(\mathcal{H}_{\{2\}}, \lambda\right)=\lambda^{\{2\}}, R\left(\mathcal{H}_{\{2\}}\right)=(0,1,0,0)$	$P\left(\mathcal{H}_1, \lambda\right)=3 \lambda^{\{3\}}+6 \lambda^{\{2\}}, R\left(\mathcal{H}_1\right)=(0,6,3,0) ; P\left(\mathcal{H}_2, \lambda\right)=\lambda^{\{2\}}, R\left(\mathcal{H}_2\right)=(0,1,0,0)$	1. $P\left(\mathcal{H}_1, \lambda\right)=3 \lambda^{\{3\}}+6 \lambda^{\{2\}}, R\left(\mathcal{H}_1\right)=(0,6,3,0) ; P\left(\mathcal{H}_2, \lambda\right)=\lambda^{\{2\}}, R\left(\mathcal{H}_2\right)=(0,1,0,0)$.
voloshin-hypergraph_F02506	291	■ 0.577	$\chi\left(\mathcal{H}_{\mathcal{D}}\right)=3, \bar{\chi}\left(\mathcal{H}_{\mathcal{C}}\right)=3$	$\chi\left(\mathcal{H}_{\mathcal{D}}\right)=3, \bar{\chi}\left(\mathcal{H}_{\mathcal{C}}\right)=3$	1. $\chi\left(\mathcal{H}_{\mathcal{D}}\right)=3, \bar{\chi}\left(\mathcal{H}_{\mathcal{C}}\right)=3$.
voloshin-hypergraph_F02507	291	■ 0.278	$\alpha_{\mathcal{C}}\left(\mathcal{H}_1\right)=3, \bar{\alpha}_{\mathcal{C}}\left(\mathcal{H}_1\right)=3 ; \alpha_{\mathcal{C}}\left(\mathcal{H}_2\right)=3, \bar{\alpha}_{\mathcal{C}}\left(\mathcal{H}_2\right)=3$	$\alpha_{\mathcal{C}}\left(\mathcal{H}_1\right)=3, \bar{\alpha}_{\mathcal{C}}\left(\mathcal{H}_1\right)=3 ; \alpha_{\mathcal{C}}\left(\mathcal{H}_2\right)=3, \bar{\alpha}_{\mathcal{C}}\left(\mathcal{H}_2\right)=3$	1. $\alpha_{\mathcal{C}}\left(\mathcal{H}_1\right)=3, \bar{\alpha}_{\mathcal{C}}\left(\mathcal{H}_1\right)=3 ; \alpha_{\mathcal{C}}\left(\mathcal{H}_2\right)=3, \bar{\alpha}_{\mathcal{C}}\left(\mathcal{H}_2\right)=3$. 2. $\mathcal{H}_1$ is perfect, $\mathcal{H}_2$ is not. 3.
voloshin-hypergraph_F02508	291	■ 1.000	$\tau_2\left(\mathcal{H}_1\right)=2, \tau_2\left(\mathcal{H}_2\right)=3$	$\tau_2\left(\mathcal{H}_1\right)=2, \tau_2\left(\mathcal{H}_2\right)=3$	$\tau_2\left(\mathcal{H}_1\right)=2, \tau_2\left(\mathcal{H}_2\right)=3$.
voloshin-hypergraph_F02509	291	■ 0.962	$S\left(\mathcal{H}_{\mathcal{C}}\right)=\{1,2,3,4,5,6\}, S\left(\mathcal{H}_{\mathcal{D}}\right)=\{2,3,4,5,6,7,8\}$	$S\left(\mathcal{H}_{\mathcal{C}}\right)=\{1,2,3,4,5,6\}, S\left(\mathcal{H}_{\mathcal{D}}\right)=\{2,3,4,5,6,7,8\}$	1. $S\left(\mathcal{H}_{\mathcal{C}}\right)=\{1,2,3,4,5,6\}, S\left(\mathcal{H}_{\mathcal{D}}\right)=\{2,3,4,5,6,7,8\}$. 3. $S\left(\mathcal{H}\right)=\{2,3,4,5\}$.
voloshin-hypergraph_F02510	291	■ 0.962	$S\left(\mathcal{H}\right)=\{2,3,4,5\}$	$S\left(\mathcal{H}\right)=\{2,3,4,5\}$	1. $S\left(\mathcal{H}_{\mathcal{C}}\right)=\{1,2,3,4,5,6\}, S\left(\mathcal{H}_{\mathcal{D}}\right)=\{2,3,4,5,6,7,8\}$. 3. $S\left(\mathcal{H}\right)=\{2,3,4,5\}$.
voloshin-hypergraph_F02511	291	■ 1.000	$n \rightarrow \infty$	$n \rightarrow \infty$	• Almost always true: a property which has asymptotic (as $n \rightarrow \infty$) probability 1
voloshin-hypergraph_F02512	292	■ 1.000	$\overline{C_k}$	$\overline{C_k}$	• Antihole: a subgraph induced by $\overline{C_k}$
voloshin-hypergraph_F02513	292	■ 0.966	(n, k, λ)	(n, k, λ)	• (n, k, λ) -configuration: a k -uniform hypergraph of order n such that every pair of
voloshin-hypergraph_F02514	292	■ 1.000	$m(G) / n(G)$	$m(G) / n(G)$	• Density: ratio $m(G) / n(G)$
voloshin-hypergraph_F02515	292	■ 1.000	$ X = \mathcal{D} =r^2-r+$	$ X = \mathcal{D} =r^2-r+$	• Finite projective plane of rank r : a hypergraph $\mathcal{H}=(X, \mathcal{D})$ with $ X = \mathcal{D} =r^2-r+$
voloshin-hypergraph_F02516	293	■ 1.000	$f: V\left(G_1\right) \rightarrow V\left(G_2\right)$	$f: V\left(G_1\right) \rightarrow V\left(G_2\right)$	• Homomorphism: a map $f: V\left(G_1\right) \rightarrow V\left(G_2\right)$ that keeps adjacency
voloshin-hypergraph_F02517	293	■ 1.000	$n > m$	$n > m$	• Pigeonhole principle: if n items are put into m pigeonholes and $n > m$, then at least
voloshin-hypergraph_F02518	293	■ 1.000	$R(p, q)$	$R(p, q)$	• Ramsey number $R(p, q)$: the minimum n such that any 2-coloring of the edges of K_n
voloshin-hypergraph_F02519	293	■ 1.000	K_p	K_p	produces either a monochromatic copy of K_p or a monochromatic copy of K_q ; for
voloshin-hypergraph_F02520	293	■ 1.000	K_q	K_q	produces either a monochromatic copy of K_p or a monochromatic copy of K_q ; for
voloshin-hypergraph_F02521	293	■ 1.000	$R(3,3)=6$	$R(3,3)=6$	example, $R(3,3)=6$
voloshin-hypergraph_F02522	294	■ 0.996	$SQS(v)$	$SQS(v)$	• Steiner Quadruple System $SQS(v)$: Steiner System $S(3,4, v)$
voloshin-hypergraph_F02523	294	■ 0.996	$S(3,4, v)$	$S(3,4, v)$	• Steiner Quadruple System $SQS(v)$: Steiner System $S(3,4, v)$
voloshin-hypergraph_F02524	294	■ 1.000	$S(t, k, v)$	$S(t, k, v)$	• Steiner System $S(t, k, v)$: a k -uniform hypergraph $\mathcal{H}=(X, \mathcal{B})$ with $ X =v$, hyper-
voloshin-hypergraph_F02525	294	■ 1.000	$\mathcal{H}=(X, \mathcal{B})$	$\mathcal{H}=(X, \mathcal{B})$	• Steiner System $S(t, k, v)$: a k -uniform hypergraph $\mathcal{H}=(X, \mathcal{B})$ with $ X =v$, hyper-
voloshin-hypergraph_F02526	294	■ 1.000	$ X =v$	$ X =v$	• Steiner System $S(t, k, v)$: a k -uniform hypergraph $\mathcal{H}=(X, \mathcal{B})$ with $ X =v$, hyper-
voloshin-hypergraph_F02527	294	■ 0.959	$\operatorname{Tr} \mathcal{H}$	$\operatorname{Tr} \mathcal{H}$	• Transversal hypergraph $Tr\mathcal{H}$: for a given hypergraph $\mathcal{H}=(X, \mathcal{E})$, $Tr\mathcal{H}=(X, \mathcal{D})$
voloshin-hypergraph_F02528	294	■ 0.959	$\mathcal{H}=(X, \mathcal{E}), \operatorname{Tr} \mathcal{H}=(X, \mathcal{D})$	$\mathcal{H}=(X, \mathcal{E}), \operatorname{Tr} \mathcal{H}=(X, \mathcal{D})$	• Transversal hypergraph $Tr\mathcal{H}$: for a given hypergraph $\mathcal{H}=(X, \mathcal{E})$, $Tr\mathcal{H}=(X, \mathcal{D})$

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voloshin-hypergraph_F02529	294	■ 1.000	$T(n, p, r)$	$T(n, p, r)$	• Turán number $T(n, p, r)$: the smallest number of edges in an r -uniform hypergraph
voloshin-hypergraph_F02530	297	■ 0.998	$x, 7$	$x, 7$	$E(x)$, set of edges containing vertex $x, 7$
voloshin-hypergraph_F02531	297	■ 0.998	$\Lambda(G)$	$\Lambda(G)$	$\Lambda(G)$, cyclomatic number, 40
voloshin-hypergraph_F02532	297	■ 0.333	$\boldsymbol{\alpha}_c(\mathcal{H}), \mathcal{C}$	$\alpha_c(\mathcal{H}), \mathcal{C}$	$\alpha_c(\mathcal{H})$, $\mathcal{C}$ -stability number, 236
voloshin-hypergraph_F02533	297	■ 0.986	$\kappa(G)$	$\kappa(G)$	$\kappa(G)$, connectivity, 34
voloshin-hypergraph_F02534	297	■ 0.933	$\tau_2\left(\mathcal{H}_{\mathcal{C}}\right)$	$\tau_2(\mathcal{H}_{\mathcal{C}})$	$\tau_2(\mathcal{H}_{\mathcal{C}})$, bitransversal number, 238
voloshin-hypergraph_F02535	297	■ 0.831	$T, 174$	$T, 174$	est $T, 174$
voloshin-hypergraph_F02536	297	■ 1.000	$m=m(\mathcal{H})$	$m = m(\mathcal{H})$	$m = m(\mathcal{H})$, the number of edges, 136
voloshin-hypergraph_F02537	297	■ 1.000	$r_i(\mathcal{H})$	$r_i(\mathcal{H})$	$r_i(\mathcal{H})$, number of feasible partitions, 203
voloshin-hypergraph_F02538	297	■ 0.587	$\alpha_c(\mathcal{H}), 236$	$\alpha_c(\mathcal{H}), 236$	$\mathcal{C}$ -stability number, $\alpha_c(\mathcal{H})$, 236
voloshin-hypergraph_F02539	298	■ 0.994	$(\mathcal{H})_2, 154$	$(\mathcal{H})_2, 154$	2-section, $(\mathcal{H})_2, 154$
voloshin-hypergraph_F02540	298	■ 0.388	$\mathbf{\tau}_2\left(\mathcal{H}_{\mathcal{C}}\right), 238$	$\tau_2(\mathcal{H}_{\mathcal{C}}), 238$	bitransversal number, $\tau_2(\mathcal{H}_{\mathcal{C}})$, 238
voloshin-hypergraph_F02541	298	■ 0.793	$\chi^{\prime}(\mathcal{H}), 185$	$\chi'(\mathcal{H}), 185$	chromatic index, $\chi'(\mathcal{H})$, 185
voloshin-hypergraph_F02542	298	■ 0.813	$\chi(G), 80$	$\chi(G), 80$	chromatic number, $\chi(G)$, 80
voloshin-hypergraph_F02543	298	■ 0.896	$\chi(\mathcal{H}), 193$	$\chi(\mathcal{H}), 193$	chromatic number, $\chi(\mathcal{H})$, 193
voloshin-hypergraph_F02544	298	■ 0.999	$P(\mathcal{H}, \lambda), 204$	$P(\mathcal{H}, \lambda), 204$	$P(\mathcal{H}, \lambda), 204$
voloshin-hypergraph_F02545	298	■ 0.658	$P(G, \lambda), 80$	$P(G, \lambda), 80$	chromatic polynomial, $P(G, \lambda)$, 80
voloshin-hypergraph_F02546	299	■ 0.803	$\mathrm{K}(G)$	$K(G)$	connectivity, $\kappa(G)$, 34
voloshin-hypergraph_F02547	299	■ 0.923	$\Lambda(\mathcal{H}), 179$	$\Lambda(\mathcal{H}), 179$	$\Lambda(\mathcal{H})$, 179
voloshin-hypergraph_F02548	300	■ 0.848	$[\mathcal{H}]_2, 174$	$[\mathcal{H}]_2, 174$	generalized 2-section, $[\mathcal{H}]_2, 174$
voloshin-hypergraph_F02549	300	■ 0.772	$\Lambda(\mathcal{H}, T), 174$	$\Lambda(\mathcal{H}, T), 174$	$\Lambda(\mathcal{H}, T), 174$
voloshin-hypergraph_F02550	301	■ 0.973	$\Delta, 6$	$\Delta, 6$	maximum degree, $\Delta, 6$
voloshin-hypergraph_F02551	302	■ 0.715	$r(\mathcal{H})$	$r(\mathcal{H})$	rank of hypergraph, $r(\mathcal{H})$, 136
voloshin-hypergraph_F02552	303	■ 0.516	$\tau(G), 31$	$\tau(G), 31$	transversal number, $\tau(G)$, 31
voloshin-hypergraph_F02553	303	■ 0.798	$\bar{\chi}^{\prime}(G), 116$	$\bar{\chi}'(G), 116$	upper chromatic index, $\bar{\chi}'(G)$, 116
voloshin-hypergraph_F02554	303	■ 0.988	$w(T), 43$	$w(T), 43$	weight of a tree, $w(T)$, 43